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Glossary

MS Memory Safety. 1, 12

sCCT Strict Cryptographic Constant Time. 1, 13, 14, 22

SMS Spatial Memory Safety. 1, 10–12, 22

SS Speculation Safety. 1, 15

TMS Temporal Memory Safety. 1, 8, 9, 12, 26, 30, 32, 33, 37–43, 51, 101

1 Preliminaries

In comparison to the paper, some notation is different. This document should be considered work-in-progress, we want to make sure both the report and the paper use similar notation. A non-exhaustive description of differences is: compilers are not γ, but $[\![\bullet]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$. Trace relations are written $\approx^*_{\delta;\mathsf{X}}$ or $\theta\,(\bullet)$.

Definition 1 (Events)**.** *$E \cup \{\lightning\}$ is the set of atomic propositions hereby called events or actions. The internal action is ε. The action $\lightning$ is program termination.*

We assume that any programming language can be enriched with a self-composition operator in the style of [**?**]. Furthermore, we also assume the existence of a low-equivalence relation that distinguishes program states only by their public memory. Two traces $\overline{a}_1$,$\overline{a}_2$ are low-equivalent $\overline{a}_1 =_L \overline{a}_2$ iff all their public events coincide.

Definition 2 (Programming Languages)**.** *A programming language is a tuple $\left(\mathcal{P}, \vdash_w, \overset{e}{\hookrightarrow}, \bowtie\right)$ s.t.:*

- $\mathcal{P}$ *: Set - is a set of admissible, partial programs.*
- $\vdash_w$ *: $\mathcal{P}$ - a judgement that holds iff a program is not partial.*
- $\overset{e}{\hookrightarrow}$ *: $\mathcal{W} \to E \cup \{\lightning\} \to \mathcal{W}$ - a step relation, where $\mathcal{W} = \{w \in \mathcal{P} \mid \vdash_w w\}$. For $e \in E \cup \{\lightning\}$ and $p, p' \in \mathcal{W}$ we say for $p \overset{e}{\hookrightarrow} p'$ that program p performs a step with action e to program p'. If $e = \varepsilon$, we write $p \hookrightarrow p'$. In case $e = \lightning$, we write $p \Downarrow$.*
- $\bowtie$ *: $\mathcal{P} \to \mathcal{P} \to \mathcal{P}$ - links two partial programs together in some way, resulting in a new partial program.*

Let S, I, and $\mathbf{T}$ be any programming language.

Definition 3 (Notation for Sequences)**.** *For any sequence of events, let $\cdot$ denote the empty sequence and $e :: \bar{t}$ the sequence that starts with e and continues with $\bar{t}$. Hereby, it does not matter whether $\bar{t}$ is finite or infinite, it's merely syntactic sugar to work on sequences of events.*

Definition 4 (Traces)**.** *A trace $\overline{a}$ is an infinite sequence of events that results from the relation $\overset{e}{\hookrightarrow}$. That is, we obtain the trace $\overline{a} = e_0 :: e_1 :: \ldots$ for the execution sequence $p \overset{e_0}{\hookrightarrow} p' \overset{e_1}{\hookrightarrow} \ldots$ and write $p \rightsquigarrow \overline{a}$. The set of all traces is Traces.*

We assume $\lightning$ to occur in traces representing terminating programs such that it occurs infinitely often in a one-by-one sequence.

Definition 5 (Once)**.** *Given an event a and a trace $\overline{a}$, we say a appears exactly once in the trace, written $a \in_! \overline{a}$, iff*

If $n \in \mathbb{N}$ such that $\overline{a}[n] = a$, then for any $m \in \mathbb{N}$ such that $\overline{a}[m] = a$ we have $n = m$.

Definition 6 (Before)**.** *Given two events a_0, a_1 and a trace $\overline{a}$, event a_0 occurs before a_1 in $\overline{a}$, written $a_0 \leq_{\overline{a}} a_1$, iff*

If $n \in \mathbb{N}$ such that $\overline{a}[n] = a_0$, then $\exists m \in \mathbb{N}, \overline{a}[m] = a_1 \wedge n < m$

Definition 7 (Finite Trace Prefixes)**.** *A finite sequence of events m is a finite trace prefix of $\overline{a}$ iff it satisfies the following judgement.*

$$\frac{}{\cdot \leq \overline{a}} \qquad \frac{m \leq \overline{a}}{e :: m \leq e :: \overline{a}}$$

Definition 8 (Behavior)**.** *The behavior of a whole program p is a set of all traces it produces, i.e. Behav$(p) = \{\overline{a} \mid p \rightsquigarrow \overline{a}\}$.*

Definition 9 (Observation)**.** *An observation is a finite set of finite trace prefixes. We say that an observation o is the prefix of a behavior b iff*

$$\forall m \in o. \exists \overline{a} \in b. m \leq t$$

.

Definition 10 (Properties)**.** *A property π is a set of admissible traces. For a program p to satisfy π it must not produce a trace that is not part of π. Thus, p satisfies π iff Behav$(p) \subseteq \pi$ and we write $p \vDash \pi$.*

Definition 11 (Hyperproperties)**.** *A hyperproperty H is a set of admissible sets of traces. Thus, if p satisfies H (also written $p \vDash H$), then Behav$(p) \in H$.*

Lemma 1 (Lifting Properties)**.** *Given a property π, there exists a unique hyperproperty $\lceil \pi \rceil$ that satisfies the exact same policy.*

Proof. We want $\forall p \in \mathcal{P}, p \vDash \pi \equiv p \vDash \lceil \pi \rceil$. Henceforth, given a $p \in \mathcal{P}$, we have Behav$(p) \subseteq \pi$ iff Behav$(p) \in \lceil \pi \rceil$. Note that if Behav$(p) \subseteq \pi$, we have Behav$(p) \in \{\Pi \mid \Pi = \text{Behav}(p) \subseteq \pi\}$. Thus, $\lceil \pi \rceil$ is the set of all possible program behaviors that are a subset of π. This is exactly the powerset of π and we conclude $\lceil \pi \rceil = \mathscr{P}(\pi)$. □

The lifting of properties to a singleton set does not suffice, since the empty behavior trivially satisfies any property $\emptyset \subseteq \pi = \{\overline{a}\}$, but if we would define $\lceil \overline{a} \rceil = \{\{\overline{a}\}\}$, then $\emptyset \notin \lceil \overline{a} \rceil$.

Lemma 2 (Property Satisfaction Refinement)**.** *For a property π that refines π', i.e. $\pi \subseteq \pi'$, if any $p \in \mathcal{P}$ satisfies $p \vDash \pi'$, then $p \vDash \pi$.*

Proof. Pick any property π' and $p \in \mathcal{P}$ such that $p \vDash \pi$ and assume $\pi \subseteq \pi'$. Simple unfolding reveals Behav$(p) \subseteq \pi \implies$ Behav$(p) \subseteq \pi'$. □

For lifted properties, this refinement property also holds on the hyperproperty level. However, it does not work for any hyperproperty [**?**].

Definition 12 (Robust Property Satisfaction)**.** *A program p robustly satisfies a property π, written $p \vDash_R \pi$, iff $\forall C \in \mathcal{P}, C \bowtie p \vDash \pi$. The same notation is used for robust hyperproperty satisfaction.*

Lemma 3 (Weakening Robust Satisfaction)**.** *Given classes $\mathbb{C}_1, \mathbb{C}_2$ and any program p such that*

(a) $\mathbb{C}_1 \subseteq \mathbb{C}_2$

(b) $p \vDash_R \mathbb{C}_2$

We show

(i) $p \vDash_R \mathbb{C}_1$

Proof. Unfolding Item (i) (Weakening Robust Satisfaction), let $\Pi \in \mathbb{C}_2$ and p be a program, we want to show that $p \vDash_R \Pi$. By Item (a) (Weakening Robust Satisfaction), we also know that $\Pi \in \mathbb{C}_1$. Thus, we can use Item (b) (Weakening Robust Satisfaction) to conclude. □

Definition 13 (Classes)**.** *A class of hyperproperties* $\mathbb{C}$ *is a set of hyperproperties. Likewise, a class of properties* $\mathbb{C}$ *is a set of hyperproperties, where every property is lifted to the hyperproperty level. From now on, we use* Π *for elements of any class* $\mathbb{C}$ *in case it does not matter whether it is a lifted property or any hyperproperty.*

Definition 14 (Compilers)**.** *A compiler between languages* S *and* $\mathbf{T}$ *is a partial function* $[\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}}$ *from* $\mathcal{P}$ *to* $\mathcal{P}$*.*

2 Compositionality of Secure Compilers

Definition 15 (Universal Image)**.**

$$\sigma_\sim(\pi) = \{\bar{\mathsf{a}} | \forall \bar{\mathbf{a}}, \bar{\mathsf{a}} \sim \bar{\mathbf{a}} \implies \bar{\mathbf{a}} \in \pi\}$$

Definition 16 (Existential Image)**.**

$$\tau_\sim(\pi) = \{\bar{\mathbf{a}} | \exists \bar{\mathsf{a}}, \bar{\mathsf{a}} \sim \bar{\mathbf{a}} \wedge \bar{\mathsf{a}} \in \pi\}$$

Definition 17 (Robust Trace-Hyperproperty Preservation with Universal Image)**.** *For a given class* $\mathbb{C}$*, a compiler from languages* S *to* $\mathbf{T}$ *robustly preserves* $\mathbb{C}$ *iff*

$$\forall \mathbf{\Pi} \in \mathbb{C}, \forall \mathsf{p} \in \mathcal{P}, \mathsf{p} \vDash_R \sigma_\sim(\mathbf{\Pi}) \implies [\![\mathsf{p}]\!]^{\mathsf{S}\to\mathbf{T}} \vDash_R \mathbf{\Pi}$$

We write $\vdash_{\sigma_\sim} [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}$*.*

Definition 18 (Robust Trace-Hyperproperty Preservation with Existential Image)**.** *For a given class* $\mathbb{C}$*, a compiler from languages* S *to* $\mathbf{T}$ *robustly preserves* $\mathbb{C}$ *iff*

$$\forall \Pi \in \mathbb{C}, \forall \mathsf{p} \in \mathcal{P}, \mathsf{p} \vDash_R \Pi \implies [\![\mathsf{p}]\!]^{\mathsf{S}\to\mathbf{T}} \vDash_R \tau_\sim(\Pi)$$

We write $\vdash_{\tau_\sim} [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}$*.*

Definition 19 (Robust Trace-Hyperproperty Preservation)**.** *For a given class* $\mathbb{C}$*, a compiler from languages* S *to* $\mathbf{T}$ *robustly preserves* $\mathbb{C}$ *iff*

$$\forall \Pi \in \mathbb{C}, \forall \mathsf{p} \in \mathcal{P}, \mathsf{p} \vDash_R \Pi \implies [\![\mathsf{p}]\!]^{\mathsf{S}\to\mathbf{T}} \vDash_R \Pi$$

We write $\vdash [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}$*.*

Definition 20 (Sequential Composition of Compilers)**.** *Given two compilers* $[\![\bullet]\!]^{\mathsf{S}\to I}$ *and* $[\![\bullet]\!]^{I\to\mathbf{T}}$*, their sequential composition is* $[\![\bullet]\!]^{\mathsf{S}\to I\to\mathbf{T}} = [\![[\![\bullet]\!]^{\mathsf{S}\to I}]\!]^{I\to\mathbf{T}}$*.*

Lemma 4 (Weakening RTP)**.** *Given classes* $\mathbb{C}_1, \mathbb{C}_2$ *such that*

(a) $\mathbb{C}_1 \subseteq \mathbb{C}_2$

(b) $\vdash [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}_2$

We show

(i) $\vdash [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}_1$

Proof. Using Definition 19 on the goal, let $\Pi \in \mathbb{C}_1$ and $\mathsf{p} \in \mathcal{P}$ such that $\mathsf{p} \vDash_R \Pi$, so what's left to prove is $[\![\mathsf{p}]\!]^{\mathsf{S}\to\mathbf{T}} \vDash_R \Pi$. Since $\mathbb{C}_1 \subseteq \mathbb{C}_2$ and $\Pi \in \mathbb{C}_1$, we know that $\Pi \in \mathbb{C}_2$. Thus, we can apply the assumption $\vdash [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}_2$ to our goal, leaving us with $\mathsf{p} \vDash_R \Pi$ to show, which was an assumption we made. □

Definition 21 (Well-formedness of $\sim$ for a Class $\mathbb{C}$)**.**

$$\vdash_{wf} \sim : \mathbb{C} := \forall \pi \in \mathbb{C}, \tau_{\sim}(\pi) \in \tau_{\sim}(\mathbb{C})$$

Definition 22 (Well-formedness of $\sim$ for a Class $\mathbb{C}$)**.**

$$\vdash_{wf} \sim : \mathbb{C} := \forall \pi \in \mathbb{C}, \sigma_{\sim}(\pi) \in \sigma_{\sim}(\mathbb{C})$$

Definition 23 (Safety Properties)**.** *The class of safety properties contains the lifting of all properties that can be refuted with a finite trace prefix:*

$$\mathit{Safety} = \{\lceil \pi \rceil \mid \forall \overline{a} \in \mathit{Traces}, t \notin \lceil \pi \rceil \ \mathit{iff}\ \exists m \geq \overline{a}, \forall \overline{a}' \in \mathit{Traces}, m \leq \overline{a}' \implies \overline{a}' \notin \lceil \pi \rceil\}$$

Definition 24 (Hypersafety Properties)**.** *The class of hypersafety properties contains all hyperpropert that can be refuted with an observation:*

$$\mathit{HyperSafety} = \left\{\Pi \mid \forall b \in 2^{\mathit{Traces}}, b \notin \Pi \ \mathit{iff}\ \exists o \geq b, \forall b' \in 2^{\mathit{Traces}}, o \leq b' \implies b' \notin \Pi\right\}$$

Definition 25 (Subset Closed Hyperproperties)**.** *The class of hyperproperties that are closed with respect to the subset relation is*

$$\mathit{SSC} = \{H \mid \forall X \in H, \forall Y \subseteq X, Y \in H\}$$

Lemma 5 (Hypersafety is entailed in SSC)**.** *HyperSafety* $\subseteq$ *SSC.*

Proof. [?] □

Definition 26 (K-Hypersafety)**.** *Exactly the same as Definition 24, but the observations o are restricted to cardinality k. 2-Hypersafety is simply $k = 2$. Definition 27 gives an example instance of a classic 2-hypersafe property.*

Definition 27 (Non-Interference (NI))**.** *We define the class containing the non-interference hyperproperty as:*

$$\mathit{NI} = \{H | \forall \overline{a}_1, \overline{a}_2 \in H. \overline{a}_1 =_L \overline{a}_2 \implies \overline{a}_1 = \overline{a}_2\}$$

Note that $=$ may not be strict equality, but some suitable trace equivalence that checks both public and private actions, instead of just public.

3 Composition of Previous Results

Lemma 6 (Sequential Composition with RTPτ). *Given* $\vdash_{wf} \sim_1 : \mathbb{C}_2$, $\vdash_{\tau_{\sim_1}} [\![\bullet]\!]^{\mathsf{S}\to I} : \mathbb{C}_1$, *and* $\vdash_{\tau_{\sim_2}} [\![\bullet]\!]^{I\to\mathbf{T}} : \tau_{\sim_1}(\mathbb{C}_2)$, *then* $\vdash_{\tau_{\sim_1\circ\sim_2}} [\![\bullet]\!]^{\mathsf{S}\to I\to\mathbf{T}} : \mathbb{C}_1 \cap \mathbb{C}_2$.

Proof. We need to show $\vdash_{\tau_{\sim_1\circ\sim_2}} [\![\bullet]\!]^{\mathsf{S}\to I\to\mathbf{T}} : \mathbb{C}_1 \cap \mathbb{C}_2$. By definition, assume $\pi \in \mathbb{C}_1 \cap \mathbb{C}_2$ and $\mathsf{p} \in \mathcal{P}$ such that $\mathsf{p} \vDash_R \pi$. What is left to show is $[\![\mathsf{p}]\!]^{\mathsf{S}\to I\to\mathbf{T}} \vDash_R \tau_{\sim_1\circ\sim_2}(\pi)$. Note that $\pi \in \mathbb{C}_1$ as well as $\pi \in \mathbb{C}_2$. Also, $[\![\mathsf{p}]\!]^{\mathsf{S}\to I} \in \mathcal{P}$ and $\tau_{\sim_1\circ\sim_2}(\pi) = \tau_{\sim_2}(\tau_{\sim_1}(\pi))$. Since $\sim_1$ is well-formed with respect to $\mathbb{C}_2$, we can apply $\vdash_{\tau_{\sim_2}} [\![\bullet]\!]^{I\to\mathbf{T}} : \tau_{\sim_1}(\mathbb{C}_2)$, changing our goal to $[\![\mathsf{p}]\!]^{\mathsf{S}\to I} \vDash_R \tau_{\sim_1}(\pi)$. Since $\pi \in \mathbb{C}_1$ also holds, we can this time apply $\vdash_{\tau_{\sim_1}} [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}_1$. What is left to show is $\mathsf{p} \vDash_R \pi$, which is an assumption of ours. □

Lemma 7 (Sequential Composition with RTPσ). *Given* $\vdash_{wf} \sim_2 : \mathbb{C}_1$, $\vdash_{\sigma_{\sim_1}} [\![\bullet]\!]^{\mathsf{S}\to I} : \sigma_{\sim_2}(\mathbb{C}_1)$, *and* $\vdash_{\sigma_{\sim_2}} [\![\bullet]\!]^{I\to\mathbf{T}} : \mathbb{C}_2$, *then* $\vdash_{\sigma_{\sim_1\circ\sim_2}} [\![\bullet]\!]^{\mathsf{S}\to I\to\mathbf{T}} : \mathbb{C}_1 \cap \mathbb{C}_2$.

Proof. Dual to Lemma 6. □

Lemma 8 (Sequential Composition with RTP). *Given* $\vdash [\![\bullet]\!]^{\mathsf{S}\to I} : \mathbb{C}_1$ *and* $\vdash [\![\bullet]\!]^{I\to\mathbf{T}} : \mathbb{C}_2$, *then* $\vdash [\![\bullet]\!]^{\mathsf{S}\to I\to\mathbf{T}} : \mathbb{C}_1 \cap \mathbb{C}_2$.

Proof. Simple consequence from either Lemma 6 or Lemma 7 by setting the cross-language trace relations to $=$. □

Definition 28 (Upper Composition). *Given two compilers* $[\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}}$ *and* $[\![\bullet]\!]^{I\to\mathbf{T}}$, *their upper composition is*

$$[\![\bullet]\!]^{\mathsf{S}+I\to\mathbf{T}} = \lambda p. \begin{cases} [\![\mathsf{p}]\!]^{\mathsf{S}\to\mathbf{T}} & \text{if } p \in \mathcal{P} \\ [\![p]\!]^{I\to\mathbf{T}} & \text{if } p \in \mathcal{P} \end{cases}$$

.

Lemma 9 (Upper Composition with RTP). *Given* $\vdash [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}_1$ *and* $\vdash [\![\bullet]\!]^{I\to\mathbf{T}} : \mathbb{C}_2$, *then* $\vdash [\![\bullet]\!]^{\mathsf{S}+I\to\mathbf{T}} : \mathbb{C}_1 \cap \mathbb{C}_2$.

Proof. Analogous argument as in Lemma 8 (Sequential Composition with RTP), but with a case distinction on whether the source program is element of S or I. □

Definition 29 (Lower Composition). *Given two compilers* $[\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}}$ *and* $[\![\bullet]\!]^{\mathsf{S}\to I}$, *their lower composition is* $[\![\bullet]\!]^{\mathsf{S}\to I+\mathbf{T}}$.

Lemma 10 (Lower Composition with RTP). *Given* $\vdash [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}_1$ *and* $\vdash [\![\bullet]\!]^{\mathsf{S}\to I} : \mathbb{C}_2$, *then* $\vdash [\![\bullet]\!]^{\mathsf{S}\to I+\mathbf{T}} : \mathbb{C}_1 \cap \mathbb{C}_2$.

Proof. Analogous argument as in Lemma 8 (Sequential Composition with RTP), but with a case distinction on whether the compiled source program is element of I or $\mathbf{T}$. □

Lemma 11 (Diamond). *Given* $\vdash [\![\bullet]\!]^{\mathsf{S}\to I+\mathsf{O}} : \mathbb{C}_1$ *and* $\vdash [\![\bullet]\!]^{I+\mathsf{O}\to\mathbf{T}} : \mathbb{C}_2$ *with* $[\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} = \lambda \mathsf{p}.[\![[\![\mathsf{p}]\!]^{\mathsf{S}\to I+\mathsf{O}}]\!]^{I+\mathsf{O}\to\mathbf{T}}$, *then* $\vdash [\![\bullet]\!]^{\mathsf{S}\to\mathbf{T}} : \mathbb{C}_1 \cap \mathbb{C}_2$.

Proof. Straightforward using Lemma 8 (Sequential Composition with RTP). □

Lemma 12 (Swappable)**.** *Given* $\vdash [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_1$ *and* $\vdash [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \mathbb{C}_2$*, then* $\vdash [\![[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)}]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_2 \cap \mathbb{C}_1$ *and* $\vdash [\![[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)}]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \mathbb{C}_1 \cap \mathbb{C}_2$.

Proof. For the first conjunct $\vdash [\![[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)}]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_2 \cap \mathbb{C}_1$: we view the composed compiler as the sequential composition of $[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)}$ followed by $[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)}$. Applying Lemma 8 (Sequential Composition with RTP) with $\vdash [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_1$ and $\vdash [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \mathbb{C}_2$ yields the result. The second conjunct follows symmetrically by applying Lemma 8 (Sequential Composition with RTP) with the compilers in reversed order. □

Lemma 13 (Swappableσ)**.** *Given* $\vdash_{wf} \sim_2 : \mathbb{C}_1$, $\vdash_{wf} \sim_1 : \mathbb{C}_2$, $\vdash_{\sigma_{\sim_1}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_1$, $\vdash_{\sigma_{\sim_1}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \sigma_{\sim_2}(\mathbb{C}_1)$, $\vdash_{\sigma_{\sim_2}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \sigma_{\sim_1}(\mathbb{C}_2)$*, and* $\vdash_{\sigma_{\sim_2}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \mathbb{C}_2$*, then* $\vdash_{\sigma_{\sim_1 \circ \sim_2}} [\![[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)}]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_2 \cap \mathbb{C}_1$ *and* $\vdash_{\sigma_{\sim_2 \circ \sim_1}} [\![[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)}]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \mathbb{C}_1 \cap \mathbb{C}_2$.

Proof. For the first conjunct $\vdash_{\sigma_{\sim_1 \circ \sim_2}} [\![[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)}]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_2 \cap \mathbb{C}_1$: apply Lemma 7 (Sequential Composition with RTPσ) with the well-formedness $\vdash_{wf} \sim_2 : \mathbb{C}_1$, the preservation $\vdash_{\sigma_{\sim_1}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \sigma_{\sim_2}(\mathbb{C}_1)$, and $\vdash_{\sigma_{\sim_2}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \mathbb{C}_2$. For the second conjunct: apply Lemma 7 (Sequential Composition with RTPσ) symmetrically using $\vdash_{wf} \sim_1 : \mathbb{C}_2$, $\vdash_{\sigma_{\sim_2}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \sigma_{\sim_1}(\mathbb{C}_2)$, and $\vdash_{\sigma_{\sim_1}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_1$. □

Lemma 14 (Swappableτ)**.** *Given* $\vdash_{wf} \sim_2 : \mathbb{C}_1$, $\vdash_{wf} \sim_1 : \mathbb{C}_2$, $\vdash_{\tau_{\sim_1}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_1$, $\vdash_{\tau_{\sim_1}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \tau_{\sim_2}(\mathbb{C}_1)$, $\vdash_{\tau_{\sim_2}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \tau_{\sim_1}(\mathbb{C}_2)$*, and* $\vdash_{\tau_{\sim_2}} [\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \mathbb{C}_2$*, then* $\vdash_{\tau_{\sim_1 \circ \sim_2}} [\![[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)}]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)} : \mathbb{C}_2 \cap \mathbb{C}_1$ *and* $\vdash_{\tau_{\sim_2 \circ \sim_1}} [\![[\![\bullet]\!]^{\mathbf{T}\to\mathbf{T}}_{(1)}]\!]^{\mathbf{T}\to\mathbf{T}}_{(2)} : \mathbb{C}_1 \cap \mathbb{C}_2$.

Proof. Dual to Lemma 13, using Lemma 6 (Sequential Composition with RTPτ) instead. □

4 Case Study

4.1 Specification Language

We introduce a set of actions that allows us to describe properties abstractly instead of a language specific manner.

$$\begin{array}{rl} \textit{Control Tag } t ::= & \textit{ctx} \mid \textit{comp} \qquad \textit{Security Tag } \underline{\sigma} ::= \ \underline{🔒} \mid \underline{🔓} \\ \textit{Pre–Events } a_b^{\mathrm{ms}} ::= & \underline{\textit{Alloc } \ell\ n} \mid \underline{\textit{Dealloc } \ell} \mid \underline{\textit{Use } \ell\ n} \mid \underline{\textit{Branch } n} \mid \underline{\textit{Binop } n} \\ \textit{Events } a^{\mathrm{ms}} ::= & \underline{\varepsilon} \mid \underline{\lightning} \mid \underline{a_b^{\mathrm{ms}}}; t; \underline{\sigma} \end{array}$$

Figure 1: Specification Events.

Definition 30 (TMS)**.**

$$\mathit{tms} := \left\{ \overline{a_{\mathrm{ms}}} \;\middle|\; \begin{array}{rcl} \underline{\textit{Alloc } \ell\ n} & \leq_{\overline{a_{\mathrm{ms}}}} & \underline{\textit{Dealloc } \ell} \\ \underline{\textit{Use } \ell\ n} & \leq_{\overline{a_{\mathrm{ms}}}} & \underline{\textit{Dealloc } \ell} \\ \underline{\textit{Dealloc } \ell} & \in_! & \overline{a_{\mathrm{ms}}} \\ \underline{\textit{Alloc } \ell\ n} & \in_! & \overline{a_{\mathrm{ms}}} \end{array} \right\}$$

4.1.1 TMS Monitor

In order to just talk about temporal memory safety, we introduce a monitor that works on more abstract monitor-actions, without any other events besides those relevant to temporal memory safety.

$$
\begin{aligned}
&\textit{Abstract Store } T_{\text{TMS}} = \{A : \underline{L} \times t, F : \underline{L} \times t\} \\
&\textit{Abstract Events } \boldsymbol{a} ::= \boldsymbol{\varepsilon} \mid \textbf{Alloc } \underline{\ell}\ t \mid \textbf{Dealloc } \underline{\ell}\ t \mid \textbf{Use } \underline{\ell}\ t \mid \lightning \\
&\qquad T_{\text{TMS}} \subseteq_F T_{\text{TMS}}{}' \text{ iff } T_{\text{TMS}}.F \subseteq T_{\text{TMS}}{}'.F \\
&\qquad \boldsymbol{a} \in T_{\text{TMS}} \text{ iff } \boldsymbol{a} \in T_{\text{TMS}}.A \wedge \boldsymbol{a} \notin T_{\text{TMS}}.F \\
&\qquad \boldsymbol{a} \notin T_{\text{TMS}} \text{ iff } \boldsymbol{a} \notin T_{\text{TMS}}.A \wedge \boldsymbol{a} \notin T_{\text{TMS}}.F \\
&\qquad \{(\underline{\ell}; t)\} \cup T_{\text{TMS}} = \{A : \{(\underline{\ell}; t)\} \cup T_{\text{TMS}}.A, F : T_{\text{TMS}}.F\} \\
&\qquad T_{\text{TMS}} \setminus \{(\underline{\ell}; t)\} = \{A : T_{\text{TMS}}.A \setminus \{(\underline{\ell}; t)\}, F : T_{\text{TMS}}.F \cup \{(\underline{\ell}; t)\}\} \\
&\qquad T_{\text{TMS}} \cup T_{\text{TMS}}{}' = \{A : T_{\text{TMS}}.A \cup T_{\text{TMS}}{}'.A, F : T_{\text{TMS}}.F \cup T_{\text{TMS}}{}'.F\}
\end{aligned}
$$

Figure 2: TMS Monitor.

As before, when doing structural induction over primitive steps (Figure 27) we may encounter the ε, for which $\theta(\varepsilon) = \underline{\varepsilon}$, which needs a „partner" in the abstract events as defined in Figure 2: $\boldsymbol{\varepsilon}$.

$\boxed{\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'}$ „Given abstract state T_{TMS}, step to abstract state $T_{\text{TMS}}{}'$ emitting an abstract event $\boldsymbol{a}$."

$\boxed{\vdash T_{\text{TMS}} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{TMS}}{}'}$ „Reflexive-Transitive closure of above judgement. $\overline{\boldsymbol{a}}$ is a list of abstract events."

$$
\frac{}{\vdash T_{\text{TMS}} \overset{\boldsymbol{\varepsilon}}{\rightsquigarrow} T_{\text{TMS}}}\ \textsf{(TMS-Uninteresting)}
\qquad
\frac{}{\vdash T_{\text{TMS}} \overset{\lightning}{\rightsquigarrow} T_{\text{TMS}}}\ \textsf{(TMS-Abort)}
\qquad
\frac{(\underline{\ell}; t) \in T_{\text{TMS}}}{\vdash T_{\text{TMS}} \overset{\textbf{Use } \underline{\ell}\ t}{\rightsquigarrow} T_{\text{TMS}}}\ \textsf{(TMS-Use)}
$$

$$
\frac{T_{\text{TMS}}{}' = \{(\underline{\ell}; t)\} \cup T_{\text{TMS}} \qquad (\underline{\ell}; t) \notin T_{\text{TMS}}}{\vdash T_{\text{TMS}} \overset{\textbf{Alloc } \underline{\ell}\ t}{\rightsquigarrow} T_{\text{TMS}}{}'}\ \textsf{(TMS-Alloc)}
$$

$$
\frac{(\underline{\ell}; t) \in T_{\text{TMS}} \qquad T_{\text{TMS}}{}' = T_{\text{TMS}} \setminus \{(\underline{\ell}; t)\}}{\vdash T_{\text{TMS}} \overset{\textbf{Dealloc } \underline{\ell}\ t}{\rightsquigarrow} T_{\text{TMS}}{}'}\ \textsf{(TMS-Dealloc)}
\qquad
\frac{T_{\text{TMS}}.A = \emptyset}{\vdash T_{\text{TMS}} \overset{[\cdot]}{\rightsquigarrow}{}^{*} T_{\text{TMS}}}\ \textsf{(TMS-Refl)}
$$

$$
\frac{\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}' \qquad \vdash T_{\text{TMS}}{}' \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{TMS}}{}''}{\vdash T_{\text{TMS}} \overset{\boldsymbol{a} \cdot \overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{TMS}}{}''}\ \textsf{(TMS-Trans)}
$$

Figure 3: Steps of TMS Monitor.

The monitor-state contains two sets of locations that represent the ones that are active and the ones that have been deallocated, respectively. As seen in Rule TMS-Dealloc, the monitor only steps if the intuitive condition is true: a location can only be deallocated if it is part of the set of allocated locations. Rule TMS-Use ensures that only allocated locations occur in events representing

usage, the monitor cannot step if the location has been deallocated before.

As before, we need a way to translate one set of actions to another:

$\boxed{a^{\text{ms}} \cong \boldsymbol{a}}$ „Memory-Safety specific event a^{ms} describes same action as $\boldsymbol{a}$."

(TMS-Alloc-Authentic)
$$\frac{}{\underline{Alloc\ \ell\ n}; t; \underline{\sigma} \cong \textbf{Alloc}\ \underline{\ell}\ t}$$

(TMS-Dealloc-Authentic)
$$\frac{}{\underline{Dealloc\ \ell}; t; \underline{\sigma} \cong \textbf{Dealloc}\ \underline{\ell}\ t}$$

(TMS-Use-Authentic)
$$\frac{}{\underline{Use\ \ell\ n}; t; \underline{\sigma} \cong \textbf{Use}\ \underline{\ell}\ t}$$

(TMS-Abort-Authentic)
$$\frac{}{\underline{\lightning} \cong \lightning}$$

(TMS-None-Authentic)
$$\frac{}{\underline{\varepsilon} \cong \boldsymbol{\varepsilon}}$$

(TMS-Branch-Authentic)
$$\frac{}{\underline{Branch\ n}; t; \underline{\sigma} \cong \boldsymbol{\varepsilon}}$$

(TMS-Binop-Authentic)
$$\frac{}{\underline{Binop\ n}; t; \underline{\sigma} \cong \boldsymbol{\varepsilon}}$$

$\boxed{\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}$ „Memory-Safety specific trace $\overline{a^{\text{ms}}}$ describes same trace as $\overline{\boldsymbol{a}}$."

(TMS-Refl-Authentic)
$$\frac{}{[\cdot] \cong^* [\cdot]}$$

(TMS-Trans-Authentic)
$$\frac{a^{\text{ms}} \cong \boldsymbol{a} \qquad \overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}{a^{\text{ms}} \cdot \overline{a^{\text{ms}}} \cong^* \boldsymbol{a} \cdot \overline{\boldsymbol{a}}}$$

(TMS-Trans-Authentic-Lign)
$$\frac{\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}{\underline{\varepsilon} \cdot \overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}$$

(TMS-Trans-Authentic-Rign)
$$\frac{\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}{\overline{a^{\text{ms}}} \cong^* \boldsymbol{\varepsilon} \cdot \overline{\boldsymbol{a}}}$$

Figure 4: Trace Agreement.

Definition 31 (Trace is temporal memory safe via monitor.)**.** *We say* $\mathtt{TMS}(\overline{a^{ms}})$ *iff* $\exists \delta_{tms}\ \overline{\boldsymbol{a}}\ T_{TMS}$ *such that:*

- $\overline{a^{ms}} \cong \overline{\boldsymbol{a}}$
- *and* $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{TMS}$

Lemma 15 ($\mathtt{TMS}(\overline{a^{\text{ms}}})$ implies $\overline{a^{\text{ms}}} \in$ tmsafe)**.** *If*

(a) $\mathtt{TMS}(\overline{a^{ms}})$

Then

(i) $\overline{a^{ms}} \in$ tmsafe

Proof. Invert Assumption (a):

(H_1) $\overline{a^{\text{ms}}} \cong \overline{\boldsymbol{a}}$

(H_2) $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{\text{TMS}}$

The proof follows with an induction on Assumption (H_2). □

4.1.2 SMS Monitor

Abstract Store $T_{\text{SMS}} = \underline{L} \times t \times \mathbb{N}$

Abstract Events $\boldsymbol{a} ::= \boldsymbol{\varepsilon} \mid \textbf{Alloc}\ \underline{\ell}\ t\ \boldsymbol{n} \mid \textbf{Use}\ \underline{\ell}\ t\ \boldsymbol{n} \mid \lightning$

Figure 5: SMS Monitor.

Definition 32 (SMS)**.**

$$\text{sms} := \left\{ \overline{a_{\text{ms}}} \;\middle|\; \textit{If } \underline{\textit{Alloc } \ell\ n} \leq_{\overline{a}_{\text{ms}}} \underline{\textit{Use } \ell\ m},\ \textit{then } m < n \right\}$$

$\boxed{\vdash T_{\text{SMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{SMS}}{}'}$ „Given abstract state T_{SMS}, step to abstract state $T_{\text{SMS}}{}'$ emitting an abstract event $\boldsymbol{a}$."

$\boxed{\vdash T_{\text{SMS}} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{SMS}}{}'}$ „Reflexive-Transitive closure of above judgement. $\overline{\boldsymbol{a}}$ is a list of abstract events."

(sms-None)
$$\frac{}{\vdash T_{\text{SMS}} \overset{\boldsymbol{\varepsilon}}{\rightsquigarrow} T_{\text{SMS}}}$$

(sms-Abort)
$$\frac{}{\vdash T_{\text{SMS}} \overset{\lightning}{\rightsquigarrow} T_{\text{SMS}}}$$

(sms-Use)
$$\frac{(\underline{\ell}; t; m) \in T_{\text{SMS}} \qquad n < m}{\vdash T_{\text{SMS}} \overset{\textbf{Use } \ell\ t\ \boldsymbol{n}}{\rightsquigarrow} T_{\text{SMS}}}$$

(sms-Alloc)
$$\frac{(\underline{\ell}; t; m) \notin \text{dom } T_{\text{SMS}}}{\vdash T_{\text{SMS}} \overset{\textbf{Alloc } \underline{\ell}\ t\ \boldsymbol{n}}{\rightsquigarrow} T_{\text{SMS}} \cup \{(\underline{\ell}; t; n)\}}$$

(sms-Refl)
$$\frac{}{\vdash T_{\text{SMS}} \overset{[\cdot]}{\rightsquigarrow}{}^{*} T_{\text{SMS}}}$$

(sms-Trans)
$$\frac{\vdash T_{\text{SMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{SMS}}{}' \qquad \vdash T_{\text{SMS}}{}' \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{SMS}}{}''}{\vdash T_{\text{SMS}} \overset{\boldsymbol{a} \cdot \overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{SMS}}{}''}$$

Figure 6: Steps of SMS Monitor.

$\boxed{a^{\text{ms}} \cong \boldsymbol{a}}$ „Memory-Safety specific event a^{ms} describes same action as $\boldsymbol{a}$."

(SMS-Alloc-Authentic)
$$\frac{}{\underline{\textit{Alloc } \ell\ n}; t; \underline{\sigma} \cong \textbf{Alloc } \underline{\ell}\ t\ \boldsymbol{n}}$$

(SMS-Dealloc-Authentic)
$$\frac{}{\underline{\textit{Dealloc } \ell}; t; \underline{\sigma} \cong \boldsymbol{\varepsilon}}$$

(SMS-Use-Authentic)
$$\frac{}{\underline{\textit{Use } \ell\ n}; t; \underline{\sigma} \cong \textbf{Use } \underline{\ell}\ t\ \boldsymbol{n}}$$

(SMS-None-Authentic)
$$\frac{}{\underline{\varepsilon} \cong \boldsymbol{\varepsilon}}$$

(SMS-Branch-Authentic)
$$\frac{}{\underline{\textit{Branch } n}; t; \underline{\sigma} \cong \boldsymbol{\varepsilon}}$$

(SMS-Binop-Authentic)
$$\frac{}{\underline{\textit{Binop } n}; t; \underline{\sigma} \cong \boldsymbol{\varepsilon}}$$

(SMS-Abort-Authentic)
$$\frac{}{\underline{\lightning} \cong \lightning}$$

$\boxed{\overline{a^{\text{ms}}} \cong^{*} \overline{\boldsymbol{a}}}$ „Memory-Safety specific trace $\overline{a^{\text{ms}}}$ describes same trace as $\overline{\boldsymbol{a}}$."

(SMS-Refl-Authentic)
$$\frac{}{[\cdot] \cong [\cdot]}$$

(SMS-Trans-Authentic)
$$\frac{a^{\text{ms}} \cong \boldsymbol{a} \qquad \overline{a^{\text{ms}}} \cong \overline{\boldsymbol{a}}}{a^{\text{ms}} \cdot \overline{a^{\text{ms}}} \cong \boldsymbol{a} \cdot \overline{\boldsymbol{a}}}$$

(SMS-Trans-Authentic-Lign)
$$\frac{\overline{a^{\text{ms}}} \cong^{*} \overline{\boldsymbol{a}}}{\underline{\varepsilon} \cdot \overline{a^{\text{ms}}} \cong^{*} \overline{\boldsymbol{a}}}$$

(SMS-Trans-Authentic-Rign)
$$\frac{\overline{a^{\text{ms}}} \cong^{*} \overline{\boldsymbol{a}}}{\overline{a^{\text{ms}}} \cong^{*} \boldsymbol{\varepsilon} \cdot \overline{\boldsymbol{a}}}$$

Figure 7: Trace Agreement.

Definition 33 (Trace is spatial memory safe via monitor.)**.** *We say* $\mathtt{SMS}(\overline{a^{ms}})$ *iff* $\exists \overline{\boldsymbol{a}}\ T_{SMS}$ *such that:*

- $\overline{a^{ms}} \cong \overline{\boldsymbol{a}}$
- *and* $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{SMS}$

Lemma 16 ($\mathtt{SMS}(\overline{a^{\text{ms}}})$ implies $\overline{a^{\text{ms}}} \in \text{smsafe}$)**.** *If*

(a) $\mathtt{SMS}(\overline{a^{ms}})$

Then

(i) $\overline{a^{ms}} \in \text{smsafe}$

Proof. Invert Assumption (a):

(H_1) $\overline{a^{\text{ms}}} \cong \overline{\boldsymbol{a}}$

(H_2) $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{\text{SMS}}$

The proof follows with an induction on Assumption (H_2). □

4.1.3 MS Monitor

$$\textit{Abstract Store } T_{\text{MS}} = T_{\text{TMS}} \times T_{\text{SMS}}$$
$$\textit{Abstract Events } \boldsymbol{a} ::= \boldsymbol{a}_{\text{tms}} \times \boldsymbol{a}_{\text{sms}}$$

Figure 8: MS Monitor.

Definition 34 (MS)**.** $\text{ms} := \text{tms} \cap \text{sms}$

$$\frac{\vdash T_{\text{SMS}} \overset{\boldsymbol{a}_{\text{tms}}}{\rightsquigarrow} T_{\text{SMS}} \qquad \vdash T_{\text{TMS}} \overset{\boldsymbol{a}_{\text{sms}}}{\rightsquigarrow} T_{\text{TMS}}}{\vdash (T_{\text{TMS}}; T_{\text{SMS}}) \overset{\boldsymbol{a}_{tms};\boldsymbol{a}_{sms}}{\rightsquigarrow} (T_{\text{TMS}}'; T_{\text{SMS}}')}\ \textsf{(tms-sms-E)}$$

$$\frac{\vdash T_{\text{TMS}} \overset{[\cdot]}{\rightsquigarrow}{}^* T_{\text{TMS}} \qquad \vdash T_{\text{SMS}} \overset{[\cdot]}{\rightsquigarrow}{}^* T_{\text{SMS}}}{\vdash (T_{\text{TMS}}; T_{\text{SMS}}) \overset{[\cdot]}{\rightsquigarrow}{}^* (T_{\text{TMS}}; T_{\text{SMS}})}\ \textsf{(tms-sms-Refl)}$$

$$\frac{\vdash T \overset{\boldsymbol{a}}{\rightsquigarrow} T' \qquad \vdash T' \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T''}{\vdash T \overset{\boldsymbol{a}\cdot\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T''}\ \textsf{(tms-sms-Trans)}$$

Figure 9: Steps of combined TMS + SMS Monitor.

$\boxed{\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}$ „Memory-Safety specific trace $\overline{a^{\text{ms}}}$ describes same trace as $\overline{\boldsymbol{a}}$."

$$\frac{}{[\cdot] \cong [\cdot]}\ \textsf{(MS-Refl-Authentic)} \qquad \frac{a^{\text{ms}} \cong \boldsymbol{a}_{\text{tms}} \qquad a^{\text{ms}} \cong \boldsymbol{a}_{\text{sms}} \qquad \overline{a^{\text{ms}}} \cong \overline{\boldsymbol{a}}}{a^{\text{ms}} \cdot \overline{a^{\text{ms}}} \cong (\boldsymbol{a}_{\text{tms}}; \boldsymbol{a}_{\text{sms}}) \cdot \overline{\boldsymbol{a}}}\ \textsf{(MS-Trans-Authentic)}$$

$$\frac{\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}{\underline{\varepsilon} \cdot \overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}\ \textsf{(MS-Trans-Authentic-Lign)} \qquad \frac{\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}{\overline{a^{\text{ms}}} \cong^* \boldsymbol{\varepsilon} \cdot \overline{\boldsymbol{a}}}\ \textsf{(MS-Trans-Authentic-Rign)}$$

Figure 10: Trace Agreement.

Definition 35 (Trace is memory safe via monitor.)**.** *We say* $\texttt{MS}(\overline{a^{ms}})$ *iff* $\exists \overline{\boldsymbol{a}}\ T_{MS}$ *such that:*

- $\overline{a^{ms}} \cong \overline{\boldsymbol{a}}$
- *and* $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{MS}$

Lemma 17 ($\texttt{MS}(\overline{a^{\text{ms}}})$ implies $\overline{a^{\text{ms}}} \in \text{msafe}$)**.** *If*

(a) $\mathit{MS}(\overline{a^{ms}})$

Then

(i) $\overline{a^{ms}} \in \text{msafe}$

Proof. Invert Assumption (a):

(H_1) $\overline{a^{\text{ms}}} \cong \overline{\boldsymbol{a}}$

(H_2) $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{MS}}$

The proof follows with Lemmas 15 and 16. □

4.1.4 sCCT Monitor

`sCCT`-Problems:

- branching or predicated execution
- comparisons are an issue: can only get the content of special flag register by branching
- timing of loads/stores changes
- mult/div is problematic. mult is safe on modern procs, but not on micro-procs
- bitshifts may translate into loops (unless bitshift by constant)

To remedy most of these real-world problems, which are dependent on the hardware, we introduce another language which has `sCCT`-hardened instructions. That is, we assume there exists an $\hat{\oplus}$ which behaves similarily to $\oplus$, but without leaking timing information. Moreover, we also assume that there are `sCCT`-versions of reading and writing. If the instruction set architecture does not support these, we argue that there is some encoding that can be done in the style of FaCT that the compiler would have to do. For example, to circumvent loading/storing and ensure `sCCT`-versions, the register allocator could ensure that this variable stays in a register. Whenever this is not possible, one could spill and, prior to re-loading the variable, invalidate the caches. Another option is to simply load the whole array sequentially and select the relevant entry using a bitmask operation.

$$\begin{aligned}
\textit{Abstract Store } T_{\text{sCCT}} &= \emptyset \\
\textit{Abstract Security Tag } \boldsymbol{\sigma} &::= \text{🔒} \mid \text{🔓} \\
\textit{Abstract Pre} - \textit{Events } \boldsymbol{a}_b &::= \textbf{Any} \\
\textit{Abstract Events } \boldsymbol{a} &::= \boldsymbol{\varepsilon} \mid \boldsymbol{a}_b; \boldsymbol{\sigma} \mid \lightning
\end{aligned}$$

Figure 11: sCCT Monitor.

The `sCCT` monitor serves as an overapproximation for cryptographic constant-time code. Here, any event whatsoever is considered bad, since this leaks information by means of a sidechannel.

Definition 36 (sCCT)**.**

$$\text{scct} := \left\{ \overline{a_{\text{ct}}} \;\middle|\; \overline{a_{\text{ct}}} = [\cdot] \;\; or \;\; \exists \overline{a'_{\text{ct}}}, \overline{a_{\text{ct}}} = a_b^{\text{ct}}; \text{🔓} \cdot \overline{a'_{\text{ct}}} \wedge \overline{a'_{\text{ct}}} \in \text{scct} \;\; \right\}$$

$\boxed{\vdash T_{\text{sCCT}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{sCCT}}{}'}$ „Given abstract state T_{sCCT}, step to abstract state $T_{\text{sCCT}}{}'$ emitting an abstract event $\boldsymbol{a}$."

$\boxed{\vdash T_{\text{sCCT}} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{sCCT}}{}'}$ „Reflexive-Transitive closure of above judgement. $\overline{\boldsymbol{a}}$ is a list of abstract events."

(scct-None)
$$\frac{}{\vdash T_{\text{sCCT}} \overset{\boldsymbol{\varepsilon}}{\rightsquigarrow} T_{\text{sCCT}}}$$

(scct-Abort)
$$\frac{}{\vdash T_{\text{sCCT}} \overset{\lightning;\text{🔓}}{\rightsquigarrow} T_{\text{sCCT}}}$$

(scct-Refl)
$$\frac{}{\vdash T_{\text{sCCT}} \overset{[\cdot]}{\rightsquigarrow}{}^{*} T_{\text{sCCT}}}$$

(scct-Trans)
$$\frac{\vdash T_{\text{sCCT}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{sCCT}}{}' \qquad \vdash T_{\text{sCCT}}{}' \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{sCCT}}{}''}{\vdash T_{\text{sCCT}} \overset{\boldsymbol{a}\cdot\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{sCCT}}{}''}$$

Figure 12: Steps of sCCT Monitor.

$\boxed{\underline{\sigma} = \boldsymbol{\sigma}}$ „Equality of Security Tags."

(secure-eq)
$$\frac{}{\underline{\text{🔒}} = \text{🔒}}$$

(insecure-eq)
$$\frac{}{\underline{\text{🔓}} = \text{🔓}}$$

$\boxed{a^{\text{ms}} \cong \boldsymbol{a}}$ „Memory-Safety specific event a^{ms} describes same action as $\boldsymbol{a}$."

(sCCT-Public-Authentic)
$$\frac{}{a_b^{\text{ms}}; t; \underline{\text{🔓}} \cong \boldsymbol{\varepsilon}}$$

(sCCT-Use-Authentic)
$$\frac{}{\underline{Use\ \ell\ n}; t; \underline{\text{🔒}} \cong \textbf{Any}; \text{🔒}}$$

(sCCT-Abort-Authentic)
$$\frac{}{\underline{\lightning} \cong \lightning}$$

(sCCT-None-Authentic)
$$\frac{}{\underline{\varepsilon} \cong \boldsymbol{\varepsilon}}$$

(sCCT-Branch-Authentic)
$$\frac{}{\underline{Branch\ n}; t; \underline{\text{🔒}} \cong \textbf{Any}; \text{🔒}}$$

(sCCT-Binop-Authentic)
$$\frac{}{\underline{Binop\ n}; t; \underline{\text{🔒}} \cong \textbf{Any}; \text{🔒}}$$

$\boxed{\overline{a^{\text{ms}}} \cong^{*} \overline{\boldsymbol{a}}}$ „Memory-Safety specific trace $\overline{a^{\text{ms}}}$ describes same trace as $\overline{\boldsymbol{a}}$."

(sCCT-Refl-Authentic)
$$\frac{}{[\cdot] \cong^{*} [\cdot]}$$

(sCCT-Trans-Authentic)
$$\frac{a^{\text{ms}} \cong \boldsymbol{a} \qquad \overline{a^{\text{ms}}} \cong^{*} \overline{\boldsymbol{a}}}{a^{\text{ms}} \cdot \overline{a^{\text{ms}}} \cong^{*} \boldsymbol{a} \cdot \overline{\boldsymbol{a}}}$$

(sCCT-Trans-Authentic-Lign)
$$\frac{\overline{a^{\text{ms}}} \cong^{*} \overline{\boldsymbol{a}}}{\underline{\varepsilon} \cdot \overline{a^{\text{sct}}} \cong^{*} \overline{\boldsymbol{a}}}$$

(sCCT-Trans-Authentic-Rign)
$$\frac{\overline{a^{\text{ms}}} \cong^{*} \overline{\boldsymbol{a}}}{\overline{a^{\text{ms}}} \cong^{*} \boldsymbol{\varepsilon} \cdot \overline{\boldsymbol{a}}}$$

Figure 13: Trace Agreement.

Definition 37 (Trace is strictly cryptographic constant time via monitor.)**.** *We say* $\texttt{sCCT}(\overline{a^{ms}})$ *iff* $\exists \overline{\boldsymbol{a}}\ T_{sCCT}$ *such that:*

- $\overline{a^{ms}} \cong \overline{\boldsymbol{a}}$
- *and* $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{sCCT}$

Lemma 18 ($\texttt{SCCT}(\overline{a^{\text{ms}}})$ implies $\overline{a^{\text{ms}}} \in \text{scct}$)**.** *If*

(a) $\texttt{SCCT}(\overline{a^{ms}})$

Then

(i) $\overline{a^{ms}} \in \text{scct}$

Proof. Invert Assumption (a):

(H_1) $\overline{a^{\text{ms}}} \cong \overline{\boldsymbol{a}}$

(H_2) $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{sCCT}}$

The proof follows with an induction on Assumption (H_2). □

4.1.5 SS Monitor

$$\begin{aligned}
\textit{Abstract Store } T_{\text{🔔}} &= \emptyset \\
\textit{Abstract SPECTRE Label } \boldsymbol{vX} &::= \textbf{NONE} \mid \textbf{PHT} \\
\textit{Abstract Security Tag } \boldsymbol{\sigma} &::= \text{🔒}_{\boldsymbol{vX}} \mid \text{🔓} \\
\textit{Abstract Pre-Events } \boldsymbol{a}_b &::= \textbf{Any} \\
\textit{Abstract Events } \boldsymbol{a} &::= \boldsymbol{\varepsilon} \mid \boldsymbol{a}_b; \boldsymbol{\sigma} \mid \lightning
\end{aligned}$$

Figure 14: SS Monitor.

The spec monitor serves as an overapproximation for speculative non-interference.

Definition 38 (SS)**.**

$$\text{spec} := \left\{ \overline{a_{\text{🔔}}} \;\middle|\; \begin{array}{l} \overline{a_{\text{🔔}}} = [\cdot] \;\; \textit{or} \;\; \exists \overline{a'_{\text{🔔}}}. \\ \left(\overline{a_{\text{🔔}}} = a_b^{\text{🔔}}; \text{🔓} \cdot \overline{a'_{\text{🔔}}} \textit{ or } \overline{a_{\text{🔔}}} = a_b^{\text{🔔}}; \text{🔒}_{NONE} \cdot \overline{a'_{\text{🔔}}} \right) \\ \textit{and} \;\; \overline{a'_{\text{🔔}}} \in \text{spec} \end{array} \right\}$$

$\boxed{\vdash T_{\text{🔔}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{🔔}}{}'}$ „Given abstract state $T_{\text{🔔}}$, step to abstract state $T_{\text{🔔}}{}'$ emitting an abstract event $\boldsymbol{a}$."

$\boxed{\vdash T_{\text{🔔}} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{🔔}}{}'}$ „Reflexive-Transitive closure of above judgement. $\overline{\boldsymbol{a}}$ is a list of abstract events."

$$\frac{}{\vdash T_{\text{🔔}} \overset{\boldsymbol{\varepsilon}}{\rightsquigarrow} T_{\text{🔔}}}\ \textsf{(spec-None)} \qquad \frac{}{\vdash T_{\text{🔔}} \overset{\boldsymbol{a}_b:\text{🔒}_{\textbf{NONE}}}{\rightsquigarrow} T_{\text{🔔}}}\ \textsf{(spec-Leak-None)} \qquad \frac{}{\vdash T_{\text{🔔}} \overset{\lightning:\text{🔓}}{\rightsquigarrow} T_{\text{🔔}}}\ \textsf{(spec-Abort)}$$

$$\frac{}{\vdash T_{\text{🔔}} \overset{[\cdot]}{\rightsquigarrow}{}^{*} T_{\text{🔔}}}\ \textsf{(spec-Refl)} \qquad \frac{\vdash T_{\text{🔔}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{🔔}}{}' \quad \vdash T_{\text{🔔}}{}' \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{🔔}}{}''}{\vdash T_{\text{🔔}} \overset{\boldsymbol{a}\cdot\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{🔔}}{}''}\ \textsf{(spec-Trans)}$$

Figure 15: Steps of SS Monitor.

$\boxed{\underline{\sigma} = \boldsymbol{\sigma}}$ „Equality of Security Tags."

(spec-secure-eq)
$$\frac{}{\underline{🔒} = 🔒_{\boldsymbol{v}\boldsymbol{X}}}$$

(spec-insecure-eq)
$$\frac{}{\underline{🔓} = 🔓}$$

$\boxed{a^{\text{ms}} \cong \boldsymbol{a}}$ „Memory-Safety specific event a^{ms} describes same action as $\boldsymbol{a}$."

(spec-Public-Authentic)
$$\frac{}{a^{\text{ms}}_b; t; \underline{🔓} \cong \boldsymbol{\varepsilon}}$$

(spec-Use-Authentic)
$$\frac{}{\underline{Use\ \ell\ n}; t; \underline{🔒} \cong \mathbf{Any}; 🔒}$$

(spec-Abort-Authentic)
$$\frac{}{\underline{\lightning} \cong \lightning}$$

(spec-None-Authentic)
$$\frac{}{\underline{\varepsilon} \cong \boldsymbol{\varepsilon}}$$

(spec-Branch-Authentic)
$$\frac{}{\underline{Branch\ n}; t; \underline{🔒} \cong \mathbf{Any}; 🔒}$$

(spec-Binop-Authentic)
$$\frac{}{\underline{Binop\ n}; t; \underline{🔒} \cong \mathbf{Any}; 🔒}$$

$\boxed{\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}$ „Memory-Safety specific trace $\overline{a^{\text{ms}}}$ describes same trace as $\overline{\boldsymbol{a}}$."

(spec-Refl-Authentic)
$$\frac{}{[\cdot] \cong^* [\cdot]}$$

(spec-Trans-Authentic)
$$\frac{a^{\text{ms}} \cong \boldsymbol{a} \qquad \overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}{a^{\text{ms}} \cdot \overline{a^{\text{ms}}} \cong^* \boldsymbol{a} \cdot \overline{\boldsymbol{a}}}$$

(spec-Trans-Authentic-Lign)
$$\frac{\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}{\underline{\varepsilon} \cdot \overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}$$

(spec-Trans-Authentic-Rign)
$$\frac{\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}}{\overline{a^{\text{ms}}} \cong^* \boldsymbol{\varepsilon} \cdot \overline{\boldsymbol{a}}}$$

Figure 16: Trace Agreement.

Definition 39 (Trace is strictly cryptographic constant time via monitor.)**.** *We say* $\mathtt{SPEC}(\overline{a^{ms}})$ *iff* $\exists \overline{\boldsymbol{a}}\ T_{🔔}$ *such that:*

- $\overline{a^{ms}} \cong \overline{\boldsymbol{a}}$
- *and* $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{🔔}$

Lemma 19 ($\mathtt{SPEC}(\overline{a^{\text{ms}}})$ implies $\overline{a^{\text{ms}}} \in \text{spec}$)**.** *If*

(a) $\mathtt{spec}(\overline{a^{ms}})$

Then

(i) $\overline{a^{ms}} \in \text{spec}$

Proof. Invert Assumption (a):

(H_1) $\overline{a^{\text{ms}}} \cong \overline{\boldsymbol{a}}$

(H_2) $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{🔔}$

The proof follows with an induction on Assumption (H_2). □

4.2 Source Language

4.2.1 Syntax

$$
\begin{array}{rl}
\textit{Final Result}\ \mathsf{f} ::= & \mathsf{v} \mid \mathsf{x} \qquad \textit{May be a Result}\ \mathsf{f}_{\lightning} ::= \mathsf{f} \mid \mathsf{stuck} \\
\textit{Expressions}\ \mathsf{e} ::= & \mathsf{f}_{\lightning} \mid \mathsf{e}_1 \oplus \mathsf{e}_2 \mid \mathsf{x}[\mathsf{e}] \mid \mathsf{let}\ \mathsf{x} = \mathsf{e}_1\ \mathsf{in}\ \mathsf{e}_2 \mid \mathsf{x}[\mathsf{e}_1] \leftarrow \mathsf{e}_2 \\
& \mid \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{e}_1\ \mathsf{in}\ \mathsf{e}_2 \mid \mathsf{delete}\ \mathsf{x} \mid \mathsf{return}\ \mathsf{e} \mid \mathsf{call}\ \mathsf{foo}\ \mathsf{e} \\
& \mid \mathsf{ifz}\ \mathsf{e}_1\ \mathsf{then}\ \mathsf{e}_2\ \mathsf{else}\ \mathsf{e}_3 \qquad \text{where } \oplus \in \{+, -, \times, <\} \\
\textit{Functions}\ \mathsf{F} ::= & \mathsf{let}\ \mathsf{foo}\ \mathsf{x} : \tau_\lambda := \mathsf{e} \\
\textit{Expr. Types}\ \tau_\mathsf{e} ::= & \mathbb{N} \mid \mathsf{ref}_\mathsf{q}\ \mathbb{N} \quad \textit{Qualifier}\ \mathsf{q} ::= 1 \mid 1/2 \quad \textit{Values}\ \mathsf{v} ::= \mathsf{n} \in \mathbb{N} \\
\textit{Ectx Types}\ \tau_\lambda ::= & \tau_\mathsf{e} \to \tau_\mathsf{e} \quad \textit{Types}\ \tau ::= \tau_\mathsf{e} \mid \tau_\lambda \mid \tau_\mathsf{e} \to \bot \quad \textit{References}\ \ell \in \mathbb{N} \\
\textit{Eval.Ctx.}\ \mathsf{K} ::= & [\cdot] \mid \mathsf{K} \oplus \mathsf{e} \mid \mathsf{v} \oplus \mathsf{K} \mid \mathsf{x}[\mathsf{K}] \mid \mathsf{let}\ \mathsf{x} = \mathsf{K}\ \mathsf{in}\ \mathsf{e} \\
& \mid \mathsf{x}[\mathsf{K}] \leftarrow \mathsf{e} \mid \mathsf{x}[\mathsf{v}] \leftarrow \mathsf{K} \mid \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{K}\ \mathsf{in}\ \mathsf{e} \\
& \mid \mathsf{ifz}\ \mathsf{K}\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2 \mid \mathsf{call}\ \mathsf{foo}\ \mathsf{K} \mid \mathsf{return}\ \mathsf{K} \\
\textit{Variables}\ \mathsf{x} \mid \mathsf{y} \mid \mathsf{foo} \mid \ldots & \quad \textit{Poison}\ \rho ::= \square \mid \text{☣} \\
\textit{Sandbox Tag}\ \mathsf{t} ::= & \mathsf{ctx} \mid \mathsf{comp} \\
\textit{Typing.Env.}\ \Gamma ::= & [\cdot] \mid \Gamma, \mathsf{x} : \tau \quad \textit{Store}\ \Delta ::= [\cdot] \mid \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), \Delta \\
\textit{Communication}\ \mathbf{c} ::= & ? \mid ! \mid \varnothing \quad \textit{Heaps}\ \mathsf{H} ::= [\cdot] \mid \mathsf{H} :: \mathsf{n} \\
\textit{Cont. Stack}\ \overline{\mathsf{K}} ::= & [\cdot] \mid (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}} \quad \textit{Library}\ \Xi ::= [\cdot] \mid \mathsf{F}, \Xi \\
\textit{Relevant}\ \xi ::= & [\cdot] \mid \mathsf{foo}, \xi \quad \textit{State}\ \Omega ::= \Phi; \mathsf{t}; \Psi \\
\textit{Flow State}\ \Phi ::= & \xi; \Xi; \overline{\mathsf{K}} \quad \textit{Memory State}\ \Psi ::= \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \\
\textit{Programs}\ \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} & \quad \textit{Substitutions}\ \gamma ::= [\mathsf{v}/\mathsf{x}], \gamma \mid [\cdot]
\end{array}
$$

Figure 17: Syntax of $\mathrm{L}_{\mathrm{tms}}$

Most of the syntax is more or less standard. Qualifiers q can be attached to pointers to signal ownership. Hereby, the qualifier 1 means "fully owned", meaning we may and also must delete the pointer at some point, while we cannot do the same for $1/2$, which forbids us to delete. Poison ρ marks locations in the execution-context Ω as „to-be-deleted" ($\square$) or „deleted" (☣). ξ contains a list of identifiers referencing what functions are considered a component. Anything not present in ξ is considered to be a context. We describe the act of calling or returning from component to context or vice versa as „crossing the boundary". The heap is split into two parts, such that any component listed in ξ will allocate in $\mathsf{H}^{\mathsf{comp}}$ and anything else in $\mathsf{H}^{\mathsf{ctx}}$. Objects pointed at by a location ℓ are marshalled when they are passed across the boundary. Without the heap sandboxing, contexts can arbitrarily rewrite a component's memory, which incurs compiler correctness issues. We chose to distinguish between successful final results (f) and potentially crashed results ($\mathsf{f}_{\lightning}$). Note that a final result may also be an identifier, which we keep around as abstract representation for locations of pointers. However, we still distinct them from ordinary values (v), since they don't behave like normal values. For example, we cannot make canonical typing lemmas for them without the additional information contained in a non-empty typing-context. When discussing secure compilation, there is a notion of „context" and „component", where the latter is usually the part one cares about. We use the term „context-switching" with its usual meaning, i.e. change in control

flow to some other procedure. For $\mathbf{c}$, the representation ? signalizes a context switch from context to component, ! from component to context, and $\varnothing$ signals an internal change, either inside the context or inside the component. The state Ω carries information on what the current context is and the continuation stack $\overline{\mathsf{K}}$ is used in the semantics to mark events with above policy accordingly. Continuations K in the stack are annotated with the name of the function foo the continuation originates from. Programs $\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}}$ contain two lists of top-level definitions Ξ_{ctx} and Ξ_{comp}. Ξ_{ctx} takes the role as attacker code. We reason explicitly about substitutions in the dynamic semantics.

4.2.2 Static Semantics

$\boxed{\vdash \tau\ \mathrm{int}}$ „τ is an interface type."

$$\frac{}{\vdash \mathbb{N}\ \mathrm{int}}\ (\text{int-}\mathbb{N})$$

Figure 18: Interface types of $\mathrm{L}_{\mathrm{tms}}$.

We introduce interface types to explicitly disallow passing owned pointers.

$\boxed{\Gamma \equiv \Gamma_1 \circ \Gamma_2}$ „Environment Γ can be split into Γ_1 and Γ_2."

$$\frac{}{[\cdot] \equiv [\cdot] \circ [\cdot]}\ (\text{splitEmpty}) \qquad \frac{}{\Gamma \equiv [\cdot] \circ \Gamma}\ (\text{splitEmptyL}) \qquad \frac{}{\Gamma \equiv \Gamma \circ [\cdot]}\ (\text{splitEmptyR})$$

$$\frac{\Gamma \equiv \Gamma_1 \circ \Gamma_2}{\mathsf{x} : \mathbb{N}, \Gamma \equiv \mathsf{x} : \mathbb{N}, \Gamma_1 \circ \mathsf{x} : \mathbb{N}, \Gamma_2}\ (\mathbb{N}\text{split})$$

$$\frac{\Gamma \equiv \Gamma_1 \circ \Gamma_2}{\mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N}, \Gamma \equiv \mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N}, \Gamma_1 \circ \mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N}, \Gamma_2}\ (\text{weakPtrSplit})$$

$$\frac{\Gamma \equiv \Gamma_1 \circ \Gamma_2}{\mathsf{x} : \mathsf{ref}_1\ \mathbb{N}, \Gamma \equiv \mathsf{x} : \mathsf{ref}_1\ \mathbb{N}, \Gamma_1 \circ \Gamma_2}\ (\text{ptrLSplit})$$

$$\frac{\Gamma \equiv \Gamma_1 \circ \Gamma_2}{\mathsf{x} : \mathsf{ref}_1\ \mathbb{N}, \Gamma \equiv \mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N}, \Gamma_1 \circ \mathsf{x} : \mathsf{ref}_1\ \mathbb{N}, \Gamma_2}\ (\text{ptrRSplit})$$

$$\frac{\vdash \tau_{\mathsf{e}}^{(1)}\ \mathrm{int} \qquad \vdash \tau_{\mathsf{e}}^{(2)}\ \mathrm{int} \qquad \Gamma \equiv \Gamma_1 \circ \Gamma_2}{\mathsf{foo} : \tau_{\mathsf{e}}^{(1)} \to \tau_{\mathsf{e}}^{(2)}, \Gamma \equiv \mathsf{foo} : \tau_{\mathsf{e}}^{(1)} \to \tau_{\mathsf{e}}^{(2)}, \Gamma_1 \circ \mathsf{foo} : \tau_{\mathsf{e}}^{(1)} \to \tau_{\mathsf{e}}^{(2)}, \Gamma_2}\ (\text{arrowSplit})$$

Figure 19: Context Splitting of $\mathrm{L}_{\mathrm{tms}}$ typing contexts.

The splitting of contexts takes care to propagate owned-pointers towards the end. This way, no non-owned pointer occurs in the context after an owned one, where both have the same identifier. Note that non-owned pointers and values may be freely duplicated and that we can generate non-owned pointers if we have ownership.

Definition 40 (NoOwnedPtr)**.** *We write* $\mathit{NoOwnedPtr}\ \Gamma$ *iff for any* x, τ*, if* $\mathsf{x} : \tau \in \Gamma$*, then* $\tau \neq \mathsf{ref}_1\ \mathbb{N}$*.*

$\boxed{\Gamma \vdash \mathsf{e} : \tau}$ „Under environment Γ the expression e has type τ."

$$\frac{\text{NoOwnedPtr } \Gamma_1, \mathsf{x} : \tau, \Gamma_2}{\Gamma_1, \mathsf{x} : \tau, \Gamma_2 \vdash \mathsf{x} : \tau}\ (t - \mathsf{var}) \qquad \frac{\text{NoOwnedPtr } \Gamma}{\Gamma \vdash \mathsf{n} : \mathbb{N}}\ (t - \mathbb{N})$$

$$\frac{\Gamma_1 \vdash \mathsf{e}_1 : \mathbb{N} \quad \Gamma_2 \vdash \mathsf{e}_2 : \mathbb{N}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{e}_1 \oplus \mathsf{e}_2 : \mathbb{N}}\ (t - \oplus) \qquad \frac{\Gamma_2 \vdash \mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N} \quad \Gamma_1 \vdash \mathsf{e} : \mathbb{N}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{x}[\mathsf{e}] : \mathbb{N}}\ (t\text{–get})$$

$$\frac{\Gamma_3 \vdash \mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N} \quad \Gamma_1 \vdash \mathsf{e}_1 : \mathbb{N} \quad \Gamma_2 \vdash \mathsf{e}_2 : \mathbb{N}}{(\Gamma_1 \circ \Gamma_2) \circ \Gamma_3 \vdash \mathsf{x}[\mathsf{e}_1] \leftarrow \mathsf{e}_2 : \mathbb{N}}\ (t\text{–set})$$

$$\frac{\Gamma_1 \vdash \mathsf{e}_1 : \tau_{\mathsf{e}}^{(1)} \quad \mathsf{x} : \tau_{\mathsf{e}}^{(1)}, \Gamma_2 \vdash \mathsf{e}_2 : \tau^{(2)}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{let}\ \mathsf{x} = \mathsf{e}_1\ \mathsf{in}\ \mathsf{e}_2 : \tau^{(2)}}\ (t - \mathsf{let})$$

$$\frac{\Gamma_1 \vdash \mathsf{e}_1 : \mathbb{N} \quad \mathsf{x} : \mathsf{ref}_1\ \mathbb{N}, \Gamma_2 \vdash \mathsf{e}_2 : \mathbb{N}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{e}_1\ \mathsf{in}\ \mathsf{e}_2 : \mathbb{N}}\ (t - \mathsf{new})$$

$$\frac{\text{NoOwnedPtr } \Gamma_1 \quad \text{NoOwnedPtr } \Gamma_2}{\Gamma_1, \mathsf{x} : \mathsf{ref}_1\ \mathbb{N}, \Gamma_2 \vdash \mathsf{delete}\ \mathsf{x} : \mathbb{N}}\ (t - \mathsf{delete})$$

$$\frac{\vdash \tau_{\mathsf{e}}^{(1)}\ \text{int} \quad \vdash \tau_{\mathsf{e}}^{(2)}\ \text{int} \quad \Gamma \vdash \mathsf{foo} : \tau_{\mathsf{e}}^{(1)} \to \tau_{\mathsf{e}}^{(2)} \quad \Gamma \vdash \mathsf{e} : \tau_{\mathsf{e}}^{(1)}}{\Gamma \vdash \mathsf{call}\ \mathsf{foo}\ \mathsf{e} : \tau_{\mathsf{e}}^{(2)}}\ (t - \mathsf{call}) \qquad \frac{\tau_{\mathsf{e}}\ \text{int} \quad \Gamma \vdash \mathsf{e} : \tau_{\mathsf{e}}}{\Gamma \vdash \mathsf{return}\ \mathsf{e} : \tau_{\mathsf{e}} \to \bot}\ (t - \mathsf{return})$$

$$\frac{\Gamma_1 \vdash \mathsf{e}_1 : \mathbb{N} \quad \Gamma_2 \vdash \mathsf{e}_2 : \tau \quad \Gamma_2 \vdash \mathsf{e}_3 : \tau}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{ifz}\ \mathsf{e}_1\ \mathsf{then}\ \mathsf{e}_2\ \mathsf{else}\ \mathsf{e}_3 : \tau}\ (t - \mathsf{ifz}) \qquad \frac{}{\Gamma \vdash \mathsf{abort}() : \tau}\ (t - \mathsf{abort})$$

Figure 20: Checking of $\mathsf{L}_{\mathsf{tms}}$ expressions.

The context splitting takes care that for e.g. Rule $t - \oplus$ it cannot happen that we do something like $(\mathsf{delete}\ \mathsf{x}) + \mathsf{x}[0]$. For Rules $t - \mathsf{var}$ and $t - \mathbb{N}$ we require that the contexts do not contain any owned pointer. Intuitively, an owned pointer is useless upto getting a non-owned version from context splitting, because we cannot do anything with it besides deleting. This is inspired by linear logic.

$\boxed{\Gamma \vdash \mathsf{K} : \tau^{(1)} \to \tau^{(2)} \mathbin{@} \Gamma_{[\cdot]}}$ „Under total environment Γ the evaluation context K requires an object of type $\tau^{(1)}$ and yields one of type $\tau^{(2)}$."

$$\frac{}{\Gamma \vdash [\cdot] : \tau \to \tau \mathbin{@} \Gamma}\ (\text{et-}[\cdot]) \qquad \frac{\Gamma_1 \vdash \mathsf{v} : \mathbb{N} \quad \Gamma_2 \vdash \mathsf{K} : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{v} \oplus \mathsf{K} : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-}\oplus\text{R})$$

$$\frac{\Gamma_1 \vdash \mathsf{K} : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]} \quad \Gamma_2 \vdash \mathsf{e} : \mathbb{N}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{K} \oplus \mathsf{e} : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-}\oplus\text{L})$$

$$\frac{\Gamma_1 \vdash \mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N} \quad \Gamma_2 \vdash \mathsf{K} : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{x}[\mathsf{K}] : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-get})$$

$$\frac{\Gamma_1 \vdash \mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N} \quad \Gamma_2 \vdash \mathsf{v} : \mathbb{N} \quad \Gamma_3 \vdash \mathsf{K} : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]}}{(\Gamma_1 \circ \Gamma_2) \circ \Gamma_3 \vdash \mathsf{x}[\mathsf{v}] \leftarrow \mathsf{K} : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-setR})$$

$$\frac{\Gamma_1 \vdash \mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N} \quad \Gamma_2 \vdash \mathsf{K} : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]} \quad \Gamma_3 \vdash \mathsf{e} : \mathbb{N}}{(\Gamma_1 \circ \Gamma_2) \circ \Gamma_3 \vdash \mathsf{x}[\mathsf{K}] \leftarrow \mathsf{e} : \tau \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-setL})$$

$$\frac{\Gamma_1 \vdash \mathsf{K} : \tau^{(1)} \to \tau'_{\mathsf{e}} \mathbin{@} \Gamma_{[\cdot]} \quad \mathsf{x} : \tau'_{\mathsf{e}}, \Gamma_2 \vdash \mathsf{e} : \tau^{(2)}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{let}\ \mathsf{x} = \mathsf{K}\ \mathsf{in}\ \mathsf{e} : \tau^{(1)} \to \tau^{(2)} \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-let})$$

$$\frac{\Gamma_1 \vdash \mathsf{K} : \tau^{(1)} \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]} \quad \mathsf{x} : \mathsf{ref}_1\ \mathbb{N}, \Gamma_2 \vdash \mathsf{e} : \tau^{(2)}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{K}\ \mathsf{in}\ \mathsf{e} : \tau^{(1)} \to \tau^{(2)} \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-new})$$

$$\frac{\Gamma_1 \vdash \mathsf{K} : \tau^{(1)} \to \mathbb{N} \mathbin{@} \Gamma_{[\cdot]} \quad \Gamma_2 \vdash \mathsf{e}_2 : \tau^{(2)} \quad \Gamma_2 \vdash \mathsf{e}_3 : \tau^{(2)}}{\Gamma_1 \circ \Gamma_2 \vdash \mathsf{ifz}\ \mathsf{K}\ \mathsf{then}\ \mathsf{e}_2\ \mathsf{else}\ \mathsf{e}_3 : \tau^{(1)} \to \tau^{(2)} \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-ifz})$$

$$\frac{\Gamma \vdash \mathsf{K} : \tau^{(1)} \to \tau' \mathbin{@} \Gamma_{[\cdot]} \quad \Gamma \vdash \mathsf{foo} : \tau' \to \tau^{(2)}}{\Gamma \vdash \mathsf{call}\ \mathsf{foo}\ \mathsf{K} : \tau^{(1)} \to \tau^{(2)} \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-call})$$

$$\frac{\Gamma \vdash \mathsf{K} : \tau^{(1)} \to \tau^{(2)} \mathbin{@} \Gamma_{[\cdot]}}{\Gamma \vdash \mathsf{return}\ \mathsf{K} : \tau^{(1)} \to (\tau^{(2)} \to \bot) \mathbin{@} \Gamma_{[\cdot]}}\ (\text{et-ret})$$

Figure 21: Checking of $\mathrm{L}_{\mathrm{tms}}$ evaluation contexts.

$\boxed{\Xi \downarrow\!= \Gamma}$ „Project $\mathrm{L_{tms}}$ library to static typing context."

$$\frac{}{[\cdot] \downarrow\!= [\cdot]}\ (\Xi\text{-proj-}[\cdot]) \qquad \frac{\Xi \downarrow\!= \Gamma}{(\mathsf{let\ foo\ x} : \tau_\lambda := \mathsf{e}), \Xi \downarrow\!= (\mathsf{foo} \mapsto \tau_\lambda), \Gamma}\ (\Xi\text{-proj-cons})$$

$\boxed{\mathrm{dom}\ \Xi = \mathsf{foo}, \ldots, \mathsf{bar}}$ „Collect function names."

$$\frac{}{\mathrm{dom}\ [\cdot] = [\cdot]}\ (\Xi\text{-dom-}[\cdot]) \qquad \frac{\mathrm{dom}\ \Xi = \mathsf{D}}{\mathrm{dom}\ (\mathsf{let\ foo\ x} : \tau_\lambda := \mathsf{e}), \Xi = \mathsf{foo}, \mathsf{D}}\ (\Xi\text{-dom-cons})$$

Figure 22: Extracting type annotations and function names.

$\boxed{\Xi \equiv \Xi_1 \cup \Xi_2}$ „Merging $\mathrm{L_{tms}}$ libraries."

$$\frac{\Xi = \Xi_1}{\Xi \equiv \Xi_1 \cup [\cdot]}\ (\text{lib-merge-empty}) \qquad \frac{\Xi \equiv \Xi_1 \cup \Xi_2}{\mathsf{F}, \Xi \equiv \Xi_1 \cup \mathsf{F}, \Xi_2}\ (\text{lib-merge-cons})$$

$\boxed{\Xi_1 \bowtie \Xi_2 = \Xi}$ „Syntactically linking $\mathrm{L_{tms}}$ libraries."

$$\frac{\mathrm{dom}\,\Xi_1 \cap \mathrm{dom}\,\Xi_2 = \emptyset \qquad \Xi \equiv \Xi_1 \cup \Xi_2}{\Xi_1 \bowtie \Xi_2 = \Xi}\ (\text{syntactic-plugging})$$

Figure 23: $\mathrm{L_{tms}}$ plugging of libraries.

$\boxed{\Gamma \vdash \Xi\ \mathrm{ok}}$ „$\mathrm{L_{tms}}$ library typechecks."

$$\frac{}{\Gamma \vdash [\cdot]\ \mathrm{ok}}\ (\text{t-}\Xi\text{-empty})$$

$$\frac{\tau_\mathsf{e}^{(1)}\ \mathrm{int} \qquad \tau_\mathsf{e}^{(2)}\ \mathrm{int} \qquad \Gamma \vdash \Xi\ \mathrm{ok} \qquad \mathsf{x} : \tau_\mathsf{e}^{(1)}, \Gamma \vdash \mathsf{e} : \tau_\mathsf{e}^{(2)} \to \bot}{\Gamma \vdash (\mathsf{let\ foo\ x} : \tau_\mathsf{e}^{(1)} \to \tau_\mathsf{e}^{(2)} := \mathsf{e}), \Xi\ \mathrm{ok}}\ (\text{t-}\Xi\text{-cons})$$

$\boxed{\vdash \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \dashv \Xi, \xi}$ „$\mathrm{L_{tms}}$ program $\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}}$ typechecks. Ξ is the result of linking Ξ_{ctx} and Ξ_{comp}. ξ is $\mathrm{dom}\ \Xi_{\mathsf{comp}}$."

$$\frac{\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}} \qquad \Xi \downarrow\!= \Gamma_0 \qquad \mathsf{main} \in \mathrm{dom}\,\Gamma_0 \qquad \Gamma_0 \vdash \Xi\ \mathrm{ok} \qquad \Gamma_0 \vdash \mathsf{call\ main\ 0} : \mathbb{N} \to \mathbb{N}}{\vdash \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \dashv \Xi, \mathrm{dom}\ \Xi_{\mathsf{comp}}}\ (t - \mathsf{prog})$$

$\boxed{\vdash \Omega \triangleright \mathsf{e} : \tau}$ „$\mathrm{L_{tms}}$ runtime program $\Omega \triangleright \mathsf{e}$ typechecks."

$$\frac{\Xi \vdash \Delta : \Gamma \qquad \Gamma \vdash \mathsf{e} : \tau}{\vdash \xi; \Xi; \overline{\mathsf{K}}; \mathsf{t}; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \triangleright \mathsf{e} : \tau}\ (t - \mathsf{prog} - \mathsf{runtime})$$

Figure 24: Checking of $\mathrm{L_{tms}}$ contexts, components, programs, and whole programs.

Perhaps most interesting is Rule $t - \mathsf{prog} - \mathsf{runtime}$ which generates a suitable typing context from the execution context. Figure 25 shows how this works exactly.

$\boxed{\Xi \vdash \Delta : \Gamma}$ „$\mathrm{L}_{\mathrm{tms}}$ location map Δ yields static typing environment Γ."

$$\frac{\Xi \downarrow = \Gamma}{\Xi \vdash [\cdot] : \Gamma}\ (T\mathsf{empty}\Delta) \qquad \frac{\Xi \vdash \Delta : \Gamma}{\Xi \vdash \mathsf{x} \mapsto (\ell; \mathsf{t}; \square; \mathsf{n}), \Delta : \mathsf{x} : \mathsf{ref}_1\ \mathbb{N}, \Gamma}\ (T\mathsf{ref}_1\ \mathbb{N})$$

$$\frac{\Xi \vdash \Delta : \Gamma}{\Xi \vdash \mathsf{x} \mapsto (\ell; \mathsf{t}; \circledast; \mathsf{n}), \Delta : \Gamma}\ (T\mathsf{ref}_1\ \mathbb{N}\mathsf{poison})$$

Figure 25: $\mathrm{L}_{\mathrm{tms}}$ store typing.

Here, we want to populate the typing context with all *valid* pointers, which are those that are not poisoned. Pointers of type $\mathsf{ref}_{1/2}\ \mathbb{N}$ are implicitly generated during elaboration, see Figure 19.

4.2.3 Dynamic Semantics

The language trivially satisfies sCCT. It does not satisfy SMS, as seen in Section 4.2.4 (Translation to Specification Events).

$$\begin{aligned} \textit{Base Events}\ \mathsf{a_b} &::= \mathsf{Alloc}\ \ell\ \mathsf{v} \mid \mathsf{Dealloc}\ \ell \mid \mathsf{Get}\ \ell\ \mathsf{v} \mid \mathsf{Set}\ \ell\ \mathsf{v}\ \mathsf{v}' \\ &\quad \mid \mathsf{Call}\ \mathbf{c}\ \mathsf{foo}\ \mathsf{v} \mid \mathsf{Ret}\ \mathbf{c}\ \mathsf{v} \mid \mathsf{Start} \mid \mathsf{End}\ \mathsf{v} \\ \textit{Events}\ \mathsf{a} &::= \varepsilon \mid \mathsf{a_b}; \mathsf{t} \mid \lightning \end{aligned}$$

Figure 26: Events of $\mathrm{L}_{\mathrm{tms}}$.

$\boxed{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'}$ „Expression e applied evaluates under configuration Ω to e' and new configuration Ω', emitting event a."

$$\frac{\mathsf{n}_1 \oplus \mathsf{n}_2 = \mathsf{n}_3}{\Omega \triangleright \mathsf{n}_1 \oplus \mathsf{n}_2 \xrightarrow{\varepsilon} \Omega \triangleright \mathsf{n}_3} \quad (e - \oplus)$$

$$\frac{\Psi = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2 \qquad \ell + \mathsf{n} \in \mathrm{dom}\ \Psi.\mathsf{H}^{\mathsf{t}}}{\Phi; \mathsf{t}'; \Psi \triangleright \mathsf{x}[\mathsf{n}] \xrightarrow{\mathsf{Get}\ \ell\ \mathsf{n};\mathsf{t}} \Phi; \mathsf{t}'; \Psi \triangleright \mathsf{H}^{\mathsf{t}}(\ell + \mathsf{n})} \quad (e - \mathsf{get} - \in)$$

$$\frac{\Psi.\Delta = \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2 \qquad \ell + \mathsf{n} \notin \mathrm{dom}\ \Psi.\mathsf{H}^{\mathsf{t}}}{\Phi; \mathsf{t}'; \Psi \triangleright \mathsf{x}[\mathsf{n}] \xrightarrow{\mathsf{Get}\ \ell\ \mathsf{n};\mathsf{t}} \Phi; \mathsf{t}'; \Psi \triangleright 1729} \quad (e - \mathsf{get} - \notin)$$

$$\frac{\begin{array}{c}\Psi.\Delta = \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2 \qquad \ell + \mathsf{n} \in \mathrm{dom}\ \Psi.\mathsf{H}^{\mathsf{t}} \\ \mathsf{H}^{\mathsf{t}}_1 = \Psi.\mathsf{H}^{\mathsf{t}}(\ell + \mathsf{n} \mapsto \mathsf{v}) \qquad \Psi' = \Psi[\mathsf{H}^{\mathsf{t}} \leftarrow \mathsf{H}^{\mathsf{t}}_1]\end{array}}{\Phi; \mathsf{t}_0; \Psi \triangleright \mathsf{x}[\mathsf{n}] \leftarrow \mathsf{v} \xrightarrow{\mathsf{Set}\ \ell\ \mathsf{n}\ \mathsf{v};\mathsf{t}_0} \Phi; \mathsf{t}_0; \Psi' \triangleright \mathsf{v}} \quad (e - \mathsf{set} - \in)$$

$$\frac{\Psi.\Delta = \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2 \qquad \ell + \mathsf{n} \notin \mathrm{dom}\ \Psi.\mathsf{H}^{\mathsf{t}}}{\Phi; \mathsf{t}_0; \Psi \triangleright \mathsf{x}[\mathsf{n}] \leftarrow \mathsf{v} \xrightarrow{\mathsf{Set}\ \ell\ \mathsf{n}\ \mathsf{v};\mathsf{t}_0} \Phi; \mathsf{t}_0; \Psi' \triangleright \mathsf{v}} \quad (e - \mathsf{set} - \notin)$$

$$\frac{}{\Omega \triangleright \mathsf{let}\ \mathsf{x} = \mathsf{f}\ \mathsf{in}\ \mathsf{e} \xrightarrow{\varepsilon} \Omega \triangleright \mathsf{e}[\mathsf{f}/\mathsf{x}]} \quad (e - \mathsf{let} - \mathsf{f})$$

$$\frac{\begin{array}{c}\Psi = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \square; \mathsf{n}), \Delta_2 \\ \Psi = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \text{☣}; \mathsf{n}), \Delta_2\end{array}}{\Phi; \mathsf{t}; \Psi \triangleright \mathsf{delete}\ \mathsf{x} \xrightarrow{\mathsf{Dealloc}\ \ell;\mathsf{t}} \Phi; \mathsf{t}; \Psi' \triangleright 0} \quad (e - \mathsf{delete})$$

$$\frac{\begin{array}{c}\Psi.\Delta \vdash \ell\ \mathit{fresh} \qquad \Psi.\Delta \vdash \mathsf{z}\ \mathit{fresh} \qquad \mathsf{H}^{\mathsf{t}}_1 = \Psi.\mathsf{H}^{\mathsf{t}} \ll \mathsf{n} \\ \Psi' = \Psi[\mathsf{H}^{\mathsf{t}} \leftarrow \mathsf{H}^{\mathsf{t}}_1][\Delta \leftarrow \mathsf{z} \mapsto (\ell, \mathsf{t}, \square; \mathsf{n}), \Psi.\Delta]\end{array}}{\Phi; \mathsf{t}; \Psi \triangleright \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{n}\ \mathsf{in}\ \mathsf{e} \xrightarrow{\mathsf{Alloc}\ \ell\ \mathsf{n};\mathsf{t}} \Phi; \mathsf{t}; \Psi' \triangleright \mathsf{e}[\mathsf{z}/\mathsf{x}]} \quad (e - \mathsf{ifz} - \mathsf{true})$$

$$\frac{}{\Omega \triangleright \mathsf{ifz}\ 0\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2 \xrightarrow{\varepsilon} \Omega \triangleright \mathsf{e}_1} \quad (e - \mathsf{ifz} - \mathsf{false})$$

$$\frac{}{\Omega \triangleright \mathsf{ifz}\ \mathsf{S}(\mathsf{n})\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2 \xrightarrow{\varepsilon} \Omega \triangleright \mathsf{e}_2} \qquad \frac{}{\Omega \triangleright \mathsf{abort}() \xrightarrow{\lightning} \lightning \triangleright \mathsf{stuck}} \quad (e - \mathsf{abort})$$

Figure 27: Primitive Evaluation of $\mathrm{L}_{\mathrm{tms}}$ expressions.

Evaluation is mostly straightforward, the only interesting cases involve the pointers. Specifically, Rule $e - \mathsf{delete}$ demonstrates why we need the poison-tag on the locations: Regardless of the current tag, emit a $\mathsf{Dealloc}\ \ell$ event and mark ℓ as poisoned (☣). The intuitive solution, removing the mapping on deletion, doesn't allow to run programs that delete twice. However, we want to model memory effects and show that such situations never happen, even though they *could*. Given a poisoned location, we can still do everything with it: reading, writing, or deletion. When generating the static context from the execution context, Rule $T\mathsf{ref}_1\ \mathbb{N}$poison allows us to disregard deleted locations which will help us reason that the given execution could not have been happening if the program

was well-typed to begin with. For sake of readablity, we will omit $\gamma = [\cdot]$ when writing down evaluation steps. That is, whenever we have $\Omega \triangleright \mathsf{e}\,[\cdot] \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'\,[\cdot]$, it's written $\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'$. Moreover, γ as postfix binds anything before it up to $\triangleright$, so $\mathsf{let\ x = e_1\ in\ e_2}\gamma$ is $(\mathsf{let\ x = e_1\ in\ e_2})\gamma$.

$\boxed{\Omega \triangleright \mathsf{K[e]} \xrightarrow{\mathsf{a}}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{K'[e']}}$ „Given an evaluation context K and an expression e it evaluates under configuration Ω to e' and new configuration Ω' in context K, emitting event a."

$$\frac{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'}{\Omega \triangleright \mathsf{K[e]} \xrightarrow{\mathsf{a}}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{K[e']}}\ (e-\mathsf{ctx}) \qquad \frac{\Omega \triangleright \mathsf{e} \xrightarrow{\lightning} \Omega' \triangleright \mathsf{stuck}}{\Omega \triangleright \mathsf{K[e]} \xrightarrow{\lightning}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{stuck}}\ (e-\mathsf{ctx}-\mathsf{stuck})$$

$$\frac{\begin{array}{c}\Omega = \xi; \Xi; [\cdot]\,; \mathsf{comp}; \Psi \qquad \Xi = \Xi_1, (\mathsf{let\ main\ x} : \tau_\lambda := \mathsf{e}), \Xi_2 \\ \Omega' = \xi; \Xi; \mathsf{ctx}; (\mathsf{K}; \mathsf{main}), [\cdot]\,; \Psi\end{array}}{\Omega \triangleright \mathsf{K[call\ main\ v]} \xrightarrow{\mathsf{Start;comp}}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e[x/v]}}\ (e-\mathsf{ctx}-\mathsf{call}-\mathsf{main})$$

$$\frac{\begin{array}{c}\Omega = \xi; \Xi; \overline{\mathsf{K}}; \mathsf{t}; \Psi \qquad \Xi = \Xi_1, (\mathsf{let\ foo\ x} : \tau_\lambda := \mathsf{e}), \Xi_2 \\ \mathsf{foo} \in_{\neg \mathsf{t}} \xi \qquad \rho(\mathsf{t}) = \mathbf{c} \qquad \Omega' = \xi; \Xi; (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}}; \neg\mathsf{t}; \Psi\end{array}}{\Omega \triangleright \mathsf{K[call\ foo\ v]} \xrightarrow{\mathsf{Call\ c\ foo\ v;t}}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e[x/v]}}\ (e-\mathsf{ctx}-\mathsf{call}-\mathsf{same})$$

$$\frac{\begin{array}{c}\Omega = \xi; \Xi; \overline{\mathsf{K}}; \mathsf{t}; \Psi \qquad \Xi = \Xi_1, (\mathsf{let\ foo\ x} : \tau_\lambda := \mathsf{e}), \Xi_2 \\ \mathsf{foo} \in_{\mathsf{t}} \xi \qquad \Omega' = \xi; \Xi; (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}}; \mathsf{t}; \Psi\end{array}}{\Omega \triangleright \mathsf{K[call\ foo\ v]} \xrightarrow{\mathsf{Call\ \varnothing\ foo\ v;t}}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e[x/v]}}\ (e-\mathsf{ctx}-\mathsf{call}-\mathsf{same})$$

$$\frac{}{\xi; \Xi; [\cdot]^{\mathsf{main}}, [\cdot]\,; \mathsf{ctx}; \Psi \triangleright \mathsf{K'[return\ v]} \xrightarrow{\mathsf{End\ v;t}}_{\mathrm{ctx}} \xi; \Xi; [\cdot]\,; \mathsf{comp}; \Psi \triangleright \mathsf{v}}\ (e-\mathsf{ctx}-\mathsf{return}-\mathsf{notsame})$$

$$\frac{\mathsf{foo} \in_{\neg \mathsf{t}} \xi \qquad \rho(\mathsf{t}) = \mathbf{c}}{\xi; \Xi; (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}}; \mathsf{t}; \Psi \triangleright \mathsf{K'[return\ v]} \xrightarrow{\mathsf{Ret\ c\ v;t}}_{\mathrm{ctx}} \xi; \Xi; \overline{\mathsf{K}}; \neg\mathsf{t}; \Psi \triangleright \mathsf{K[v]}}\ (e-\mathsf{ctx}-\mathsf{return}-\mathsf{same})$$

$$\frac{\mathsf{foo} \in_{\mathsf{t}} \xi}{\xi; \Xi; (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}}; \mathsf{t}; \Psi \triangleright \mathsf{K'[return\ v]} \xrightarrow{\mathsf{Ret\ \varnothing\ v;t}}_{\mathrm{ctx}} \xi; \Xi; \overline{\mathsf{K}}; \mathsf{t}; \Psi \triangleright \mathsf{K[v]}}$$

$\boxed{\rho(\mathsf{t}) = \mathbf{c}}$ „Returns either ? or ! depending on t."

$$\frac{}{\rho(\mathsf{ctx}) = ?}\ (\mathsf{comm\text{-}ctxtocomp}) \qquad \frac{}{\rho(\mathsf{comp}) = !}\ (\mathsf{comm\text{-}comptoctx})$$

$\boxed{\neg\mathsf{t} = \mathsf{t}'}$ „Negation of t."

$$\frac{}{\neg\mathsf{ctx} = \mathsf{comp}}\ (\mathsf{neg\text{-}ctx}) \qquad \frac{}{\neg\mathsf{comp} = \mathsf{ctx}}\ (\mathsf{neg\text{-}comp})$$

Figure 28: Contextual Evaluation of $\mathrm{L}_{\mathrm{tms}}$ expressions.

$\boxed{\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}'}$ „Expression e evaluates under configuration Ω to e' and new configuration Ω', emitting list of events $\bar{\mathsf{a}}$."

$$\frac{}{\Omega \triangleright \mathsf{f} \xrightarrow{[\cdot]}{}^{*}_{\mathrm{ctx}} \Omega \triangleright \mathsf{f}}\ (es - \mathsf{refl})$$

$$\frac{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}' \qquad \Omega' \triangleright \mathsf{e}' \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}'' \qquad \mathsf{a} \neq \varepsilon}{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}\cdot\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}''}\ (es - \mathsf{trans-important})$$

$$\frac{\Omega \triangleright \mathsf{e} \xrightarrow{\varepsilon}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}' \qquad \Omega' \triangleright \mathsf{e}' \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}''}{\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}''}\ (es - \mathsf{trans-unimportant})$$

$\boxed{\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}{}^{n}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}'}$ „Expression e evaluates in n steps under configuration Ω to e' and new configuration Ω', emitting list of events $\bar{\mathsf{a}}$."

$$\frac{}{\Omega \triangleright \mathsf{e} \xrightarrow{\varepsilon}{}^{0}_{\mathrm{ctx}} \Omega \triangleright \mathsf{e}}\ (en - \mathsf{refl})$$

$$\frac{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}' \qquad \Omega' \triangleright \mathsf{e}' \xrightarrow{\bar{\mathsf{a}}}{}^{n}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}'' \qquad \mathsf{a} \neq \varepsilon}{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}\cdot\bar{\mathsf{a}}}{}^{(n+1)}_{\mathrm{ctx}} \quad \Omega'' \triangleright \mathsf{e}''}\ (en - \mathsf{trans-important})$$

$$\frac{\Omega \triangleright \mathsf{e} \xrightarrow{\varepsilon}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}' \qquad \Omega' \triangleright \mathsf{e}' \xrightarrow{\bar{\mathsf{a}}}{}^{n}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}''}{\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}{}^{(n+1)}_{\mathrm{ctx}} \quad \Omega'' \triangleright \mathsf{e}''}\ (en - \mathsf{trans-unimportant})$$

Figure 29: Trace prefix generation given a $\mathrm{L_{tms}}$ program using the reflexive-transitive closure.

$\boxed{\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \overset{\bar{\mathsf{a}}}{\Rightarrow} \Omega \triangleright \mathsf{f}}$ „Run $\mathrm{L_{tms}}$ program $\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}}$, giving dynamic state Ω and emitting trace $\bar{\mathsf{a}}$."

$$\frac{\vdash \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \dashv \xi, \Xi \qquad \mathsf{main} \notin \xi \qquad \xi; \Xi; [\cdot]\,; \mathsf{comp}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{call\ main\ 0} \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega \triangleright \mathsf{f}}{\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \overset{\bar{\mathsf{a}}}{\Rightarrow} \Omega \triangleright \mathsf{f}}\ (e - \mathsf{wprog})$$

$$\frac{\vdash \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \dashv \xi, \Xi \qquad \mathsf{main} \notin \xi \qquad \xi; \Xi; [\cdot]\,; \mathsf{comp}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{call\ main\ 0} \xrightarrow{\bar{\mathsf{a}}\cdot\lightning;\mathsf{t}}{}^{*}_{\mathrm{ctx}} \Omega \triangleright \mathsf{f}_{\lightning}}{\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \overset{\bar{\mathsf{a}}\cdot\lightning;\mathsf{t}}{\Rightarrow} \Omega \triangleright \mathsf{f}_{\lightning}}\ (e - \mathsf{wprog} - \lightning)$$

Figure 30: Running a whole $\mathrm{L_{tms}}$ program.

4.2.4 Translation to Specification Events

We need a way to translate from concrete actions, as emitted by a program's execution, to abstract actions. The translation is, however, standard. The empty action $\underline{\varepsilon}$ is necessary. When proving lemmas by structural induction on the primitive steps (Figure 27), we get a concrete event ε that needs to be related to some a^{ms}. The natural candidate is $\underline{\varepsilon}$. We also use it to project the call/return events onto it.

$\boxed{\delta_{MS}(\ell) = \underline{\ell}}$ „A map from $\mathrm{L}_{\mathrm{tms}}$ memory locations ℓ to specification locations $\underline{\ell}$."

$\boxed{\theta_{\delta_{MS}}(\mathsf{a}) = a^{\mathrm{ms}}}$ „Project an $\mathrm{L}_{\mathrm{tms}}$ event to specification events."

(filter-context)
$$\frac{\mathsf{a_b} \neq \lightning}{\theta_{\delta_{MS}}(\mathsf{a_b}; \mathsf{ctx}) = \underline{\varepsilon}}$$

(filter-comp-start)
$$\frac{}{\theta_{\delta_{MS}}(\mathsf{Start}; \mathsf{comp}) = \underline{\varepsilon}}$$

(filter-comp-alloc)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathsf{n}}{\theta_{\delta_{MS}}(\mathsf{Alloc}\ \ell\ \mathsf{n}; \mathsf{comp}) = \underline{\mathit{Alloc}\ \ell\ n}}$$

(filter-comp-dealloc)
$$\frac{\delta_{MS}(\ell) = \underline{\ell}}{\theta_{\delta_{MS}}(\mathsf{Dealloc}\ \ell; \mathsf{comp}) = \underline{\mathit{Dealloc}\ \ell}}$$

(filter-comp-get)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathsf{n}}{\theta_{\delta_{MS}}(\mathsf{Get}\ \ell\ \mathsf{n}; \mathsf{comp}) = \underline{\mathit{Use}\ \ell\ n}}$$

(filter-comp-set)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathsf{n}}{\theta_{\delta_{MS}}(\mathsf{Set}\ \ell\ \mathsf{n}\ \mathsf{v}; \mathsf{comp}) = \underline{\mathit{Use}\ \ell\ n}}$$

(filter-comp-call)
$$\frac{}{\theta_{\delta_{MS}}(\mathsf{Call}\ \mathbf{c}\ \mathsf{foo}\ \mathsf{v}; \mathsf{comp}) = \underline{\varepsilon}}$$

(filter-comp-ret)
$$\frac{}{\theta_{\delta_{MS}}(\mathsf{Ret}\ \mathbf{c}\ \mathsf{v}; \mathsf{comp}) = \underline{\varepsilon}}$$

(filter-abort)
$$\frac{}{\theta_{\delta_{MS}}(\lightning; \mathsf{t}) = \underline{\lightning}}$$

$\boxed{\theta^*_{\delta_{MS}}(\overline{\mathsf{a}}) = \overline{a^{\mathrm{ms}}}}$ „Project an $\mathrm{L}_{\mathrm{tms}}$ trace to specification traces."

(filter-empty)
$$\frac{}{\theta^*_{\delta_{MS}}([\cdot]) = \underline{[\cdot]}}$$

(filter-cons-relevant)
$$\frac{\theta_{\delta_{MS}}(\mathsf{a}) = \underline{a} \qquad \theta^*_{\delta_{MS}}(\overline{\mathsf{a}}) = \overline{a^{\mathrm{ms}}} \qquad \underline{a} \neq \underline{\varepsilon}}{\theta^*_{\delta_{MS}}(\mathsf{a} \cdot \overline{\mathsf{a}}) = \underline{a} \cdot \overline{a^{\mathrm{ms}}}}$$

(filter-cons-relevant)
$$\frac{\theta_{\delta_{MS}}(\mathsf{a}) = \underline{\varepsilon} \qquad \theta^*_{\delta_{MS}}(\overline{\mathsf{a}}) = \overline{a^{\mathrm{ms}}}}{\theta^*_{\delta_{MS}}(\mathsf{a} \cdot \overline{\mathsf{a}}) = \overline{a^{\mathrm{ms}}}}$$

Figure 31: Projection of $\mathrm{L}_{\mathrm{tms}}$ events to specification events.

$\boxed{\delta_{MS}(\ell) = \underline{\ell}}$ „Map $\mathrm{L}_{\mathrm{tms}}$ locations ℓ to abstract locations $\underline{\ell}$."

$\boxed{T_{\mathrm{TMS}} \simeq_{\delta_{MS}} \Delta}$ „Abstract memory state T_{TMS} describes the concrete state Δ."

(Empty-Agree)
$$\frac{}{\emptyset \simeq_{\delta_{MS}} [\cdot]}$$

(Abort-Agree)
$$\frac{}{\emptyset \simeq_{\delta_{MS}} \lightning}$$

(Cons-Agree)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{\ell} \notin T_{\mathrm{TMS}} \qquad T_{\mathrm{TMS}} \simeq_{\delta_{MS}} \Delta}{\{\underline{\ell}\} \cup T_{\mathrm{TMS}} \simeq_{\delta_{MS}} \mathsf{x} \mapsto (\ell; \mathsf{t}; \square; \mathsf{n}), \Delta}$$

(Poison-Agree)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad T_{\mathrm{TMS}} \simeq_{\delta_{MS}} \Delta}{T_{\mathrm{TMS}} \simeq_{\delta_{MS}} \mathsf{x} \mapsto (\ell; \mathsf{t}; \circledast; \mathsf{n}), \Delta}$$

Figure 32: Store Agreement.

Statement of Theorem 2 ($\mathrm{L}_{\mathrm{tms}}$ is TMS via Monitor)**.** If

(a) $\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \overset{\overline{\mathsf{a}}}{\Rightarrow} \Omega \triangleright \mathsf{f}_{\lightning}$

Then

(i) $\theta^*_{\delta_{ms}}(\bar{\mathsf{a}}) \in$ tmsafe

.

Theorem 1 ($\mathsf{L}_{\mathrm{tms}}$ is not spatially memory safe)**.** *There is a well-typed L_{tms} component that does not robustly satisfy Definition 32: $\Xi \vDash_R smsafe$*

Proof. We pick:

$\Xi_{\mathsf{comp}} = \mathsf{let\ foo\ x} : \mathbb{N} \to \mathbb{N} := \mathsf{let\ z = new\ x\ in\ let\ w = z[1337]\ in\ let\ _ = delete\ z\ in\ w}, [\cdot]$

and $\Xi_{\mathsf{ctx}} = \mathsf{let\ main\ z} : \mathbb{N} \to \mathbb{N} := \mathsf{call\ foo\ 42}, [\cdot]$. Let $\Xi \equiv \Xi_{\mathsf{comp}} \cup \Xi_{\mathsf{ctx}}$. We show $\vdash \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \dashv \Xi, \{\mathsf{foo}\}$. First note that $\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}}$, since dom $\Xi_{\mathsf{ctx}} \cap$ dom $\Xi_{\mathsf{comp}} = \emptyset$ and $\Xi \equiv \Xi_{\mathsf{ctx}} \cup \Xi_{\mathsf{comp}}$ by definition. Now, dom $\Xi_{\mathsf{comp}} = \{\mathsf{foo}\}$ and $\Xi \downarrow = \mathsf{foo} \mapsto \mathbb{N} \to \mathbb{N}, \mathsf{main} \mapsto \mathbb{N} \to \mathbb{N}, [\cdot] = \Gamma_0$ by Rules Ξ-proj-$[\cdot]$ and Ξ-proj-cons. Consequently, $\mathsf{main} \in \mathrm{dom}\,\Gamma_0 = \{\mathsf{foo}, \mathsf{main}\}$. Note that Rule int-$\mathbb{N}$ gives $\mathbb{N}$ int. Using Rules t-Ξ-empty and t-Ξ-cons, what is left to show are the following:

1. $\mathsf{x} : \mathbb{N}, \Gamma_0 \vdash \mathsf{let\ z = new\ x\ in\ let\ w = z[1337]\ in\ delete\ z; w} : \mathbb{N}$
2. $\mathsf{z} : \mathbb{N}, \Gamma_0 \vdash \mathsf{call\ foo\ 42} : \mathbb{N}$

Let $\mathsf{e_w} = \mathsf{let\ w = z[1337]\ in\ delete\ z; w}$, $\Gamma = \mathsf{z} : \mathsf{ref}_1\ \mathbb{N}, \mathsf{x} : \mathbb{N}, [\cdot]$, and $\Gamma' = \mathsf{z} : \mathsf{ref}_{1/2}\ \mathbb{N}, \mathsf{x} : \mathbb{N}, [\cdot]$ due to space constraints.

$$\dfrac{\dfrac{\mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N}}{\mathsf{x} : \mathbb{N}, [\cdot] \vdash \mathsf{x} : \mathbb{N}}\text{Rule } t - \mathsf{var} \qquad \dfrac{\Gamma' \vdash \mathsf{z[1337]} : \mathbb{N} \quad \mathsf{w} : \mathbb{N}, \Gamma \vdash \mathsf{delete\ z; w} : \mathbb{N}}{\Gamma' \circ \Gamma \vdash \mathsf{e_w} : \mathbb{N}}\text{Rule } t - \mathsf{let}}{\mathsf{x} : \mathbb{N}, [\cdot] \circ \mathsf{x} : \mathbb{N}, [\cdot] \vdash \mathsf{let\ z = new\ x\ in\ let\ w = z[1337]\ in\ delete\ z; w} : \mathbb{N}}\text{Rule } t - \mathsf{new}$$

Observe that:

$$\dfrac{\dfrac{\mathsf{ref}_{1/2}\ \mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N} \quad \mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N}}{\Gamma' \vdash \mathsf{z} : \mathsf{ref}_{1/2}\ \mathbb{N}}\text{Rule } t - \mathsf{var} \qquad \dfrac{\mathsf{ref}_{1/2}\ \mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N} \quad \mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N}}{\Gamma' \vdash \mathsf{x} : \mathbb{N}}\text{Rule } t - \mathsf{var}}{\Gamma' \circ \Gamma' \vdash \mathsf{z[1337]} : \mathbb{N}}\text{Rule } t\text{–get}$$

And:

$$\dfrac{\dfrac{\mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N} \quad \mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N}}{\mathsf{w} : \mathbb{N}, \mathsf{z} : \mathsf{ref}_1\ \mathbb{N}, \mathsf{x} : \mathbb{N}, [\cdot] \vdash \mathsf{delete\ z} : \mathbb{N}}\text{Rule } t - \mathsf{delete} \qquad \dfrac{\mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N} \quad \mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N}}{\mathsf{w} : \mathbb{N}, \mathsf{x} : \mathbb{N}, [\cdot] \vdash \mathsf{w} : \mathbb{N}}\text{Rule } t - \mathsf{var}}{\mathsf{w} : \mathbb{N}, \mathsf{z} : \mathsf{ref}_1\ \mathbb{N}, \mathsf{x} : \mathbb{N}, [\cdot] \circ \mathsf{w} : \mathbb{N}, \mathsf{x} : \mathbb{N}, [\cdot] \vdash \mathsf{delete\ z; w} : \mathbb{N}}\text{Rule } t - \mathsf{let}$$

Thus, it typechecks.

Now consider the other case:

$$\dfrac{\dfrac{}{\mathsf{z} : \mathbb{N}, \Gamma_0 \vdash \mathsf{foo} : \mathbb{N} \to \mathbb{N}}\text{Rule } t - \mathsf{var} \qquad \dfrac{}{\mathsf{z} : \mathbb{N}, \Gamma_0 \vdash 42 : \mathbb{N}}\text{Rule } t - \mathbb{N} \qquad \dfrac{}{\mathbb{N}\ \text{int}}\text{Rule int-}\mathbb{N}}{\mathsf{z} : \mathbb{N}, \Gamma_0 \vdash \mathsf{call\ foo\ 42} : \mathbb{N}}\text{Rule } t - \mathsf{call}$$

Note that $\mathsf{NoOwnedPtr}\ \mathsf{z} : \mathbb{N}, \Gamma_0$ holds by $\mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N}$ and $\mathbb{N} \to \mathbb{N} \neq \mathsf{ref}_1\ \mathbb{N}$

Running the whole program via Rule $e - \mathrm{wprog}$ yields trace prefix $\bar{\mathsf{a}} = \mathsf{Start}; \mathsf{comp}{\cdot}\mathsf{Call}\ !\mathsf{foo}\ 42; \mathsf{ctx}{\cdot}\mathsf{Alloc}\ \ell\ 42; \mathsf{comp}{\cdot}\mathsf{Get}\ \ell\ 1337; \mathsf{comp}{\cdot}\mathsf{Dealloc}\ \ell; \mathsf{comp}{\cdot}\mathsf{Ret}\ ?1729; \mathsf{comp} \cdot \mathsf{End}\ 1729; \mathsf{ctx}$. We omit the precise derivation for brevity, but note that Lemma 41 (Top-Level Progress) gives us the necessary execution.

Let $\delta_{ms} = \{\ell \mapsto \underline{\ell}\}$, by Figure 31, $\theta^*_{\delta_{ms}}(\bar{\mathsf{a}}) = \underline{Alloc\ \ell\ 42} \cdot \underline{Use\ \ell\ 1337} \cdot \underline{Dealloc\ \ell}$.

For Definition 32, note that $\underline{Alloc\ \ell\ 42} \in \theta^*_{\delta_{ms}}(\bar{\mathsf{a}})$ and $\underline{Use\ \ell\ 1337} \in \theta^*_{\delta_{ms}}(\bar{\mathsf{a}})$, but $1337 \not< 42$, hence $\Xi_{\mathsf{comp}} \not\models_R smsafe$. □

4.2.5 Auxiliary Definitions and Lemmas

Lemma 20 ($\to^n_{ctx}$ and $\to^*_{ctx}$ yield $\to^*_{ctx}$)**.** *If*

(a) $\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}{}^n_{ctx}\ \Omega' \triangleright \mathsf{e}'$

(b) $\Omega' \triangleright \mathsf{e}' \xrightarrow{\bar{\mathsf{a}'}}{}^*_{ctx}\ \Omega'' \triangleright \mathsf{f}_{\lightning}$

Then

(i) $\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}\cdot\bar{\mathsf{a}'}}{}^*_{ctx}\ \Omega'' \triangleright \mathsf{f}_{\lightning}$

Proof. Induction on Assumption (a). □

Lemma 21 ($\to^*_{ctx}$ splits into $\to^n_{ctx}$ and $\to^*_{ctx}$)**.** *If*

(a) $\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}_0}\cdot\bar{\mathsf{a}_1}}{}^*_{ctx}\ \Omega' \triangleright \mathsf{f}_{\lightning}$

Then $\exists \Omega_0\ \mathsf{e}_0\ n$,

(i) $\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}_0}}{}^n_{ctx}\ \Omega_0 \triangleright \mathsf{e}_0$

(ii) $\Omega_0 \triangleright \mathsf{e}_0 \xrightarrow{\bar{\mathsf{a}_1}}{}^*_{ctx}\ \Omega' \triangleright \mathsf{f}_{\lightning}$

Proof. Induction on Assumption (a). □

Lemma 22 (Static Typing implies Runtime Typing (Toplevel))**.** *If*

(a) $\Xi \downarrow = \Gamma_0$

(b) $\Gamma_0 \vdash \mathsf{call}\ \mathsf{main}\ 0 : \mathbb{N}$

Then

(i) $\vdash \xi; \Xi; [\cdot]; \mathsf{comp}; [\cdot]^{\mathsf{ctx}}; [\cdot]^{\mathsf{comp}}; [\cdot] \triangleright \mathsf{call}\ \mathsf{main}\ 0 : \mathbb{N}$

Proof. Use Rules int-$\mathbb{N}$, $t - \mathsf{var}$, $t - \mathbb{N}$, $t - \mathsf{call}$ and $t - \mathrm{prog} - \mathrm{runtime}$. □

Lemma 23 (Typed Linking Recomposition)**.** *If*

(a) $\Xi = \Xi_1 \bowtie \Xi_2$

(b) $\Gamma \vdash \Xi_1\ \mathrm{ok}$

(c) $\Gamma \vdash \Xi_2\ \mathrm{ok}$

Then

(i) $\Gamma \vdash \Xi$ ok

Proof. By induction on Ξ_1.

Case $\Xi_1 = [\cdot]$**:** Then $\Xi = \Xi_2$, and the goal follows directly from the assumption $\Gamma \vdash \Xi_2$ ok.

Case $\Xi_1 = \mathsf{F}, \Xi_1'$**:** By inversion on $\Gamma \vdash \mathsf{F}, \Xi_1'$ ok using Rule t-Ξ-cons, we obtain $\Gamma \vdash \Xi_1'$ ok and typing of the function F. By the inductive hypothesis, $\Gamma \vdash \Xi_1' \bowtie \Xi_2$ ok. Applying Rule t-Ξ-cons with the typing of F yields $\Gamma \vdash \mathsf{F}, (\Xi_1' \bowtie \Xi_2)$ ok, which equals $\Gamma \vdash \Xi$ ok by definition of linking.

□

Lemma 24 (Typing Decomposition)**.** *If*

(a) $\Gamma \vdash \mathsf{K}[\mathsf{e}] : \tau$

Then $\exists \tau_{\mathsf{e}}, \Gamma_{[\cdot]}$,

(i) $\Gamma \vdash \mathsf{K} : \tau_{\mathsf{e}} \to \tau \mathbin{@} \Gamma_{[\cdot]}$

(ii) $\Gamma_{[\cdot]} \vdash \mathsf{e} : \tau_{\mathsf{e}}$

Proof. By induction on K.

Case et-$[\cdot]$**:** $\mathsf{K} = [\cdot]$, so $\mathsf{K}[\mathsf{e}] = \mathsf{e}$. We have $\Gamma \vdash \mathsf{e} : \tau$. By Rule et-$[\cdot]$, $\Gamma \vdash [\cdot] : \tau \to \tau \mathbin{@} \Gamma$. Take $\Gamma_{[\cdot]} = \Gamma$ and $\tau_e = \tau$.

Case et-let**:** $\mathsf{K} = \mathsf{let}\ \mathsf{x} = \mathsf{K}'\ \mathsf{in}\ \mathsf{e}'$, so $\mathsf{K}[\mathsf{e}] = \mathsf{let}\ \mathsf{x} = \mathsf{K}'[\mathsf{e}]\ \mathsf{in}\ \mathsf{e}'$. By inversion of Rule $t - \mathsf{let}$, $\Gamma \equiv \Gamma_1 \circ \Gamma_2$ with:

- $\Gamma_1 \vdash \mathsf{K}'[\mathsf{e}] : \tau_{\mathsf{e}}'$
- $\mathsf{x} : \tau_{\mathsf{e}}', \Gamma_2 \vdash \mathsf{e}' : \tau$

By IH on K': $\Gamma_1 \vdash \mathsf{K}' : \tau_{\mathsf{e}} \to \tau_{\mathsf{e}}' \mathbin{@} \Gamma_{[\cdot]}$ and $\Gamma_{[\cdot]} \vdash \mathsf{e} : \tau_{\mathsf{e}}$. By Rule et-let: $\Gamma_1 \circ \Gamma_2 \vdash \mathsf{let}\ \mathsf{x} = \mathsf{K}'\ \mathsf{in}\ \mathsf{e}' : \tau_{\mathsf{e}} \to \tau \mathbin{@} \Gamma_{[\cdot]}$.

Other cases follow similarly, propagating $\Gamma_{[\cdot]}$ through the recursive structure.

□

Lemma 25 (Typing Composition)**.** *If*

(a) $\Gamma \vdash \mathsf{K} : \tau_{\mathsf{e}} \to \tau \mathbin{@} \Gamma_{[\cdot]}$

(b) $\Gamma_{[\cdot]} \vdash \mathsf{e} : \tau_{\mathsf{e}}$

Then

(i) $\Gamma \vdash \mathsf{K}[\mathsf{e}] : \tau$

Proof. By induction on Assumption (a).

Case et-$[\cdot]$**:** $\mathsf{K} = [\cdot]$ with $\Gamma \vdash [\cdot] : \tau \to \tau \mathbin{@} \Gamma$. So $\Gamma_{[\cdot]} = \Gamma$ and $\tau_e = \tau$. We have $\Gamma \vdash \mathsf{e} : \tau$, and $\mathsf{K}[\mathsf{e}] = \mathsf{e}$. Done.

Case et-let**:** $\mathsf{K} = \mathsf{let}\ \mathsf{x} = \mathsf{K}'\ \mathsf{in}\ \mathsf{e}'$ with $\Gamma_1 \circ \Gamma_2 \vdash \mathsf{K} : \tau_{\mathsf{e}} \to \tau \mathbin{@} \Gamma_{[\cdot]}$. By inversion:

- $\Gamma_1 \vdash \mathsf{K}' : \tau_\mathsf{e} \to \tau'_\mathsf{e} \mathbin{@} \Gamma_{[\cdot]}$
- $\mathsf{x} : \tau'_\mathsf{e}, \Gamma_2 \vdash \mathsf{e}' : \tau$

By assumption, $\Gamma_{[\cdot]} \vdash \mathsf{e} : \tau_\mathsf{e}$. By IH: $\Gamma_1 \vdash \mathsf{K}'[\mathsf{e}] : \tau'_\mathsf{e}$. By Rule $t-\mathsf{let}$ with split $\Gamma_1 \circ \Gamma_2$: $\Gamma_1 \circ \Gamma_2 \vdash \mathsf{let}\ \mathsf{x} = \mathsf{K}'[\mathsf{e}]\ \mathsf{in}\ \mathsf{e}' : \tau$.

Other cases follow similarly. □

Lemma 26 (Runtime Typing Decomposition)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{K}[\mathsf{e}] : \tau$

Then $\exists \tau_\mathsf{e}, \Gamma_{[\cdot]}$,

(i) $\vdash \Omega \triangleright \mathsf{K} : \tau_\mathsf{e} \to \tau \mathbin{@} \Gamma_{[\cdot]}$

(ii) $\Gamma_{[\cdot]} \vdash \mathsf{e} : \tau_\mathsf{e}$

Proof. By Rule $t-\mathrm{prog}-\mathrm{runtime}$, extract context Γ from Ω. Apply Lemma 24 to get $\Gamma_{[\cdot]}$ and τ_e. □

Lemma 27 (Runtime Typing Composition)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{K} : \tau_\mathsf{e} \to \tau \mathbin{@} \Gamma_{[\cdot]}$

(b) $\Gamma_{[\cdot]} \vdash \mathsf{e} : \tau_\mathsf{e}$

Then

(i) $\vdash \Omega \triangleright \mathsf{K}[\mathsf{e}] : \tau$

Proof. Use Lemma 25. □

Lemma 28 (Store Agree Weaken)**.** *If*

(a) $T_{TMS} \simeq_{\delta_{MS}} \Delta$

(b) $\delta_{MS} \subseteq \delta'_{MS}$

Then

(i) $T_{TMS} \simeq_{\delta'_{MS}} \Delta$

Proof. Induction on $T_{\mathrm{TMS}} \simeq_{\delta_{MS}} \Delta$. □

Lemma 29 (Filter Weaken)**.** *If*

(a) $\theta^*_{\delta_{ms}}(\bar{\mathsf{a}}) \neq \underline{[\cdot]}$

(b) $\delta_{ms} \subseteq \delta'_{ms}$

Then

(i) $\theta^*_{\delta_{ms}}(\bar{\mathsf{a}}) = \theta^*_{\delta'_{ms}}(\bar{\mathsf{a}})$

Lemma 30 (TMS-Δ-split)**.** *If*

(a) $T_{TMS} \simeq_{\delta_{MS}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), \Delta_2$

Then $\exists T_{TMS1}\ T_{TMS2}\ \underline{\ell}$,

(i) $T_{TMS1} \simeq_{\delta_{MS}} \Delta_1$

(ii) $\{\underline{\ell}\} \cup \emptyset \simeq_{\delta_{MS}} \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), [\cdot]$

(iii) $T_{TMS2} \simeq_{\delta_{MS}} \Delta_2$

(iv) $T_{TMS} = T_{TMS1} \cup \{\underline{\ell}\} \cup T_{TMS2}$

Proof. Induction on Assumption (a). □

Lemma 31 (Trace-based Transitivity Authentic)**.** *If*

(a) $\overline{a_1^{ms}} \cong^* \overline{\boldsymbol{a}}_1$

(b) $\overline{a_2^{ms}} \cong^* \overline{\boldsymbol{a}}_2$

Then

(i) $\overline{a_1^{ms}} \cdot \overline{a_2^{ms}} \cong^* \overline{\boldsymbol{a}}_1 \cdot \overline{\boldsymbol{a}}_2$

Proof. By induction on the structure of $\overline{a_1^{\text{ms}}}$.

Case empty trace: If $\overline{a_1^{\text{ms}}} = [\cdot]$, then $\overline{\boldsymbol{a}}_1 = [\cdot]$ by inversion on $[\cdot] \cong^* \overline{\boldsymbol{a}}_1$. The goal $\overline{a_2^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}_2$ follows directly from the second assumption.

Case non-empty trace: If $\overline{a_1^{\text{ms}}} = a^{\text{ms}} \cdot \overline{a^{\text{ms}}}$, by inversion on $a^{\text{ms}} \cdot \overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}_1$ we have $\overline{\boldsymbol{a}}_1 = \boldsymbol{a} \cdot \overline{\boldsymbol{a}}$ with $a^{\text{ms}} \cong \boldsymbol{a}$ and $\overline{a^{\text{ms}}} \cong^* \overline{\boldsymbol{a}}$. By the inductive hypothesis, $\overline{a^{\text{ms}}} \cdot \overline{a_2^{\text{ms}}} \cong^* \overline{\boldsymbol{a}} \cdot \overline{\boldsymbol{a}}_2$. Applying Rule TMS-Trans-Authentic with $a^{\text{ms}} \cong \boldsymbol{a}$ yields $a^{\text{ms}} \cdot (\overline{a^{\text{ms}}} \cdot \overline{a_2^{\text{ms}}}) \cong^* \boldsymbol{a} \cdot (\overline{\boldsymbol{a}} \cdot \overline{\boldsymbol{a}}_2)$, which equals the goal by associativity of trace concatenation.

□

Lemma 32 (Monitor Step Subset)**.** *If*

(a) $\vdash T_{TMS} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{TMS}'$

Then

(i) $T_{TMS} \subseteq_F T_{TMS}'$

Proof. By case analysis on the derivation of $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}'$.

Cases Rules TMS-Uninteresting to TMS-Use: In these cases, $T_{\text{TMS}}' = T_{\text{TMS}}$, so $T_{\text{TMS}}.F = T_{\text{TMS}}'.F$ and the goal holds by reflexivity of $\subseteq$.

Case Rule TMS-Alloc: By definition of $\cup$ on abstract stores (Figure 2), $T_{\text{TMS}}' = \{(\underline{\ell}; t)\} \cup T_{\text{TMS}}$ only modifies the A component, leaving $T_{\text{TMS}}'.F = T_{\text{TMS}}.F$.

Case Rule TMS-Dealloc: By definition of $\setminus$ on abstract stores (Figure 2), $T_{\text{TMS}}' = T_{\text{TMS}} \setminus \{(\underline{\ell}; t)\}$ yields $T_{\text{TMS}}'.F = T_{\text{TMS}}.F \cup \{(\underline{\ell}; t)\}$, so $T_{\text{TMS}}.F \subseteq T_{\text{TMS}}'.F$.

□

Lemma 33 (Monitor Steps Subset)**.** *If*

(a) $\vdash T_{TMS} \overset{\boldsymbol{a}}{\rightsquigarrow}{}^{*} T_{TMS}{}'$

Then

(i) $T_{TMS} \subseteq_F T_{TMS}{}'$

Proof. By induction on the derivation of $\vdash T_{\text{TMS}} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\text{TMS}}{}'$.

Case Rule TMS-Refl: Then $T_{\text{TMS}}{}' = T_{\text{TMS}}$ and $\overline{\boldsymbol{a}} = [\cdot]$. The goal $T_{\text{TMS}}.F \subseteq T_{\text{TMS}}.F$ holds by reflexivity.

Case Rule TMS-Trans: By inversion, we have $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}''$ and $\vdash T_{\text{TMS}}{}'' \overset{\overline{\boldsymbol{a}}'}{\rightsquigarrow}{}^{*} T_{\text{TMS}}{}'$ for some intermediate state $T_{\text{TMS}}{}''$. By Lemma 32 (Monitor Step Subset), $T_{\text{TMS}}.F \subseteq T_{\text{TMS}}{}''.F$. By the inductive hypothesis, $T_{\text{TMS}}{}''.F \subseteq T_{\text{TMS}}{}'.F$. By transitivity of $\subseteq$, we conclude $T_{\text{TMS}}.F \subseteq T_{\text{TMS}}{}'.F$.

□

Lemma 34 (α-conv. Typing)**.** *If*

(a) $\Gamma \vdash \mathsf{e} : \tau$

Then

(i) $\Gamma[\mathsf{z} : \tau'/\mathsf{x} : \tau'] \vdash \mathsf{e}[\mathsf{z}/\mathsf{x}] : \tau$

.

Proof. Induction on Assumption (a). □

Lemma 35 (Substitution)**.** *If*

(a) $\mathsf{x} : \tau', \Gamma_1 \vdash \mathsf{e} : \tau$

(b) $\Gamma_2 \vdash \mathsf{v} : \tau'$

(c) $\Gamma_3 = \Gamma_1 \circ \Gamma_2$

Then

(i) $\Gamma_3 \vdash \mathsf{e}[\mathsf{v}/\mathsf{x}] : \tau$

.

Proof. By induction on Assumption (a). □

Lemma 36 (Base Preservation)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(b) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'\gamma'$

(c) $\Omega' \neq \lightning$

Then

(i) $\vdash \Omega' \triangleright \mathsf{e}'\gamma' : \tau$

.

Proof. By induction on Assumption (b). In each case, we extract the typing context Γ from Ω via Rule $t-\text{prog}-\text{runtime}$.

Case $e-\oplus$**:** $\mathsf{e} = \mathsf{n}_1 \oplus \mathsf{n}_2$ and $\mathsf{e}' = \mathsf{n}_3$ where $\mathsf{n}_1 \oplus \mathsf{n}_2 = \mathsf{n}_3$. Configuration unchanged: $\Omega' = \Omega$. By inversion of Assumption (a), $\Gamma \vdash \mathsf{n}_1 \oplus \mathsf{n}_2 : \mathbb{N}$, so $\tau = \mathbb{N}$. By Rule $t-\mathbb{N}$, $\Gamma \vdash \mathsf{n}_3 : \mathbb{N}$. Reconstruct via Rule $t-\text{prog}-\text{runtime}$.

Case $e-\mathsf{get}-\in$**:** $\mathsf{e} = \mathsf{x}[\mathsf{n}]$ and $\mathsf{e}' = \mathsf{H}^{\mathsf{t}}(\ell + \mathsf{n})$. Configuration unchanged: $\Omega' = \Omega$. By inversion, $\Gamma_1 \vdash \mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N}$ and $\Gamma_2 \vdash \mathsf{n} : \mathbb{N}$ with $\Gamma = \Gamma_1 \circ \Gamma_2$. Since heap values are natural numbers, $\Gamma \vdash \mathsf{H}^{\mathsf{t}}(\ell + \mathsf{n}) : \mathbb{N}$ by Rule $t-\mathbb{N}$. Reconstruct via Rule $t-\text{prog}-\text{runtime}$.

Case $e-\mathsf{get}-\notin$**:** Similar to above; result is 1729 which has type $\mathbb{N}$.

Case $e-\mathsf{set}-\in$**:** $\mathsf{e} = \mathsf{x}[\mathsf{n}] \leftarrow \mathsf{v}$ and $\mathsf{e}' = \mathsf{v}$. By inversion, $\Gamma \vdash \mathsf{v} : \mathbb{N}$. The heap is updated but the store Δ is unchanged, so the typing context Γ remains valid. Reconstruct via Rule $t-\text{prog}-\text{runtime}$.

Case $e-\mathsf{set}-\notin$**:** Similar to above.

Case $e-\mathsf{let}-\mathsf{f}$**:** $\mathsf{e} = \mathsf{let}\ \mathsf{x} = \mathsf{f}\ \mathsf{in}\ \mathsf{e}_0$ and $\mathsf{e}' = \mathsf{e}_0[\mathsf{f}/\mathsf{x}]$. By inversion of Rule $t-\mathsf{let}$, $\Gamma = \Gamma_1 \circ \Gamma_2$ with:

- $\Gamma_1 \vdash \mathsf{f} : \tau'$
- $\mathsf{x} : \tau', \Gamma_2 \vdash \mathsf{e}_0 : \tau$

By Lemma 35 with $\Gamma_3 = \Gamma_1 \circ \Gamma_2$: $\Gamma_1 \circ \Gamma_2 \vdash \mathsf{e}_0[\mathsf{f}/\mathsf{x}] : \tau$. Reconstruct via Rule $t-\text{prog}-\text{runtime}$.

Case $e-\mathsf{delete}$**:** $\mathsf{e} = \mathsf{delete}\ \mathsf{x}$ and $\mathsf{e}' = 0$. The store is updated to mark x as poisoned. By Rule $T\mathsf{ref}_1\ \mathbb{N}\text{poison}$, poisoned locations do not contribute to Γ. Hence the new typing context Γ' from the updated store satisfies $\Gamma' \vdash 0 : \mathbb{N}$. Reconstruct via Rule $t-\text{prog}-\text{runtime}$.

Case $e-\mathsf{new}$**:** $\mathsf{e} = \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{n}\ \mathsf{in}\ \mathsf{e}_0$ and $\mathsf{e}' = \mathsf{e}_0[\mathsf{z}/\mathsf{x}]$. A fresh location z is allocated. By inversion of Rule $t-\mathsf{new}$, $\Gamma = \Gamma_1 \circ \Gamma_2$ with:

- $\Gamma_1 \vdash \mathsf{n} : \mathbb{N}$
- $\mathsf{x} : \mathsf{ref}_1\ \mathbb{N}, \Gamma_2 \vdash \mathsf{e}_0 : \tau$

By Rule $T\mathsf{ref}_1\ \mathbb{N}$, the new store yields context $\mathsf{z} : \mathsf{ref}_1\ \mathbb{N}, \Gamma'$. By α-conversion (Lemma 34) and Lemma 35: $\mathsf{z} : \mathsf{ref}_1\ \mathbb{N}, \Gamma_1 \circ \Gamma_2 \vdash \mathsf{e}_0[\mathsf{z}/\mathsf{x}] : \tau$. Reconstruct via Rule $t-\text{prog}-\text{runtime}$.

Case $e-\mathsf{ifz}-\text{true}$**:** $\mathsf{e} = \mathsf{ifz}\ 0\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2$ and $\mathsf{e}' = \mathsf{e}_1$. By inversion of Rule $t-\mathsf{ifz}$, $\Gamma \vdash \mathsf{e}_1 : \tau$. Reconstruct via Rule $t-\text{prog}-\text{runtime}$.

Case $e-\mathsf{ifz}-\text{false}$**:** Similar; by inversion $\Gamma \vdash \mathsf{e}_2 : \tau$.

Case $e-\mathsf{abort}$**:** Excluded by Assumption (c) since $\Omega' = \lightning$. □

Lemma 37 (Ctx Preservation)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(b) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\mathsf{a}}_{ctx} \Omega' \triangleright \mathsf{e}'\gamma'$

Then

(i) $\vdash \Omega' \triangleright \mathsf{e}'\gamma' : \tau$

.

Proof. By induction on Assumption (b).

Case $e - \text{ctx}$**:** We have $\mathsf{e} = \mathsf{K}[\mathsf{e}_0]$, $\mathsf{e}' = \mathsf{K}[\mathsf{e}'_0]$, and $\Omega \triangleright \mathsf{e}_0 \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'_0$.

By Lemma 26 on Assumption (a):

- $\vdash \Omega \triangleright \mathsf{K} : \tau_{\mathsf{e}} \to \tau \mathbin{@} \Gamma_{[\cdot]}$
- $\Gamma_{[\cdot]} \vdash \mathsf{e}_0 : \tau_{\mathsf{e}}$

If $\Omega' = \lightning$, then the goal holds vacuously (crashed configurations are well-typed by convention).

Otherwise, by Lemma 36: $\vdash \Omega' \triangleright \mathsf{e}'_0 : \tau_{\mathsf{e}}$.

The evaluation context K is unchanged, and the portion of the context $\Gamma_{[\cdot]}$ that flows to the hole may be updated according to the store changes in Ω'. By inspection of the base evaluation rules (Figure 27), store updates preserve the typing derivation structure.

By Lemma 27: $\vdash \Omega' \triangleright \mathsf{K}[\mathsf{e}'_0] : \tau$.

Case $e - \text{ctx} - \mathsf{stuck}$**:** We have $\Omega \triangleright \mathsf{e}_0 \xrightarrow{\lightning} \Omega' \triangleright \mathsf{stuck}$. Then $\mathsf{e}' = \mathsf{stuck}$ and Ω' represents a crashed state. Crashed configurations are well-typed by convention.

Case $e - \text{ctx} - \mathsf{call} - \text{main}$**:** We have $\mathsf{e} = \mathsf{K}[\mathsf{call\ main\ v}]$ and $\mathsf{e}' = \mathsf{e}_{\text{body}}[\mathsf{v}/\mathsf{x}]$. The library contains $\mathsf{let\ main\ x} : \tau_\lambda := \mathsf{e}_{\text{body}}$.

By Lemma 26:

- $\vdash \Omega \triangleright \mathsf{K} : \tau_{\mathsf{e}} \to \tau \mathbin{@} \Gamma_{[\cdot]}$
- $\Gamma_{[\cdot]} \vdash \mathsf{call\ main\ v} : \tau_{\mathsf{e}}$

By inversion of Rule $t - \mathsf{call}$, $\tau_e = \tau_{\text{ret}}$ where $\mathsf{main} : \tau_{\text{arg}} \to \tau_{\text{ret}} \in \Gamma_0$. By the well-formedness of the library (Rule $t - \text{prog}$): $\mathsf{x} : \tau_{\text{arg}}, \Gamma_0 \vdash \mathsf{e}_{\text{body}} : \tau_{\text{ret}}$.

By Lemma 35: $\Gamma_0 \vdash \mathsf{e}_{\text{body}}[\mathsf{v}/\mathsf{x}] : \tau_{\text{ret}}$. The continuation stack is updated to $(K; main), [\cdot]$.

Reconstruct via Rule $t - \text{prog} - \text{runtime}$.

Case $e - \text{ctx} - \mathsf{call} - \text{notsame}$**:** Similar to above. The function foo is in a different compartment. By library well-formedness and Lemma 35, the body is well-typed. The continuation stack and sandbox tag are updated accordingly. Reconstruct via Rule $t - \text{prog} - \text{runtime}$.

Case $e - \text{ctx} - \mathsf{call} - \text{same}$**:** Similar to above. The function foo is in the same compartment. Reconstruct via Rule $t - \text{prog} - \text{runtime}$.

Case $e - \text{ctx} - \mathsf{return} - \text{main}$**:** We have $\mathsf{e} = \mathsf{K}'[\mathsf{return\ v}]$ with continuation stack $[\cdot]^{\mathsf{main}}, [\cdot]$. The result is $\mathsf{e}' = \mathsf{v}$. By inversion, $\Gamma \vdash \mathsf{v} : \tau_{\text{ret}}$. Reconstruct via Rule $t - \text{prog} - \text{runtime}$.

Case $e - \text{ctx} - \mathsf{return} - \text{notsame}$**:** We have continuation stack $(\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}}$ where foo is in a different compartment. The return restores the context K with the value v plugged in. By inversion of the call typing, K expects a value of type τ_{ret}. By Lemma 27: $\vdash \Omega' \triangleright \mathsf{K}[\mathsf{v}] : \tau$.

Case $e - \text{ctx} - \mathsf{return} - \text{same}$**:** Similar to above; the function is in the same compartment. Reconstruct via Rule $t - \text{prog} - \text{runtime}$. □

Lemma 38 (N-Steps Preservation)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(b) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\bar{\mathsf{a}}}{}^{n}_{ctx} \Omega' \triangleright \mathsf{e}'\gamma'$

Then

(i) $\vdash \Omega' \triangleright \mathsf{e}'\gamma' : \tau$

.

Proof. By induction on n in Assumption (b).

Case $en - \text{refl}$**:** $(n = 0)$ We have $\Omega \triangleright \mathsf{e} \xrightarrow{\varepsilon}{}^{0}_{\text{ctx}} \Omega \triangleright \mathsf{e}$. The configuration and expression are unchanged: $\Omega' = \Omega$ and $\mathsf{e}' = \mathsf{e}$. The goal follows immediately from Assumption (a).

Case $en - \text{trans}-\text{important}$**:** $(n = k + 1)$ We have:

- $\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}}_{\text{ctx}} \Omega_1 \triangleright \mathsf{e}_1$ with $\mathsf{a} \neq \varepsilon$
- $\Omega_1 \triangleright \mathsf{e}_1 \xrightarrow{\bar{\mathsf{a}}}{}^{k}_{\text{ctx}} \Omega' \triangleright \mathsf{e}'$

  By Lemma 37 applied to the first step: $\vdash \Omega_1 \triangleright \mathsf{e}_1 : \tau$.
  By the induction hypothesis on k: $\vdash \Omega' \triangleright \mathsf{e}' : \tau$.

Case $en - \text{trans}-\text{unimportant}$**:** $(n = k + 1)$ We have:

- $\Omega \triangleright \mathsf{e} \xrightarrow{\varepsilon}_{\text{ctx}} \Omega_1 \triangleright \mathsf{e}_1$
- $\Omega_1 \triangleright \mathsf{e}_1 \xrightarrow{\bar{\mathsf{a}}}{}^{k}_{\text{ctx}} \Omega' \triangleright \mathsf{e}'$

  By Lemma 37 applied to the first step: $\vdash \Omega_1 \triangleright \mathsf{e}_1 : \tau$.
  By the induction hypothesis on k: $\vdash \Omega' \triangleright \mathsf{e}' : \tau$. □

Lemma 39 (Steps Preservation)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(b) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{ctx} \Omega' \triangleright \mathsf{e}'\gamma'$

Then

(i) $\vdash \Omega' \triangleright \mathsf{e}'\gamma' : \tau$

.

Proof. By induction on Assumption (b).

Case $es - \text{refl}$**:** We have $\Omega \triangleright \mathsf{f} \xrightarrow{[\cdot]}{}^{*}_{\text{ctx}} \Omega \triangleright \mathsf{f}$. The configuration and expression are unchanged. The goal follows immediately from Assumption (a).

Case $es - \text{trans}-\text{important}$**:** We have:

- $\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}}_{\text{ctx}} \Omega_1 \triangleright \mathsf{e}_1$ with $\mathsf{a} \neq \varepsilon$

- $\Omega_1 \triangleright \mathsf{e}_1 \xrightarrow{\bar{\mathsf{a}}}{}^*_{\text{ctx}} \Omega' \triangleright \mathsf{e}'$

  By Lemma 37 applied to the first step: $\vdash \Omega_1 \triangleright \mathsf{e}_1 : \tau$.
  By the induction hypothesis on the remaining steps: $\vdash \Omega' \triangleright \mathsf{e}' : \tau$.

Case $es - \text{trans}-\text{unimportant}$**:** We have:

- $\Omega \triangleright \mathsf{e} \xrightarrow{\varepsilon}_{\text{ctx}} \Omega_1 \triangleright \mathsf{e}_1$
- $\Omega_1 \triangleright \mathsf{e}_1 \xrightarrow{\bar{\mathsf{a}}}{}^*_{\text{ctx}} \Omega' \triangleright \mathsf{e}'$

  By Lemma 37 applied to the first step: $\vdash \Omega_1 \triangleright \mathsf{e}_1 : \tau$.
  By the induction hypothesis on the remaining steps: $\vdash \Omega' \triangleright \mathsf{e}' : \tau$. □

Lemma 40 (Progress)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

Then $\exists \Omega'\ \mathsf{f}_{\lightning}\ \bar{\mathsf{a}}$,

(i) $\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}{}^*_{ctx} \Omega' \triangleright \mathsf{f}_{\lightning}$

.

Proof. By nested induction: outer induction on the structure of e, inner induction on evaluation contexts when needed. We extract the typing context Γ from Ω via Rule $t - \text{prog} - \text{runtime}$.

Case $\mathsf{e} = \mathsf{f}$ (values or variables). The expression is already a final result. By Rule $es - \text{refl}$: $\Omega \triangleright \mathsf{f} \xrightarrow{[\cdot]}{}^*_{\text{ctx}} \Omega \triangleright \mathsf{f}$.

Case $\mathsf{e} = \mathsf{n}_1 \oplus \mathsf{n}_2$ (both operands are values). By Rule $e - \oplus$: $\Omega \triangleright \mathsf{n}_1 \oplus \mathsf{n}_2 \xrightarrow{\varepsilon} \Omega \triangleright \mathsf{n}_3$ where $\mathsf{n}_3 = \mathsf{n}_1 \oplus \mathsf{n}_2$. By Rule $e - \text{ctx}$ with $\mathsf{K} = [\cdot]$: $\Omega \triangleright \mathsf{n}_1 \oplus \mathsf{n}_2 \xrightarrow{\varepsilon}_{\text{ctx}} \Omega \triangleright \mathsf{n}_3$. By Rules $es - \text{refl}$ and $es - \text{trans}-\text{unimportant}$: $\Omega \triangleright \mathsf{n}_1 \oplus \mathsf{n}_2 \xrightarrow{[\cdot]}{}^*_{\text{ctx}} \Omega \triangleright \mathsf{n}_3$.

Case $\mathsf{e} = \mathsf{e}_1 \oplus \mathsf{e}_2$ (at least one non-value).

- If e_1 is not a value: By IH on e_1: $\Omega \triangleright \mathsf{e}_1 \xrightarrow{\bar{\mathsf{a}_1}}{}^*_{\text{ctx}} \Omega' \triangleright \mathsf{f}_{1\lightning}$. The evaluation context is $\mathsf{K} = [\cdot] \oplus \mathsf{e}_2$. By lifting through the context: $\Omega \triangleright \mathsf{e}_1 \oplus \mathsf{e}_2 \xrightarrow{\bar{\mathsf{a}}}{}^*_{\text{ctx}} \Omega'' \triangleright \mathsf{f}_{\lightning}$.
- If $\mathsf{e}_1 = \mathsf{v}_1$ is a value, e_2 is not: By IH on e_2: $\Omega \triangleright \mathsf{e}_2 \xrightarrow{\bar{\mathsf{a}_2}}{}^*_{\text{ctx}} \Omega' \triangleright \mathsf{f}_{2\lightning}$. The evaluation context is $\mathsf{K} = \mathsf{v}_1 \oplus [\cdot]$. By lifting through the context: $\Omega \triangleright \mathsf{v}_1 \oplus \mathsf{e}_2 \xrightarrow{\bar{\mathsf{a}}}{}^*_{\text{ctx}} \Omega'' \triangleright \mathsf{f}_{\lightning}$.

Case $\mathsf{e} = \mathsf{x}[\mathsf{n}]$ (index is a value). By well-typedness, $\mathsf{x} : \mathsf{ref}_{1/2}\ \mathbb{N} \in \Gamma$. By Rule $t-\text{prog} - \text{runtime}$, x is bound in $\Omega.\Delta$. Either $\ell + \mathsf{n} \in \text{dom}(\mathsf{H}^{\mathsf{t}})$ or not:

- If in domain: by Rule $e - \mathsf{get} - \in$, $\Omega \triangleright \mathsf{x}[\mathsf{n}] \xrightarrow{\cdots} \Omega \triangleright \mathsf{H}^{\mathsf{t}}(\ell + \mathsf{n})$.
- If not in domain: by Rule $e - \mathsf{get} - \notin$, $\Omega \triangleright \mathsf{x}[\mathsf{n}] \xrightarrow{\cdots} \Omega \triangleright 1729$.

Lift to multi-step via Rules $e - \text{ctx}$, $es - \text{refl}$ and $es - \text{trans}-\text{important}$.

Case $\mathsf{e} = \mathsf{x}[\mathsf{e}_1]$ (e_1 not a value). By IH on e_1: $\Omega \triangleright \mathsf{e}_1 \xrightarrow{\bar{\mathsf{a}_1}}{}^*_{\text{ctx}} \Omega' \triangleright \mathsf{f}_{1\lightning}$. Lift through context $\mathsf{K} = \mathsf{x}[[\cdot]]$.

Case $\mathsf{e} = \mathsf{x}[\mathsf{n}] \leftarrow \mathsf{v}$ (both index and value are values). Similar to get case. By Rule $e - \mathsf{set} - \in$ or Rule $e - \mathsf{set} - \notin$.

Case $\mathsf{e} = \mathsf{x}[\mathsf{e}_1] \leftarrow \mathsf{e}_2$ (at least one non-value). By IH on the leftmost non-value sub-expression, lift through appropriate context.

Case $\mathsf{e} = \mathsf{let}\ \mathsf{x} = \mathsf{f}\ \mathsf{in}\ \mathsf{e}_0$**.** By Rule $e - \mathsf{let} - \mathsf{f}$: $\Omega \triangleright \mathsf{let}\ \mathsf{x} = \mathsf{f}\ \mathsf{in}\ \mathsf{e}_0 \xrightarrow{\varepsilon} \Omega \triangleright \mathsf{e}_0[\mathsf{f}/\mathsf{x}]$. Lift to multi-step.

Case $\mathsf{e} = \mathsf{let}\ \mathsf{x} = \mathsf{e}_1\ \mathsf{in}\ \mathsf{e}_2$ (e_1 not final). By IH on e_1: $\Omega \triangleright \mathsf{e}_1 \xrightarrow{\overline{\mathsf{a}_1}}{}^*_{\mathsf{ctx}} \Omega' \triangleright \mathsf{f}_{1\lightning}$. Lift through context $\mathsf{K} = \mathsf{let}\ \mathsf{x} = [\cdot]\ \mathsf{in}\ \mathsf{e}_2$.

Case $\mathsf{e} = \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{n}\ \mathsf{in}\ \mathsf{e}_0$**.** By Rule $e - \mathsf{new}$: a fresh location is allocated. $\Omega \triangleright \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{n}\ \mathsf{in}\ \mathsf{e}_0 \xrightarrow{\cdots} \Omega' \triangleright \mathsf{e}_0[\mathsf{z}/\mathsf{x}]$. Lift to multi-step.

Case $\mathsf{e} = \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{e}_1\ \mathsf{in}\ \mathsf{e}_0$ (e_1 not a value). By IH on e_1, lift through context $\mathsf{K} = \mathsf{let}\ \mathsf{x} = \mathsf{new}\ [\cdot]\ \mathsf{in}\ \mathsf{e}_0$.

Case $\mathsf{e} = \mathsf{delete}\ \mathsf{x}$**.** By Rule $e - \mathsf{delete}$: $\Omega \triangleright \mathsf{delete}\ \mathsf{x} \xrightarrow{\cdots} \Omega' \triangleright 0$. Lift to multi-step.

Case $\mathsf{e} = \mathsf{ifz}\ 0\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2$**.** By Rule $e - \mathsf{ifz} - \text{true}$: $\Omega \triangleright \mathsf{ifz}\ 0\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2 \xrightarrow{\varepsilon} \Omega \triangleright \mathsf{e}_1$. By IH on e_1: $\Omega \triangleright \mathsf{e}_1 \xrightarrow{\overline{\mathsf{a}}}{}^*_{\mathsf{ctx}} \Omega' \triangleright \mathsf{f}_{\lightning}$. Combine steps.

Case $\mathsf{e} = \mathsf{ifz}\ \mathsf{S}(\mathsf{n})\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2$**.** By Rule $e - \mathsf{ifz} - \text{false}$: step to e_2. By IH on e_2: complete evaluation.

Case $\mathsf{e} = \mathsf{ifz}\ \mathsf{e}_0\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2$ (e_0 not a value). By IH on e_0, lift through context $\mathsf{K} = \mathsf{ifz}\ [\cdot]\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2$.

Case $\mathsf{e} = \mathsf{call}\ \mathsf{foo}\ \mathsf{v}$**.** By well-typedness, foo is defined in $\Omega.\Xi$. By Rule $e - \text{ctx} - \mathsf{call} - \text{main}$, Rule $e - \text{ctx} - \mathsf{call} - \text{notsame}$, or Rule $e - \text{ctx} - \mathsf{call} - \text{same}$: the call steps to the function body with argument substituted. By IH on the function body: complete evaluation.

Case $\mathsf{e} = \mathsf{call}\ \mathsf{foo}\ \mathsf{e}_0$ (e_0 not a value). By IH on e_0, lift through context $\mathsf{K} = \mathsf{call}\ \mathsf{foo}\ [\cdot]$.

Case $\mathsf{e} = \mathsf{return}\ \mathsf{v}$**.** By Rule $e - \text{ctx} - \mathsf{return} - \text{main}$, Rule $e - \text{ctx} - \mathsf{return} - \text{notsame}$, or Rule $e - \text{ctx} - \mathsf{return} - \text{same}$: the return pops the continuation stack and resumes the caller. By IH: complete evaluation of the resumed computation.

Case $\mathsf{e} = \mathsf{return}\ \mathsf{e}_0$ (e_0 not a value). By IH on e_0, lift through context $\mathsf{K} = \mathsf{return}\ [\cdot]$.

Case $\mathsf{e} = \mathsf{abort}()$**.** By Rule $e - \mathsf{abort}$: $\Omega \triangleright \mathsf{abort}() \xrightarrow{\lightning} \lightning \triangleright \mathsf{stuck}$. Lift to multi-step with crash.

□

Lemma 41 (Top-Level Progress)**.** *If*

(a) $\vdash \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \dashv \Xi, \xi$

Then $\exists \Omega\ \mathsf{f}_{\lightning}\ \overline{\mathsf{a}}$,

(i) $\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \overset{\bar{\mathsf{a}}}{\Longrightarrow} \Omega \triangleright \mathsf{f}_{\lightning}$

(ii) $\Omega.\Xi = \Xi$

(iii) $\Omega.\xi = \xi$

.

Proof. By Assumption (a) and inversion of Rule $t-\mathrm{prog}$:

- $\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}}$
- $\Xi \downarrow = \Gamma_0$
- $\mathsf{main} \in \mathrm{dom}(\Gamma_0)$
- $\Gamma_0 \vdash \mathsf{call\ main\ 0} : \mathbb{N} \to \mathbb{N}$

Construct the initial configuration: $\Omega_0 = \xi; \Xi; [\cdot]\,; \mathsf{comp}; [\cdot]\,; [\cdot]\,; [\cdot]$.

By Rule $T\mathrm{empty}\Delta$: $\Xi \vdash [\cdot] : \Gamma_0$. By Rule $t-\mathrm{prog}-\mathrm{runtime}$: $\vdash \Omega_0 \triangleright \mathsf{call\ main\ 0} : \mathbb{N} \to \mathbb{N}$.

By Lemma 40: $\exists \Omega'\ \mathsf{f}_{\lightning}\ \bar{\mathsf{a}}$ such that $\Omega_0 \triangleright \mathsf{call\ main\ 0} \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathsf{ctx}} \Omega' \triangleright \mathsf{f}_{\lightning}$.

By Rule $e-\mathrm{wprog}$ or Rule $e-\mathrm{wprog}-\lightning$: $\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \overset{\bar{\mathsf{a}}'}{\Longrightarrow} \Omega' \triangleright \mathsf{f}_{\lightning}$.

The library and component library are preserved throughout execution: $\Omega'.\Xi = \Xi$ and $\Omega'.\xi = \xi$. □

Lemma 42 (Base TMS via Monitor)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(b) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'\gamma'$

(c) $T_{TMS} \simeq_{\delta_{MS}} \Omega.\Delta$

Then $\exists \boldsymbol{a}\ \delta'_{MS}\ T_{TMS}{}'$,

(i) $\delta_{MS} \subseteq \delta'_{MS}$

(ii) $\theta_{\delta'_{MS}}(\mathsf{a}) \cong \boldsymbol{a}$

(iii) $\vdash T_{TMS} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{TMS}{}'$

(iv) $T_{TMS}{}' \simeq_{\delta'_{MS}} \Omega'.\Delta$

.

Proof. First, we note

- $\Omega = \Phi; \mathsf{t}; \Psi$
- $\Phi = \xi; \Xi; \overline{\mathsf{K}}$
- $\Psi = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta$

Induction on $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'\gamma'$.

Case $e-\mathsf{delete}$**:** If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\vdash \Phi; \mathsf{t}; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), \Delta_2 \triangleright \mathsf{delete}\ \mathsf{x} : \tau$

(H_3) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), \Delta_2$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\underline{\textit{Dealloc}}\ \delta'_{\text{MS}}(\ell) \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \text{☣}; \mathsf{n}), \Delta_2$

Apply Lemma 30 (TMS-Δ-split) on Assumption (H_3):

(F_1) $T_{\text{TMS1}} \simeq_{\delta_{MS}} \Delta_1$

(F_2) $\{\underline{\ell}\} \cup \emptyset \simeq_{\delta_{MS}} \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), [\cdot]$

(F_3) $T_{\text{TMS2}} \simeq_{\delta_{MS}} \Delta_2$

(F_4) $T_{\text{TMS}} = T_{\text{TMS1}} \cup \{\underline{\ell}\} \cup T_{\text{TMS2}}$

Invert Assumption (F_2) to conclude $\delta_{\text{MS}}(\ell) = \underline{\ell}$. Choose $\boldsymbol{a} = \textbf{Dealloc}\ \underline{\ell}$ and $\delta'_{\text{MS}} = \delta_{\text{MS}}$.

Goal (i), $\delta_{\text{MS}} \subseteq \delta_{\text{MS}}$, follows trivially.

For $\underline{\textit{Dealloc}}\ \delta'_{\text{MS}}(\ell) \cong \textbf{Dealloc}\ \underline{\ell}$ (Goal (ii)), apply Rule TMS-Dealloc-Authentic.

The inversion yields two cases:

Case Cons-Agree**:** If

(F_1) $\vdash \Phi; \mathsf{t}; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \square; \mathsf{n}), \Delta_2 \triangleright \mathsf{delete}\ \mathsf{x} : \tau$

(F_2) $T_{\text{TMS1}} \simeq_{\delta_{MS}} \Delta_1$

(F_3) $\delta_{\text{MS}}(\ell) = \underline{\ell}$

(F_4) $T_{\text{TMS2}} \simeq_{\delta_{MS}} \Delta_2$

(F_5) $T_{\text{TMS}} = T_{\text{TMS1}} \cup \{\underline{\ell}\} \cup T_{\text{TMS2}}$

then $\exists T_{\text{TMS}}{}'$,

(i) $\vdash T_{\text{TMS}} \overset{\textbf{Dealloc}\ \underline{\ell}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(ii) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \text{☣}; \mathsf{n}), \Delta_2$

Choose $T_{\text{TMS}}{}' = T_{\text{TMS1}} \cup T_{\text{TMS2}}$.

Goal (i), $\vdash T_{\text{TMS}} \overset{\textbf{Dealloc}\ \underline{\ell}}{\rightsquigarrow} T_{\text{TMS}}{}'$, follows immediately by Rule TMS-Dealloc.

Goal (ii), $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \text{☣}; \mathsf{n}), \Delta_2$, follows immediately by Rule Poison-Agree.

Case Poison-Agree**:** If

(F_1) $\vdash \Phi; \mathsf{t}; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \text{☣}; \mathsf{n}), \Delta_2 \triangleright \mathsf{delete}\ \mathsf{x} : \tau$

(F_2) $T_{\text{TMS1}} \simeq_{\delta_{MS}} \Delta_1$

(F_3) $\delta_{\text{MS}}(\ell) = \underline{\ell}$

(F_4) $T_{\text{TMS2}} \simeq_{\delta_{MS}} \Delta_2$

(F_5) $T_{\text{TMS}} = T_{\text{TMS1}} \cup T_{\text{TMS2}}$

then $\exists T_{\text{TMS}}{}'$,

(i) $\vdash T_{\text{TMS}} \overset{\textbf{Dealloc}\ \underline{\ell}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(ii) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \circledast; \mathsf{n}), \Delta_2$

.

Invert Assumption (F_1) to get

(F_7) $\Xi \vdash \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \circledast; \mathsf{n}), \Delta_2 : \Gamma$

(F_8) $\Gamma \vdash \mathsf{delete}\ \mathsf{x} : \mathbb{N}$

Due to Assumption (F_7), we know $\mathsf{x} : \mathsf{ref}_1\ \mathbb{N} \notin \Gamma$.

But, that contradicts Assumption (F_8).

Case $e - \oplus$**:** If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\vdash \Omega \triangleright \mathsf{n}_1 + \mathsf{n}_2 : \tau$

(H_3) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\underline{\varepsilon} \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Omega$

Choose $\boldsymbol{a} = \boldsymbol{\varepsilon}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$.

All goals are completely trivial.

Case $e - \mathsf{let} - \mathsf{f}$**:** If

(H_1) $\gamma = [\cdot]$

(H_2) $\gamma' = [\mathsf{f}/\mathsf{x}], [\cdot]$

(H_3) $\vdash \Omega \triangleright \mathsf{let}\ \mathsf{x} = \mathsf{f}\ \mathsf{in}\ \mathsf{e}' : \tau$

(H_4) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\underline{\varepsilon} \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Omega$

Choose $\boldsymbol{a} = \boldsymbol{\varepsilon}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$.

All goals are completely trivial.

Case $e - \mathsf{ifz}-\text{true}$**:** If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\vdash \Omega \triangleright \mathsf{ifz\ 0\ then\ e_1\ else\ e_2} : \tau$

(H_3) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\underline{\varepsilon} \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Omega$

Choose $\boldsymbol{a} = \boldsymbol{\varepsilon}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$.

All goals are completely trivial.

Case $e - \mathsf{ifz}-$false**:** If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\vdash \Omega \triangleright \mathsf{ifz\ S(n)\ then\ e_1\ else\ e_2} : \tau$

(H_3) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\underline{\varepsilon} \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Omega$

Choose $\boldsymbol{a} = \boldsymbol{\varepsilon}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$.

All goals are completely trivial.

Case $e - \mathsf{abort}$**:** If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\vdash \Omega \triangleright \mathsf{abort()} : \tau$

(H_3) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\underline{\lightning} \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \lightning$

Choose $\boldsymbol{a} = \lightning$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = \emptyset$.

All goals are completely trivial.

Case $e - \mathsf{new}$**:** If

(H_1) $\gamma = [\cdot]$

(H_2) $\gamma' = [\mathsf{z/x}], [\cdot]$

(H_3) $\vdash \Omega : \tau$

(H_4) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega$

(H_5) $\Delta \vdash \ell$ *fresh*

(H_6) $\Delta \vdash \mathsf{z}$ *fresh*

(H_7) $\mathsf{H}^{\mathsf{t}'} = \mathsf{H}^{\mathsf{t}} \ll \mathsf{n}$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\underline{\mathit{Alloc}}\ \delta'_{\text{MS}}(\ell)\ \underline{n} \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \mathsf{z} \mapsto (\ell; \mathsf{t}; \square; \mathsf{n}), \Delta$

Let $\underline{\ell}$ be such that $\operatorname{dom} \delta_{\text{MS}} \vdash \underline{\ell}$ *fresh*. Choose $\boldsymbol{a} = \boldsymbol{Alloc}\ \underline{\ell}$, $\delta'_{\text{MS}} = \delta_{\text{MS}} \cup \{\ell \mapsto \underline{\ell}\}$, and $T_{\text{TMS}}{}' = \{\underline{\ell}\} \cup T_{\text{TMS}}$. The goals become:

(i) $\delta_{\text{MS}} \subseteq \delta_{\text{MS}} \cup \{\ell \mapsto \underline{\ell}\}$

(ii) $\underline{\mathit{Alloc}\ \ell\ n} \cong \boldsymbol{Alloc}\ \underline{\ell}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{Alloc}\ \underline{\ell}}{\rightsquigarrow} \{A : T_{\text{TMS}}.A \cup \{\underline{\ell}\}, F : T_{\text{TMS}}.F\}$

(iv) $\{\underline{\ell}\} \cup T_{\text{TMS}} \simeq_{\delta_{MS} \cup \{\ell \mapsto \underline{\ell}\}} \mathsf{z} \mapsto (\ell; \mathsf{t}; \square; \mathsf{n}), \Delta$

Goal (i) is trivial.

Goal (ii) follows immediately by Rule TMS-Alloc-Authentic.

Goal (iii) follows by Rule TMS-Alloc.

Goal (iv) follows by Rule Cons-Agree.

Case $e - \mathsf{get} - \in$**:** The proof is analogous to the next, up to the nested case analysis.

Case $e - \mathsf{get} - \notin$**:** If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\vdash \Phi; \mathsf{t}; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2 \triangleright \mathsf{x}[\mathsf{n}] : \tau$

(H_3) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2$

(H_4) $\ell \in \operatorname{dom} \mathsf{H}^{\mathsf{t}} \implies \mathsf{v} = \mathsf{H}^{\mathsf{t}}(\ell + \mathsf{n})$

(H_5) $\ell \notin \operatorname{dom} \mathsf{H}^{\mathsf{t}} \implies \mathsf{v} = 1729$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\underline{\mathit{Use}}\ \delta'_{\text{MS}}(\ell)\ \underline{n} \cong_{\delta'_{\text{MS}}} \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2$

Apply Lemma 30 (TMS-Δ-split) on Assumption (H_3):

(F_1) $T_{\text{TMS}1} \simeq_{\delta_{MS}} \Delta_1$

(F_2) $\{\underline{\ell}\} \cup \emptyset \simeq_{\delta_{MS}} \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), [\cdot]$

(F_3) $T_{\text{TMS2}} \simeq_{\delta_{MS}} \Delta_2$

(F_4) $T_{\text{TMS}} = T_{\text{TMS1}} \cup \{\underline{\ell}\} \cup T_{\text{TMS2}}$

Invert Assumption (F_2) to conclude $\delta_{\text{MS}}(\ell) = \underline{\ell}$. Choose $\boldsymbol{a} = \boldsymbol{Use}\ \underline{\ell}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$. The goals become:

(i) $\delta_{\text{MS}} \subseteq \delta_{\text{MS}}$

(ii) $\underline{Use\ \ell\ n} \cong \boldsymbol{Use}\ \underline{\ell}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{Use}\ \underline{\ell}}{\rightsquigarrow} T_{\text{TMS}}$

(iv) $T_{\text{TMS}} \simeq_{\delta_{\text{MS}}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2$

All goals follow easily.

Case $e - \mathsf{set} - \in$**:** The proof is analogous to the next, up to the nested case analysis.

Case $e - \mathsf{set} - \notin$**:** If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\vdash \Phi; \mathsf{t}; \mathsf{H}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2 \triangleright \mathsf{x}[\mathsf{n}] : \tau$

(H_3) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2$

(H_4) $\ell + \mathsf{n} \in \operatorname{dom} \mathsf{H}^{\mathsf{t}} \implies \mathsf{H}^{\mathsf{t}'} = \mathsf{H}^{\mathsf{t}}\,(\ell + \mathsf{n} \mapsto \mathsf{v})$

(H_5) $\ell + \mathsf{n} \notin \operatorname{dom} \mathsf{H}^{\mathsf{t}} \implies \mathsf{H}^{\mathsf{t}'} = \mathsf{H}^{\mathsf{t}}$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\underline{Use\ \ell\ n} \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2$

Apply Lemma 30 (TMS-Δ-split) on Assumption (H_3):

(F_1) $T_{\text{TMS1}} \simeq_{\delta_{MS}} \Delta_1$

(F_2) $\{\underline{\ell}\} \cup \emptyset \simeq_{\delta_{MS}} \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), [\cdot]$

(F_3) $T_{\text{TMS2}} \simeq_{\delta_{MS}} \Delta_2$

(F_4) $T_{\text{TMS}} = T_{\text{TMS1}} \cup \{\underline{\ell}\} \cup T_{\text{TMS2}}$

Invert Assumption (F_2) to conclude $\delta_{\text{MS}}(\ell) = \underline{\ell}$. Choose $\boldsymbol{a} = \boldsymbol{Use}\ \underline{\ell}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$. The goals become:

(i) $\delta_{\text{MS}} \subseteq \delta_{\text{MS}}$

(ii) $\underline{Use\ \ell\ n} \cong_{\delta'_{\text{MS}}} \boldsymbol{Use}\ \underline{\ell}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{Use}\ \underline{\ell}}{\rightsquigarrow} T_{\text{TMS}}$

(iv) $T_{\text{TMS}} \simeq_{\delta'_{\text{MS}}} \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2$

All goals follow easily.

□

Lemma 43 (Ctx TMS via Monitor)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(b) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\mathsf{a}}_{ctx} \Omega' \triangleright \mathsf{e}'\gamma'$

(c) $T_{TMS} \simeq_{\delta_{MS}} \Omega.\Delta$

Then $\exists \boldsymbol{a}\ \delta'_{MS}\ T_{TMS}{}'$,

(i) $\delta_{MS} \subseteq \delta'_{MS}$

(ii) $\theta_{\delta'_{MS}}(\mathsf{a}) \cong \boldsymbol{a}$

(iii) $\vdash T_{TMS} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{TMS}{}'$

(iv) $T_{TMS}{}' \simeq_{\delta'_{MS}} \Omega'.\Delta$

.

Proof. Induction on Assumption (b).

Case $e - \text{ctx}$**:** Analogous to next case.

Case $e - \text{ctx} - \mathsf{stuck}$**:** If

(H_1) $\mathsf{e}\gamma = \mathsf{K}[\mathsf{e}_0]$

(H_2) $\mathsf{e}'\gamma' = \mathsf{K}[\mathsf{e}'_0]$

(H_3) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(H_4) $\Omega \triangleright \mathsf{e}_0 \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'_0$

(H_5) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega.\Delta$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\theta_{\delta'_{\text{MS}}}(\mathsf{a}) \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Omega'.\Delta$

Rewrite Assumption (H_3) using Assumption (H_1):

(H_6) $\vdash \Omega \triangleright \mathsf{K}[\mathsf{e}_0] : \tau$

Apply Lemma 26 (Runtime Typing Decomposition) on Assumption (H_6):

(H_7) $\vdash \Omega \triangleright \mathsf{K} : \tau_{\mathsf{e}} \to \tau \ @\ \Gamma_{[\cdot]}$

(H_8) $\Gamma_{[\cdot]} \vdash \mathsf{e}_0 : \tau_{\mathsf{e}}$

Apply Lemma 42 (Base TMS via Monitor) on Assumptions (H_4), (H_5) and (H_8), giving us exactly the assumptions necessary to prove the goals.

Case $e - \text{ctx} - \mathsf{call}-\text{main}$**:** If

(H_1) $\mathsf{e}\gamma = \mathsf{K}[\mathsf{call\ main\ 0}]$

(H_2) $\gamma' = [\mathsf{0}/\mathsf{x}]$

(H_3) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(H_4) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega.\Delta$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\theta_{\delta'_{\text{MS}}}(\mathsf{Start}; \mathsf{ctx}) \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Omega'.\Delta$

Note that $\theta_{\delta_{\text{MS}}}(\mathsf{Start}; \mathsf{ctx}) = \underline{\varepsilon}$ for any δ_{MS} whatsoever. So, instantiate $\boldsymbol{a} = \boldsymbol{\varepsilon}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$. All goals follow easily.

Case $e - \text{ctx} - \mathsf{call} - \text{same}$**:** Mostly similar to next case.

Case $e - \text{ctx} - \mathsf{call} - \text{notsame}$**:** If

(H_1) $\mathsf{e}\gamma = \mathsf{K}[\mathsf{call\ foo\ v}]$

(H_2) $\gamma' = [\mathsf{v}/\mathsf{x}]$

(H_3) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(H_4) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega.\Delta$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\theta_{\delta'_{\text{MS}}}(\mathsf{Call}\ \mathbf{c}\ \mathsf{foo\ v}; \mathsf{t}) \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Omega'.\Delta$

Note that $\theta_{\delta_{\text{MS}}}(\mathsf{Call}\ \mathbf{c}\ \mathsf{foo\ v}; \mathsf{t}) = \underline{\varepsilon}$ for any δ_{MS} and t whatsoever. So, instantiate $\boldsymbol{a} = \boldsymbol{\varepsilon}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$. All goals follow easily.

Case $e - \text{ctx} - \mathsf{return}-\text{main}$**:** If

(H_1) $\mathsf{e}\gamma = \mathsf{K}[\mathsf{return\ v}]$

(H_2) $\mathsf{e}'\gamma' = \mathsf{v}$

(H_3) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(H_4) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega.\Delta$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\theta_{\delta'_{\text{MS}}}(\mathsf{End\ v}; \mathsf{ctx}) \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Omega'.\Delta$

Note that $\theta_{\delta_{\text{MS}}}(\mathsf{End\ v; ctx}) = \underline{\varepsilon}$ for any δ_{MS} whatsoever. So, instantiate $\boldsymbol{a} = \boldsymbol{\varepsilon}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$. All goals follow easily.

Case e – ctx – return – same**:** Mostly similar to next case.

Case e – ctx – return – notsame**:** If

(H_1) $\mathsf{e}\gamma = \mathsf{K}'[\mathsf{return\ v}]$

(H_2) $\mathsf{e}\gamma' = \mathsf{K}[\mathsf{v}]$

(H_3) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(H_4) $T_{\text{TMS}} \simeq_{\delta_{MS}} \Omega.\Delta$

Then $\exists \boldsymbol{a}\ \delta'_{\text{MS}}\ T_{\text{TMS}}{}'$,

(i) $\delta_{\text{MS}} \subseteq \delta'_{\text{MS}}$

(ii) $\theta_{\delta'_{\text{MS}}}(\mathsf{Ret\ c\ v; t}) \cong \boldsymbol{a}$

(iii) $\vdash T_{\text{TMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{TMS}}{}'$

(iv) $T_{\text{TMS}}{}' \simeq_{\delta'_{\text{MS}}} \Omega'.\Delta$

Note that $\theta_{\delta_{\text{MS}}}(\mathsf{Ret\ c\ v; t}) = \underline{\varepsilon}$ for any δ_{MS} and t whatsoever. So, instantiate $\boldsymbol{a} = \boldsymbol{\varepsilon}$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$. All goals follow easily.

□

Lemma 44 (Steps TMS via Monitor)**.** *If*

(a) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(b) $\Omega \triangleright \mathsf{e}\gamma \overset{\bar{\mathsf{a}}}{\rightsquigarrow}{}^*_{ctx}\ \Omega' \triangleright \mathsf{e}'\gamma'$

(c) $T_{TMS} \simeq_{\delta_{MS}} \Omega.\Delta$

Then $\exists \overline{\boldsymbol{a}}\ \delta'_{MS}\ T_{TMS}{}'$,

(i) $\delta_{MS} \subseteq \delta'_{MS}$

(ii) $\theta_{\delta'_{MS}}(\bar{\mathsf{a}}) \cong^* \overline{\boldsymbol{a}}$

(iii) $\vdash T_{TMS} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{TMS}{}'$

(iv) $T_{TMS}{}' \simeq_{\delta'_{MS}} \Omega'.\Delta$

.

Proof. Induction on Assumption (b).

$\Omega' = \Omega$**,** $\mathsf{e}' = \mathsf{e}$**, and** $\bar{\mathsf{a}} = [\cdot]$**:** This is the base case where no execution step occurs. Instantiate $\overline{\boldsymbol{a}} = [\cdot]$, $\delta'_{\text{MS}} = \delta_{\text{MS}}$, and $T_{\text{TMS}}{}' = T_{\text{TMS}}$. Goal (i) holds since $\delta_{\text{MS}} \subseteq \delta_{\text{MS}}$ by reflexivity of subset. Goal (ii) holds since $\theta_{\delta_{\text{MS}}}([\cdot]) = [\cdot]$ by definition of trace projection on empty traces, and then $[\cdot] \cong^* [\cdot]$ follows by Rule TMS-Refl-Authentic. Goal (iii) holds by Rule TMS-Refl, noting the monitor state is unchanged. Goal (iv) follows directly from Assumption (c).

Induction step: The inductive hypothesis is as follows. If

(A_1) $\vdash \Omega_{\mathsf{H}} \triangleright \mathsf{e}_{\mathsf{H}}\gamma_{\mathsf{H}} : \tau$

(A_2) $T_{\mathrm{TMS}\,IH} \simeq_{\delta^0_{\mathrm{MS}}} \Omega_{\mathsf{H}}.\Delta$

then $\exists \overline{\boldsymbol{a}}_{IH}\ \delta^{IH}_{\mathrm{MS}}\ T_{\mathrm{TMS}}{}'_{IH}$,

(IH_1) $\delta^0_{\mathrm{MS}} \subseteq \delta^{IH}_{\mathrm{MS}}$

(IH_2) $\theta_{\delta^{IH}_{\mathrm{MS}}}(\overline{\mathsf{a}}) \cong \overline{\boldsymbol{a}}_{IH}$

(IH_3) $\vdash T_{\mathrm{TMS}\,IH} \overset{\overline{\boldsymbol{a}}_{IH}}{\rightsquigarrow}{}^{*} T_{\mathrm{TMS}}{}'_{IH}$

(IH_4) $T_{\mathrm{TMS}}{}'_{IH} \simeq_{\delta^{IH}_{\mathrm{MS}}} \Omega'.\Delta$

Note that above, anything not bound by $\exists$ is universally quantified, besides τ and Ω'. While our remaining context (excluding the inductive hypothesis) is: If

(H_1) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\mathsf{a}_0}_{\mathrm{ctx}} \Omega_0 \triangleright \mathsf{e}_0\gamma_0$

(H_2) $\Omega_0 \triangleright \mathsf{e}_0\gamma_0 \xrightarrow{\overline{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}'\gamma'$

(H_3) $\vdash \Omega \triangleright \mathsf{e}\gamma : \tau$

(H_4) $T_{\mathrm{TMS}} \simeq_{\delta_{MS}} \Omega.\Delta$

Then $\exists \overline{\boldsymbol{a}}\ \delta'_{\mathrm{MS}}\ T_{\mathrm{TMS}}{}'$,

(i) $\delta_{\mathrm{MS}} \subseteq \delta'_{\mathrm{MS}}$

(ii) $\theta_{\delta'_{\mathrm{MS}}}(\mathsf{a}_0 \cdot \overline{\mathsf{a}}) \cong \overline{\boldsymbol{a}}$

(iii) $\vdash T_{\mathrm{TMS}} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{\mathrm{TMS}}{}'$

(iv) $T_{\mathrm{TMS}}{}' \simeq_{\delta'_{\mathrm{MS}}} \Omega'.\Delta$

Use Lemma 43 (Ctx TMS via Monitor) with Assumptions (H_1), (H_3) and (H_4), obtaining:

(F_1) $\delta_{\mathrm{MS}} \subseteq \delta^0_{\mathrm{MS}}$

(F_2) $\theta_{\delta^0_{\mathrm{MS}}}(\mathsf{a}_0) \cong \boldsymbol{a}_0$

(F_3) $\vdash T_{\mathrm{TMS}} \overset{\boldsymbol{a}_0}{\rightsquigarrow} T_{\mathrm{TMS0}}$

(F_4) $T_{\mathrm{TMS0}} \simeq_{\delta^0_{\mathrm{MS}}} \Omega_0.\Delta$

Discharge the inductive hypothesis with Assumption (F_4) and the result of applying Lemma 37 (Ctx Preservation) using Assumptions (H_1) and (H_3):

(G_1) $\delta^0_{\mathrm{MS}} \subseteq \delta^{IH}_{\mathrm{MS}}$

(G_2) $\theta_{\delta^{IH}_{\mathrm{MS}}}(\overline{\mathsf{a}}) \cong \overline{\boldsymbol{a}}_{IH}$

(G_3) $\vdash T_{\mathrm{TMS0}} \overset{\overline{\boldsymbol{a}}_{IH}}{\rightsquigarrow}{}^{*} T_{\mathrm{TMS}}{}^{IH}$

(G_4) $T_{\mathrm{TMS}}{}^{IH} \simeq_{\delta^{IH}} \Omega'.\Delta$

We solve our goals by instantiating $\overline{\boldsymbol{a}} = \boldsymbol{a} \cdot \overline{\boldsymbol{a}}_{IH}$, $\delta'_{\mathrm{MS}} = \delta^{IH}_{\mathrm{MS}}$, and $T_{\mathrm{TMS}}{}' = T_{\mathrm{TMS}}{}^{IH}$. Goal (i) follows by transitivity using Assumptions (F_1) and (G_1).

Similarily for Goal (ii) using Assumptions (F_2) and (G_2) and Lemma 29 (Filter Weaken).

Goal (iii) also by transitivity using Assumptions (F_3) and (G_3).

Goal (iv), again, by transitivity using Assumptions (F_4) and (G_4), making use of Lemma 28 (Store Agree Weaken).

□

Theorem 2 ($\mathrm{L}_{\mathrm{tms}}$ is TMS via Monitor)**.** *If*

(a) $\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \overset{\overline{\mathsf{a}}}{\Longrightarrow} \Omega \triangleright \mathsf{f}_{\lightning}$

Then

(i) $\theta^*_{\delta_{ms}}(\overline{\mathsf{a}}) \in \mathrm{tmsafe}$

.

Proof. Apply Lemma 15 ($\mathtt{TMS}(\overline{a^{\mathrm{ms}}})$ implies $\overline{a^{\mathrm{ms}}} \in \mathrm{tmsafe}$) on the goal, what is left to show is:

(i) $\mathtt{TMS}(\theta_{\delta_{ms}}(\overline{\mathsf{a}}))$

Inverting Assumption (a) and omitting the spurious cases, we get

(H_1) $\vdash \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} : \xi, \Xi$

(H_2) $\xi; \Xi; \mathsf{comp}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{call\ main\ 0} \xrightarrow{\overline{\mathsf{a}}}{}^*_{\mathsf{ctx}}\ \Omega \triangleright \mathsf{f}_{\lightning}$

Invert Assumption (H_1):

(H_3) $\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}}$

(H_4) $\Xi \downarrow = \Gamma_0$

(H_5) $\mathsf{main} \in \Gamma_0$

(H_6) $\Gamma_0 \vdash \Xi\ \mathrm{ok}$

(H_7) $\Gamma_0 \vdash \mathsf{call\ main\ 0} : \mathbb{N}$

Apply Lemma 22 (Static Typing implies Runtime Typing (Toplevel)) on Assumptions (H_4) and (H_7):

(H_8) $\vdash \xi; \Xi; [\cdot]\,; \mathsf{comp}; [\cdot]^{\mathsf{ctx}}\,; [\cdot]^{\mathsf{comp}}\,; [\cdot] \triangleright \mathsf{call\ main\ 0} : \mathbb{N}$

Note that by definition, we have:

(H_9) $\emptyset \simeq_{\delta_{ms}} [\cdot]$

With Assumptions (H_2), (H_8) and (H_9) apply Lemma 44 (Steps TMS via Monitor):

(H_{10}) $\delta_{\mathrm{ms}} \subseteq \delta'_{\mathrm{ms}}$

(H_{11}) $\theta_{\delta'_{\text{MS}}}(\overline{\mathsf{a}}) \cong^* \overline{\boldsymbol{a}}$

(H_{12}) $\vdash \emptyset \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{\text{TMS}}$

(H_{13}) $T_{\text{TMS}} \simeq_{\delta'_{\text{MS}}} \Omega.\Delta$

Immediately, we instantiate our goal with $\delta'_{\text{ms}}, \overline{\boldsymbol{a}}$, and T_{TMS}. Make use of Lemma 29 (Filter Weaken) on the first part, and Lemma 28 (Store Agree Weaken) on the second part, noting that Assumption (H_{10}). Now, Assumptions (H_{11}) and (H_{12}) are precisely what is required in Definition 31 (Trace is temporal memory safe via monitor). □

4.3 Target Language

4.3.1 Syntax

$$
\begin{array}{rl}
\textit{Final Result}\ \mathbf{f} ::= & \mathbf{v} \mid \mathbf{x} \qquad \textit{May be a Result}\ \mathbf{f}_{\lightning} ::= \mathbf{f} \mid \mathbf{stuck} \\
\textit{Expressions}\ \mathbf{e} ::= & \mathbf{f}_{\lightning} \mid \mathbf{e_1} \oplus \mathbf{e_2} \mid \mathbf{x}[\mathbf{e}] \mid \mathbf{let}\ \mathbf{x} = \mathbf{e_1}\ \mathbf{in}\ \mathbf{e_2} \mid \mathbf{x}[\mathbf{e_1}] \leftarrow \mathbf{e_2} \\
& \mid \mathbf{let}\ \mathbf{x} = \mathbf{new}\ \mathbf{e_1}\ \mathbf{in}\ \mathbf{e_2} \mid \mathbf{delete}\ \mathbf{x} \mid \mathbf{return}\ \mathbf{e} \mid \mathbf{call}\ \mathbf{foo}\ \mathbf{e} \\
& \mid \mathbf{ifz}\ \mathbf{e_1}\ \mathbf{then}\ \mathbf{e_2}\ \mathbf{else}\ \mathbf{e_3} \mid \mathbf{abort}() \mid \mathbf{x}\ \mathbf{is}\ \text{☣} \\
& \mid \langle \mathbf{e_1}; \mathbf{e_2} \rangle \mid \pi_1\ \mathbf{e} \mid \pi_2\ \mathbf{e} \mid \mathbf{e}\ \mathbf{has}\ \tau \qquad \text{where } \oplus \in \{+, -, \times, <\} \\
\textit{Functions}\ \mathbf{F} ::= & \mathbf{let}\ \mathbf{foo}\ \mathbf{x} := \mathbf{e} \qquad \textit{Types}\ \tau ::= \mathbb{N} \mid \mathbb{N} \times \mathbb{N} \\
\textit{Values}\ \mathbf{v} ::= & \langle \mathbf{n_1}, \mathbf{n_2} \rangle \mid \mathbf{n} \in \mathbb{N} \qquad \textit{References}\ \ell \in \mathbb{N} \\
\textit{Eval.Ctx.}\ \mathbf{K} ::= & [\cdot] \mid \mathbf{K} \oplus \mathbf{e} \mid \mathbf{v} \oplus \mathbf{E} \mid \mathbf{x}[\mathbf{K}] \mid \mathbf{let}\ \mathbf{x} = \mathbf{K}\ \mathbf{in}\ \mathbf{e} \\
& \mid \mathbf{x}[\mathbf{K}] \leftarrow \mathbf{e} \mid \mathbf{x}[\mathbf{v}] \leftarrow \mathbf{K} \mid \mathbf{let}\ \mathbf{x} = \mathbf{new}\ \mathbf{K}\ \mathbf{in}\ \mathbf{e} \\
& \mid \langle \mathbf{K}; \mathbf{e} \rangle \mid \langle \mathbf{n}; \mathbf{K} \rangle \mid \mathbf{ifz}\ \mathbf{K}\ \mathbf{then}\ \mathbf{e_1}\ \mathbf{else}\ \mathbf{e_2} \mid \pi_1\ \mathbf{K} \mid \pi_2\ \mathbf{K} \\
& \mid \mathbf{K}\ \mathbf{has}\ \tau \mid \mathbf{call}\ \mathbf{foo}\ \mathbf{K} \mid \mathbf{return}\ \mathbf{K} \\
\textit{Variables}\ \mathbf{x} \mid \mathbf{y} \mid \mathbf{foo} \mid \ldots & \qquad \textit{Poison}\ \rho ::= \square \mid \text{☣} \\
\textit{Sandbox Tag}\ \mathbf{t} ::= & \mathbf{ctx} \mid \mathbf{comp} \\
\textit{Store}\ \boldsymbol{\Delta} ::= & [\cdot] \mid \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}), \boldsymbol{\Delta} \qquad \textit{Communication}\ \mathbf{c} ::= ? \mid ! \mid \varnothing \\
\textit{Cont. Stack}\ \overline{\mathbf{K}} ::= & [\cdot] \mid (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}} \qquad \textit{Library}\ \boldsymbol{\Xi} ::= [\cdot] \mid \mathbf{F}, \boldsymbol{\Xi} \\
\textit{Relevant}\ \xi ::= & [\cdot] \mid \mathbf{foo}, \xi \qquad \textit{Heaps}\ \mathbf{H} ::= [\cdot] \mid \mathbf{H} :: \mathbf{n} \qquad \textit{State}\ \boldsymbol{\Omega} ::= \boldsymbol{\Phi}; \mathbf{t}; \boldsymbol{\Psi} \\
\textit{Flow State}\ \boldsymbol{\Phi} ::= & \xi; \boldsymbol{\Xi}; \overline{\mathbf{K}} \qquad \textit{Memory State}\ \boldsymbol{\Psi} ::= \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \boldsymbol{\Delta} \\
\textit{Programs}\ \mathbf{prog}\ \boldsymbol{\Xi}_{\mathbf{ctx}}\ \boldsymbol{\Xi}_{\mathbf{comp}} & \qquad \textit{Substitutions}\ \gamma ::= [\mathbf{v}/\mathbf{x}], \gamma \mid [\cdot]
\end{array}
$$

Figure 33: Syntax of L_{ms}

The target language is very similar to the source language presented in the previous section. However, it does contain simple, non-nested pairs and dynamic type checks.

4.3.2 Dynamic Semantics

$$\boxed{\text{dom } \Xi = \mathbf{foo}, \ldots, \mathbf{bar}} \quad \text{„Collect function names."}$$

$$\frac{}{\text{dom } [\cdot] = [\cdot]}\ (\Xi\text{-dom-}[\cdot]) \qquad \frac{\text{dom } \Xi = \mathbf{D}}{\text{dom } (\mathbf{let\ foo\ x} : \tau_\lambda := \mathbf{e}), \Xi = \mathbf{foo}, \mathbf{D}}\ (\Xi\text{-dom-cons})$$

$$\boxed{\Xi \equiv \Xi_1 \cup \Xi_2} \quad \text{„Merging } \mathrm{L_{ms}} \text{ libraries."}$$

$$\frac{\Xi = \Xi_1}{\Xi \equiv \Xi_1 \cup [\cdot]}\ (\text{lib-merge-empty}) \qquad \frac{\Xi \equiv \Xi_1 \cup \Xi_2}{\mathbf{F}, \Xi \equiv \Xi_1 \cup \mathbf{F}, \Xi_2}\ (\text{lib-merge-cons})$$

$$\boxed{\Xi_1 \bowtie \Xi_2 = \Xi} \quad \text{„Syntactically linking } \mathrm{L_{ms}} \text{ libraries."}$$

$$\frac{\text{dom } \Xi_1 \cap \text{dom } \Xi_2 = \emptyset \qquad \Xi \equiv \Xi_1 \cup \Xi_2}{\Xi_1 \bowtie \Xi_2 = \Xi}\ (\text{syntactic-plugging})$$

Figure 34: $\mathrm{L_{ms}}$ plugging of libraries and collecting of function names.

$$\begin{aligned} \textit{Base Events } \mathbf{a_b} &::= \mathbf{Alloc}\ \ell\ \mathbf{v} \mid \mathbf{Dealloc}\ \ell \mid \mathbf{Get}\ \ell\ \mathbf{v} \mid \mathbf{Set}\ \ell\ \mathbf{v}\ \mathbf{v}' \mid \lightning \\ &\quad \mid \mathbf{Call\ c\ foo\ v} \mid \mathbf{Ret\ c\ v} \mid \mathbf{Start} \mid \mathbf{End\ v} \\ \textit{Events } \mathbf{a} &::= \varepsilon \mid \mathbf{a_b}; \mathbf{t} \end{aligned}$$

Figure 35: Events of $\mathrm{L_{ms}}$.

$\boxed{\Omega \triangleright e \xrightarrow{a} \Omega' \triangleright e'}$ „Expression e evaluates under configuration Ω to e' and new configuration Ω', emitting event a."

$$\frac{n_1 \oplus n_2 = n_3}{\Omega \triangleright n_1 \oplus n_2 \xrightarrow{\varepsilon} \Omega \triangleright n_3} \quad (e - \oplus)$$

$$\frac{\Psi = H^{ctx}; H^{comp}; \Delta_1, x \mapsto (\ell; t; \rho; m), \Delta_2 \qquad \ell + n \in \text{dom } H^t}{\Phi; t'; \Psi \triangleright x[n] \xrightarrow{\text{Get } \ell \text{ n;t}} \Phi; t'; \Psi \triangleright H^t(\ell + n)} \quad (e - \text{get} - \in)$$

$$\frac{\Psi = H^{ctx}; H^{comp}; \Delta_1, x \mapsto (\ell; t; \rho; m), \Delta_2 \qquad \ell \notin \text{dom } H^t}{\Phi; t'; \Psi \triangleright x[n] \xrightarrow{\text{Get } \ell \text{ n;t}} \Phi; t'; \Psi \triangleright 1729} \quad (e - \text{get} - \notin)$$

$$\frac{\begin{array}{c}\Psi = H^{ctx}; H^{comp}; \Delta_1, x \mapsto (\ell; ctx; \rho; n), \Delta_2 \qquad H^{ctx'} = H^{ctx}(\ell + n \mapsto v) \\ \Psi' = H^{ctx'}; H^{comp}; \Delta_1, x \mapsto (\ell; ctx; \rho; n), \Delta_2\end{array}}{\Phi; ctx; \Psi \triangleright x[n] \leftarrow v \xrightarrow{\text{Set } \ell \text{ n v;ctx}} \Phi; ctx; \Psi' \triangleright v} \quad (e - \text{set} - ctx)$$

$$\frac{\begin{array}{c}\Psi = H^{ctx}; H^{comp}; \Delta_1, x \mapsto (\ell; comp; \rho; n), \Delta_2 \qquad H^{comp'} = H^{comp}(\ell + n \mapsto v) \\ \Psi' = H^{ctx}; H^{comp'}; \Delta_1, x \mapsto (\ell; comp; \rho; n), \Delta_2\end{array}}{\Phi; comp; \Psi \triangleright x[n] \leftarrow v \xrightarrow{\text{Set } \ell \text{ n v;comp}} \Phi; comp; \Psi' \triangleright v} \quad (e - \text{set} - comp)$$

$$\frac{}{\Omega \triangleright \text{let } x = f \text{ in } e \xrightarrow{\varepsilon} \Omega \triangleright e[f/x]} \quad (e - \text{let} - f)$$

$$\frac{\begin{array}{c}\Psi = H^{ctx}; H^{comp}; \Delta_1, x \mapsto (\ell; t; \square; n), \Delta_2 \\ \Psi = H^{ctx}; H^{comp}; \Delta_1, x \mapsto (\ell; t; \circledast; n), \Delta_2\end{array}}{\Phi; t; \Psi \triangleright \text{delete } x \xrightarrow{\text{Dealloc } \ell;t} \Phi; t; \Psi' \triangleright 0} \quad (e - \text{delete})$$

$$\frac{\begin{array}{c}\Delta \vdash \ell \textit{ fresh} \qquad \Delta \vdash z \textit{ fresh} \qquad H^{ctx'} = H^{ctx} \ll n \\ \Psi = H^{ctx'}; H^{comp}; z \mapsto (\ell; ctx; \square; n), \Delta\end{array}}{\Phi; ctx; H^{ctx}; H^{comp}; \Delta \triangleright \text{let } x = \text{new } n \text{ in } e \xrightarrow{\text{Alloc } \ell \text{ n;ctx}} \Phi; ctx; \Psi \triangleright e[z/x]} \quad (e - \text{new} - ctx)$$

$$\frac{\begin{array}{c}\Delta \vdash \ell \textit{ fresh} \qquad \Delta \vdash z \textit{ fresh} \qquad H^{comp'} = H^{comp} \ll n \\ \Psi = H^{ctx}; H^{comp'}; z \mapsto (\ell; ctx; \square; n), \Delta\end{array}}{\Phi; comp; H^{ctx}; H^{comp}; \Delta \triangleright \text{let } x = \text{new } n \text{ in } e \xrightarrow{\text{Alloc } \ell \text{ n;comp}} \Phi; comp; \Psi \triangleright e[z/x]} \quad (e - \text{new} - comp)$$

$$\frac{}{\Omega \triangleright \text{ifz } 0 \text{ then } e_1 \text{ else } e_2 \xrightarrow{\varepsilon} \Omega \triangleright e_1} \quad (e - \text{ifz} - \text{true})$$

$$\frac{n \neq 0}{\Omega \triangleright \text{ifz } n \text{ then } e_1 \text{ else } e_2 \xrightarrow{\varepsilon} \Omega \triangleright e_2} \quad (e - \text{ifz} - \text{false})$$

$$\frac{}{\Omega \triangleright \text{abort}() \xrightarrow{\lightning} \Omega \triangleright \text{stuck}} \quad (e - \text{abort})$$

Figure 36: Evaluation of L_{ms} expressions.

$$\frac{}{\Omega \triangleright \pi_1\langle n_1; n_2\rangle \xrightarrow{\varepsilon} \Omega \triangleright n_1}\ (e-\pi_1) \qquad \frac{}{\Omega \triangleright \pi_2\langle n_1; n_2\rangle \xrightarrow{\varepsilon} \Omega \triangleright n_2}\ (e-\pi_2)$$

$$\frac{\Psi = H; \Delta \qquad \Delta = \Delta_1, x \mapsto (\ell; t; ☣; n), \Delta_2}{\Phi; t'; \Psi \triangleright x \text{ is } ☣ \xrightarrow{\varepsilon} \Phi; t'; \Psi \triangleright 0}\ (e - x \text{ is } ☣ - \text{yes})$$

$$\frac{\Psi = H; \Delta \qquad \Delta = \Delta_1, x \mapsto (\ell; t; \square; n), \Delta_2}{\Phi; t'; \Psi \triangleright x \text{ is } ☣ \xrightarrow{\varepsilon} \Phi; t'; \Psi \triangleright 1}\ (e - x \text{ is } ☣ - \text{no})$$

$$\frac{}{\Omega \triangleright n \text{ has } \mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 0}\ (e - n \text{ has } \mathbb{N}) \qquad \frac{}{\Omega \triangleright \langle n_1, n_2\rangle \text{ has } \mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 1}\ (e - pair - \text{has } \mathbb{N})$$

$$\frac{}{\Omega \triangleright x \text{ has } \mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 1}\ (e - x \text{ has } \mathbb{N}) \qquad \frac{}{\Omega \triangleright n \text{ has } \mathbb{N} \times \mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 1}\ (e - n \text{ has } \mathbb{N} \times \mathbb{N})$$

$$\frac{}{\Omega \triangleright \langle n_1, n_2\rangle \text{ has } \mathbb{N} \times \mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 0}\ (e - pair - \text{has } \mathbb{N} \times \mathbb{N}) \qquad \frac{}{\Omega \triangleright x \text{ has } \mathbb{N} \times \mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 1}\ (e - x \text{ has } \mathbb{N} \times \mathbb{N})$$

Figure 37: Evaluation of L_{ms} expressions, continued.

γ acts as postfix and binds anything before it up to $\triangleright$, so $\text{let } x = e_1 \text{ in } e_2\gamma$ is $(\text{let } x = e_1 \text{ in } e_2)\gamma$.

$\boxed{\Omega \triangleright K[e] \xrightarrow{a}_{ctx} \Omega' \triangleright K[e']}$ „Given an evaluation context K and an expression e, it evaluates under configuration Ω to e' and new configuration Ω' in context K, emitting event a."

$$\frac{\Omega \triangleright e \xrightarrow{a} \Omega' \triangleright e'}{\Omega \triangleright K[e] \xrightarrow{a}_{ctx} \Omega' \triangleright K[e']}\ (e-\text{ctx}) \qquad \frac{\Omega \triangleright e \xrightarrow{\lightning} \Omega' \triangleright \text{stuck}}{\Omega \triangleright K[e] \xrightarrow{\lightning}_{ctx} \Omega' \triangleright \text{stuck}}\ (e-\text{ctx}-\text{stuck})$$

$$\frac{\begin{array}{c}\Omega = \xi; \Xi; [\cdot]; \text{comp}; \Psi \qquad \Xi = \Xi_1, (\text{let main } x : \tau_\lambda := e), \Xi_2 \\ \Omega' = \xi; \Xi; \text{ctx}; K^{\text{main}}, [\cdot]; \Psi\end{array}}{\Omega \triangleright K[\text{call main } v] \xrightarrow{\text{Start;comp}}_{ctx} \Omega' \triangleright e[x/v]}\ (e-\text{ctx}-\text{call}-\text{main})$$

$$\frac{\begin{array}{c}\Omega = \xi; \Xi; \overline{K}; t; \Psi \qquad \Xi = \Xi_1, (\text{let foo } x : \tau_\lambda := e), \Xi_2 \\ \text{foo} \in_{\neg t} \xi \qquad \rho(\neg t) = c \qquad \Omega' = \xi; \Xi; K^{\text{foo}}, \overline{K}; \neg t; \Psi\end{array}}{\Omega \triangleright K[\text{call foo } v] \xrightarrow{\text{Call c foo v;t}}_{ctx} \Omega' \triangleright e[x/v]}\ (e-\text{ctx}-\text{call}-\text{notsame})$$

$$\frac{\begin{array}{c}\Omega = \xi; \Xi; \overline{K}; t; \Psi \qquad \Xi = \Xi_1, (\text{let foo } x : \tau_\lambda := e), \Xi_2 \\ \text{foo} \in_{t} \xi \qquad \Omega' = \xi; \Xi; K^{\text{foo}}, \overline{K}; t; \Psi\end{array}}{\Omega \triangleright K[\text{call foo } v] \xrightarrow{\text{Call } \varnothing \text{ foo v;t}}_{ctx} \Omega' \triangleright e[x/v]}\ (e-\text{ctx}-\text{call}-\text{same})$$

$$\frac{}{\xi; \Xi; [\cdot]^{\text{main}}, [\cdot]; \text{ctx}; \Psi \triangleright K'[\text{return } v] \xrightarrow{\text{End v;t}}_{ctx} \xi; \Xi; [\cdot]; \text{comp}; \Psi \triangleright v}\ (e-\text{ctx}-\text{return}-\text{notsame})$$

$$\frac{\text{foo} \in_{\neg t} \xi \qquad \rho(\neg t) = c; t}{\xi; \Xi; K^{\text{foo}}, \overline{K}; t; \Psi \triangleright K'[\text{return } v] \xrightarrow{\text{Ret c v;t}}_{ctx} \xi; \Xi; \overline{K}; \neg t; \Psi \triangleright K[v]}\ (e-\text{ctx}-\text{return}-\text{same})$$

$$\frac{\text{foo} \in_{t} \xi}{\xi; \Xi; K^{\text{foo}}, \overline{K}; t; \Psi \triangleright K'[\text{return } v] \xrightarrow{\text{Ret } \varnothing \text{ v;t}}_{ctx} \xi; \Xi; \overline{K}; t; \Psi \triangleright K[v]}$$

$\boxed{\rho(t) = c}$ „Returns either ? or ! depending on t."

$$\frac{}{\rho(\text{ctx}) = ?}\ (\text{comm-ctxtocomp}) \qquad \frac{}{\rho(\text{comp}) = !}\ (\text{comm-comptoctx})$$

$\boxed{\neg t = t'}$ „Negation of t."

$$\frac{}{\neg\text{ctx} = \text{comp}}\ (\text{neg-ctx}) \qquad \frac{}{\neg\text{comp} = \text{ctx}}\ (\text{neg-comp})$$

Figure 38: Contextual Evaluation of L_{ms} expressions.

$\boxed{\Omega \triangleright \mathbf{e} \xrightarrow{\bar{\mathbf{a}}}{}^{*}_{\text{ctx}} \Omega' \triangleright \mathbf{e}'}$ „Expression $\mathbf{e}$ evaluates under configuration Ω to $\mathbf{e}'$ and new configuration Ω', emitting list of events $\bar{\mathbf{a}}$."

$$\frac{}{\Omega \triangleright \mathbf{f}_{\lightning} \xrightarrow{[\cdot]}{}^{*}_{\text{ctx}} \Omega \triangleright \mathbf{f}_{\lightning}} \quad (et - \text{refl})$$

$$\frac{\Omega \triangleright \mathbf{e} \xrightarrow{\mathbf{a}}_{\text{ctx}} \Omega' \triangleright \mathbf{e}' \qquad \Omega' \triangleright \mathbf{e}' \xrightarrow{\bar{\mathbf{a}}}{}^{*}_{\text{ctx}} \Omega'' \triangleright \mathbf{e}'' \qquad \mathbf{a} \neq \varepsilon}{\Omega \triangleright \mathbf{e} \xrightarrow{\mathbf{a}\cdot\bar{\mathbf{a}}}{}^{*}_{\text{ctx}} \Omega'' \triangleright \mathbf{e}''} \quad (et - \text{trans-important})$$

$$\frac{\Omega \triangleright \mathbf{e} \xrightarrow{\varepsilon}_{\text{ctx}} \Omega' \triangleright \mathbf{e}' \qquad \Omega' \triangleright \mathbf{e}' \xrightarrow{\bar{\mathbf{a}}}{}^{*}_{\text{ctx}} \Omega'' \triangleright \mathbf{e}''}{\Omega \triangleright \mathbf{e} \xrightarrow{\bar{\mathbf{a}}}{}^{*}_{\text{ctx}} \Omega'' \triangleright \mathbf{e}''} \quad (et - \text{trans-unimportant})$$

$\boxed{\Omega \triangleright \mathbf{e} \xrightarrow{\bar{\mathbf{a}}}{}^{n}_{\text{ctx}} \Omega' \triangleright \mathbf{e}'}$ „Expression $\mathbf{e}$ evaluates in n steps under configuration Ω to $\mathbf{e}'$ and new configuration Ω', emitting list of events $\bar{\mathbf{a}}$."

$$\frac{}{\Omega \triangleright \mathbf{e} \xrightarrow{\varepsilon}_{\text{ctx}}{}^{0} \Omega \triangleright \mathbf{e}} \quad (en - \text{refl})$$

$$\frac{\Omega \triangleright \mathbf{e} \xrightarrow{\mathbf{a}}_{\text{ctx}} \Omega' \triangleright \mathbf{e}' \qquad \Omega' \triangleright \mathbf{e}' \xrightarrow{\bar{\mathbf{a}}}_{\text{ctx}}{}^{n} \Omega'' \triangleright \mathbf{e}'' \qquad \mathbf{a} \neq \varepsilon}{\Omega \triangleright \mathbf{e} \xrightarrow{\mathbf{a}\cdot\bar{\mathbf{a}}}_{\text{ctx}}{}^{(n+1)} \Omega'' \triangleright \mathbf{e}''} \quad (en - \text{trans-important})$$

$$\frac{\Omega \triangleright \mathbf{e} \xrightarrow{\varepsilon}_{\text{ctx}} \Omega' \triangleright \mathbf{e}' \qquad \Omega' \triangleright \mathbf{e}' \xrightarrow{\bar{\mathbf{a}}}_{\text{ctx}}{}^{n} \Omega'' \triangleright \mathbf{e}''}{\Omega \triangleright \mathbf{e} \xrightarrow{\bar{\mathbf{a}}}_{\text{ctx}}{}^{(n+1)} \Omega'' \triangleright \mathbf{e}''} \quad (en - \text{trans-unimportant})$$

Figure 39: Trace prefix generation given a L_{ms} program using the reflexive-transitive closure.

$\boxed{\mathbf{prog}\ \Xi_{\mathbf{ctx}}\ \Xi_{\mathbf{comp}} \overset{\bar{\mathbf{a}}}{\Longrightarrow} \Omega \triangleright \mathbf{f}}$ „Run L_{ms} program $\mathbf{prog}\ \Xi_{\mathbf{ctx}}\ \Xi_{\mathbf{comp}}$, giving dynamic state Ω and emitting trace $\bar{\mathbf{a}}$."

$$\frac{\Xi = \Xi_{\mathbf{ctx}} \bowtie \Xi_{\mathbf{comp}} \qquad \mathbf{main} \notin \xi \qquad \xi = \mathrm{dom}\ \Xi_{\mathbf{comp}} \qquad \xi; \Xi; [\cdot]\,; \mathbf{t}; [\cdot]\,; [\cdot] \triangleright \mathbf{call\ main\ 0} \xrightarrow{\bar{\mathbf{a}}}{}^{*}_{\text{ctx}} \Omega \triangleright \mathbf{f}_{\lightning}}{\mathbf{prog}\ \Xi_{\mathbf{ctx}}\ \Xi_{\mathbf{comp}} \overset{\bar{\mathbf{a}}}{\Longrightarrow} \Omega \triangleright \mathbf{f}_{\lightning}} \quad (e - \text{wprog})$$

$$\frac{\Xi = \Xi_{\mathbf{ctx}} \bowtie \Xi_{\mathbf{comp}} \qquad \mathbf{main} \notin \xi \qquad \xi = \mathrm{dom}\ \Xi_{\mathbf{comp}} \qquad \xi; \Xi; [\cdot]\,; \mathbf{t}; [\cdot]\,; [\cdot] \triangleright \mathbf{call\ main\ 0} \xrightarrow{\bar{\mathbf{a}}\cdot\lightning\,;\mathbf{t}}{}^{*}_{\text{ctx}} \Omega \triangleright \mathbf{f}_{\lightning}}{\mathbf{prog}\ \Xi_{\mathbf{ctx}}\ \Xi_{\mathbf{comp}} \overset{\bar{\mathbf{a}}\cdot\lightning\,;\mathbf{t}}{\Longrightarrow} \Omega \triangleright \mathbf{f}_{\lightning}} \quad (e - \text{wprog} - \lightning)$$

Figure 40: Running a whole L_{ms} program.

4.3.3 Proofs and Auxiliary Lemmas

Lemma 45 (Determinism of Step)**.** *If*

(a) $\Omega \triangleright \mathbf{e}\gamma \xrightarrow{\mathbf{a}_1} \Omega' \triangleright \mathbf{e}_1\gamma'$

(b) $\mathbf{\Omega} \triangleright \mathbf{e}\gamma \xrightarrow{\mathbf{a_2}} \mathbf{\Omega}'' \triangleright \mathbf{e_2}\gamma''$

Then

(i) $\mathbf{e_1} = \mathbf{e_2}$

(ii) $\gamma' = \gamma''$

(iii) $\mathbf{\Omega}' = \mathbf{\Omega}''$

(iv) $\mathbf{a_1} = \mathbf{a_2}$

Proof. By induction on Assumption (a). □

Lemma 46 (Determinism of Ctx-Step)**.** *If*

(a) $\mathbf{\Omega} \triangleright \mathbf{e}\gamma \xrightarrow{\mathbf{a_2}}_{ctx} \mathbf{\Omega}' \triangleright \mathbf{e_1}\gamma'$

(b) $\mathbf{\Omega} \triangleright \mathbf{e}\gamma \xrightarrow{\mathbf{a_1}}_{ctx} \mathbf{\Omega}'' \triangleright \mathbf{e_2}\gamma''$

Then

(i) $\mathbf{e_1} = \mathbf{e_2}$

(ii) $\gamma' = \gamma''$

(iii) $\mathbf{\Omega}' = \mathbf{\Omega}''$

(iv) $\mathbf{a_1} = \mathbf{a_2}$

Proof. By induction on Assumption (a) making use of Lemma 45 (Determinism of Step). □

Lemma 47 (Determinism of n-Steps)**.** *If*

(a) $\mathbf{\Omega} \triangleright \mathbf{e}\gamma \xrightarrow{\overline{\mathbf{a_1}}}^{n}_{ctx} \mathbf{\Omega}' \triangleright \mathbf{e_1}\gamma'$

(b) $\mathbf{\Omega} \triangleright \mathbf{e}\gamma \xrightarrow{\overline{\mathbf{a_2}}}^{n}_{ctx} \mathbf{\Omega}'' \triangleright \mathbf{e_2}\gamma''$

Then

(i) $\mathbf{e_1} = \mathbf{e_2}$

(ii) $\gamma' = \gamma''$

(iii) $\mathbf{\Omega}' = \mathbf{\Omega}''$

(iv) $\overline{\mathbf{a_1}} = \overline{\mathbf{a_2}}$

Proof. By induction on Assumption (a) making use of Lemma 46 (Determinism of Ctx-Step). □

Lemma 48 (Determinism of Steps)**.** *If*

(a) $\mathbf{\Omega} \triangleright \mathbf{e}\gamma \xrightarrow{\overline{\mathbf{a_1}}}^{*}_{ctx} \mathbf{\Omega}' \triangleright \mathbf{f}^{\mathbf{1}}_{\lightning}\gamma'$

(b) $\mathbf{\Omega} \triangleright \mathbf{e}\gamma \xrightarrow{\overline{\mathbf{a_2}}}^{*}_{ctx} \mathbf{\Omega}'' \triangleright \mathbf{f}^{\mathbf{2}}_{\lightning}\gamma''$

Then

(i) $\mathbf{f}^1_{\ell} = \mathbf{f}^2_{\ell}$

(ii) $\gamma' = \gamma''$

(iii) $\mathbf{\Omega}' = \mathbf{\Omega}''$

(iv) $\overline{\mathbf{a_1}} = \overline{\mathbf{a_2}}$

Proof. By induction on Assumption (a) making use of Lemma 46 (Determinism of Ctx-Step). □

4.4 Robust TMS Preserving Compiler

4.4.1 Compiler

$\boxed{[\![\mathsf{e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} = \mathbf{e}}$ „Compile $\mathsf{L_{tms}}$ expression e to $\mathsf{L_{ms}}$ expression $\mathbf{e}$.

$\boxed{[\![[\mathsf{v}/\mathsf{x}]]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} = [\mathbf{v}/\mathbf{x}]}$ „Compile $\mathsf{L_{tms}}$ substitution to $\mathsf{L_{ms}}$ substitution."

$\boxed{[\![\xi]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} = \xi}$ „Compile $\mathsf{L_{tms}}$ component library to $\mathsf{L_{ms}}$ component library."

$$
\begin{array}{rcl}
[\![\mathsf{f}_{\ell}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & \mathbf{f}_{\ell} \\
[\![\mathsf{call\ foo\ e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & \mathbf{call}\ [\![\mathsf{foo}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}\ [\![\mathsf{e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
[\![\mathsf{return\ e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & \mathbf{return}\ [\![\mathsf{e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
[\![\mathsf{e_1} \oplus \mathsf{e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & [\![\mathsf{e_1}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \oplus [\![\mathsf{e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
[\![\mathsf{x}[\mathsf{e}]]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & [\![\mathsf{x}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}\,[\,[\![\mathsf{e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}\,] \\
[\![\mathsf{let\ x = e_1\ in\ e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & \mathbf{let}\ [\![\mathsf{x}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} = [\![\mathsf{e_1}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}\ \mathbf{in}\ [\![\mathsf{e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
[\![\mathsf{x}[\mathsf{e_1}] \leftarrow \mathsf{e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & [\![\mathsf{x}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}\,[\,[\![\mathsf{e_1}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}\,] \leftarrow [\![\mathsf{e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
[\![\mathsf{let\ x = new\ e_1\ in\ e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & \mathbf{let}\ [\![\mathsf{x}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} = \mathbf{new}\ [\![\mathsf{e_1}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}\ \mathbf{in}\ [\![\mathsf{e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
[\![\mathsf{delete\ x}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & \mathbf{delete}\ [\![\mathsf{x}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
[\![\mathsf{if z\ e_1\ then\ e_2\ else\ e_3}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & \mathbf{if z}\ [\![\mathsf{e_1}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
 & & \mathbf{then}\ [\![\mathsf{e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
 & & \mathbf{else}\ [\![\mathsf{e_3}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \\
 & & \\
[\![[\mathsf{v}/\mathsf{x}]]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & [\,[\![\mathsf{v}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} / [\![\mathsf{x}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}\,] \\
 & & \\
[\![[\cdot]]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & [\cdot] \\
[\![\mathsf{foo}, \xi]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} & = & \mathbf{foo}, [\![\xi]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}
\end{array}
$$

Figure 41: Compiler from $\mathsf{L_{tms}}$ to $\mathsf{L_{ms}}$.

$$\boxed{[\![\mathsf{F}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} = \mathbf{F}}$$ „Compile $\mathsf{L}_{\mathsf{tms}}$ procedures to L_{ms} procedures."

$$\begin{array}{rcl}
[\![\mathsf{let\ foo\ x} : \mathbb{N} \to \tau_{\mathsf{e}} := \mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & \mathbf{let}\ [\![\mathsf{foo}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}\ [\![\mathsf{x}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} := \\
& & \mathbf{ifz}\ [\![\mathsf{x}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}\ \mathbf{has}\ \mathbb{N}\ \mathbf{then} \\
& & \quad [\![\mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
& & \mathbf{else} \\
& & \quad \mathbf{abort}()
\end{array}$$

$$\boxed{[\![\Xi]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} = \Xi}$$ „Compile $\mathsf{L}_{\mathsf{tms}}$ libraries to L_{ms} libraries."

$$\begin{array}{rcl}
[\![[\cdot]]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & [\cdot] \\
[\![\mathsf{F}, \Xi]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & [\![\mathsf{F}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}, [\![\Xi]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}
\end{array}$$

Figure 42: Compiler from $\mathsf{L}_{\mathsf{tms}}$ components to L_{ms} components.

$$\boxed{[\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} = \mathbf{K}}$$ „Compile $\mathsf{L}_{\mathsf{tms}}$ evaluation contexts to L_{ms} evaluation contexts."

$$\begin{array}{rcl}
[\![[\cdot]]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & [\cdot] \\
[\![\mathsf{K} \oplus \mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & [\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \oplus [\![\mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
[\![\mathsf{v} \oplus \mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & [\![\mathsf{v}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \oplus [\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
[\![\mathsf{x}[\mathsf{K}]]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & [\![\mathsf{x}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} [\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}] \\
[\![\mathsf{let\ x} = \mathsf{K\ in\ e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & \mathbf{let}\ [\![\mathsf{x}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} = [\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}\ \mathbf{in}\ [\![\mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
[\![\mathsf{x}[\mathsf{K}] \leftarrow \mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & [\![\mathsf{x}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} [[\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}] \leftarrow [\![\mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
[\![\mathsf{x}[\mathsf{v}] \leftarrow \mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & [\![\mathsf{x}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} [[\![\mathsf{v}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}] \leftarrow [\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
[\![\mathsf{let\ x} = \mathsf{new\ K\ in\ e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & \mathbf{let}\ [\![\mathsf{x}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} = \mathbf{new}\ [\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}\ \mathbf{in}\ [\![\mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
[\![\mathsf{ifz\ K\ then\ e_1\ else\ e_2}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & \mathbf{ifz}\ [\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}\ \mathbf{then} \\
& & \quad [\![\mathsf{e_1}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
& & \mathbf{else} \\
& & \quad [\![\mathsf{e_2}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
[\![\mathsf{call\ foo\ K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & \mathbf{call}\ [\![\mathsf{foo}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}\ [\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} \\
[\![\mathsf{return\ K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}} & = & \mathbf{return}\ [\![\mathsf{K}]\!]^{\mathsf{L}_{\mathsf{tms}}\to\mathsf{L}_{\mathsf{ms}}}
\end{array}$$

Figure 43: Compiling $\mathsf{L}_{\mathsf{tms}}$ evaluation contexts to L_{ms} evaluation contexts.

Compiling components requires a wrapper to ensure that target contexts invoke the compiled component with the right runtime terms. For example, by adding the dynamic type check we prevent contexts from binding $\langle \mathbf{42}, \mathbf{1729} \rangle$ to $\mathbf{x}$, which is never valid for $\mathsf{L}_{\mathsf{tms}}$ components.

$\boxed{\theta_\bullet(\mathbf{a}) = \mathbf{a}}$ „Filter an L_{ms} event."

(trg-filter-context)

$$\frac{\mathbf{a_b} = \lightning \lor \mathbf{a_b} = \mathbf{Alloc}\ \ell\ \mathbf{n} \lor \mathbf{a_b} = \mathbf{Dealloc}\ \ell \lor \mathbf{a_b} = \mathbf{Get}\ \ell\ \mathbf{n} \lor \mathbf{a_b} = \mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{m} \lor \mathbf{a_b} = \mathbf{Call}\ \emptyset\ \mathbf{foo}\ \mathbf{n} \lor \mathbf{a_b} = \mathbf{Ret}\ \emptyset\ \mathbf{n}}{\theta_\bullet(\mathbf{a_b}; \mathbf{ctx}) = \varepsilon}$$

(trg-filter-unimportant)

$$\frac{}{\theta_\bullet(\varepsilon; \mathbf{t}) = \varepsilon}$$

(trg-filter-comp-start)

$$\frac{}{\theta_\bullet(\mathbf{Start}; \mathbf{comp}) = \varepsilon}$$

(trg-filter-comp-alloc)

$$\frac{}{\theta_\bullet(\mathbf{Alloc}\ \ell\ \mathbf{n}; \mathbf{comp}) = \mathbf{Alloc}\ \ell\ \mathbf{n}}$$

(trg-filter-comp-dealloc)

$$\frac{}{\theta_\bullet(\mathbf{Dealloc}\ \ell; \mathbf{comp}) = \mathbf{Dealloc}\ \ell}$$

(trg-filter-comp-get)

$$\frac{}{\theta_\bullet(\mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{comp}) = \mathbf{Get}\ \ell\ \mathbf{n}}$$

(trg-filter-comp-set)

$$\frac{}{\theta_\bullet(\mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{v}; \mathbf{comp}) = \mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{v}}$$

(trg-filter-comp-call)

$$\frac{}{\theta_\bullet(\mathbf{Call}\ \mathbf{c}\ \mathbf{foo}\ \mathbf{v}; \mathbf{t}) = \mathbf{Call}\ \mathbf{c}\ \mathbf{foo}\ \mathbf{v}; \mathbf{t}}$$

(trg-filter-comp-ret)

$$\frac{}{\theta_\bullet(\mathbf{Ret}\ \mathbf{c}\ \mathbf{v}; \mathbf{t}) = \mathbf{Ret}\ \mathbf{c}\ \mathbf{v}; \mathbf{t}}$$

(trg-filter-abort)

$$\frac{}{\theta_\bullet(\lightning; \mathbf{t}) = \lightning}$$

$\boxed{\theta^*_\bullet(\overline{\mathbf{a}}) = \overline{\mathbf{a}}'}$ „Filter an L_{ms} trace."

(trg-filter-empty)

$$\frac{}{\theta^*_\bullet([\cdot]) = [\cdot]}$$

(trg-filter-cons-relevant)

$$\frac{\theta_\bullet(\mathbf{a}) = \mathbf{a} \neq \varepsilon \qquad \theta^*_\bullet(\overline{\mathbf{a}}) = \overline{\mathbf{a}}}{\theta^*_\bullet(\mathbf{a} \cdot \overline{\mathbf{a}}) = \mathbf{a} \cdot \overline{\mathbf{a}}}$$

(trg-filter-cons-relevant)

$$\frac{\theta_\bullet(\mathbf{a}) = \varepsilon \qquad \theta^*_\bullet(\overline{\mathbf{a}}) = \overline{\mathbf{a}}}{\theta^*_\bullet(\mathbf{a} \cdot \overline{\mathbf{a}}) = \overline{\mathbf{a}}}$$

Figure 44: Filtering of L_{ms} events, getting rid of unimportant ones, for back-translation.

$\boxed{\mathsf{t} = \mathbf{t}}$ „Equality of sandboxtags between $\mathsf{L}_{\mathrm{tms}}$ and L_{ms}."

$$\frac{}{\mathsf{ctx} = \mathbf{ctx}}\ \text{(sandboxtag-ctx)} \qquad \frac{}{\mathsf{comp} = \mathbf{comp}}\ \text{(sandboxtag-comp)}$$

$\boxed{\delta(\ell) = \ell}$ „Map from $\mathsf{L}_{\mathrm{tms}}$ locations ℓ to L_{ms} locations ℓ."

$\boxed{\bar{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \bar{\mathbf{a}}}$ „The $\mathsf{L}_{\mathrm{tms}}$ trace $\bar{\mathsf{a}}$ describes the same actions as L_{ms} trace $\bar{\mathbf{a}}$. Any action in X is ignored, these are generated by the backtranslation wrapper."

$$\frac{}{[\cdot] \approx^*_{\delta;\mathsf{X}} [\cdot]}\ \text{(empty-trace-eq)} \qquad \frac{\mathsf{a_b} \notin \mathsf{X} \quad \mathsf{t} = \mathbf{t} \quad \mathsf{a_b} \approx_\delta \mathbf{a_b} \quad \bar{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \bar{\mathbf{a}}}{\mathsf{a_b}; \mathsf{t} \cdot \bar{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \mathbf{a_b}; \mathbf{t} \cdot \bar{\mathbf{a}}}\ \text{(cons-trace-eq)}$$

$$\frac{\mathsf{a_b} \in \mathsf{X} \quad \bar{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \bar{\mathbf{a}}}{\mathsf{a_b}; \mathsf{t} \cdot \bar{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \bar{\mathbf{a}}}\ \text{(ignore-cons-trace-eq)}$$

$\boxed{\mathsf{a_b} \approx_\delta \mathbf{a_b}}$ „The $\mathsf{L}_{\mathrm{tms}}$ event $\mathsf{a_b}$ describes the same action as L_{ms} event $\mathbf{a_b}$."

$$\frac{}{\mathsf{Start} \approx_\delta \mathbf{Start}}\ \text{(start-event-eq)} \qquad \frac{\mathsf{v} = \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle_{\mathcal{D}}^{\mathsf{L}_{\mathrm{ms}} \to \mathsf{L}_{\mathrm{tms}}}}{\mathsf{End}\ \mathsf{v} \approx_\delta \mathbf{End}\ \mathbf{v}}\ \text{(end-event-eq)}$$

$$\frac{[\![\mathsf{n}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{n} \quad \delta(\ell) = \ell}{\mathsf{Alloc}\ \ell\ \mathsf{n} \approx_\delta \mathbf{Alloc}\ \ell\ \mathbf{n}}\ \text{(alloc-event-eq)} \qquad \frac{\delta(\ell) = \ell}{\mathsf{Dealloc}\ \ell \approx_\delta \mathbf{Dealloc}\ \ell}\ \text{(dealloc-event-eq)}$$

$$\frac{[\![\mathsf{n}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{n} \quad \delta(\ell) = \ell}{\mathsf{Get}\ \ell\ \mathsf{n} \approx_\delta \mathbf{Get}\ \ell\ \mathbf{n}}\ \text{(get-event-eq)}$$

$$\frac{[\![\mathsf{n}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{n} \quad [\![\mathsf{v}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{v} \quad \delta(\ell) = \ell}{\mathsf{Set}\ \ell\ \mathsf{n}\ \mathsf{v} \approx_\delta \mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{v}}\ \text{(set-event-eq)}$$

$$\frac{[\![\mathsf{n}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{n} \quad [\![\mathsf{foo}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{foo} \quad \mathbf{c} = \mathbf{c}}{\mathsf{Call}\ \mathbf{c}\ \mathsf{foo}\ \mathsf{n} \approx_\delta \mathbf{Call}\ \mathbf{c}\ \mathbf{foo}\ \mathbf{n}}\ \text{(call-event-eq)}$$

$$\frac{[\![\mathsf{n}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{n} \quad \mathbf{c} = \mathbf{c}}{\mathsf{Ret}\ \mathbf{c}\ \mathsf{n} \approx_\delta \mathbf{Ret}\ \mathbf{c}\ \mathbf{n}}\ \text{(ret-event-eq)} \qquad \frac{}{\varepsilon \approx_\delta \varepsilon}\ \text{(ε-event-eq)} \qquad \frac{}{\lightning \approx_\delta \lightning}\ \text{($\lightning$-event-eq)}$$

$\boxed{\mathbf{c} = \mathbf{c}}$ „Communications are equal."

$$\frac{}{? = ?}\ \text{(? = ?)} \qquad \frac{}{! = !}\ \text{(! = !)} \qquad \frac{}{\varnothing = \varnothing}\ \text{($\varnothing = \varnothing$)}$$

Figure 45: Trace Relation from $\mathsf{L}_{\mathrm{tms}}$ to L_{ms}.

$\boxed{\delta(\ell) = \ell}$ „The $\mathsf{L}_{\mathrm{tms}}$ memory location ℓ corresponds to the L_{ms} memory location ℓ."

$\boxed{\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}}$ „The $\mathsf{L}_{\mathrm{tms}}$ state Ω agrees with L_{ms} state $\mathbf{\Omega}$. L contains locations introduced by the backtranslation wrapper, which are subsequently ignored."

(state-eq)

$$\frac{\Omega = \Phi; \mathsf{t}; \Psi \qquad \mathbf{\Omega} = \mathbf{\Phi}; \mathbf{t}; \mathbf{\Psi} \qquad \mathsf{t} = \mathbf{t} \qquad \Phi \approx \mathbf{\Phi} \qquad \Psi \approx_{\delta;\mathsf{L}} \mathbf{\Psi}}{\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}}$$

(abort-state-eq)

$$\frac{}{\lightning \approx_{\delta;\mathsf{L}} \lightning}$$

$\boxed{\Psi \approx_{\delta;\mathsf{L}} \mathbf{\Psi}}$ „The $\mathsf{L}_{\mathrm{tms}}$ memory-state Ψ agrees with L_{ms} one $\mathbf{\Psi}$. L contains locations introduced by the backtranslation wrapper."

(empty-memstate-eq)

$$\frac{}{[\cdot]\,;[\cdot]\,;[\cdot] \approx_{\delta;\mathsf{L}} [\cdot]\,;[\cdot]\,;[\cdot]}$$

(comp-cons-memstate-eq)

$$\frac{\ell \notin \mathsf{L} \qquad \mathsf{n} = \mathbf{n} \qquad \rho = \rho \qquad \delta(\ell) = \ell \qquad \ell, \mathsf{n} \vdash_{\delta} \mathsf{H}^{\mathsf{comp}} \approx \mathbf{H}^{\mathbf{comp}} \qquad \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{\Delta_2}}{\mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \mathsf{x} \mapsto (\ell; \mathsf{comp}; \rho; \mathsf{n}), \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{comp}; \rho; \mathbf{n}), \mathbf{\Delta_2}}$$

(ctx-cons-memstate-eq)

$$\frac{\ell \notin \mathsf{L} \qquad \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}}{\mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \mathsf{x} \mapsto (\ell; \mathsf{ctx}; \rho; \mathsf{n}), \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}}$$

(whatever-cons-memstate-eq)

$$\frac{\ell \in \mathsf{L} \qquad \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}}{\mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}}$$

$\boxed{\ell, \mathsf{n} \vdash_{\delta} \mathsf{H}^{\mathsf{comp}} \approx \mathbf{H}^{\mathbf{comp}}}$ „The heaps $\mathsf{H}^{\mathsf{comp}}$ and $\mathbf{H}^{\mathbf{comp}}$ are related at ℓ for n memory cells."

(heap-related)

$$\frac{[\![\mathsf{n}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{n} \qquad \delta(\ell) = \ell \qquad \forall 0 \leq \mathsf{m} < \mathsf{n},\ \mathbf{0} \leq \mathbf{m} < \mathbf{n}, \mathsf{H}^{\mathsf{comp}}(\ell + \mathsf{m}) = \mathbf{H}^{\mathbf{comp}}(\ell + \mathbf{m})}{\ell, \mathsf{n} \vdash_{\delta} \mathsf{H}^{\mathsf{comp}} \approx \mathbf{H}^{\mathbf{comp}}}$$

$\boxed{\Phi \approx \mathbf{\Phi}}$ „The $\mathsf{L}_{\mathrm{tms}}$ control-flow-state Φ agrees with L_{ms} one $\mathbf{\Phi}$."

(cfstate-eq)

$$\frac{\Phi = \xi; \Xi; \overline{\mathsf{K}} \qquad \mathbf{\Phi} = \xi; \mathbf{\Xi}; \overline{\mathbf{K}} \qquad \Xi \approx \mathbf{\Xi} \qquad \overline{\mathsf{K}} \approx_{\xi} \overline{\mathbf{K}} \qquad \xi = [\![\xi]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}}}{\Phi \approx \mathbf{\Phi}}$$

$\boxed{\rho = \rho}$ „$\mathsf{L}_{\mathrm{tms}}$ poison equals L_{ms} one ρ."

(☣-equal)

$$\frac{}{☣ = ☣}$$

(□-equal)

$$\frac{}{\square = \square}$$

Figure 46: State Relation from $\mathsf{L}_{\mathrm{tms}}$ to L_{ms}. This is meant to relate the states whenever we are „inside" a component.

$\boxed{\Xi \approx \Xi}$ „The procedures of $\mathsf{L}_{\mathrm{tms}}$ agree with L_{ms} ones."

$$\frac{}{[\cdot] \approx \Xi}\ \text{(empty-commlib-lib-eq)}$$

$$\frac{[\![\mathsf{let\ foo\ x} : \tau_\lambda := \mathsf{e}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{let\ foo\ x := e} \qquad \Xi \approx \Xi_1, \Xi_2}{(\mathsf{let\ foo\ x} : \tau_\lambda := \mathsf{e}), \Xi \approx \Xi_1, (\mathbf{let\ foo\ x := e}), \Xi_2}\ \text{(cons-commlib-eq)}$$

$\boxed{\overline{\mathsf{K}} \approx_\xi \overline{\mathbf{K}}}$ „The stack of $\mathsf{L}_{\mathrm{tms}}$ continuations $\overline{\mathsf{K}}$ agrees with L_{ms} one $\overline{\mathbf{K}}$. Only checks $\mathsf{foo} \in \xi$"

$$\frac{}{[\cdot] \approx_\xi [\cdot]}\ \text{(empty-kontstack-eq)}$$

$$\frac{\mathsf{foo} \notin \xi \qquad \mathbf{foo} = [\![\mathsf{foo}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} \qquad \overline{\mathsf{K}} \approx_\xi \overline{\mathbf{K}}}{\overline{\mathsf{K}} \approx_\xi (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}}\ \text{(ign-cons-kontstack-eq)}$$

$$\frac{\mathsf{foo} \in \xi \qquad \mathsf{foo} = \mathbf{foo} \qquad \overline{\mathsf{K}} \approx_\xi \overline{\mathbf{K}}}{(\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}} \approx_\xi (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}}\ \text{(cons-kontstack-eq)}$$

Figure 47: Memory-State Relations from $\mathsf{L}_{\mathrm{tms}}$ to L_{ms}.

$\boxed{\delta(\ell) = \ell}$ „The $\mathsf{L}_{\mathrm{tms}}$ memory location ℓ corresponds to the L_{ms} memory location ℓ."

$\boxed{\boldsymbol{\Omega} \multimap_{\delta;\mathsf{L}} \Omega}$ „The $\mathsf{L}_{\mathrm{tms}}$ state Ω agrees with L_{ms} state $\boldsymbol{\Omega}$. L contains locations introduced by the backtranslation wrapper, which are subsequently ignored."

$$\frac{\begin{array}{cc}\Omega = \Phi; \mathsf{t}; \Psi & \boldsymbol{\Omega} = \boldsymbol{\Phi}; \mathbf{t}; \boldsymbol{\Psi} \\ \boldsymbol{\Phi} \multimap \Phi & \boldsymbol{\Psi} \multimap_{\delta;\mathsf{L}} \Psi\end{array}}{\boldsymbol{\Omega} \multimap_{\delta;\mathsf{L}} \Omega}\ \text{(state-qe)} \qquad \frac{}{\lightning \multimap_{\delta;\mathsf{L}} \lightning}\ \text{(abort-state-qe)}$$

$\boxed{\boldsymbol{\Psi} \multimap_{\delta;\mathsf{L}} \Psi}$ „The $\mathsf{L}_{\mathrm{tms}}$ memory-state Ψ agrees with L_{ms} one $\boldsymbol{\Psi}$. L contains locations introduced by the backtranslation wrapper."

$$\frac{\Psi \approx_{\delta;\mathsf{L}} \boldsymbol{\Psi}}{\boldsymbol{\Psi} \multimap_{\delta;\mathsf{L}} \Psi}\ \text{(memstate-qe)}$$

$\boxed{\boldsymbol{\Phi} \multimap \Phi}$ „The $\mathsf{L}_{\mathrm{tms}}$ control-flow-state Φ agrees with L_{ms} one $\boldsymbol{\Phi}$."

$$\frac{\Phi \approx \boldsymbol{\Phi}}{\boldsymbol{\Phi} \multimap \Phi}\ \text{(cfstate-qe)}$$

Figure 48: State Relation from L_{ms} to $\mathsf{L}_{\mathrm{tms}}$. This is meant to relate the states whenever we are „inside" a context.

Lemma 49 (Compiler Injective)**.** *If*

(a) $[\![\mathsf{e}]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}} = \mathbf{e}$

(b) $[\![\mathsf{e}]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}} = \mathbf{e}'$

Then

(i) $\mathbf{e} = \mathbf{e}'$

Proof. Simple induction on e. □

Lemma 50 (Substitution Commutes with $[\![\bullet]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$).

(i) $[\![\mathsf{e}[\mathsf{v}/\mathsf{x}]]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}} = [\![\mathsf{e}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}} [\![[\mathsf{v}/\mathsf{x}]]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}$

Proof. Induction on e. □

Lemma 51 (Step Forward Simulation)**.** *If*

(a) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\mathsf{a}} \Omega' \triangleright \mathsf{e}'\gamma'$

(b) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(c) $\vdash \Omega \triangleright \mathsf{e}\gamma : \mathbb{N}$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{e}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}} [\![\gamma]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}} \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright [\![\mathsf{e}']\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}} [\![\gamma']\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}$

(iii) $\mathsf{a} \approxeq_{\delta'} \mathbf{a}$

(iv) $\Omega' \approx_{\delta'} \mathbf{\Omega}'$

Proof. We proceed by induction on Assumption (a).

Case $e - \oplus$**:**

If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(H_3) $\vdash \Omega \triangleright \mathsf{n}_1 + \mathsf{n}_2 : \mathbb{N}$

(H_4) $n_3 = n_1 + n_2$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{n}_1 + \mathsf{n}_2]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright [\![\mathsf{n}_3]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(iii) $\varepsilon \approxeq_{\delta} \mathbf{a}$

(iv) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}'$

Instantiate the goal with $\delta' = \delta$, $\mathbf{\Omega}' = \mathbf{\Omega}$, and $\mathbf{a} = \varepsilon$, so that what is left to prove is:

(i) $\delta \subseteq \delta$

(ii) $\mathbf{\Omega} \triangleright \mathbf{n_1} + \mathbf{n_2} \xrightarrow{\varepsilon} \mathbf{\Omega} \triangleright \mathbf{n_3}$

(iii) $\varepsilon \approxeq_{\delta} \varepsilon$

(iv) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

Goal (i) follows immediately from reflexivity of the subset relation.

Goal (ii) follows using Rule $e - \oplus$ and Assumption (H_4).

Goal (iii) follows using Rule ε-event-eq.

Goal (iv) follows using Assumption (H_2).

Case $e - \mathsf{delete}$**:**

If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), \Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(H_3) $\vdash \Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), \Delta_2 \triangleright \mathsf{delete}\ \mathsf{x} : \mathbb{N}$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{delete}\ \mathsf{x}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright [\![0]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(iii) $\mathsf{Dealloc}\ \ell \approxeq_{\delta} \mathbf{a}$

(iv) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \text{☣}; \mathsf{n}), \Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Omega}'$

First note that $\mathbf{\Omega} = \mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}), \mathbf{\Delta_2}$, because of Assumption (H_2).

Instantiate the goal with $\delta' = \delta$, $\mathbf{\Omega}' = \mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \text{☣}; \mathbf{n}), \mathbf{\Delta_2}$, and $\mathbf{a} = \mathbf{Dealloc}\ \ell; \mathbf{t}'$ so that what is left to prove is:

(i) $\delta \subseteq \delta$

(ii) $\mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}), \mathbf{\Delta_2} \triangleright \mathbf{delete}\ \mathbf{x} \xrightarrow{\mathbf{Dealloc}\ \ell} \mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}$
$\mathbf{0}$

(iii) $\mathsf{Dealloc}\ \ell; \mathsf{t}' \approxeq_{\delta} \mathbf{Dealloc}\ \ell; \mathbf{t}'$

(iv) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \text{☣}; \mathsf{n}), \Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \text{☣}; \mathbf{n}), \mathbf{\Delta_2}$

Goal (i) follows immediately from reflexivity of the subset relation.

Goal (ii) follows using Rule $e - \mathbf{delete}$.

For Goal (iii), apply Rule dealloc-event-eq, what is left to show is $\delta(\ell) = \ell$.

Similarily for Goal (iv), apply Rule state-eq and Rule comp-cons-memstate-eq or Rule ctx-cons-memstate-eq (depending on the shape of t) „suitably" so that what is left to show is $\Phi \approx_{\delta;\mathsf{L}} \mathbf{\Phi}$, $\mathsf{H} \approx_{\delta;\mathsf{L}} \mathbf{H}$, $\Delta_1 \approx_{\delta;\mathsf{L}} \mathbf{\Delta_1}$, $\Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Delta_2}$, $[\![\mathsf{x}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} = \mathbf{x}$, and, like in the previous case, $\delta(\ell) = \ell$.

$[\![\mathsf{x}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} = \mathbf{x}$ follows by definition of the compiler, anything else follows immediately by inverting Assumption (H_2) „suitably".

Case $e - \mathsf{get} - \in$**:** Analogous to next case, up to nested case analysis.

Case $e - \mathsf{get} - \notin$**:**

If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}'), \Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(H_3) $\vdash \Phi; \mathsf{t}; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}'), \Delta_2 \triangleright \mathsf{x}[\mathsf{n}] : \mathbb{N}$

(H_4) $\ell \in \mathrm{dom}\, \mathsf{H}^{\mathsf{t}'} \implies \mathsf{v} = \mathsf{H}^{\mathsf{t}'}(\ell + \mathsf{n})$

(H_5) $\ell \notin \mathrm{dom}\, \mathsf{H}^{\mathsf{t}'} \implies \mathsf{v} = 1729$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{x}[\mathsf{n}]]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}} \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright [\![\mathsf{v}]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}}$

(iii) $\mathsf{Get}\ \ell\ \mathsf{n}; \mathsf{t}' \approxeq_{\delta} \mathbf{a}$

(iv) $\Phi; \mathsf{t}; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}'), \Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Omega}'$

First note that $\mathbf{\Omega} = \mathbf{\Phi}; \mathbf{t}; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'), \mathbf{\Delta_2}$, because of Assumption (H_2).

Instantiate the goal with $\delta' = \delta$, $\mathbf{\Omega}' = \mathbf{\Phi}; \mathbf{t}; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'), \mathbf{\Delta_2}$, and $\mathbf{a} = \mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{t}'$, so that what is left to prove is:

(i) $\delta \subseteq \delta$

(iii) $\mathsf{Get}\ \ell\ \mathsf{n}; \mathsf{t}' \approxeq_{\delta} \mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{t}'$

(iv) $\Phi; \mathsf{t}; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}'), \Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Phi}; \mathbf{t}; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'), \mathbf{\Delta_2}$

(ii) $\mathbf{\Phi}; \mathbf{t}; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'), \mathbf{\Delta_2} \triangleright \mathbf{x}[\mathbf{n}] \xrightarrow{\mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{t}'} \mathbf{\Phi}; \mathbf{t}; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'$ $\mathbf{v}$

Goal (i) follows immediately from reflexivity of the subset relation.

For Goal (iii), invert Assumption (H_2) „suitably" to obtain the assumption $\delta(\ell) = \ell$. The claim then follows by applying Rule get-event-eq.

Goal (iv) follows immediately from Assumption (H_2).

Goal (ii) is a bit technical. First apply Rules $e - \mathbf{get} - \in$ and $e - \mathbf{get} - \notin$, what is left to show is:

(v) $\ell \in \operatorname{dom} \mathbf{H}^{\mathbf{t}'} \implies \mathbf{v} = \mathbf{H}^{\mathbf{t}'}(\ell + \mathbf{n})$

(vi) $\ell \notin \operatorname{dom} \mathbf{H}^{\mathbf{t}'} \implies \mathbf{v} = \mathbf{1729}$

Now, note that $\ell \in \operatorname{dom} \mathsf{H}^{\mathsf{t}'} \Leftrightarrow \delta(\ell) = \ell \in \operatorname{dom} \mathbf{H}^{\mathbf{t}'}$. We continue with a case distinction.

Case $\ell \in \operatorname{dom} \mathsf{H}^{\mathsf{t}'}$**:**

Assumption (H_4) gives $\mathsf{v} = \mathsf{H}^{\mathsf{t}'}(\ell + \mathsf{n})$. Since $\mathbf{v} = [\![\mathsf{v}]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}}$, $\mathbf{v} = [\![\mathsf{H}^{\mathsf{t}'}(\ell + \mathsf{n})]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}}$. Using Assumption (H_2), $\mathbf{v} = \mathbf{H}^{\mathbf{t}'}(\ell + \mathbf{n})$, done.

Case $\ell \notin \operatorname{dom} \mathsf{H}^{\mathsf{t}'}$**:**

Assumption (H_5) gives $\mathsf{v} = 1729$. Since $\mathbf{v} = [\![\mathsf{v}]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}}$, $\mathbf{v} = [\![1729]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}} = \mathbf{1729}$, done.

Case $e - \mathsf{set} - \in$**:** Analogous to next case, up to nested case analysis.

Case $e - \mathsf{set} - \notin$**:**

If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}'), \Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(H_3) $\vdash \Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}'), \Delta_2 \triangleright \mathsf{x}[\mathsf{n}] : \mathbb{N}$

(H_4) $\mathsf{H}' = \mathsf{H}(\ell + \mathsf{n} \mapsto \mathsf{v})$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{x}[\mathsf{n}] \leftarrow \mathsf{v}]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}} \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright [\![\mathsf{v}]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}}$

(iii) $\mathsf{Set}\ \ell\ \mathsf{n}; \mathsf{t}' \approxeq_{\delta} \mathbf{a}$

(iv) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}'}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}'), \Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Omega}'$

First note that $\mathbf{\Omega} = \mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'), \mathbf{\Delta_2}$, because of Assumption (H_2).

Instantiate the goal with $\delta' = \delta$, $\mathbf{\Omega}' = \mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}'}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'), \mathbf{\Delta_2}$, and $\mathbf{a} = \mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{m}$, so that what is left to prove is:

(i) $\delta \subseteq \delta$

(ii) $\mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'), \mathbf{\Delta_2} \triangleright \mathbf{x}[\mathbf{n}] \leftarrow \mathbf{v} \xrightarrow{\mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{m}; \mathbf{t}'}$
$\mathbf{\Phi}; \mathbf{t}; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}'}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'), \mathbf{\Delta_2} \triangleright \mathbf{v}$

(iii) $\mathsf{Set}\ \ell\ \mathsf{n}\ \mathsf{m}; \mathsf{t}' \approxeq_{\delta} \mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{m}; \mathbf{t}'$

(iv) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}'}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}'), \Delta_2 \approx_{\delta;\mathsf{L}} \mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}'}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{t}; \rho; \mathbf{n}'), \mathbf{\Delta_2}$

Goal (i) follows immediately from reflexivity of the subset relation.

For Goal (ii), apply Rules $e - \mathbf{set} - ctx$ and $e - \mathbf{set} - comp$ depending on the shape of $\mathbf{t}$. What is left to prove is $\mathbf{H}^{\mathbf{t}'} = \mathbf{H}^{\mathbf{t}}(\ell + \mathbf{n} \mapsto \mathbf{v})$. This follows by Assumption (H_2).

Apply Rule set-event-eq on Goal (iii), what is left to show is $[\![\mathsf{n}]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}} = \mathbf{n}$, which is trivial, and $\delta(\ell) = \ell$. The latter can be obtained by a „suitable" inversion of Assumption (H_2).

Goal (iv) is assumed in Assumption (H_2).

Case $e - \mathsf{new}$**:**

If

(H_1) $\gamma = [\cdot]$

(H_2) $\gamma' = [\mathsf{z}/\mathsf{x}]$

(H_3) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(H_4) $\vdash \Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \triangleright \mathsf{let}\ \mathsf{x}\ =\ \mathsf{new}\ \mathsf{n}\ \mathsf{in}\ \mathsf{e} : \mathbb{N}$

(H_5) $\mathsf{H}^{\mathsf{t}'} = \mathsf{H}^{\mathsf{t}'} \ll \mathsf{n}$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{let}\ \mathsf{x}\ =\ \mathsf{new}\ \mathsf{n}\ \mathsf{in}\ \mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}} \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright [\![\mathsf{e}]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}} [\![[\mathsf{z}/\mathsf{x}]]\!]^{\mathsf{L}_{\mathsf{tms}} \to \mathsf{L}_{\mathsf{ms}}}$

(iii) $\mathsf{Alloc}\ \ell\ \mathsf{n}; \mathsf{t}' \approxeq_{\delta} \mathbf{a}$

(iv) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}'}; \mathsf{z} \mapsto (\ell; \mathsf{t}; \square; \mathsf{n}), \Delta \approx_{\delta;\mathsf{L}} \mathbf{\Omega}'$

First note that $\mathbf{\Psi} = \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{z} \mapsto (\ell; \mathbf{t}; \square; \mathbf{n}), \mathbf{\Delta}$, because of Assumption (H_3).

Instantiate the goal with $\delta' = \delta \cup \{\ell \mapsto \ell\}$, $\mathbf{\Omega}' = \mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{z} \mapsto (\ell; \mathbf{t}; \square; \mathbf{n}), \mathbf{\Delta}$, and $\mathbf{a} = \mathbf{Alloc}\ \ell\ \mathbf{n}; \mathbf{t}'$, so that what is left to prove is:

(i) $\delta \subseteq \delta \cup \{\ell \mapsto \ell\}$

(ii) $\mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta} \triangleright \mathbf{let\ x\ =\ new\ n\ in}\ [\![\mathsf{e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \xrightarrow{\mathbf{Alloc}\ \ell\ \mathbf{n}; \mathbf{t}'}$
$\mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}'}; \mathbf{z} \mapsto (\ell; \mathbf{t}; \square; \mathbf{n}), \mathbf{\Delta} \triangleright [\![\mathsf{e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}[\mathbf{z}/\mathbf{x}]$

(iii) $\mathsf{Alloc}\ \ell\ \mathsf{n}; \mathsf{t}' \approxeq_{\delta \cup \{\ell \mapsto \ell\}} \mathbf{Alloc}\ \ell\ \mathbf{n}; \mathbf{t}'$

(iv) $\Phi; \mathsf{t}'; \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}'}; \mathsf{z} \mapsto (\ell; \mathsf{t}; \square; \mathsf{n}), \Delta \approx_{\delta \cup \{\ell \mapsto \ell\}} \mathbf{\Phi}; \mathbf{t}'; \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}'}; \mathbf{z} \mapsto (\ell; \mathbf{t}; \square; \mathbf{n}), \mathbf{\Delta}$

Goal (i) follows by rewriting it as $\forall x, x \in \delta \implies x \in \delta \vee x \in \{\ell \mapsto \ell\}$, then just choose the left side of the disjunction.

Apply Rules $e - \mathbf{new} - ctx$ and $e - \mathbf{new} - comp$ depending on the shape of t on Goal (ii), what is left to show is $\mathbf{H^t} = \mathbf{H^t} \ll \mathbf{n}$. This follows by Assumption (H_3).

Apply Rule set-event-eq on Goal (iii), what is left to show is $[\![\mathsf{n}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} = \mathbf{n}$, follows immediately by definition of the compiler, and $(\delta \cup \{\ell \mapsto \ell\})(\ell) = \ell$, which follows by definition.

Apply Rules state-eq, comp-cons-memstate-eq and ctx-cons-memstate-eq depending on the shape of t on Goal (iv), what is left to show is $[\![\mathsf{x}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} = \mathbf{x}$, follows immediately by definition of the compiler, and $(\delta \cup \{\ell \mapsto \ell\})(\ell) = \ell$, which follows by definition.

Case $e - \mathsf{abort}$**:**

If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\Omega \approx_{\delta; \mathsf{L}} \mathbf{\Omega}$

(H_3) $\vdash \Omega \triangleright \mathsf{abort}() : \mathbb{N}$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{abort}()]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright [\![\mathsf{stuck}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}$

(iii) $\lightning; \mathsf{t} \approxeq_{\delta} \mathbf{a}$

(iv) $\lightning \approx_{\delta; \mathsf{L}} \mathbf{\Omega}'$

Instantiate the goal with $\delta' = \delta$, $\mathbf{\Omega}' = \lightning$, and $\mathbf{a} = \lightning; \mathbf{t}$. All goals go through easily.

Case $e - \mathsf{let} - \mathsf{f}$**:**

If

(H_1) $\gamma = [\cdot]$

(H_2) $\gamma' = [\mathsf{v}/\mathsf{x}]$

(H_3) $\Omega \approx_{\delta; \mathsf{L}} \mathbf{\Omega}$

(H_4) $\vdash \Omega \triangleright \mathsf{let\ x\ =\ v\ in\ e} : \mathbb{N}$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{let\ x\ =\ v\ in\ e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright [\![\mathsf{e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} [\![\mathsf{[v/x]}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}$

(iii) $\varepsilon \approxeq_{\delta} \mathbf{a}$

(iv) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}'$

Instantiate the goal with $\delta' = \delta$, $\mathbf{\Omega}' = \mathbf{\Omega}$, and $\mathbf{a} = \varepsilon$, so that what is left to prove is:

(i) $\delta \subseteq \delta$

(ii) $\mathbf{\Omega} \triangleright \mathbf{let\ x\ =\ v\ in}\ [\![\mathsf{e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \xrightarrow{\varepsilon} \mathbf{\Omega} \triangleright [\![\mathsf{e}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \mathbf{[v/x]}$

(iii) $\varepsilon \approxeq_{\delta} \varepsilon$

(iv) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

Goal (i) follows by reflexivity of the subset relation.

Goal (ii) follows by definition of Rule $e - \mathbf{let} - \mathbf{f}$.

Goal (iii) follows by definition of Rule ε-event-eq.

Goal (iv) follows by Assumption (H_3).

Case $e - \mathsf{ifz}-$true**:**

If

(H_1) $\gamma = \gamma' = [\cdot]$

(H_2) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(H_3) $\vdash \Omega \triangleright \mathsf{ifz\ 0\ then\ e_1\ else\ e_2} : \mathbb{N}$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{ifz\ 0\ then\ e_1\ else\ e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright [\![\mathsf{e_1}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}$

(iii) $\varepsilon \approxeq_{\delta} \mathbf{a}$

(iv) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}'$

Instantiate the goal with $\delta' = \delta$, $\mathbf{\Omega}' = \mathbf{\Omega}$, and $\mathbf{a} = \varepsilon$, so that what is left to prove is:

(i) $\delta \subseteq \delta$

(ii) $\mathbf{\Omega} \triangleright \mathbf{ifz\ 0\ then}\ [\![\mathsf{e_1}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}\ \mathbf{else}\ [\![\mathsf{e_2}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}} \xrightarrow{\varepsilon} \mathbf{\Omega} \triangleright [\![\mathsf{e_1}]\!]^{\mathsf{L_{tms}} \to \mathsf{L_{ms}}}$

(iii) $\varepsilon \approxeq_{\delta} \varepsilon$

(iv) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

Goal (i) follows by reflexivity of the subset relation.

Goal (ii) follows by definition of Rule $e - \mathbf{ifz}-$true.

Goal (iii) follows by definition of Rule ε-event-eq.

Goal (iv) follows by Assumption (H_2).

Case $e - \mathsf{ifz}-$false**:**

This case is completely analogous to the previous case, the only difference is the use of Rule $e - \mathbf{ifz}-$false instead of Rule $e - \mathbf{ifz}-$true.

□

Lemma 52 (Ctx Step Forward Simulation)**.** *If*

(a) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\mathsf{a}}_{ctx} \Omega' \triangleright \mathsf{e}'\gamma'$

(b) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(c) $\vdash \Omega \triangleright \mathsf{e}\gamma : \mathbb{N}$

Then $\exists \delta'\ \mathbf{\Omega}'\ \mathbf{a}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{e}\gamma]\!]^{\mathsf{L}_{tms}\to\mathsf{L}_{ms}} \xrightarrow{\mathbf{a}}_{ctx} \mathbf{\Omega}' \triangleright [\![\mathsf{e}'\gamma']\!]^{\mathsf{L}_{tms}\to\mathsf{L}_{ms}}$

(iii) $\mathsf{a} \approxeq_{\delta} \mathbf{a}$

(iv) $\Omega' \approx_{\delta;\mathsf{L}} \mathbf{\Omega}'$

Proof. Induction on Assumption (a). □

Lemma 53 (Forward Simulation)**.** *If*

(a) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\bar{\mathsf{a}}}^{*}_{ctx} \Omega' \triangleright \mathsf{f}_{\lightning}\,\gamma'$

(b) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(c) $\vdash \Omega \triangleright \mathsf{e}\gamma : \mathbb{N}$

Then $\exists \delta'\ \mathbf{\Omega}'\ \bar{\mathbf{a}}$,

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright [\![\mathsf{e}]\!]^{\mathsf{L}_{tms}\to\mathsf{L}_{ms}} [\![\gamma]\!]^{\mathsf{L}_{tms}\to\mathsf{L}_{ms}} \xrightarrow{\bar{\mathbf{a}}}^{*}_{ctx} \mathbf{\Omega}' \triangleright [\![\mathsf{f}_{\lightning}]\!]^{\mathsf{L}_{tms}\to\mathsf{L}_{ms}} [\![\gamma']\!]^{\mathsf{L}_{tms}\to\mathsf{L}_{ms}}$

(iii) $\bar{\mathsf{a}} \approxeq^{*}_{\delta'} \bar{\mathbf{a}}$

(iv) $\Omega' \approx_{\delta'} \mathbf{\Omega}'$

Proof. Induction on Assumption (a).

Case *es* − refl**:** Trivial. Witnesses are $\delta, \mathbf{\Omega}$, and $[\cdot]$, goals follow immediately by reflexivity of the respective relation.

Case *es* − trans−important**:** Very similar to next case.

Case *es* − trans−unimportant**:**

(H_1) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\mathsf{a}}_{\mathrm{ctx}} \Omega_0 \triangleright \mathsf{e}_0\gamma_0$

(H_2) $\Omega_0 \triangleright \mathsf{e}_0\gamma_0 \xrightarrow{\bar{\mathsf{a}}}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{f}_{\lightning}\,\gamma'$

With the inductive hypothesis: $\forall \delta_{\forall}\ \mathbf{\Omega}_{\forall}$, If

$\left(IH_1^{(as)}\right)$ $\Omega_0 \approx_{\delta_{\forall}} \mathbf{\Omega}_{\forall}$

$\left(IH_2^{(as)}\right)$ $\vdash \Omega_0 \triangleright \mathsf{e}_0\gamma_0 : \mathbb{N}$

Then $\exists \delta_{IH}\ \Omega_{\mathbb{H}}\ \overline{\mathbf{a}_{\mathbb{H}}}$,

(IH_1) $\delta_\forall \subseteq \delta_{IH}$

(IH_2) $\Omega_\forall \triangleright [\![\mathsf{e}_0]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} [\![\gamma_0]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \xrightarrow{\overline{\mathbf{a}_{\mathbb{H}}}}{}^*_{\mathrm{ctx}}\ \Omega_{\mathbb{H}} \triangleright [\![\mathsf{f}_{\lightning}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} [\![\gamma']\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(IH_3) $\overline{\mathsf{a}} \approxeq^*_{\delta_{IH}} \overline{\mathbf{a}_{\mathbb{H}}}$

(IH_4) $\Omega' \approx_{\delta_{IH}} \Omega_{\mathbb{H}}$

Now apply Lemma 52 (Ctx Step Forward Simulation) on Assumption (H_1) using Assumptions (b) and (c), giving us witnesses $\delta_0\ \Omega_0\ \mathbf{a}$,

(F_1) $\delta \subseteq \delta_0$

(F_2) $\Omega \triangleright [\![\mathsf{e}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} [\![\gamma]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \xrightarrow{\mathbf{a}}{}_{\mathrm{ctx}}\ \Omega_0 \triangleright [\![\mathsf{e}_0]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} [\![\gamma_0]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(F_3) $\mathsf{a} \approxeq_{\delta_0} \mathbf{a}$

(F_4) $\Omega_0 \approx_{\delta_0} \Omega_0$

Apply Lemma 37 (Ctx Preservation) on Assumptions (H_1) and (c):

(F_5) $\vdash \Omega_0 \triangleright \mathsf{e}_0\gamma_0 : \mathbb{N}$

Instantiate the inductive hypothesis with $\delta_\forall = \delta_0$ and $\Omega_\forall = \Omega_0$, provide Assumption (F_4) and Assumption (F_5), and obtain witnesses $\delta_{IH}, \Omega_{\mathbb{H}}$, and $\overline{\mathbf{a}_{\mathbb{H}}}$ such that:

(IH'_1) $\delta_0 \subseteq \delta_{IH}$

(IH'_2) $\Omega_0 \triangleright [\![\mathsf{e}_0]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} [\![\gamma_0]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \xrightarrow{\overline{\mathbf{a}_{\mathbb{H}}}}{}^*_{\mathrm{ctx}}\ \Omega_{\mathbb{H}} \triangleright [\![\mathsf{f}_{\lightning}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} [\![\gamma']\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(IH'_3) $\overline{\mathsf{a}} \approxeq^*_{\delta_{IH}} \overline{\mathbf{a}_{\mathbb{H}}}$

(IH'_4) $\Omega' \approx_{\delta_{IH}} \Omega_{\mathbb{H}}$

Our goal looks as follows: $\exists \delta'\ \Omega'\ \overline{\mathbf{a}'}$,

(i) $\delta \subseteq \delta'$

(ii) $\Omega \triangleright [\![\mathsf{e}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \xrightarrow{\overline{\mathbf{a}}}{}^*_{\mathrm{ctx}}\ \Omega' \triangleright [\![\mathsf{f}_{\lightning}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(iii) $\mathsf{a} \cdot \overline{\mathsf{a}} \approxeq^*_{\delta;\mathsf{X}} \overline{\mathbf{a}'}$

(iv) $\Omega' \approx_{\delta;\mathsf{L}} \Omega'$

Case $\mathsf{a} \neq \varepsilon$**:**

(H_1) $\mathsf{a} \neq \varepsilon$

Instantiate the goal with $\delta_{IH}, \Omega_{\mathbb{H}}, \mathbf{a} \cdot \overline{\mathbf{a}}$. Note that $\delta \subseteq \delta_{IH}$ (Goal (i)) follows by transitivity using Assumptions (F_1) and (IH'_1). Similarily for Goal (iv).

Apply Rule $et - \mathrm{trans} - \mathrm{important}$ on Goal (ii). So, what is left to show is

(iii) $\mathsf{a} \cdot \overline{\mathsf{a}} \approxeq^*_{\delta;\mathsf{X}} \mathbf{a} \cdot \overline{\mathbf{a}}$

(v) $\Omega \triangleright [\![\mathsf{e}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \xrightarrow{\mathbf{a}}{}_{\mathrm{ctx}}\ \Omega_0 \triangleright [\![\mathsf{e}_0]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(vi) $\Omega_0 \triangleright [\![\mathsf{e}_0]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \xrightarrow{\overline{\mathbf{a}}}{}^*_{\mathrm{ctx}}\ \Omega_{\mathbb{H}} \triangleright [\![\mathsf{f}_{\lightning}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

Goal (iii) follows from Assumptions (IH_3) and (F_3) using Rule constrace-eq. Goal (v) is proven by Assumption (F_2). Goal (vi) is proven by Assumption (IH_2').

Case $\mathsf{a} = \varepsilon$**:** Similar to the other case, but instantiate with $\overline{\mathbf{a}_{\mathbb{H}}}$ instead of $\mathbf{a} \cdot \overline{\mathbf{a}_{\mathbb{H}}}$ and use Rule $et - \text{trans} - \text{unimportant}$

□

Lemma 54 (Different Reduction)**.** *If*

(a) $\neg\left(\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{ctx} \Omega' \triangleright \mathsf{f}_{\lightning}\right)$

(b) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{ctx} \Omega' \triangleright \mathsf{f}'_{\lightning}$

Then

(i) $\mathsf{f}_{\lightning} \neq \mathsf{f}'_{\lightning}$

Proof. Assume $\mathsf{f}_{\lightning} = \mathsf{f}'_{\lightning}$, apply Assumption (a) to the goal $\bot$, rewrite in the goal using $\mathsf{f}_{\lightning} = \mathsf{f}'_{\lightning}$ and solve the goal by Assumption (b). □

Lemma 55 (Expression Correctness)**.** *If*

(a) $\mathbf{\Omega} \triangleright [\![\mathsf{e}]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}} [\![\gamma]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}} \xrightarrow{\bar{\mathbf{a}}}{}^{*}_{ctx} \mathbf{\Omega}' \triangleright [\![\mathsf{f}_{\lightning}]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}} [\![\gamma']\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}}$

(b) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(c) $\vdash \Omega \triangleright \mathsf{e}\gamma : \mathbb{N}$

Then $\exists \delta'\ \Omega'\ \bar{\mathsf{a}}$*,*

(i) $\delta \subseteq \delta'$

(ii) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{ctx} \Omega' \triangleright \mathsf{f}_{\lightning}\,\gamma'$

(iii) $\Omega' \approx_{\delta';\mathsf{L}} \mathbf{\Omega}'$

(iv) $\bar{\mathsf{a}} \approxeq^{*}_{\delta';\mathsf{X}} \bar{\mathbf{a}}$

Proof. We prove this by contradiction. So, we get the assumption:

(I_1) $\forall \delta'\ \Omega'\ \bar{\mathsf{a}}, \delta' \subset \delta \vee \neg\left(\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{f}_{\lightning}\,\gamma'\right) \vee \Omega' \not\approx_{\delta'} \mathbf{\Omega}' \vee \bar{\mathsf{a}} \not\approxeq^{*}_{\delta'} \bar{\mathbf{a}}$

and need to prove $\bot$. Apply Lemma 40 (Progress) on Assumption (c), so:

(I_2) $\exists \Omega'\ \mathsf{f}_{\lightning}\ \bar{\mathsf{a}}, \Omega \triangleright \mathsf{e}\gamma \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{f}_{\lightning}$

First extract the witnesses and obtain:

(H_1) $\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{f}'_{\lightning}$

Apply Lemma 53 (Forward Simulation) to Assumption (H_1) with Assumptions (b) and (c) to get:

(F_1) $\delta \subseteq \delta_v$

(F_2) $\mathbf{\Omega} \triangleright [\![\mathsf{e}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} [\![\gamma]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} \xrightarrow{\overline{\mathbf{a_v}}}{}^{*}_{\mathrm{ctx}} \mathbf{\Omega'_v} \triangleright [\![\mathsf{f}'_{\lightning}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}}$

(F_3) $\Omega' \approx_{\delta_v} \mathbf{\Omega'_v}$

(F_4) $\overline{\mathsf{a}} \approx^{*}_{\delta_v} \overline{\mathbf{a_v}}$

Use Lemma 48 (Determinism of Steps) on Assumptions (a) and (F_2), giving us:

(K_1) $[\![\mathsf{f}_{\lightning}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = [\![\mathsf{f}'_{\lightning}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}}$

(K_2) $\mathbf{\Omega'} = \mathbf{\Omega'_v}$

(K_3) $\overline{\mathsf{a}} = \overline{\mathbf{a_v}}$

Rewrite using Assumption (K_2) in Assumption (F_3), similarily Assumption (K_3) in Assumption (F_4):

(F'_3) $\Omega' \approx_{\delta_v} \mathbf{\Omega'}$

(F'_4) $\overline{\mathsf{a}} \approx^{*}_{\delta_v} \overline{\mathbf{a}}$

Specialize Assumption (I_1) for $\delta' = \delta_v$, $\Omega' = \Omega'$, and $\overline{\mathsf{a}} = \overline{\mathsf{a}}$. We proceed by case analysis:

Case $\neg(\delta \subseteq \delta_v)$**:** Apply $\neg(\delta \subseteq \delta_v)$ on our goal $\bot$, then $\delta \subseteq \delta_v$ immediately follows from Assumption (F_1).

Case $\neg\left(\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\overline{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{f}_{\lightning}\right)$**:** Using Assumption (H_1) and the assumption $\neg\left(\Omega \triangleright \mathsf{e}\gamma \xrightarrow{\overline{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{f}'_{\lightning}\right)$, we conclude $\mathsf{f}_{\lightning} \neq \mathsf{f}'_{\lightning}$ using Lemma 54 (Different Reduction). Apply Lemma 49 (Compiler Injective) on Assumption (K_1), so $\mathsf{f}_{\lightning} = \mathsf{f}'_{\lightning}$, contradicting the above.

Case $\Omega' \not\approx_{\delta_v} \mathbf{\Omega'}$**:** Immediate contradiction with Assumption (F'_3).

Case $\overline{\mathsf{a}} \not\approx^{*}_{\delta_v} \overline{\mathbf{a}}$**:** Immediate contradiction with Assumption (F'_4).

□

Lemma 56 (Component Correctness)**.** *If*

(a) $\mathbf{\Omega} \triangleright [\![\mathsf{e}]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}} \xrightarrow{\overline{\mathsf{a}}}{}^{n}_{ctx} \mathbf{\Omega'} \triangleright \mathbf{K[return\ v]}$

(b) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(c) $\vdash \Omega \triangleright \mathsf{e} : \mathbb{N}$

Then $\exists \delta'\ \Omega'\ \overline{\mathsf{a}}$

(i) $\delta \subseteq \delta'$

(ii) $\mathbf{\Omega} \triangleright \mathbf{e} \xrightarrow{\overline{\mathsf{a}}}{}^{n}_{ctx} \mathbf{\Omega'} \triangleright \mathbf{K[return\ v]}$

(iii) $\Omega' \approx_{\delta;\mathsf{L}} \mathbf{\Omega'}$

(iv) $\overline{\mathsf{a}} \approx^{*}_{\delta;\mathsf{X}} \overline{\mathbf{a}}$

Proof. We prove this by contradiction. Assume:

(I_1) $\forall \delta' \; \Omega' \; \overline{\mathsf{a}}, \neg(\delta \subseteq \delta') \vee \neg \left(\mathbf{\Omega} \triangleright \mathbf{e} \xrightarrow{\overline{\mathsf{a}}}{}^{n}_{\mathrm{ctx}} \mathbf{\Omega'} \triangleright \mathbf{K[return\ v]} \right) \vee \Omega' \not\approx_{\delta'} \mathbf{\Omega'} \vee \overline{\mathsf{a}} \not\approxeq^{*}_{\delta'} \overline{\mathbf{a}}$

and need to prove $\bot$.

Apply Lemma 40 (Progress) on the assumption that the expression is well-typed. Since the component expression e is well-typed and the configuration Ω is well-formed, there exist Ω', $\mathsf{f}_{\lightning}$, and $\overline{\mathsf{a}}$ such that:

(H_1) $\Omega \triangleright \mathsf{e} \xrightarrow{\overline{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{f}_{\lightning}$

Apply Lemma 53 (Forward Simulation) on Assumption (H_1) using the state relation and typing assumptions:

(F_1) $\delta \subseteq \delta_v$

(F_2) $\mathbf{\Omega} \triangleright [\![\mathsf{e}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \xrightarrow{\overline{\mathbf{a_v}}}{}^{*}_{\mathrm{ctx}} \mathbf{\Omega'_v} \triangleright [\![\mathsf{f}_{\lightning}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(F_3) $\Omega' \approx_{\delta_v} \mathbf{\Omega'_v}$

(F_4) $\overline{\mathsf{a}} \approxeq^{*}_{\delta_v} \overline{\mathbf{a_v}}$

Apply Lemma 48 (Determinism of Steps) on the assumed target execution and Assumption (F_2):

(D_1) $[\![\mathsf{f}_{\lightning}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} = \mathbf{K[return\ v]}$

(D_2) $\mathbf{\Omega'_v} = \mathbf{\Omega'}$

(D_3) $\overline{\mathbf{a_v}} = \overline{\mathbf{a}}$

By Assumption (D_1) and injectivity of compilation (Lemma 49 (Compiler Injective)):

(H_7) $\mathsf{f}_{\lightning} = \mathsf{K[return\ v]}$

Rewrite using Assumption (D_2) in Assumption (F_3) and Assumption (D_3) in Assumption (F_4):

(F'_3) $\Omega' \approx_{\delta_v} \mathbf{\Omega'}$

(F'_4) $\overline{\mathsf{a}} \approxeq^{*}_{\delta_v} \overline{\mathbf{a}}$

Rewrite using Assumption (H_7) in Assumption (H_1):

(H_8) $\Omega \triangleright \mathsf{e} \xrightarrow{\overline{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{K[return\ v]}$

Specialize Assumption (I_1) with $\delta' = \delta_v$, $\Omega' = \Omega'$, and $\overline{\mathsf{a}} = \overline{\mathsf{a}}$. We proceed by case analysis:

Case $\neg(\delta \subseteq \delta_v)$**:** Immediate contradiction with Assumption (F_1).

Case $\neg \left(\Omega \triangleright \mathsf{e} \xrightarrow{\overline{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{K[return\ v]} \right)$**:** This contradicts Assumption (H_8).

Case $\Omega' \not\approx_{\delta_v} \mathbf{\Omega'}$**:** Immediate contradiction with Assumption (F'_3).

Case $\overline{\mathsf{a}} \not\approxeq^{*}_{\delta_v} \overline{\mathbf{a}}$**:** Immediate contradiction with Assumption (F'_4).

□

4.4.2 Trace-based Backtranslation

$\boxed{\langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{v}}$ „Map an $\mathrm{L_{ms}}$ value to an $\mathrm{L_{tms}}$ value."

(backtrans-value)

$$\frac{}{\langle\!\langle\!\langle \mathbf{n} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{n}}$$

Figure 49: Backtranslation of $\mathrm{L_{ms}}$ values to $\mathrm{L_{tms}}$ values.

$\boxed{\langle\!\langle \bar{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}}$ „Backtranslate a non-interface trace $\bar{\mathbf{a}}$ to an $\mathrm{L_{tms}}$ expression."

(backtrans-empty)

$$\frac{}{\langle\!\langle [\cdot] \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = 42}$$

(backtrans-alloc)

$$\frac{\langle\!\langle \bar{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}}{\langle\!\langle \mathbf{Alloc}\ \ell\ \mathbf{s}; \mathbf{t} \cdot \bar{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}}$$

(backtrans-dealloc)

$$\frac{\langle\!\langle \bar{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}}{\langle\!\langle \mathbf{Dealloc}\ \ell; \mathbf{t} \cdot \bar{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}}$$

(backtrans-set)

$$\frac{\langle\!\langle \bar{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}}{\langle\!\langle \mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{v}; \mathbf{t} \cdot \bar{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}}$$

(backtrans-get)

$$\frac{\langle\!\langle \bar{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}}{\langle\!\langle \mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{t} \cdot \bar{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}}$$

(backtrans-abort)

$$\frac{}{\langle\!\langle \lightning; \mathbf{t} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{abort}()}$$

Figure 50: Trace-Based Backtranslation from $\mathrm{L_{ms}}$ backtranslation-events to $\mathrm{L_{tms}}$ terms.

$\boxed{\eth(\ell) = \mathsf{x}}$ „Map an $\mathrm{L_{tms}}$ location to an $\mathrm{L_{tms}}$ identifier."

$\boxed{{}^{!}\langle\!\langle \mathbf{a} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;\bar{\ell}} = \bar{\ell'}; \mathsf{e}}$ „Given a `VarMap` $\eth$, δ, and $\bar{\ell}$, yield another stack of locations $\bar{\ell'}$, and an $\mathrm{L_{tms}}$ expression from target !-interaction-event $\mathbf{a}$."

(backtrans-start)

$$\frac{}{{}^{!}\langle\!\langle \mathbf{Start} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\emptyset;\emptyset;[\cdot]} = [\cdot]; 42}$$

(backtrans-ret-comptoctx)

$$\frac{\delta(\ell) = \ell \qquad \mathsf{z} = \eth(\ell) \qquad \mathsf{e} = \mathsf{delete\ z}}{{}^{!}\langle\!\langle \mathbf{Ret}\ !\ \mathbf{v} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;\ell,\bar{\ell}} = \bar{\ell}; \mathsf{e}; \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}}$$

$\boxed{{}^{?}\langle\!\langle \mathbf{a} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;\bar{\ell}} = \eth'; \delta'; \bar{\ell'}; \mathsf{e}}$ „Given a `VarMap` $\eth$, δ, and $\bar{\ell}$, construct a `VarMap` $\eth'$, a δ', another stack of locations $\bar{\ell'}$, and an $\mathrm{L_{tms}}$ expression from target ?-interaction-event $\mathbf{a}$."

(backtrans-call-ctxtocomp)

$$\frac{\begin{array}{ccc} \eth, \delta \vdash \ell\ \mathit{fresh} & \eth \vdash \mathsf{z}\ \mathit{fresh} & \delta \vdash \ell\ \mathit{fresh} \\ \eth_z = \eth \cup \{\ell \mapsto \mathsf{z}\} & & \delta' = \delta \cup \{\ell \mapsto \ell\} \\ \mathsf{e} = \mathsf{let\ z = new\ 42\ in\ call}\ \langle\!\langle \mathbf{foo} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} & & \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} \end{array}}{{}^{?}\langle\!\langle \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;\bar{\ell}} = \eth_z; \delta'; \ell, \bar{\ell}; \mathsf{e}}$$

(backtrans-end-ctxtocomp)

$$\frac{\begin{array}{ccc} \eth, \delta \vdash \ell\ \mathit{fresh} & \eth \vdash \mathsf{z}\ \mathit{fresh} & \delta \vdash \ell\ \mathit{fresh} \\ \eth_z = \eth \cup \{\ell \mapsto \mathsf{z}\} & & \delta' = \delta \cup \{\ell \mapsto \ell\} \\ \mathsf{e} = \mathsf{let\ z = new\ 42\ in\ delete\ z; return}\ \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} & & \end{array}}{{}^{?}\langle\!\langle \mathbf{End}\ \mathbf{v} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;[\cdot]} = \eth_z; \delta'; [\cdot]; \mathsf{e}}$$

Figure 51: Backtranslation of interaction-events from $\mathrm{L_{ms}}$ backtranslation-events to $\mathrm{L_{tms}}$ terms.

$$\frac{}{\vdash \varepsilon \text{ non-int-}\mathbf{a}}\ \text{(non-interface-}\varepsilon\text{)} \qquad \frac{}{\vdash \mathbf{Alloc}\ \ell\ \mathbf{n}; \mathbf{t} \text{ non-int-}\mathbf{a}}\ \text{(non-interface-Alloc)}$$

$$\frac{}{\vdash \mathbf{Dealloc}\ \ell; \mathbf{t} \text{ non-int-}\mathbf{a}}\ \text{(non-interface-Dealloc)} \qquad \frac{}{\vdash \mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{t} \text{ non-int-}\mathbf{a}}\ \text{(non-interface-Get)}$$

$$\frac{}{\vdash \mathbf{Set}\ \ell\ \mathbf{n}; \mathbf{t} \text{ non-int-}\mathbf{a}}\ \text{(non-interface-Set)} \qquad \frac{}{\vdash \lightning; \mathbf{t} \text{ non-int-}\mathbf{a}}\ \text{(non-interface-}\lightning\text{)} \qquad \frac{}{\vdash [\cdot] \text{ non-int-}\overline{\mathbf{a}}}\ \text{(non-interface-trace-empty)}$$

$$\frac{\vdash \mathbf{a} \text{ non-int-}\overline{\mathbf{a}} \qquad \vdash \overline{\mathbf{a}} \text{ non-int-}\overline{\mathbf{a}}}{\vdash \mathbf{a} \cdot \overline{\mathbf{a}} \text{ non-int-}\overline{\mathbf{a}}}\ \text{(non-interface-trace-cons)}$$

Figure 52: Non-Interfacing events.

$$\boxed{{}^{!?}\langle\!\langle \mathbf{a_1} \cdot \overline{\mathbf{a}} \cdot \mathbf{a_2} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;\overline{\ell}} = \eth'; \delta'; \overline{\ell'}; \mathsf{e}_1; \mathsf{e}_2; \mathsf{e}_3}$$

„Given a `VarMap` $\eth$, δ, and $\overline{\ell}$, construct a `VarMap` $\eth'$, a δ', another stack of locations $\overline{\ell'}$, and an $\mathrm{L_{tms}}$ ensemble of expressions from target-trace $\overline{\mathbf{a}^{\mathrm{ms}}}$ which starts with a ? event and ends in a !."

$$\frac{\vdash \overline{\mathbf{a}} \text{ non-int-}\overline{\mathbf{a}} \qquad {}^{!}\langle\!\langle \mathbf{a_1} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;\overline{\ell}} = \overline{\ell'}; \mathsf{e}_1 \qquad \langle\!\langle \overline{\mathbf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}_2 \qquad {}^{?}\langle\!\langle \mathbf{a_2} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;\overline{\ell'}} = \eth'; \delta'; \overline{\ell''}; \mathsf{e}_3}{{}^{!?}\langle\!\langle \mathbf{a_1} \cdot \overline{\mathbf{a}} \cdot \mathbf{a_2} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;\overline{\ell}} = \eth'; \delta'; \overline{\ell''}; \mathsf{e}_1; \mathsf{e}_2; \mathsf{e}_3}\ \text{(subtoplevel-backtrans)}$$

Figure 53: Interaction-Trace-Based Backtranslation from $\mathrm{L_{ms}}$ backtranslation-events to $\mathrm{L_{tms}}$ terms.

$$\boxed{\langle\!\langle\!\langle \overline{\mathbf{a}^{\mathrm{ms}}} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta} = \eth'; \delta; \Xi}$$

„Top-Level backtranslation of $\overline{\mathbf{a}}$ to $\mathrm{L_{tms}}$ library Ξ."

$$\frac{\begin{array}{c} \overline{\mathbf{a}} = \overline{\mathbf{a_0}} \cdot \overline{\mathbf{a}^{\mathbf{comp}}} \cdot \overline{\mathbf{a_1}} \qquad \vdash \overline{\mathbf{a}^{\mathbf{comp}}} \text{ non-int-}\overline{\mathbf{a}} \\ \overline{\mathbf{a_0}} = \mathbf{Start} \cdot \overline{\mathbf{a}'_0} \cdot \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_1} \qquad \overline{\mathbf{a_1}} = \mathbf{Ret}\ !\ \mathbf{v_2} \cdot \overline{\mathbf{a}'_0} \cdot \mathbf{End}\ \mathbf{v_3} \\ {}^{!?}\langle\!\langle \theta^*_\bullet(\overline{\mathbf{a_0}}) \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\emptyset;\emptyset;[\cdot]} = \eth; \delta; \overline{\ell}; \mathsf{e}_0; \mathsf{e}'_0; \mathsf{let}\ \mathsf{x}_0 = \mathsf{new}\ \mathsf{n}_0\ \mathsf{in}\ \mathsf{e}''_0 \\ {}^{!?}\langle\!\langle \theta^*_\bullet(\overline{\mathbf{a_1}}) \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\eth;\delta;\overline{\ell}} = \eth'; \delta'; [\cdot]; \mathsf{e}_1; \mathsf{e}'_1; \mathsf{e}''_1 \\ \mathsf{e} = \mathsf{e}_0; \mathsf{e}'_0; \mathsf{let}\ \mathsf{x}_0 = \mathsf{new}\ \mathsf{n}_0\ \mathsf{in}\ (\mathsf{e}''_0; \mathsf{e}_1); \mathsf{e}'_1; \mathsf{e}''_1 \end{array}}{\langle\!\langle\!\langle \overline{\mathbf{a}} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\emptyset;\emptyset} = \eth'; \delta'; \mathsf{let}\ \mathsf{main}\ \mathsf{x} : \mathbb{N} \to \mathbb{N} := \mathsf{e}, [\cdot]}\ \text{(backtrans-top-level)}$$

Figure 54: Top-Level trace-based Backtranslation from $\mathrm{L_{ms}}$ trace $\overline{\mathbf{a}}$ to $\mathrm{L_{tms}}$ context Ξ.

4.4.3 Proofs and Auxiliary Lemmas

Lemma 57 ($\theta^*_\bullet$ distributive)**.**

(i) $\theta^*_\bullet(\overline{\mathbf{a_1}} \cdot \overline{\mathbf{a_2}}) = \theta^*_\bullet(\overline{\mathbf{a_1}}) \cdot \theta^*_\bullet(\overline{\mathbf{a_2}})$

Proof. Induction on $\overline{\mathbf{a_1}}$. □

Lemma 58 ($\theta_\bullet$ invert concrete)**.** *If*

(i) $\mathbf{a} = \mathbf{a_b}; \mathbf{t}$

(ii) $\forall \ell\ \mathbf{n}\ \mathbf{m}, \mathbf{t} = \mathbf{ctx} \implies \mathbf{event_b} \notin \{\mathbf{Alloc}\ \ell\ \mathbf{n}, \mathbf{Dealloc}\ \ell, \mathbf{Get}\ \ell\ \mathbf{n}, \mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{m}\}$

Then

(a) $\theta(\mathbf{a}) = \mathbf{a}$

Proof. Induction on $\overline{\mathbf{a_b}}$. □

Lemma 59 ($\xi; \Xi; [\cdot] \approx_{\delta;\mathsf{L}} \xi; \Xi; [\cdot]$ holds)**.** *If*

(a) $\Xi \approx_{\delta;\mathsf{L}} \Xi$

Then

(i) $\xi; \Xi; [\cdot] \approx_{\delta;\mathsf{L}} \xi; \Xi; [\cdot]$

Proof. By definition of $\cdot \approx_{\delta;\mathsf{L}} \cdot$ on control-flow states, we need to show that each component is related. The component library ξ is trivially equal on both sides. The library relation $\Xi \approx_{\delta;\mathsf{L}} \Xi$ holds by assumption. The empty continuation stack $[\cdot]$ is trivially related to itself. Combining these, the goal follows by the constructor for control-flow state equality. □

Lemma 60 (if $\ell \in \mathsf{L}$, then ρ doesn't matter)**.** *If*

(a) $\Psi = \mathsf{H}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2$

(b) $\Psi' = \mathsf{H}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho'; \mathsf{m}), \Delta_2$

(c) $\ell \in \mathsf{L}$

(d) $\Psi \multimap_{\delta;\mathsf{L}} \Psi$

Then

(i) $\Psi \multimap_{\delta;\mathsf{L}} \Psi'$

Proof. The set L contains locations introduced by the backtranslation wrapper, which are ignored by the state relation $\cdot \multimap_{\delta;\mathsf{L}} \cdot$. Since $\ell \in \mathsf{L}$, the binding $\mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m})$ is not compared against any target binding. Therefore, changing ρ to ρ' does not affect the relation. All other bindings in Δ_1 and Δ_2, as well as the heap H, remain unchanged, so the goal follows from the assumption $\Psi \multimap_{\delta;\mathsf{L}} \Psi$. □

Lemma 61 ($\ell \in \operatorname{dom} \delta$ decomposes Δ)**.** *If*

(a) $\ell \in \operatorname{dom} \delta$

(b) $\mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \boldsymbol{\Delta}$

Then $\exists \Delta_1\ \mathsf{x}\ \mathsf{t}\ \rho\ \mathsf{m}\ \Delta_2$,

(i) $\Delta = \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2$

Proof. Since $\ell \in \operatorname{dom} \delta$, by definition of δ there exists a corresponding target location ℓ such that $\delta(\ell) = \ell$. The state relation $\mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \boldsymbol{\Delta}$ requires that for each $\ell \in \operatorname{dom} \delta$, there is a binding in Δ containing that location. Therefore, we can decompose Δ into $\Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{m}), \Delta_2$ where the binding for x contains ℓ. □

Lemma 62 (MS-Location Generator)**.** *If*

(a) there is a set of locations L

Then $\exists \delta_{MS}$

(i) left-total, injective $\delta_{MS} : \mathsf{L} \to \underline{L}$

Proof. This can be done by just generating fresh names $\underline{\ell}$ and assign them to each $\ell \in \mathsf{L}$. □

Lemma 63 (Filters Equal)**.** *If*

(a) $\forall \ell\ \ell, \delta(\ell) = \ell \implies \delta_{MS}(\ell) = \delta_{MS}(\delta^{-1}(\ell))$

(b) $\overline{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \overline{\mathbf{a}}$

Then

(i) $\theta^*_{\delta_{MS}}(\overline{\mathsf{a}}) = \theta^*_{\delta'_{MS}}(\overline{\mathbf{a}})$

Proof. Induction on Assumption (b). □

Lemma 64 (Cross-Language Relation Preserves SMS)**.** *The cross-language trace relation* $\sim$ *defined by* $\overline{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \overline{\mathbf{a}}$ *is well-formed with respect to* smsafe*, i.e.,* $\vdash \sim :$ smsafe*.*

Equivalently: if $\overline{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \overline{\mathbf{a}}$ *and the filtered source trace* $\theta^*_{\delta_{MS}}(\overline{\mathsf{a}}) \in$ smsafe*, then* $\theta^*_{\delta'_{MS}}(\overline{\mathbf{a}}) \in$ smsafe*.*

Proof. By induction on $\overline{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \overline{\mathbf{a}}$.

The key cases are the memory events:

Case Rule alloc-event-eq: From $\mathsf{Alloc}\ \ell\ \mathsf{n} \approx_\delta \mathbf{Alloc}\ \ell\ \mathbf{n}$, we have $[\![\mathsf{n}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{n}$. Since compilation preserves numeric values, the allocation size is identical. The SMS monitor records $(\delta_{MS}(\ell), \mathsf{n})$ for source and $(\delta'_{MS}(\ell), \mathbf{n})$ for target. By the location correspondence δ, these represent the same abstract allocation.

Case Rule get-event-eq: From $\mathsf{Get}\ \ell\ \mathsf{n} \approx_\delta \mathbf{Get}\ \ell\ \mathbf{n}$, we have $[\![\mathsf{n}]\!]^{\mathsf{L}_{\mathrm{tms}} \to \mathsf{L}_{\mathrm{ms}}} = \mathbf{n}$. The access offset is identical in source and target. If the source access is in-bounds (satisfies smsafe), then the target access at the same offset into an allocation of the same size is also in-bounds.

Case Rule set-event-eq: Analogous to Get case.

Case Rule dealloc-event-eq: Deallocation events are related by location correspondence. The SMS monitor removes the same abstract allocation from both monitors.

By induction, the entire trace preserves smsafe. □

Lemma 65 (Backtranslation for Non-Interacting Traces is Well-Typed)**.** *If*

(a) $\vdash \overline{\mathbf{a}}$ non-int-$\overline{\mathbf{a}}$

(b) $\langle\!\langle \theta^*_\bullet(\overline{\mathbf{a}}) \rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}} = \mathsf{e}$

Then

(i) $\vdash \mathsf{e} : \mathbb{N}$

Proof. Note that $\theta^*_\bullet(\overline{\mathbf{a}}) = [\cdot] \vee \theta^*_\bullet(\overline{\mathbf{a}}) = \lightning$ because of Assumption (a). So, the only left cases are $\mathsf{e} = 42$ and $\mathsf{e} = \mathsf{abort}()$. Both terms easily typecheck under an empty gamma. □

Lemma 66 (Backtranslation for !-Interacting Events is Well-Typed)**.** *If*

(a) $^{!}\langle\!\langle \theta^*_\bullet(\mathbf{a}) \rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}_{\eth;\delta;\overline{\ell}} = \mathsf{e}$

Then $\exists \Gamma$,

(i) $\Gamma \vdash \mathsf{e} : \mathbb{N}$

Proof. Inverting Assumption (a) yields two cases, so we continue with them:

Case $\mathbf{a} = \mathbf{Start}$**:**

Here, $\mathsf{e} = 42$. Let $\Gamma = [\cdot]$, it's easy to see that $\vdash 42 : \mathbb{N}$ holds by Rule $t-\mathbb{N}$.

Case $\mathbf{a} = \mathbf{Ret}\ !\ \mathbf{v}$**:**

$\mathsf{e} = \mathsf{delete}\ \mathsf{z}; \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}$, where $\delta(\ell) = \ell$, $\mathsf{z} = \eth(\ell)$ and ℓ is at the top of the stack of locations $\overline{\ell}$. Choose $\Gamma = \mathsf{z} : \mathsf{ref}_1\ \mathbb{N}, [\cdot]$, the typing follows using Rule $t - \mathsf{delete}$.

□

Lemma 67 (Backtranslation for Call-? is Well-Typed)**.** *If*

(a) $^{?}\langle\!\langle \theta^*_\bullet(\mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v}) \rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}_{\eth;\delta;\ell,\overline{\ell}} = \eth'; \delta; \ell, \overline{\ell'}; \mathsf{e}'$

(b) $\delta(\ell) = \ell$

(c) $\mathsf{z} = \eth'(\ell)$

(d) $\mathsf{e} = \mathsf{e}';\ \mathsf{delete}\ \mathsf{z}; \langle\!\langle\!\langle \mathbf{v}' \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}$

(e) $\Gamma = \Gamma_1, \mathsf{foo} : \mathbb{N} \to \mathbb{N}, \Gamma_2$

Then,

(i) $\Gamma \vdash \mathsf{e} : \mathbb{N}$

Proof. Invert Assumption (a) giving
$\mathsf{e} = \mathsf{let}\ \mathsf{z} = \mathsf{new}\ 42\ \mathsf{in}\ \mathsf{call}\ \langle\!\langle\!\langle \mathbf{foo} \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}\ \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}};\ \mathsf{delete}\ \mathsf{z}; \langle\!\langle\!\langle \mathbf{v}' \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}$.
The claim follows using Rules $t - \mathsf{new}$ and $t - \mathsf{delete}$ and Rules $t - \mathsf{let}$ and $t - \mathsf{call}$, where the latter uses Assumption (e). □

Lemma 68 (Backtranslation for End-? is Well-Typed)**.** *If*

(a) $^{?}\langle\!\langle \theta^*_\bullet(\mathbf{End}\ \mathbf{v}) \rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}_{\eth;\delta;[\cdot]} = \eth'; \delta'; [\cdot]; \mathsf{e}$

Then,

(i) $\vdash \mathsf{e} : \mathbb{N} \to \bot$

Proof. Invert Assumption (a) giving $\mathsf{e} = \mathsf{let}\ \mathsf{z} = \mathsf{new}\ 42\ \mathsf{in}\ \mathsf{delete}\ \mathsf{z};\ \mathsf{return}\ \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}_{\eth}$, where $\eth \vdash \mathsf{z}$ *fresh*. The claim follows using Rules $t - \mathsf{new}$, $t - \mathsf{delete}$ and $t - \mathsf{return}$. □

Lemma 69 (Backtranslation is Well-Typed)**.** *If*

(a) $\langle\!\langle\!\langle \overline{\mathbf{a}} \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}_{\emptyset;\emptyset} = \eth; \delta; \Xi_{\mathsf{ctx}}$

(b) $\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}}$

(c) $\Gamma \vdash \Xi_{\mathsf{comp}}$ ok

Then,

(i) $\Gamma \vdash \Xi$ ok

Proof. First up, note that $\Gamma \vdash \Xi_{\mathsf{comp}}$ ok, so it suffices to check $\Gamma \vdash \Xi_{\mathsf{ctx}}$ ok by Lemma 23 (Typed Linking Recomposition). By inverting Assumption (a) we get $\overline{\mathbf{a}} = \overline{\mathbf{a_0}} \cdot \mathbf{a^{comp}} \cdot \overline{\mathbf{a_1}}$ and
$\mathsf{e} = \mathsf{e}_0; \mathsf{e}'_0; \mathsf{let}\ \mathsf{x}_0 = \mathsf{new}\ \mathsf{n}_0\ \mathsf{in}\ (\mathsf{e}''_0; \mathsf{e}_1); \mathsf{e}'_1; \mathsf{e}''_1$, where:

(H_1) $\vdash \overline{\mathbf{a^{comp}}}$ non-int-$\overline{\mathbf{a}}$

(H_2) $\langle\!\langle \theta^*_{\bullet}(\overline{\mathbf{a_0}}) \rangle\!\rangle^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}_{\mathrm{tms}}}_{\emptyset;\emptyset;[\cdot]} = \eth'; \delta'; \overline{\ell}; \mathsf{e}_0; \mathsf{e}'_0; \mathsf{let}\ \mathsf{x}_0 = \mathsf{new}\ \mathsf{n}_0\ \mathsf{in}\ \mathsf{e}''_0$

(H_3) $\langle\!\langle \theta^*_{\bullet}(\overline{\mathbf{a_1}}) \rangle\!\rangle^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}_{\mathrm{tms}}}_{\eth';\delta';\overline{\ell}} = \eth''; \delta''; [\cdot]; \mathsf{e}_1; \mathsf{e}'_1; \mathsf{e}''_1$.

Inverting Assumptions (H_2) and (H_3), first take note that
$\overline{\mathbf{a_0}} = \mathbf{Start} \cdot \overline{\mathbf{a'_0}} \cdot \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_2}$ and $\overline{\mathbf{a_1}} = \mathbf{Ret}\ ?\ \mathbf{v_3} \cdot \overline{\mathbf{a'_1}} \cdot \mathbf{End}\ \mathbf{v_4}$. Furthermore, $\mathsf{e}_0 = 42$, $\langle\!\langle \overline{\mathbf{a'_0}} \rangle\!\rangle^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}_{\mathrm{tms}}} = \mathsf{e}'_0$, and $\mathsf{e}''_0 = \mathsf{let}\ \mathsf{z} = \mathsf{new}\ 42\ \mathsf{in}\ \mathsf{call}\ \mathsf{foo}\ \mathsf{n}$, where $\mathsf{z} = \eth'(\ell)$ and $\delta(\ell) = \ell$, $\mathsf{n} = \langle\!\langle\!\langle \mathbf{v_2} \rangle\!\rangle\!\rangle^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}_{\mathrm{tms}}}$, and $[\![\mathsf{foo}]\!]^{\mathrm{L}_{\mathrm{tms}} \to \mathrm{L}_{\mathrm{ms}}} = \mathbf{foo}$ with $\mathsf{foo} \in \mathrm{dom}\ \Xi_{\mathsf{comp}}$. Also, $\mathsf{e}_1 = \mathsf{delete}\ \mathsf{z}$, $\langle\!\langle \overline{\mathbf{a'_1}} \rangle\!\rangle^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}_{\mathrm{tms}}} = \mathsf{e}'_1$, and $\mathsf{e}''_1 = \mathsf{let}\ \mathsf{x} = \mathsf{new}\ 42\ \mathsf{in}\ \mathsf{delete}\ \mathsf{x}; \mathsf{return}\ \mathsf{m}$, where $\mathsf{m} = \langle\!\langle\!\langle \mathbf{v_4} \rangle\!\rangle\!\rangle^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}_{\mathrm{tms}}}$ and $\eth'' \vdash \mathsf{x}$ *fresh*. Therefore:

$$\mathsf{e} = 42; \mathsf{e}'_0; \mathsf{let}\ \mathsf{z} = \mathsf{new}\ 42\ \mathsf{in}\ (\mathsf{call}\ \mathsf{foo}\ \mathsf{n}; \mathsf{delete}\ \mathsf{z}); \mathsf{e}'_1; \mathsf{let}\ \mathsf{x} = \mathsf{new}\ 42\ \mathsf{in}\ \mathsf{delete}\ \mathsf{x}; \mathsf{return}\ \mathsf{m}$$

This typechecks easily, making use of Rules $t - \mathsf{new}$ and $t - \mathsf{delete}$ and Rules $t - \mathsf{let}$, $t - \mathsf{call}$ and $t - \mathsf{return}$, as well as Lemma 65 (Backtranslation for Non-Interacting Traces is Well-Typed). □

Lemma 70 (Backtranslation Correctness of **Start**)**.** *If*

(a) $\Xi = \Xi_{\mathbf{ctx}} \bowtie [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}$

(b) $\xi = [\![\xi]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}} = \mathrm{dom}\ [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}$

(c) $\xi; \Xi; [\cdot]; \mathbf{comp}; [\cdot]; [\cdot]; [\cdot] \triangleright \mathbf{call\ main\ 0} \xrightarrow{\mathbf{Start;comp}}_{ctx} \xi; \Xi; ([\cdot]; \mathbf{main}), [\cdot]; \mathbf{ctx}; [\cdot]; [\cdot]; [\cdot] \triangleright \mathbf{e_{main}}[\mathbf{0/x}]$

(d) $\langle\!\langle \theta_{\bullet}(\mathbf{Start}) \rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}_{\emptyset;\emptyset;[\cdot]} = [\cdot]; 42$

(e) $\xi; \Xi; [\cdot]; \mathsf{comp}; [\cdot]; [\cdot]; [\cdot] \approx_{\emptyset;[\cdot]} \xi; \Xi; [\cdot]; \mathbf{comp}; [\cdot]; [\cdot]; [\cdot]$

(f) $\Xi = \{\mathsf{main} \mapsto 42; \mathsf{e}_{\mathsf{main}}\} \bowtie \Xi_{\mathsf{comp}}$

Then $\exists n$

(i) $\xi; \Xi; [\cdot]; \mathsf{comp}; [\cdot]; [\cdot]; [\cdot] \triangleright \mathsf{call\ main\ 0} \xrightarrow{\mathsf{Start;comp}}_{ctx}{}^{n} \xi; \Xi; ([\cdot]; \mathsf{main}), [\cdot]; \mathsf{ctx}; [\cdot]; [\cdot]; [\cdot] \triangleright \mathsf{e_{main}}[\mathsf{0/x}]$

(ii) $\xi; \Xi; ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \multimap_{\emptyset;[\cdot]} \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]$

(iii) $\mathsf{Start} \approx^{*}_{\emptyset;[\cdot]} \mathbf{Start}$

Proof. Note using Rule $e-\mathrm{ctx}-\mathsf{call}-\mathrm{main}$ we can easily conclude:

(H_1) $\xi; \Xi; [\cdot]\,; \mathsf{comp}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{call\ main\ 0} \xrightarrow{\mathsf{Start;comp}}{}^{1}_{\mathrm{ctx}}\ \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright$ $\mathsf{let\ _ = 42\ in\ e_{main}[0/x]}$

On Assumption (H_1) apply Rules $e-\mathsf{let}-\mathsf{f}$ and $e-\mathrm{ctx}$

(H_2) $\xi; \Xi; ([\cdot]\,, \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{let\ _ = 42\ in\ e_{main}[0/x]} \xrightarrow{\varepsilon}{}^{1}_{\mathrm{ctx}}\ \xi; \Xi; ([\cdot]\,, \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright$ $\mathsf{e_{main}[0/x]}$

Use Assumptions (H_1) and (H_2) on Rules $en-\mathrm{refl}$ and $en-\mathrm{trans}-$important:

(H_3) $\xi; \Xi; [\cdot]\,; \mathsf{comp}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{call\ main\ 0} \xrightarrow{\mathsf{Start}}{}^{2}_{\mathrm{ctx}}\ \xi; \Xi; ([\cdot]\,, \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright$ $\mathsf{e_{main}[0/x]}$

So, instantiating $n = 2$ concludes Goal (i).

Goal (ii) by Rule state-qe, Rules memstate-qe and cfstate-qe and Rules empty-memstate-eq, empty-commlib-lib-eq and empty-kontstack-eq. Goal (iii) follow Rule start-event-eq. □

Lemma 71 (Backtranslation Correctness of **Ret**)**.** *If*

(a) $\mathbf{\Omega} = \xi; \mathbf{\Xi}; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}; \mathbf{comp}; \mathbf{\Psi}$

(b) $\mathbf{\Xi} = \mathbf{\Xi_{ctx}} \bowtie [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}$

(c) $\xi = [\![\xi]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}} = \mathrm{dom}\ [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}$

(d) $\mathbf{\Omega} \triangleright \mathbf{K_{component}[return\ v]} \xrightarrow{\mathbf{Ret\ !\ v;comp}}_{ctx} \mathbf{\Omega'} \triangleright \mathbf{K[v]}$

(e) $\mathbf{K_{component}} = [\![\mathsf{K_{component}}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}$

(f) $\llbracket\!\langle\!\langle \theta_{\bullet}\ (\mathbf{Ret\ !\ v}) \rangle\!\rangle\!\rrbracket^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}_{\eth;\delta;\overline{\ell}} = \overline{\ell'}; \mathsf{delete\ z}; \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}$

(g) $\Omega \approx_{\delta;\mathsf{L}} \mathbf{\Omega}$

(h) $\Omega = \xi; \Xi; (\mathsf{K}, \mathsf{foo}), \overline{\mathsf{K}}; \mathsf{comp}; \Psi$

(i) $\mathsf{K} = [\cdot]\,; \mathsf{delete\ z}; \mathsf{K}[\langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}]$

(j) $\Xi = \{\mathsf{main} \mapsto \mathsf{e_{main}}\} \bowtie \Xi_{\mathsf{comp}}$

(k) $\delta(\ell) = \ell$

(l) $\mathsf{z} = \eth(\ell)$

(m) $\ell \in \mathsf{L}$

(n) $\mathbf{foo} \in \mathbf{\Xi_{comp}}$

then $\exists n\ \overline{\mathsf{a}}\ \Psi'$,

(i) $\mathsf{X'} = \mathsf{X} \cup \{\mathsf{Dealloc}\ \ell\}$

(ii) $\Omega \triangleright \mathsf{K}_{\mathsf{component}}[\mathsf{return}\ \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}}] \xrightarrow{\bar{\mathsf{a}}}_{ctx} n\ \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi' \triangleright \mathsf{K}[\langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}}]$

(iii) $\mathbf{\Omega'} \multimap_{\delta;\mathsf{L}} \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; \Psi'$

(iv) $\bar{\mathsf{a}} \approx^*_{\delta;\mathsf{X}'} \mathbf{Ret}\ !\ \mathbf{v}$

Proof. From Assumption (k) it is clear that:

(H_1) $\ell \in \operatorname{dom} \delta$

Invert Assumption (g):

(H_2) $\Omega = \Phi; \mathsf{comp}; \Psi$

(H_3) $\mathbf{\Omega} = \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi}$

(H_4) $\Phi \approx \mathbf{\Phi}$

(H_5) $\Psi \approx_{\delta;\mathsf{L}} \mathbf{\Psi}$

Decompose Ψ into $\mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta$. Make use of Assumptions (H_1) and (H_5) with Lemma 61 ($\ell \in \operatorname{dom} \delta$ decomposes Δ):

(H_6) $\Delta = \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}'; \rho; \mathsf{m}), \Delta_2$

For the goals, instantiate the existentials with:

(E_1) $n = 3$

(E_2) $\mathsf{X}' = \mathsf{X} \cup \{\mathsf{Dealloc}\ \ell\}$

(E_3) $\bar{\mathsf{a}} = \mathsf{Dealloc}\ \ell \cdot \mathsf{Ret}\ !\ \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathsf{L}_{\mathrm{ms}} \to \mathsf{L}_{\mathrm{tms}}}_{\mathfrak{O}}$

(E_4) $\Psi' = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}'; \cancel{\rho}; \mathsf{m}), \Delta_2$

Goal (i) follows immediately by reflexivity, since we instantiated X' to be exactly $\mathsf{X} \cup \{\mathsf{Dealloc}\ \ell\}$. Let $\mathsf{n} = \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathsf{L}_{\mathrm{ms}} \to \mathsf{L}_{\mathrm{tms}}}$. For Goal (ii), first note Assumption (h), use Rule $e-\mathrm{ctx}-\mathsf{return}-\mathrm{notsame}$ and rewrite K using Assumption (i):

(H_7) $\xi; \Xi; (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}}; \mathsf{comp}; \Psi \triangleright \mathsf{K}_{\mathsf{component}}\,[\mathsf{return}\ \mathsf{n}] \xrightarrow{\mathsf{Ret}\ !\ \mathsf{n}}_{\mathrm{ctx}} \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi \triangleright \mathsf{n}; \mathsf{delete}\ \mathsf{z}; \mathsf{K}[\mathsf{n}]$

This works, because of Assumption (n), then $\rho(\mathsf{comp}) = !$. Note, Assumption (H_7), continue with Rules $e-\mathsf{let}-\mathsf{f}$ and $e-\mathrm{ctx}$:

(H_8) $\xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi \triangleright \mathsf{K}_{\mathsf{component}}\,[\mathsf{return}\ \mathsf{n}] \xrightarrow{\varepsilon}_{\mathrm{ctx}} \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi \triangleright \mathsf{delete}\ \mathsf{z}; \mathsf{K}[\mathsf{n}]$

Finally, use Rules $e-\mathsf{delete}$ and $e-\mathrm{ctx}$ on Assumption (H_8):

(H_9) $\xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi \triangleright \mathsf{delete}\ \mathsf{z}; \mathsf{K}[\mathsf{n}] \xrightarrow{\mathsf{Dealloc}\ \ell}_{\mathrm{ctx}} \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi' \triangleright \mathsf{K}[\mathsf{n}]$

Use Assumptions (H_7) to (H_9) on Rule $en-\mathrm{trans}-\mathrm{important}$, Rules $en-\mathrm{refl}$ and $en-\mathrm{trans}-\mathrm{unimportant}$ to get exactly what Goal (ii) wants us to prove.

With Assumption (h), invert Assumption (g):

(H_{10}) $\mathbf{\Omega} = \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi}$

(H_{11}) $\mathbf{\Phi} = \xi; \mathbf{\Xi}; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}$

(H_{12}) $\Phi \approx \mathbf{\Phi}$

(H_{13}) $\Psi \approx_{\delta;\mathsf{L}} \mathbf{\Psi}$

Invert Assumption (d) to extract:

(H_{14}) $\mathbf{\Omega}' = \mathbf{\Phi}'; \mathbf{ctx}; \mathbf{\Psi}$

(H_{15}) $\mathbf{\Phi}' = \xi; \mathbf{\Xi}; \overline{\mathbf{K}}$

Let $\Phi' = \xi; \Xi; \overline{\mathsf{K}}$.

Use Rule state-qe on Goal (iii) giving goals:

(v) $\Phi' \approx \mathbf{\Phi}'$

(vi) $\Psi' \approx_{\delta;\mathsf{L}'} \mathbf{\Psi}$

Goal (v) follows by Assumption (H_{12}).

Note that $\Psi' = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}'; \cancel{\rho}; \mathsf{m}), \Delta_2$ and remember that $\Psi = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta_1, \mathsf{x} \mapsto (\ell; \mathsf{t}'; \rho; \mathsf{m}), \Delta_2$, of which we know Assumption (H_{13}). So, Goal (vi) follows from Lemma 60 (if $\ell \in \mathsf{L}$, then ρ doesn't matter), which needs Assumption (m). Now Goal (iv): First apply Rule ignore-cons-trace-eq, given that $\mathsf{Dealloc}\ \ell \in \mathsf{X}'$. What is left to show is:

(vii) $\mathsf{Ret}\ !\ \mathsf{v} \approx^*_{\delta;\mathsf{X}'} \mathbf{Ret}\ !\ \mathbf{v}$

Goal (vii) follows by Rules empty-trace-eq, cons-trace-eq and ret-event-eq. □

Lemma 72 (Middle of Backtranslation Correctness)**.** *If*

(a) $\mathbf{\Omega} = \xi; \mathbf{\Xi}; \overline{\mathbf{K}}; \mathbf{ctx}; \mathbf{\Psi}$

(b) $\mathbf{\Xi} = \mathbf{\Xi}_{\mathbf{ctx}} \bowtie [\![\Xi_{\mathsf{comp}}]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}}$

(c) $\xi = [\![\xi]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}} = \mathrm{dom}\ [\![\Xi_{\mathsf{comp}}]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}}$

(d) $\mathbf{\Omega} \triangleright \mathbf{e}\gamma \xrightarrow{\overline{\mathbf{a}}}_{ctx}{}^{n} \mathbf{\Omega}' \triangleright \mathbf{e}'\gamma'$

(e) $\langle\!\langle \theta^*_{\bullet}(\overline{\mathsf{a}}) \rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}} = \mathsf{e}$

(f) $\vdash \overline{\mathbf{a}}$ non-int- $\overline{\mathbf{a}}$

(g) $\mathbf{\Omega} \multimap_{\delta;\mathsf{L}} \Omega$

(h) $\Omega = \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi$

(i) $\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}}$

then $\exists n'\ \overline{\mathsf{a}}$,

(i) $\Omega \triangleright \mathsf{K}[\mathsf{e}] \xrightarrow{\overline{\mathsf{a}}}_{ctx}{}^{n'} \xi; \Xi; \overline{\mathsf{K}}; \Psi' \triangleright \mathsf{K}[42]$

(ii) $\mathbf{\Omega}' \multimap_{\delta;\mathsf{L}} \xi; \Xi; \overline{\mathsf{K}}; \Psi'$

(iii) $\overline{\mathsf{a}} \approx^*_{\delta;\emptyset} [\cdot]$

Proof. Induction on Assumption (f).

Case $\overline{\mathsf{a}} = [\cdot]$

Invert Assumption (e) to obtain:

(H_1) $\mathsf{e} = 42$

$\mathsf{e} = \mathsf{abort}()$ is not an option, since it contradicts Assumption (d).

Instantiate our goals with:

(H_2) $n = 0$

(H_3) $\overline{\mathsf{a}} = [\cdot]$

Goal (i) follows by reflexivity.

Use Assumption (g) to solve Goal (ii).

For Goal (iii), use Rule empty-trace-eq.

Case $\overline{\mathsf{a}} = \mathsf{a} \cdot \overline{\mathsf{a}}$**:**

Regardless of the shape of a (noting that it cannot be $\lightning$) note that $\langle\!\langle \mathsf{a} \cdot \overline{\mathsf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \langle\!\langle \overline{\mathsf{a}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}$. Thus, all cases follow immediately by the induction hypothesis.

□

Lemma 73 (Backtranslation Correctness of Call ?)**.** *If*

(a) $\Omega = \xi; \Xi; \overline{\mathbf{K}}; \mathbf{ctx}; \Psi$

(b) $\overline{\mathbf{K}} = (\mathbf{K}'; \mathbf{bar}), \overline{\mathbf{K}}'$

(c) $\Xi = \Xi_{\mathbf{ctx}} \bowtie [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}$

(d) $\xi = [\![\xi]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}} = \mathrm{dom}\ [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}$

(e) $\Omega \triangleright \mathbf{K}\,[\mathbf{call\ foo\ v}] \xrightarrow{\mathbf{Call\ ?\ foo\ v;ctx}}_{ctx} \xi; \Xi; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}; \mathbf{comp}; \Psi \triangleright [\![\mathbf{e_{foo}}]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}[\mathbf{v}/\mathbf{y}]$

(f) ${}^{?}\langle\!\langle \theta_{\bullet}\,(\mathbf{Call\ ?\ foo\ v}) \rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}_{\Im;\delta;\overline{\ell}} = \Im'; \delta'; \overline{\ell'}; \mathsf{let\ z = new\ 42\ in\ call\ foo}\ \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}$

(g) $\Omega \multimap_{\delta;\mathsf{L}} \Omega$

(h) $\Omega = \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi$

(i) $\mathsf{let\ foo\ y} : \tau_\lambda := \mathsf{e_{foo}} \in \Xi_{\mathsf{comp}}$

(j) $\Im, \delta \vdash \ell$ *fresh*

(k) $\Im \vdash \mathsf{z}$ *fresh*

(l) $\delta \vdash \ell$ *fresh*

(m) $\delta'(\ell) = \ell$

(n) $\Im'(\ell) = \mathsf{z}$

(o) $\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}}$

then $\exists n\ \overline{\mathsf{a}}\ \Psi'$,

(i) $\mathsf{L}' = \mathsf{L} \cup \{\ell\}$

(ii) $\mathsf{X}' = \mathsf{X} \cup \{\mathsf{Alloc}\ \ell\ 42\}$

(iii) $\Omega \triangleright \mathsf{K}[\mathsf{let\ z = new\ 42\ in\ call\ foo}\ \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L}_{ms} \to \mathrm{L}_{tms}}] \xrightarrow{\bar{\mathsf{a}}}_{ctx}{}^{n}\ \xi; \Xi; (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}}; \mathsf{comp}; \Psi' \triangleright \mathsf{e}_{\mathsf{foo}}\gamma$

(iv) $\xi; \Xi; (\mathsf{K}, \mathsf{foo}), \overline{\mathsf{K}}; \mathsf{comp}; \Psi' \approx_{\delta';\mathsf{L}'} \xi; \boldsymbol{\Xi}; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}; \mathbf{comp}; \boldsymbol{\Psi}$

(v) $\bar{\mathsf{a}} \approx^{*}_{\delta';\mathsf{X}'} \mathbf{Call\ ?\ foo\ v; ctx}$

Proof. First, we unfold the source memory state:

(H_1) $\Psi = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta$

Let $\mathsf{v} = \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}$. Next, we instantiate the existentials in the goals as follows:

(H_6) $n = 2$

(H_7) $\mathsf{L}' = \mathsf{L} \cup \{\ell\}$

(H_8) $\mathsf{X}' = \mathsf{X} \cup \{\mathsf{Alloc}\ \ell\ 42\}$

(H_9) $\bar{\mathsf{a}} = \mathsf{Alloc}\ \ell\ 42; \mathsf{ctx} \cdot \mathsf{Call\ ?\ foo\ v; ctx}$

(H_{10}) $\Psi' = \underbrace{0, \cdots, 0}_{42-times}, \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \mathsf{z} \mapsto (\ell; \mathsf{ctx}; \square; 42), \Delta$

(H_{11}) $\gamma = [\langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}/\mathsf{y}]$

Goals (i) and (ii) follow easily. For Goal (iii), first use Rules $e-\mathsf{new}$ and $e-\mathrm{ctx}$, noting Assumptions (j) and (k):

(H_{11}) $\Omega \triangleright \mathsf{K}[\mathsf{let\ z = new\ 42\ in\ call\ foo\ v}] \xrightarrow{\mathsf{Alloc}\ \ell\ 42;\mathsf{ctx}}_{\mathrm{ctx}}{}^{1}\ \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi' \triangleright \mathsf{K}[\mathsf{call\ foo\ v}]$

Now use Rule $e - \mathrm{ctx} - \mathsf{return} - \mathrm{notsame}$:

(H_{12}) $\xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi' \triangleright \mathsf{K}[\mathsf{call\ foo\ v}] \xrightarrow{\mathsf{Call\ ?\ foo\ z;ctx}}_{\mathrm{ctx}}{}^{1}\ \xi; \Xi; (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}}; \mathsf{ctx}; \Psi' \triangleright \mathsf{e}_{\mathsf{foo}}[\mathsf{v}/\mathsf{y}]$

Use Assumptions (H_{11}) and (H_{12}) on Rules $en-\mathrm{refl}$ and $en-\mathrm{trans}-$important to get exactly what Goal (iii) wants us to prove.

Invert Assumption (g):

(H_{12}) $\Omega = \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi$

(H_{13}) $\boldsymbol{\Omega} = \xi; \boldsymbol{\Xi}; \overline{\mathbf{K}}; \mathbf{ctx}; \boldsymbol{\Psi}$

(H_{14}) $\boldsymbol{\Phi} \multimap \Phi$

(H_{15}) $\boldsymbol{\Psi} \multimap_{\delta;\mathsf{L}} \Psi$

Invert Assumption (H_{14}):

(H_{16}) $\Xi \approx \boldsymbol{\Xi}$

(H_{17}) $\overline{\mathbf{K}} \multimap \overline{\mathsf{K}}$

Invert Assumption (H_{15})

(H_{18}) $\mathbf{\Psi} = \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}$

(H_{19}) $\mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}$

Apply Rule state-eq on Goal (iv):

(v) $\underbrace{0, \ldots, 0}_{42-\mathsf{times}}, \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \mathsf{z} \mapsto (\ell; \mathsf{ctx}; \square; 42), \Delta \approx_{\delta';\mathsf{L}'} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}$

(vi) $\xi; \Xi; (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}} \approx \xi; \mathbf{\Xi}; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}$

Use Rule whatever-cons-memstate-eq on Goal (v), since $\ell \in \mathsf{L}'$:

(vii) $\mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \approx_{\delta';\mathsf{L}'} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}$

Goal (vii) can be resolved by Assumption (H_{19}).
Apply Rule cfstate-eq on Goal (vi):

(ix) $\Xi \approx \mathbf{\Xi}$

(x) $(\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}} \approx (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}$

Goal (ix) is resolved by Assumption (H_{16}).
Apply Rule cons-kontstack-eq to Goal (x), leaving us with:

(xi) $\overline{\mathsf{K}} \approx \overline{\mathbf{K}}$

Invert Assumption (H_{17}):

(H_{20}) $\overline{\mathsf{K}} \approx \overline{\mathbf{K}}$

Assumption (H_{20}) can now be used to solve Goal (xi).
Lastly, apply Rule ignore-cons-trace-eq on Goal (v), given that $\mathsf{Alloc}\ \ell\ 42 \in \mathsf{X}'$.
What is left to show is:

(vii) $\mathsf{Call}\ ?\ \mathsf{foo}\ \mathsf{v} \approx^*_{\delta;\mathsf{X}'} \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v}$

Goal (vii) follows by Rules empty-trace-eq, cons-trace-eq and call-event-eq. $\square$

Lemma 74 (Backtranslation Correctness of **End**)**.** *If*

(a) $\mathbf{\Xi} = \mathbf{\Xi}_{\mathbf{ctx}} \bowtie [\![\Xi_{\mathsf{comp}}]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}}$

(b) $\xi = [\![\xi]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}} = \mathrm{dom}\ [\![\Xi_{\mathsf{comp}}]\!]^{\mathsf{L}_{tms} \to \mathsf{L}_{ms}}$

(c) $\xi; \mathbf{\Xi}; ([\cdot]; \mathbf{main}), [\cdot]; \mathbf{ctx}; \mathbf{\Psi} \triangleright \mathbf{K}\ [\mathbf{return}\ \mathbf{n}] \xrightarrow{\mathbf{End}\ \mathbf{n};\mathbf{ctx}}_{ctx} \xi; \mathbf{\Xi}; [\cdot]; \mathbf{comp}; \mathbf{\Psi} \triangleright \mathbf{n}$

(d) $^{?}\langle\!\langle \theta_{\bullet}\ (\mathbf{End}\ \mathbf{n}) \rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}}_{\Im;\delta;[\cdot]} = \Im'; \delta'; [\cdot]; \mathsf{let}\ \mathsf{z} = \mathsf{new}\ 42\ \mathsf{in}\ \mathsf{delete}\ \mathsf{z}; \mathsf{return}\ \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}}$

(e) $\xi; \mathbf{\Xi}; ([\cdot]; \mathbf{main}), [\cdot]; \mathbf{ctx}; \mathbf{\Psi} \multimap_{\delta;\mathsf{L}} \xi; \Xi; ([\cdot]; \mathsf{main}), [\cdot]; \mathsf{ctx}; \Psi$

(f) $\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}}$

(g) $\Im, \delta \vdash \ell$ *fresh*

(h) $\Im \vdash \mathsf{z}$ *fresh*

(i) $\delta \vdash \ell$ *fresh*

(j) $\delta'(\ell) = \ell$

(k) $\Im'(\ell) = \mathsf{z}$

then $\exists n\ \bar{\mathsf{a}}\ \Psi'$,

(i) $\mathsf{L}' = \mathsf{L} \cup \{\ell\}$

(ii) $\mathsf{X}' = \mathsf{X} \cup \{\mathsf{Alloc}\ \ell\ 42, \mathsf{Dealloc}\ \ell\}$

(iii) $\Omega \triangleright \mathsf{K}[\mathsf{let\ z = new\ 42\ in\ delete\ z; return}\ \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}}] \xrightarrow{\bar{\mathsf{a}}}_{ctx}{}^{n}\ \xi; \Xi; [\cdot]; \mathsf{comp}; \Psi' \triangleright \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}}$

(iv) $\xi; \Xi; [\cdot]; \mathsf{comp}; \Psi' \approx_{\delta'; \mathsf{L}'} \xi; \boldsymbol{\Xi}; [\cdot]; \mathbf{comp}; \boldsymbol{\Psi}$

(v) $\bar{\mathsf{a}} \approx^{*}_{\delta'; \mathsf{X}'} \mathbf{End\ v; ctx}$

Proof. First, we unfold the source memory state:

(H_1) $\Psi = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta$

Let $\mathsf{v} = \langle\!\langle\!\langle \mathbf{v} \rangle\!\rangle\!\rangle^{\mathsf{L}_{\mathrm{ms}} \to \mathsf{L}_{\mathrm{tms}}}$. Next, we instantiate the existentials in the goals as follows:

(H_6) $n = 3$

(H_7) $\mathsf{L}' = \mathsf{L} \cup \{\ell\}$

(H_8) $\mathsf{X}' = \mathsf{X} \cup \{\mathsf{Alloc}\ \ell\ 42, \mathsf{Dealloc}\ \ell\}$

(H_9) $\bar{\mathsf{a}} = \mathsf{Alloc}\ \ell\ 42; \mathsf{ctx} \cdot \mathsf{Dealloc}\ \ell; \mathsf{ctx} \cdot \mathsf{End\ v; ctx}$

(H_{10}) $\Psi' = \underbrace{0, \cdots, 0}_{42-times}, \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \mathsf{z} \mapsto (\ell; \mathsf{ctx}; \circledast; 42), \Delta$

Let $\Psi_{\square} = \underbrace{0, \cdots, 0}_{42-times}, \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \mathsf{z} \mapsto (\ell; \mathsf{ctx}; \square; 42), \Delta$.

Let $\overline{\mathsf{K}} = ([\cdot]; \mathsf{main}), [\cdot]$. Goals (i) and (ii) follow easily. For Goal (iii), first use Rules $e-\mathsf{new}$ and $e-\mathrm{ctx}$, noting Assumptions (g) and (h):

(H_{10}) $\Omega \triangleright \mathsf{K}[\mathsf{let\ z = new\ 42\ in\ delete\ z; return\ v}] \xrightarrow{\mathsf{Alloc}\ \ell\ 42; \mathsf{ctx}}_{\mathrm{ctx}} \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx};\ \Psi_{\square} \triangleright \mathsf{K}[\mathsf{delete\ z; return\ v}]$

Next, use Rules $e-\mathsf{let-f}$, $e-\mathsf{delete}$, $e-\mathrm{ctx}$, $en-\mathrm{refl}$ and $en-\mathrm{trans}-\mathrm{unimportant}$ on Assumption (H_{10}):

(H_{11}) $\xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi_{\square} \triangleright \mathsf{K}[\mathsf{let\ _ = delete\ z\ in\ return\ v}] \xrightarrow{\mathsf{Dealloc}\ \ell; \mathsf{ctx}}_{\mathrm{ctx}}{}^{2}\ \xi; \Xi; \overline{\mathsf{K}}; \mathsf{ctx}; \Psi' \triangleright \mathsf{K}[\mathsf{return\ v}]$

Now, apply Rules $e-\mathrm{ctx}$ and $e-\mathrm{ctx}-\mathsf{return}-\mathrm{main}$ on Assumption (H_{11}).

(H_{12}) $\xi; \Xi; ([\cdot], \mathsf{main}), [\cdot]; \mathsf{ctx}; \Psi' \triangleright \mathsf{K}[\mathsf{return\ v}] \xrightarrow{\mathsf{End\ v; ctx}}_{\mathrm{ctx}} \xi; \Xi; [\cdot]; \mathsf{comp}; \Psi' \triangleright \mathsf{v}$

Use Assumptions (H_{10}) to (H_{12}) on Rules $en-$refl and $en-$trans$-$important to get exactly what Goal (iii) wants us to prove.

Invert Assumption (e):

(H_{13}) $\Omega = \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; \Psi$

(H_{14}) $\mathbf{\Omega} = \xi; \mathbf{\Xi}; ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \mathbf{\Psi}$

(H_{15}) $\xi; \mathbf{\Xi}; ([\cdot]\,; \mathbf{main}), [\cdot] \multimap \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]$

(H_{16}) $\mathbf{\Psi} \multimap_{\delta;\mathsf{L}} \Psi_0$

Invert Assumption (H_{15}):

(H_{16}) $\Xi \approx \mathbf{\Xi}$

(H_{17}) $([\cdot]\,; \mathbf{main}), [\cdot] \multimap_{\xi} ([\cdot]\,; \mathsf{main}), [\cdot]$

Invert Assumption (H_{16})

(H_{19}) $\Psi_0 = \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta$

(H_{20}) $\mathbf{\Psi} = \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}$

(H_{21}) $\mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \approx_{\delta;\mathsf{L}} \mathbf{H}^{\mathbf{ctx}}; \mathbf{H}^{\mathbf{comp}}; \mathbf{\Delta}$

Use Assumptions (H_{16}) and (H_{21}) and rule empty-kontstack-eq to solve Goal (iv). Lastly, apply Rule ignore-cons-trace-eq on Goal (v), given that $\{\mathsf{Alloc}\ \ell\ 42, \mathsf{Dealloc}\ \ell\} \subseteq \mathsf{X}'$. What is left to show is:

(vii) $\mathsf{End}\ \mathsf{v} \approx^*_{\delta;\mathsf{X}'} \mathbf{End}\ \mathbf{v}$

Goal (vii) follows by Rules empty-trace-eq, cons-trace-eq and end-event-eq. □

Lemma 75 (Backtranslation Correctness)**.** *If*

(a) $\mathbf{\Omega} \triangleright \mathbf{call\ main\ 0} \xrightarrow{\overline{\mathbf{a}}}{}^*_{ctx} \mathbf{\Omega}' \triangleright \mathbf{f}_{\lightning}$

(b) $\langle\!\langle\!\langle \overline{\mathbf{a}} \rangle\!\rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}}_{\emptyset;\emptyset} = \mathfrak{I}; \delta; \Xi_{\mathsf{ctx}}$

(c) $\Omega \approx_{\emptyset;\emptyset} \mathbf{\Omega}$

(d) $\mathbf{\Omega} = \mathbf{foo}, [\cdot]\,; \mathbf{\Xi}; [\cdot]\,; \mathbf{comp}; [\cdot]\,; [\cdot]\,; [\cdot]$

(e) $\Omega = \xi; \Xi; \overline{\mathsf{K}}; \mathsf{comp}; \Psi$

(f) $\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}}$

(g) $\Gamma \vdash \Xi_{\mathsf{comp}}$ ok

then $\exists \delta_e\ \mathsf{L}_e\ \mathsf{X}_e\ \Omega'_e\ \overline{\mathsf{a}}_e$,

(i) $\Omega \triangleright \mathsf{call\ main\ 0} \xrightarrow{\overline{\mathsf{a}}_e}{}^*_{ctx} \Omega'_e \triangleright \langle\!\langle\!\langle \mathbf{f}_{\lightning} \rangle\!\rangle\!\rangle^{\mathsf{L}_{ms} \to \mathsf{L}_{tms}}$

(ii) $\Omega'_e \approx_{\delta_e;\mathsf{L}_e} \mathbf{\Omega}'$

(iii) $\overline{\mathsf{a}}_e \approx^*_{\delta_e;\mathsf{X}_e} \overline{\mathbf{a}}$

Proof. Inverting Assumption (b) gives:

(H_1) $\theta^*_\bullet(\overline{\mathsf{a}}) = \mathbf{Start}; \mathbf{comp} \cdot \overline{\mathbf{a_0}} \cdot \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_0}; \mathbf{ctx} \cdot \overline{\mathbf{a_{comp}}} \cdot \mathbf{Ret}\ !\ \mathbf{v_1}; \mathbf{comp} \cdot \overline{\mathbf{a_1}} \cdot \mathbf{End}\ \mathbf{v_2}; \mathbf{ctx}$

(H_2) $\vdash \overline{\mathbf{a_{comp}}}$ non-int-$\overline{\mathbf{a}}$

(H_3) $^{?}\langle\!\langle \mathbf{Start} \cdot \overline{\mathbf{a_0}} \cdot \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_0} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\emptyset;\emptyset;[\cdot]} = \mathcal{D}; \delta; \overline{\ell}; \mathsf{e}_0; \mathsf{e}'_0; \mathsf{let}\ \mathsf{x}_0 = \mathsf{new}\ \mathsf{n}_0\ \mathsf{in}\ \mathsf{e}''_0$

(H_4) $^{!}\langle\!\langle \mathbf{Ret}\ !\ \mathbf{v_1} \cdot \overline{\mathbf{a_1}} \cdot \mathbf{End}\ \mathbf{v_2} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\mathcal{D};\delta;\overline{\ell}} = \mathcal{D}'; \delta'; [\cdot]; \mathsf{e}_1; \mathsf{e}'_1; \mathsf{let}\ \mathsf{x}_1 = \mathsf{new}\ \mathsf{n}_1\ \mathsf{in}\ \mathsf{e}''_1$

(H_5) $\mathsf{e} = \mathsf{e}_0; \mathsf{e}'_0; \mathsf{let}\ \mathsf{x}_0 = \mathsf{new}\ \mathsf{n}_0\ \mathsf{in}\ (\mathsf{e}''_0; \mathsf{e}_1); \mathsf{e}'_1; \mathsf{let}\ \mathsf{x}_1 = \mathsf{new}\ \mathsf{n}_1\ \mathsf{in}\ \mathsf{e}''_1$

(H_6) $\Xi_{\mathsf{ctx}} = \mathsf{let}\ \mathsf{main}\ \mathsf{x} : \mathbb{N} \to \mathbb{N} := \mathsf{e}, [\cdot]$

Inverting Assumption (H_3):

(H_7) $\vdash \overline{\mathbf{a_0}}$ non-int-$\overline{\mathbf{a}}$

(H_8) $^{!}\langle\!\langle \mathbf{Start} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\emptyset;\emptyset;[\cdot]} = \overline{\ell_0}; \mathsf{e}_0$

(H_9) $\langle\!\langle \overline{\mathbf{a_0}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}'_0$

(H_{10}) $^{?}\langle\!\langle \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_0} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\emptyset;\emptyset;\overline{\ell_0}} = \mathcal{D}; \delta; \overline{\ell}, \mathsf{let}\ \mathsf{x}_0 = \mathsf{new}\ \mathsf{n}_0\ \mathsf{in}\ \mathsf{e}''_0$

Inverting Assumption (H_4):

(H_{11}) $\vdash \overline{\mathbf{a_1}}$ non-int-$\overline{\mathbf{a}}$

(H_{12}) $^{!}\langle\!\langle \mathbf{Ret}\ !\ \mathbf{v_1} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\mathcal{D};\delta;\overline{\ell}} = \overline{\ell_1}; \mathsf{e}_1$

(H_{13}) $\langle\!\langle \overline{\mathbf{a_1}} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}} = \mathsf{e}'_1$

(H_{14}) $^{?}\langle\!\langle \mathbf{End}\ \mathbf{v_2} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\mathcal{D};\delta;\overline{\ell_1}} = \mathcal{D}'; \delta'; [\cdot]; \mathsf{let}\ \mathsf{x}_1 = \mathsf{new}\ \mathsf{n}_1\ \mathsf{in}\ \mathsf{e}''_1$

By Lemmas 57 and 58 and assumptions (H_7) and (H_{11}) and noting that $\overline{\mathbf{a_0}}$ and $\overline{\mathbf{a_1}}$ are all tagged with $\mathbf{ctx}$, we can conclude:

(H_{15}) $\overline{\mathbf{a_0}} = [\cdot]$

(H_{16}) $\overline{\mathbf{a_1}} = [\cdot]$

Invert Assumption (a) and noting Assumption (H_1):

(H_{15}) $\overline{\mathbf{a}} = \mathbf{Start}; \mathbf{comp} \cdot \overline{\mathbf{a_a}}$

(H_{16}) $\mathbf{\Omega} \triangleright \mathbf{call}\ \mathbf{main}\ \mathbf{0} \xrightarrow{\mathbf{Start};\mathbf{comp}}_{\mathrm{ctx}} \mathbf{foo}, [\cdot]; \mathbf{\Xi}; ([\cdot]; \mathbf{main}), [\cdot]; \mathbf{ctx}; [\cdot]; [\cdot]; [\cdot] \triangleright \mathbf{e_0}[\mathbf{0}/\mathbf{x}]$

(H_{17}) $\mathbf{foo}, [\cdot]; \mathbf{\Xi}; ([\cdot]; \mathbf{main}), [\cdot]; \mathbf{ctx}; [\cdot]; [\cdot]; [\cdot] \triangleright \mathbf{e_0}[\mathbf{0}/\mathbf{x}] \xrightarrow{\overline{\mathbf{a_a}}}{}^{*}_{\mathrm{ctx}} \mathbf{\Omega}' \triangleright \mathbf{f}$

Invert Assumption (c):

(H_{18}) $\Omega = \Phi; \mathsf{comp}; \Psi$

(H_{19}) $\mathbf{\Omega} = \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi}$

(H_{20}) $\Phi \approx \mathbf{\Phi}$

(H_{21}) $\Psi \approx_{\emptyset;\emptyset} \mathbf{\Psi}$

Subsequently inverting Assumption (H_{20}):

(H_{22}) $\Phi = \xi; \Xi; \overline{\mathsf{K}}$

(H_{23}) $\mathbf{\Phi} = \mathbf{foo}, [\cdot]\,; \mathbf{\Xi}; [\cdot]$

(H_{24}) $\Xi \approx \mathbf{\Xi}$

(H_{25}) $\overline{\mathsf{K}} \approx [\cdot]$

Inverting Assumption (H_{25}):

(H_{26}) $\overline{\mathsf{K}} = [\cdot]$

Similarly, inverting Assumption (H_{21}) and noting that $\mathbf{\Psi} = [\cdot]\,; [\cdot]\,; [\cdot]$:

(H_{27}) $\Psi = [\cdot]\,; [\cdot]\,; [\cdot]$

Use Assumption (H_{24}) to conclude:

(H_{28}) $\mathbf{\Xi} = \mathbf{\Xi_{ctx}} \blacktriangleright\!\!\blacktriangleleft [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(H_{29}) $\xi = [\![\xi]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} = \mathrm{dom}[\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

Note by inversion on Assumption (H_8) we have:

(H_{30}) $\ell_{\mathbf{0}} = [\cdot]$

(H_{31}) $\mathsf{e}_0 = 42$

(H_{32}) $^{!}\langle\!\langle \mathbf{Start} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\emptyset;\emptyset;[\cdot]} = [\cdot]; 42$

Thus also:

(H_{33}) $^{?}\langle\!\langle \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_0} \rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\emptyset;\emptyset;[\cdot]} = \eth; \delta; \overline{\ell}; \mathsf{let}\ \mathsf{x}_0 = \mathsf{new}\ \mathsf{n}_0\ \mathsf{in}\ \mathsf{e}_0''$

From previous assumptions, we also know:

(H_{34}) $\mathsf{foo}, [\cdot]\,; \Xi; [\cdot]\,; \mathsf{comp}; [\cdot]\,; [\cdot]\,; [\cdot] \approx_{\emptyset;\emptyset} \mathbf{foo}, [\cdot]\,; \mathbf{\Xi}; [\cdot]\,; \mathbf{comp}; [\cdot]\,; [\cdot]\,; [\cdot]$

(H_{35}) $\Xi_{\mathsf{ctx}} = \mathsf{let}\ \mathsf{main}\ \mathsf{x} : \mathbb{N} \to \mathbb{N} := 42; \mathsf{e}_{01}, [\cdot]$

Now use Lemma 70 (Backtranslation Correctness of **Start**) on Assumptions (H_{16}), (H_{28}), (H_{29}), (H_{32}), (H_{34}) and (H_{35}) to get

(H_{36}) $\mathsf{foo}, [\cdot]\,; \Xi; [\cdot]\,; \mathsf{comp}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{call}\ \mathsf{main}\ 0 \xrightarrow{\mathsf{Start};\mathsf{comp}}{}^{n_0'}_{\mathsf{ctx}}\ \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{e}_{01}[0/\mathsf{x}]$

(H_{37}) $\xi; \mathbf{\Xi}; ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \multimap_{\emptyset;\emptyset} \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot]$

(H_{38}) $\mathsf{Start}; \mathsf{comp} \approx^{*}_{\emptyset;\emptyset} \mathbf{Start}; \mathbf{comp}$

By transitivity, we can split Assumption (H_{17}) arbitrarily, noting Assumption (H_7). So let n_1 be such that:

(H_{39}) $\mathbf{foo}, [\cdot]\,; \mathbf{\Xi}; ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathbf{e_0}[\mathbf{0}/\mathbf{x}] \xrightarrow{\overline{\mathsf{a}}_0}{}^{n_0}_{\mathsf{ctx}}\ \mathbf{foo}, [\cdot]\,; \mathbf{\Xi}; \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \mathbf{H_0^{ctx}}; [\cdot]\,; \mathbf{\Delta_0}$
$\mathbf{e_1}$

(H_{40}) $\mathbf{foo}, [\cdot]\,; \mathbf{\Xi}; ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \mathbf{H^{ctx}}; [\cdot]\,; \mathbf{\Delta_0} \triangleright \mathbf{e_1} \xrightarrow{\overline{\mathsf{a}}_b}{}^{*}_{\mathsf{ctx}}\ \mathbf{\Omega'} \triangleright \mathbf{f}$

(H_{41}) $\overline{\mathbf{a_a}} = \overline{\mathbf{a_0}} \cdot \overline{\mathbf{a_b}}$

With Assumptions (H_6), (H_7), (H_9), (H_{28}), (H_{29}), (H_{37}) and (H_{39}), we can use Lemma 72 (Middle of Backtranslation Correctness) to obtain:

(H_{42}) $\mathsf{foo}, [\cdot]\,; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{e}_{01}[0/\mathsf{x}] \xrightarrow{\overline{\mathsf{a}_0}}{}^{n'_1}_{\mathrm{ctx}} \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{K}\,[42]$

(H_{43}) $\mathbf{foo}, [\cdot]\,; \Xi; ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \multimap_{\emptyset;\emptyset} \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot]$

(H_{44}) $\overline{\mathsf{a}_0} \approx^*_{\emptyset;\emptyset} \overline{\mathbf{a_0}}$

By previous assumptions, it is clear that $\overline{\mathbf{a_b}} = \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_0}; \mathbf{ctx} \cdot \overline{\mathbf{a_c}}$, so by inversion on the execution Assumption (H_{40}), we know $\mathbf{e_1} = \mathbf{K'}\,[\mathbf{call}\ \mathbf{foo}\ \mathbf{v_0}]$.

By transitivity, we can split Assumption (H_{40}) arbitrarily:

(H_{45}) $\mathbf{foo}, [\cdot]\,; \Xi; \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \mathbf{H_0^{ctx}}; [\cdot]\,; \mathbf{\Delta_0} \triangleright \mathbf{K}\,[\mathbf{call}\ \mathbf{foo}\ \mathbf{v_0}] \xrightarrow{\mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_0}; \mathbf{ctx}}{}_{\mathrm{ctx}}\ \mathbf{foo}, [\cdot]\,; \Xi; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{comp}; \mathbf{H_0^{ctx}}; [\cdot]\,; \mathbf{\Delta_0} \triangleright \mathbf{e_{foo}}[\mathbf{v_0}/\mathbf{y}]$

(H_{46}) $\mathbf{foo}, [\cdot]\,; \Xi; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{comp}; \mathbf{H_0^{ctx}}; [\cdot]\,; \mathbf{\Delta_0} \triangleright \mathbf{e_{foo}} \xrightarrow{\overline{\mathbf{a_c}}}{}^*_{\mathrm{ctx}}\ \mathbf{\Omega'} \triangleright \mathbf{f}$

(H_{47}) $\overline{\mathbf{a_b}} = \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_0}; \mathbf{ctx} \cdot \overline{\mathbf{a_c}}$

By the fact that $\mathbf{foo} \in \xi$, we have $\mathbf{foo} \in \mathrm{dom}[\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$ and thus $\mathbf{e_{foo}} = [\![\mathsf{e_{foo}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$, where $\mathsf{let\ foo\ x} : \tau_\lambda := \mathsf{e_{foo}} \in \Xi_{\mathsf{comp}}$. Now, let:

(H_{45}) $\vdash \ell$ *fresh*

(H_{46}) $\Delta_0 \vdash \ell$ *fresh*

(H_{47}) $\vdash \mathsf{z}$ *fresh*

(H_{48}) $\delta = \{\ell \mapsto \ell\}$

(H_{49}) $\eth = \{\ell \mapsto \mathsf{z}\}$

We are ready to apply Lemma 73 (Backtranslation Correctness of **Call ?**) with Assumptions (H_{28}), (H_{29}), (H_{33}), (H_{43}) and (H_{45}) to (H_{49}):

(H_{50}) $\mathsf{L} = \{\ell\}$

(H_{51}) $\mathsf{X} = \{\mathsf{Alloc}\ \ell\ 42\}$

(H_{52}) $\mathsf{foo}, [\cdot]\,; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{K'}[\mathsf{let\ x} = \mathsf{new\ 42\ in\ call\ foo}\ \langle\!\langle\!\langle \mathsf{v_0} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}] \xrightarrow{\overline{\mathsf{a_1}}}{}^{n'_2}_{\mathrm{ctx}}\ \mathsf{foo}, [\cdot]\,; \Xi; (\mathsf{K}, \mathsf{foo}), ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{comp}; \Psi \triangleright \mathsf{e_{foo}}[\langle\!\langle\!\langle \mathsf{v_0} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}/\mathsf{y}]$

(H_{53}) $\mathsf{foo}, [\cdot]\,; \Xi; (\mathsf{K}; \mathsf{foo}), ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{comp}; \Psi \approx_{\delta;\mathsf{L}} \mathbf{foo}, [\cdot]\,; \Xi; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{comp}; \mathbf{H_0^{ctx}}; [\cdot]\,; \mathbf{\Delta}$

(H_{54}) $\overline{\mathsf{a_1}} \approx^*_{\delta;\mathsf{X}} \mathbf{Call}\ ?\ \mathbf{foo}\ \mathbf{v_0}; \mathbf{ctx}$

By transitivity, we can split Assumption (H_{46}) arbitrarily, noting that the immediate next action after $\overline{\mathbf{a_{comp}}}$ is $\mathbf{Ret}\ !\ \mathbf{v_1}$. So let n_3 be such that:

(H_{55}) $\mathbf{foo}, [\cdot]\,; \Xi; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{comp}; \mathbf{H_0^{ctx}}; [\cdot]\,; \mathbf{\Delta_0} \triangleright \mathbf{e_{foo}}[\mathbf{v_0}/\mathbf{y}] \xrightarrow{\overline{\mathbf{a_{comp}}}}{}_{\mathrm{ctx}}^{n_3}\ \mathbf{foo}, [\cdot]\,; \Xi; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{comp}; \mathbf{H_0^{ctx}}; \mathbf{H_0^{comp}}; \mathbf{\Delta_1} \triangleright \mathbf{K_c}[\mathbf{return}\ \mathbf{v_1}]$

(H_{56}) $\mathbf{foo}, [\cdot]\,; \Xi; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{comp}; \mathbf{H_0^{ctx}}; \mathbf{H_0^{comp}}; \mathbf{\Delta_1} \triangleright \mathbf{K_c}[\mathbf{return}\ \mathbf{v_1}] \xrightarrow{\overline{\mathbf{a_d}}}{}^{*}_{\mathrm{ctx}}\ \mathbf{\Omega'} \triangleright \mathbf{f}$

(H_{57}) $\overline{\mathbf{a_c}} = \overline{\mathbf{a_{comp}}} \cdot \overline{\mathbf{a_d}}$

Remember that $[\![\mathsf{e_{foo}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} = \mathbf{e_{foo}}$.
With Assumption (g) use Lemma 69 (Backtranslation is Well-Typed):

(H_{58}) $\Gamma \vdash \Xi\ \mathrm{ok}$

Now note that Lemma 22 (Static Typing implies Runtime Typing (Toplevel)) with Assumption (H_{58}) gives:

(H_{59}) $\vdash \mathsf{foo}, [\cdot]\,; \Xi; [\cdot]\,; \mathsf{comp}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{call\ main\ 0} : \mathbb{N}$

By Lemma 39 (Steps Preservation) with Assumptions (H_{36}) and (H_{59}):

(H_{60}) $\vdash \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{e_{01}}[0/\mathsf{x}] : \mathbb{N}$

By Lemma 39 (Steps Preservation) with Assumptions (H_{42}) and (H_{60}):

(H_{61}) $\vdash \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright \mathsf{K}[42] : \mathbb{N}$

By Lemma 39 (Steps Preservation) with Assumptions (H_{52}) and (H_{61}):

(H_{62}) $\vdash \mathsf{foo}, [\cdot]\,; \Xi; (\mathsf{K}, \mathsf{foo}), ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{comp}; \Psi \triangleright \mathsf{e_{foo}}[\langle\!\langle\!\langle \mathsf{v_0} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}/\mathsf{y}] : \mathbb{N}$

Using Assumption (H_{12}):

(H_{58}) $\mathsf{K} = [\cdot]\,; \mathsf{delete\ z}; \mathsf{K_r}[\langle\!\langle\!\langle \mathbf{v_1} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}]$

Use Assumptions (H_{53}), (H_{55}) and (H_{62}) to apply Lemma 56 (Component Correctness):

(H_{60}) $\delta \subseteq \delta'$

(H_{61}) $\mathsf{foo}, [\cdot]\,; \Xi; (\mathsf{K}; \mathsf{foo}), ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{comp}; \Psi \triangleright \mathsf{e_{foo}}[\mathsf{v_0}/\mathsf{y}] \xrightarrow{\overline{\mathsf{a_{comp}}}}{}^{n'_{\mathrm{a}}}_{\mathrm{ctx}}\ \mathsf{foo}, [\cdot]\,; \Xi; (\mathsf{K}; \mathsf{foo}), ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{comp}; \Psi' \triangleright \mathsf{K_c}[\mathsf{return\ v_1}]$

(H_{62}) $\mathsf{foo}, [\cdot]\,; \Xi; (\mathsf{K}; \mathsf{foo}), ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{comp}; \Psi' \approx_{\delta';\mathsf{L}} \mathbf{foo}, [\cdot]\,; \Xi; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{comp}; \mathbf{H_0^{ctx}}; \mathbf{H_0^{co}}$

(H_{63}) $\overline{\mathsf{a_{comp}}} \approx^{*}_{\delta';\mathsf{X}} \overline{\mathbf{a_{comp}}}$

By transitivity, we can split Assumption (H_{56}) arbitrarily:

(H_{64}) $\mathbf{foo}, [\cdot]\,; \Xi; (\mathbf{K}; \mathbf{foo}), \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{comp}; \Psi' \triangleright \mathbf{K_c}[\mathbf{return}\ \mathbf{v_1}] \xrightarrow{\mathbf{Ret}\ !\ \mathbf{v_1}}{}_{\mathrm{ctx}}\ \mathbf{foo}, [\cdot]\,; \Xi; \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \Psi' \triangleright \mathbf{K}[\mathbf{v_1}]$

(H_{65}) $\mathbf{foo}, [\cdot]\,; \Xi; \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \Psi' \triangleright \mathbf{K}[\mathbf{v_1}] \xrightarrow{\overline{\mathbf{a_e}}}{}^{*}_{\mathrm{ctx}}\ \mathbf{\Omega'} \triangleright \mathbf{f}$

(H_{66}) $\overline{\mathbf{a_d}} = \mathbf{Ret}\ !\ \mathbf{v_1} \cdot \overline{\mathbf{a_e}}$

Remember that $[\![\mathsf{e_{foo}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} = \mathbf{e_{foo}}$, so:

(H_{67}) $[\![\mathsf{K_c}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} = \mathbf{K_c}$.

Using Assumptions (H_{48}) and (H_{60}):

(H_{68}) $\delta'(\ell) = \ell$

We are ready to apply Lemma 71 (Backtranslation Correctness of **Ret**) with Assumptions (H_{12}), (H_{28}), (H_{29}), (H_{49}), (H_{50}), (H_{58}), (H_{62}), (H_{64}), (H_{67}) and (H_{68}):

(H_{69}) $\mathsf{X}' = \mathsf{X} \cup \{\mathsf{Dealloc}\ \ell\}$

(H_{70}) $\mathsf{foo}, [\cdot]\,; \Xi; (\mathsf{K}; \mathsf{foo}), ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{comp}; \Psi' \triangleright \mathsf{K_c}[\mathsf{return}\ \mathsf{v_1}] \xrightarrow{\overline{\mathsf{a_2}}}{}^{n'_4}_{\mathsf{ctx}}\ \mathsf{foo}, [\cdot]\,; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; \Psi' \triangleright \mathsf{K}[\mathsf{v_1}]$

(H_{71}) $\mathbf{foo}, [\cdot]\,; \Xi; \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \Psi' \multimap_{\delta';\mathsf{X}'} \mathsf{foo}, [\cdot]\,; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; \Psi'$

(H_{72}) $\overline{\mathsf{a_2}} \approx^*_{\delta';\mathsf{L}} \mathbf{Ret}\ !\ \mathbf{v_1}$

By transitivity, we can split Assumption (H_{65}) arbitrarily, noting that the immediate next action after $\overline{\mathbf{a_1}}$ is $\mathbf{End}\ \mathbf{v_2}$. So, let n_5 be such that:

(H_{73}) $\mathbf{foo}, [\cdot]\,; \Xi; \overline{\mathbf{K}}, ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \Psi' \triangleright \mathbf{K}[\mathbf{v_1}] \xrightarrow{\overline{\mathbf{a_1}}}{}_{\mathrm{ctx}}{}^{n_5}\ \mathbf{foo}, [\cdot]\,; \Xi; ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \Psi'' \triangleright \mathbf{K_e}[\mathbf{v_2}]$

(H_{74}) $\mathbf{foo}, [\cdot]\,; \Xi; ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \Psi'' \triangleright \mathbf{K_e}[\mathbf{v_2}] \xrightarrow{\mathbf{End}\ \mathbf{v_2}}{}^{*}_{\mathrm{ctx}}\ \Omega' \triangleright \mathbf{f}$

(H_{75}) $\overline{\mathbf{a_e}} = \overline{\mathbf{a_1}} \cdot \mathbf{End}\ \mathbf{v_2}$

With Assumptions (H_6), (H_{11}), (H_{13}), (H_{28}), (H_{29}), (H_{71}) and (H_{73}), we can use Lemma 72 (Middle of Backtranslation Correctness) to obtain:

(H_{76}) $\mathsf{foo}, [\cdot]\,; \Xi; \overline{\mathsf{K}}, ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; \Psi' \triangleright \mathsf{K}[\mathsf{v_1}] \xrightarrow{\overline{\mathsf{a_3}}}{}^{n'_5}_{\mathsf{ctx}}\ \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; \Psi' \triangleright \mathsf{K_e}\,[42]$

(H_{77}) $\mathbf{foo}, [\cdot]\,; \Xi; ([\cdot]\,; \mathbf{main}), [\cdot]\,; \mathbf{ctx}; \Psi'' \multimap_{\delta';\emptyset} \xi; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; \Psi'$

(H_{78}) $\overline{\mathsf{a_3}} \approx^*_{\delta';\emptyset} \overline{\mathbf{a_1}}$

Let:

(H_{79}) $\Im, \delta' \vdash \ell'$ *fresh*

(H_{80}) $\Im \vdash \mathsf{w}$ *fresh*

(H_{81}) $\delta \vdash \ell'$ *fresh*

(H_{82}) $\delta'' = \delta' \cup \{\ell' \mapsto \ell'\}$

(H_{83}) $\Im' = \Im \cup \{\ell' \mapsto \mathsf{w}\}$

Finally, use Lemma 74 (Backtranslation Correctness of **End**) with Assumptions (H_{14}), (H_{28}), (H_{29}), (H_{74}), (H_{77}) and (H_{79}) to (H_{83}) to acquire:

(H_{79}) $\mathsf{L}' = \mathsf{L} \cup \{\ell'\}$

(H_{80}) $\mathsf{X}'' = \mathsf{X}' \cup \{\mathsf{Alloc}\ \ell'\ 42, \mathsf{Dealloc}\ \ell'\}$

(H_{81}) $\mathsf{foo}, [\cdot]\,; \Xi; ([\cdot]\,; \mathsf{main}), [\cdot]\,; \mathsf{ctx}; \Psi'' \triangleright \mathsf{K_e}[42] \xrightarrow{\overline{\mathsf{a}}_4}{}^{n'_6}_{\mathsf{ctx}}\ \xi; \Xi; [\cdot]\,; \mathsf{ctx}; \Psi'' \triangleright \mathsf{v_2}$

(H_{82}) $\xi; \Xi; [\cdot]\,; \mathsf{ctx}; \Psi'' \approx_{\delta'';\mathsf{L}'} \mathbf{foo}, [\cdot]\,; \Xi; [\cdot]\,; \mathbf{ctx}; \Psi''$

(H_{83}) $\overline{\mathsf{a}}_4 \approx^*_{\delta'';\mathsf{X}''}$ **End $\mathbf{v_2}$**

Finally, instantiate the existentials in our goals as follows:

(H_{84}) $\delta_e = \delta''$

(H_{85}) $\mathsf{L_e} = \mathsf{L}'$

(H_{86}) $\mathsf{X_e} = \mathsf{X}''$

(H_{87}) $\Omega'_\mathsf{e} = \xi; \Xi; [\cdot]\,; \mathsf{ctx}; \Psi''$

(H_{88}) $\overline{\mathsf{a}}_\mathsf{e} = \mathsf{Start}; \mathsf{comp} \cdot \overline{\mathsf{a}_0} \cdot \overline{\mathsf{a}_1} \cdot \overline{\mathsf{a}_{\mathsf{comp}}} \cdot \overline{\mathsf{a}_2} \cdot \overline{\mathsf{a}_3} \cdot \overline{\mathsf{a}_4}$

We solve Goal (i) by transitivity, making use of Assumptions (H_{36}), (H_{42}), (H_{52}), (H_{61}), (H_{70}), (H_{76}) and (H_{81}) and Lemma 20 ($\rightarrow^n_{ctx}$ and $\rightarrow^*_{ctx}$ yield $\rightarrow^*_{ctx}$).

Goal (ii) by Assumption (H_{82}).

Lastly, Goal (iii) by transitivity, making use of Assumptions (H_{38}), (H_{44}), (H_{54}), (H_{63}), (H_{72}), (H_{78}) and (H_{83}). □

Definition 41 ($\mathsf{L}_{\mathrm{tms}}$ Robust Satisfaction)**.** *We write* $\Xi_{\mathsf{comp}} \vDash_R \pi$ *for If*

(a) $\mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \overset{\overline{\mathsf{a}}}{\Longrightarrow} \Omega \triangleright \mathsf{f}_{\lightning}$

Then $\exists \delta_{MS}$

(i) $\theta_{\delta_{MS}}(\overline{\mathsf{a}}) \in \pi$

Definition 42 (L_{ms} Robust Satisfaction)**.** *We write* $\mathbf{\Xi_{comp}} \vDash_R \pi$ *for If*

(a) $\mathbf{prog}\ \mathbf{\Xi_{ctx}}\ \mathbf{\Xi_{comp}} \overset{\overline{\mathsf{a}}}{\Longrightarrow} \mathbf{\Omega} \triangleright \mathbf{f}_{\lightning}$

Then $\exists \delta_{MS}$

(i) $\theta_{\delta_{MS}}(\overline{\mathsf{a}}) \in \pi$

Theorem 3 (Robust Memory Safety Preservation)**.**

(i) $\vdash_{\sigma_\sim} [\![\bullet]\!]^{\mathsf{L}_{tms} \rightarrow \mathsf{L}_{ms}} : \mathrm{msafe}$

Proof. Unfolding the definition of $\vdash_{\sigma_\sim} [\![\bullet]\!]^{\mathsf{L}_{\mathrm{tms}} \rightarrow \mathsf{L}_{\mathrm{ms}}} : \mathrm{msafe}$: If

(a) $\Xi_{\mathsf{comp}} \vDash_R \sigma_\sim(\mathrm{msafe})$

then

(i) $[\![\Xi_{\mathsf{comp}}]\!]^{\mathsf{L}_{\mathrm{tms}} \rightarrow \mathsf{L}_{\mathrm{ms}}} \vDash_R \mathrm{msafe}$

By Definition 34, $\mathrm{msafe} = \mathrm{tmssafe} \cap \mathrm{smsafe}$.

By distributivity of σ over intersection:

(H_1) $\sigma_\sim(\mathrm{msafe}) = \sigma_\sim(\mathrm{tmssafe}) \cap \sigma_\sim(\mathrm{smsafe})$

So Assumption (a) gives us:

(H_2) $\Xi_{\mathsf{comp}} \vDash_R \sigma_\sim(\mathrm{tmssafe})$

(H_3) $\Xi_{\mathsf{comp}} \vDash_R \sigma_\sim(\text{smsafe})$

Note: Assumption (H_2) is satisfiable because well-typed source programs satisfy tmssafe by type safety, and tmssafe $\subseteq \sigma_\sim(\text{tmssafe})$ follows from Lemma 63 (Filters Equal) (related traces have equal filtered forms, so if the source trace satisfies tmssafe, all related target traces do as well).

Unfold Goal (i): If

(F_1) $\mathbf{prog}\ \Xi_{\mathbf{ctx}}\ [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}} \overset{\bar{\mathbf{a}}}{\Longrightarrow} \mathbf{\Omega} \triangleright \mathbf{f}_{\lightning}$

then $\exists \delta_{\mathrm{MS}}$

(i) $\theta_{\delta_{\mathrm{MS}}}(\bar{\mathbf{a}}) \in \text{msafe}$

We must show $\bar{\mathbf{a}} \in \text{tmssafe} \cap \text{smsafe}$, i.e., two sub-goals.

Suppose $\mathbf{f}_{\lightning} = \mathbf{v}$.Invert Assumption (F_1):

(H_4) $\Xi = \Xi_{\mathbf{ctx}} \bowtie [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(H_5) $\mathbf{main} \notin \xi$

(H_6) $\xi = \mathrm{dom}\ [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$

(H_7) $[\![\xi]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}; \Xi; \emptyset; \mathbf{comp}; \emptyset; \emptyset; \emptyset \triangleright \mathbf{call\ main\ 0} \xrightarrow{\bar{\mathbf{a}}}{}^*_{\mathrm{ctx}} \mathbf{\Omega} \triangleright \mathbf{v}$

We backtranslate:

(H_8) $\langle\!\langle\!\langle \bar{\mathbf{a}} \rangle\!\rangle\!\rangle^{\mathrm{L_{ms}} \to \mathrm{L_{tms}}}_{\emptyset;\emptyset} = \supset; \delta; \mathsf{let\ main\ z\ := e}, [\cdot].$

Invert Assumption (H_8), so we can decompose $\bar{\mathbf{a}}$ into $\mathbf{Start} \cdot \overline{\mathbf{a}_0^{\mathbf{ctx}}} \cdot \mathbf{Call\ ?\ foo\ n} \cdot \overline{\mathbf{a}^{\mathbf{comp}}} \cdot \mathbf{Ret\ !\ m} \cdot \overline{\mathbf{a}_1^{\mathbf{ctx}}} \cdot \mathbf{End\ n'}$ such that $\vdash \overline{\mathbf{a}^{\mathbf{comp}}}$ non-int-**trace**, $\vdash \overline{\mathbf{a}_0^{\mathbf{ctx}}}$ non-int-**trace**, and $\vdash \overline{\mathbf{a}_1^{\mathbf{ctx}}}$ non-int-**trace**.

By backtranslation correctness, there is a source execution producing $\bar{\mathsf{a}}$ such that $\bar{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \bar{\mathbf{a}}$.

Sub-goal 1: $\bar{\mathbf{a}} \in \text{tmssafe}$

By Assumption (H_2), since $\bar{\mathsf{a}}$ comes from linking Ξ_{comp} with the backtranslated context, we have $\bar{\mathsf{a}} \in \sigma_\sim(\text{tmssafe})$.

By Definition 15, since $\bar{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \bar{\mathbf{a}}$, we conclude $\bar{\mathbf{a}} \in \text{tmssafe}$.

Sub-goal 2: $\bar{\mathbf{a}} \in \text{smsafe}$

By Assumption (H_3), $\bar{\mathsf{a}} \in \sigma_\sim(\text{smsafe})$.

By Lemma 64 (Cross-Language Relation Preserves SMS), the cross-language trace relation $\sim$ is well-formed with respect to smsafe.

By Definition 15, since $\bar{\mathsf{a}} \approx^*_{\delta;\mathsf{X}} \bar{\mathbf{a}}$, we conclude $\bar{\mathbf{a}} \in \text{smsafe}$.

Conclusion: Combining sub-goals 1 and 2, $\bar{\mathbf{a}} \in \text{tmssafe} \cap \text{smsafe} = \text{msafe}$.

What remains is to show the filtered trace satisfies the property. By Lemma 62 (MS-Location Generator) using $\mathrm{dom}\,\delta$, there is a $\delta_{MS} : \bullet \to \underline{\bullet}$ injective.

Instantiate the goal:

(H_9) $\delta'_{MS} : \bullet \to \underline{\bullet}, \ell : \mathbf{L} \mapsto \delta_{MS}(\delta^{-1}(\ell))$

From the shape of the trace $\bar{\mathbf{a}}$, $\mathbf{foo} \in [\![\Xi_{\mathsf{comp}}]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$.

So $\mathsf{foo} \in \Xi_{\mathsf{comp}}$ and $\mathbf{main} \in \Xi_{\mathbf{ctx}}$.

Let $\Xi_{\mathsf{ctx}} = (\mathsf{let\ main\ z\ := e}), [\cdot]$.

Let $\Xi = \Xi_{\mathsf{ctx}} \bowtie \Xi_{\mathsf{comp}}$ which follows by inverting Assumption (H_4).

Let $\Gamma_0 = \Xi \downarrow$ and note that $\mathsf{main} \in \operatorname{dom} \Gamma_0$.

We need to verify that $\Gamma_0 \vdash \Xi$ ok to conclude $\vdash \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} \dashv \Xi, \operatorname{dom} \Xi_{\mathsf{comp}}$. This follows from Lemma 69 (Backtranslation is Well-Typed).

By Rules state-eq, empty-memstate-eq, cfstate-eq and empty-commlib-lib-eq to empty-kontstack-eq, we have $\xi; \Xi; \emptyset; \mathsf{comp}; \emptyset; \emptyset; \emptyset \approx_{\delta;\mathsf{L}} [\![\xi]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}; \mathbf{\Xi}; \emptyset; \mathbf{comp}; \emptyset; \emptyset; \emptyset$.

Finally, use Lemma 63 (Filters Equal), whose assumptions are satisfied by the results we got from applying backtranslation correctness. □

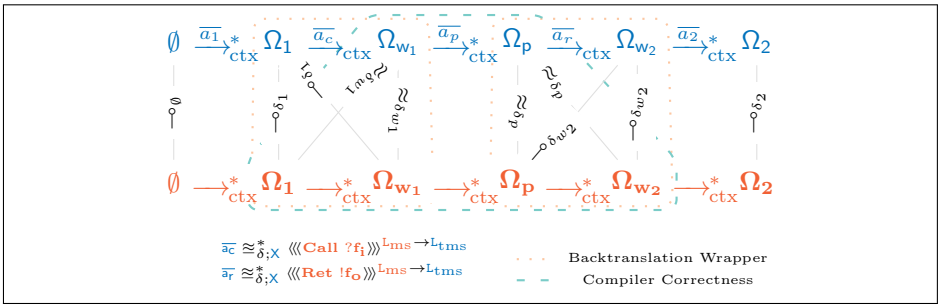

Figure 55: Sketch of the proof of Theorem 3 (Robust Memory Safety Preservation). Arrows in the horizontal direction are the usual step relations, but the term is omitted for clarity. Likewise for the emitted target-level traces. Trace $\overline{\mathsf{a_1}}$ starts with Start and $\overline{\mathsf{a_2}}$ ends with $\mathsf{End}\ \mathsf{v}$, where v is the final result of the program. Target states $\mathbf{\Omega_1}$ and $\mathbf{\Omega_{w_1}}$ are exactly equal, so are $\mathbf{\Omega_p}$ and $\mathbf{\Omega_{w_2}}$. They are drawn for aesthetic purposes.

4.5 Robust SMS Preserving Compiler

For instrumented $\mathtt{L}_{\mathrm{ms}}$ code, we'll write $\mathtt{L}$. The syntax and semantics are entirely identical.

4.5.1 Translation to Specification Events

$\boxed{\delta_{MS}(\ell) = \underline{\ell}}$ „Map $\mathtt{L}_{\mathrm{ms}}$ locations ℓ to abstract locations $\underline{\ell}$."

$\boxed{T_{\mathrm{SMS}} \simeq_{\delta_{MS}} \Delta}$ „Abstract memory state T_{TMS} describes the concrete state Δ."

$$\frac{}{\emptyset \simeq_{\delta_{MS}} [\cdot]}\ \text{(SMS-Empty-Agree)} \qquad \frac{}{\emptyset \simeq_{\delta_{MS}} \lightning}\ \text{(SMS-Abort-Agree)}$$

$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{\ell} \notin T_{\mathrm{SMS}} \qquad T_{\mathrm{SMS}} \simeq_{\delta_{MS}} \Delta}{\{(\underline{\ell}, \boldsymbol{m})\} \cup T_{\mathrm{SMS}} \simeq_{\delta_{MS}} \mathbf{x} \mapsto (\ell; \mathbf{comp}; \rho; \mathbf{n}), \Delta}\ \text{(SMS-Cons-Agree)}$$

$$\frac{T_{\mathrm{SMS}} \simeq_{\delta_{MS}} \Delta}{T_{\mathrm{SMS}} \simeq_{\delta_{MS}} \mathbf{x} \mapsto (\ell; \mathbf{ctx}; \rho; \mathbf{n}), \Delta}\ \text{(SMS-Ignore-Agree)}$$

Figure 56: Store Agreement.

$\boxed{\delta_{MS}(\ell) = \underline{\ell}}$ „A map from $\mathrm{L_{ms}}$ memory locations ℓ to specification locations $\underline{\ell}$."

$\boxed{\theta_{\delta_{MS}}(\mathbf{a}) = a^{\mathrm{ms}}}$ „Project an $\mathrm{L_{ms}}$ event to specification events."

(t-filter-context)
$$\frac{\mathbf{a_b} \neq \lightning}{\theta_{\delta_{MS}}(\mathbf{a_b}; \mathbf{ctx}) = \underline{\varepsilon}}$$

(t-filter-comp-start)
$$\frac{}{\theta_{\delta_{MS}}(\mathbf{Start}; \mathbf{comp}) = \underline{\varepsilon}}$$

(t-filter-comp-alloc)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathbf{n}}{\theta_{\delta_{MS}}(\mathbf{Alloc}\ \ell\ \mathbf{n}; \mathbf{comp}) = \underline{\mathit{Alloc}\ \ell\ n}}$$

(t-filter-comp-dealloc)
$$\frac{\delta_{MS}(\ell) = \underline{\ell}}{\theta_{\delta_{MS}}(\mathbf{Dealloc}\ \ell; \mathbf{comp}) = \underline{\mathit{Dealloc}\ \ell}}$$

(t-filter-comp-get)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathbf{n}}{\theta_{\delta_{MS}}(\mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{comp}) = \underline{\mathit{Use}\ \ell\ n}}$$

(t-filter-comp-set)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathbf{n}}{\theta_{\delta_{MS}}(\mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{v}; \mathbf{comp}) = \underline{\mathit{Use}\ \ell\ n}}$$

(t-filter-comp-call)
$$\frac{}{\theta_{\delta_{MS}}(\mathbf{Call}\ \mathbf{c}\ \mathbf{foo}\ \mathbf{v}; \mathbf{comp}) = \underline{\varepsilon}}$$

(t-filter-comp-ret)
$$\frac{}{\theta_{\delta_{MS}}(\mathbf{Ret}\ \mathbf{c}\ \mathbf{v}; \mathbf{comp}) = \underline{\varepsilon}}$$

(t-filter-abort)
$$\frac{}{\theta_{\delta_{MS}}(\lightning; \mathbf{t}) = \underline{\lightning}}$$

$\boxed{\theta^*_{\delta_{MS}}(\overline{\mathbf{a}}) = \overline{a^{\mathrm{ms}}}}$ „Project an $\mathrm{L_{ms}}$ trace to specification traces."

(t-filter-empty)
$$\frac{}{\theta^*_{\delta_{MS}}([\cdot]) = \underline{[\cdot]}}$$

(t-filter-cons-relevant)
$$\frac{\theta_{\delta_{MS}}(\mathbf{a}) = \underline{a} \qquad \theta^*_{\delta_{MS}}(\overline{\mathbf{a}}) = \overline{a^{\mathrm{ms}}} \qquad \underline{a} \neq \underline{\varepsilon}}{\theta^*_{\delta_{MS}}(\mathbf{a} \cdot \overline{\mathbf{a}}) = \underline{a} \cdot \overline{a^{\mathrm{ms}}}}$$

(t-filter-cons-relevant)
$$\frac{\theta_{\delta_{MS}}(\mathbf{a}) = \underline{\varepsilon} \qquad \theta^*_{\delta_{MS}}(\overline{\mathbf{a}}) = \overline{a^{\mathrm{ms}}}}{\theta^*_{\delta_{MS}}(\mathbf{a} \cdot \overline{\mathbf{a}}) = \overline{a^{\mathrm{ms}}}}$$

Figure 57: Projection of $\mathrm{L_{ms}}$ events to specification events.

4.5.2 Compiler

$\boxed{[\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} = e'}$ „Compile $\mathtt{L}_{\mathrm{ms}}$ expression $\mathbf{e}$ to $\mathtt{L}_{\mathrm{ms}}$ expression e'."

$\boxed{[\![[\mathbf{v}/\mathbf{x}]]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} = [v/x]}$ „Compile $\mathtt{L}_{\mathrm{ms}}$ substitution to $\mathtt{L}$ substitution."

$\boxed{[\![\xi]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} = \xi'}$ „Compile $\mathtt{L}_{\mathrm{ms}}$ component library to $\mathtt{L}$ component library."

$$
\begin{array}{rcl}
[\![\mathbf{f}_i]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & f_i \\
[\![\mathbf{call\ foo\ e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \mathit{call}\ [\![\mathbf{foo}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{return\ e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \mathit{return}\ [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{e_1} \oplus \mathbf{e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & [\![\mathbf{e_1}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \oplus [\![\mathbf{e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{x}[\mathbf{e}]]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \mathit{let}\ x_{ACCESS} = [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{in} \\
 & & \mathit{ifz}\ 0 \leq x_{ACCESS} < x_{SIZE}\ \mathit{then} \\
 & & \quad x[x_{ACCESS}] \\
 & & \mathit{else}\ \mathit{abort}() \\
[\![\mathbf{let\ x = e_1\ in\ e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \mathit{let}\ x = [\![\mathbf{e_1}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{in}\ [\![\mathbf{e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{x}[\mathbf{e_1}] \leftarrow \mathbf{e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \mathit{let}\ x_{ACCESS} = [\![\mathbf{e_1}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{in} \\
 & & \mathit{ifz}\ 0 \leq x_{ACCESS} < x_{SIZE}\ \mathit{then} \\
 & & \quad x[x_{ACCESS}] \leftarrow [\![\mathbf{e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
 & & \mathit{else}\ \mathit{abort}() \\
[\![\mathbf{let\ x = new\ e_1\ in\ e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \mathit{let}\ x_{\mathtt{SIZE}} = [\![\mathbf{e_1}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{in} \\
 & & \mathit{let}\ x = \mathit{new}\ x_{\mathtt{SIZE}}\ \mathit{in}\ [\![\mathbf{e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{delete\ x}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \mathit{delete}\ x \\
[\![\mathbf{x\ is}\ \text{☣}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & x\ \mathit{is}\ \text{☣} \\
[\![(\mathbf{e_1}; \mathbf{e_2})]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & ([\![\mathbf{e_1}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}; [\![\mathbf{e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}) \\
[\![\pi_1\ \mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \pi_1\ [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\pi_2\ \mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \pi_2\ [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{e\ has}\ \tau]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{has}\ \tau \\
[\![\mathbf{ifz\ e_1\ then\ e_2\ else\ e_3}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \mathit{ifz}\ [\![\mathbf{e_1}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
 & & \mathit{then}\ [\![\mathbf{e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
 & & \mathit{else}\ [\![\mathbf{e_3}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
\\
[\![[\mathbf{v}/\mathbf{x}]]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & [\,[\![\mathbf{v}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}/[\![\mathbf{x}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\,] \\
\\
[\![[\cdot]]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & [\cdot] \\
[\![\mathbf{foo}, \xi]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} & = & \mathit{foo}, [\![\xi]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}
\end{array}
$$

Figure 58: Compiler from $\mathtt{L}_{\mathrm{ms}}$ to $\mathtt{L}$.

$$\boxed{[\![\mathbf{F}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} = F}$$ „Compile $\mathtt{L}_{\mathrm{ms}}$ procedures to $\mathtt{L}$ procedures."

$$[\![\mathbf{let\ foo\ x := e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \quad = \quad \mathit{let\ foo\ x :=} [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}$$

$$\boxed{[\![\Xi]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} = \Xi}$$ „Compile $\mathtt{L}_{\mathrm{ms}}$ libraries to $\mathtt{L}$ libraries."

$$\begin{aligned}
[\![[\cdot]]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= [\cdot] \\
[\![\mathbf{F}, \Xi]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= [\![\mathbf{F}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}, [\![\Xi]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}
\end{aligned}$$

Figure 59: Compiler from $\mathtt{L}_{\mathrm{ms}}$ components to $\mathtt{L}$ components.

$$\boxed{[\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} = K}$$ „Compile $\mathtt{L}_{\mathrm{ms}}$ evaluation contexts to $\mathtt{L}$ evaluation contexts."

$$\begin{aligned}
[\![[\cdot]]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= [\cdot] \\
[\![\mathbf{K} \oplus \mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \oplus [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{v} \oplus \mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= [\![\mathbf{v}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \oplus [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{x}[\mathbf{K}]]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= \mathit{let}\ x_{ACCESS} = [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{in} \\
&\quad \mathit{if}\ 0 \leq x_{ACCESS} < x_{SIZE}\ \mathit{then} \\
&\quad\ x[x_{ACCESS}] \\
&\quad \mathit{else}\ \mathit{abort}() \\
[\![\mathbf{let\ x\ = K\ in\ e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= \mathit{let}\ x\ = [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{in}\ [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{x}[\mathbf{K}] \leftarrow \mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= \mathit{let}\ x_{ACCESS} = [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{in} \\
&\quad \mathit{if}\ 0 \leq x_{ACCESS} < x_{SIZE}\ \mathit{then} \\
&\quad\ x[x_{ACCESS}] \leftarrow [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
&\quad \mathit{else}\ \mathit{abort}() \\
[\![\mathbf{x}[\mathbf{n}] \leftarrow \mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= \mathit{if}\ 0 \leq n < x_{SIZE}\ \mathit{then} \\
&\quad\ x[n] \leftarrow [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
&\quad \mathit{else}\ \mathit{abort}() \\
[\![\mathbf{let\ x\ = new\ K\ in\ e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= \mathit{let}\ x_{SIZE} = [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{in} \\
&\quad \mathit{let}\ x = \mathit{new}\ x_{SIZE}\ \mathit{in}\ [\![\mathbf{e}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{ifz\ K\ then\ e_1\ else\ e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= \mathit{ifz}\ [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}\ \mathit{then} \\
&\quad [\![\mathbf{e_1}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
&\quad \mathit{else} \\
&\quad [\![\mathbf{e_2}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{call\ foo\ K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= \mathit{call\ foo}\ [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \\
[\![\mathbf{return\ K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} &= \mathit{return}\ [\![\mathbf{K}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}
\end{aligned}$$

Figure 60: Compiling $\mathtt{L}_{\mathrm{ms}}$ evaluation contexts to $\mathtt{L}$ evaluation contexts.

$$\frac{\Delta_1, x \mapsto (\ell; \mathit{comp}; \rho; m), \Delta_2 \Vdash \gamma}{\Delta_1, x \mapsto (\ell; \mathit{comp}; \rho; m), \Delta_2 \Vdash [m/x_{SIZE}], \gamma}\ (\gamma\text{-}x_{SIZE}\text{-preserve}) \qquad \frac{\Delta \Vdash \gamma}{\Delta \Vdash [v/x], \gamma}\ (\gamma\text{-}x_{SIZE}\text{-skip})$$

$$\frac{}{\Delta \Vdash [\cdot]}\ (\gamma\text{-}x_{SIZE}\text{-empty})$$

Figure 61: Well formed $\mathtt{L}$ substitutions.

$$\frac{}{[\cdot] \prec [\cdot]}\ \text{(subst-subset-empy)} \qquad \frac{\mathbf{x} = x \quad \mathbf{v} = v \quad \gamma \prec \gamma}{[\mathbf{v}/\mathbf{x}], \gamma \prec [v/x], \gamma}\ \text{(subst-subset-cons)} \qquad \frac{\gamma \prec \gamma}{\gamma \prec [v/x_{SIZE}], \gamma}\ \text{(subst-subset-}x_{SIZE}\text{)}$$

Figure 62: Cross-Language Substitutions subset.

$\boxed{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell}}$ „A map from L memory locations ℓ to specification locations $\underline{\ell}$."

$\boxed{\theta_{\delta_{\mathtt{sCCT}}}(a) = a^{\mathrm{scct}}}$ „Project an L event to specification events."

$$\frac{a_b \neq \lightning}{\theta_{\delta_{\mathtt{sCCT}}}(a_b; ctx) = \underline{\varepsilon}}\ \text{(i-scct-filter-context)} \qquad \frac{}{\theta_{\delta_{\mathtt{sCCT}}}(Start; comp) = \underline{\varepsilon}}\ \text{(i-scct-filter-comp-start)}$$

$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(End\ v; comp) = \underline{\varepsilon}}\ \text{(i-scct-filter-comp-end)} \qquad \frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell} \quad \underline{n} = n}{\theta_{\delta_{\mathtt{sCCT}}}(Alloc\ \ell\ n; comp) = \underline{Alloc\ \ell\ n; 🔓}}\ \text{(i-scct-filter-comp-alloc)}$$

$$\frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell}}{\theta_{\delta_{\mathtt{sCCT}}}(Dealloc\ \ell; comp) = \underline{Dealloc\ \ell; 🔓}}\ \text{(i-scct-filter-comp-dealloc)} \qquad \frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell} \quad \underline{n} = n}{\theta_{\delta_{\mathtt{sCCT}}}(Get\ \ell\ n; comp) = \underline{\varepsilon}}\ \text{(i-scct-filter-comp-get-noleak)}$$

$$\frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell} \quad \underline{n} = n}{\theta_{\delta_{\mathtt{sCCT}}}(Set\ \ell\ n\ v; comp) = \underline{\varepsilon}}\ \text{(i-scct-filter-comp-set-noleak)} \qquad \frac{}{\theta_{\delta_{\mathtt{sCCT}}}(Call\ \mathbf{c}\ foo\ v; comp) = \underline{\varepsilon}}\ \text{(i-scct-filter-comp-call)}$$

$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(Ret\ \mathbf{c}\ v; comp) = \underline{\varepsilon}}\ \text{(i-scct-filter-comp-ret)} \qquad \frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\lightning; t) = \underline{\lightning; 🔓}}\ \text{(i-scct-filter-abort)} \qquad \frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\varepsilon) = \underline{\varepsilon}}\ \text{(i-scct-filter-emptyevent)}$$

$\boxed{\theta^*_{\delta_{\mathtt{sCCT}}}(\overline{a}) = \overline{a^{\mathrm{scct}}}}$ „Project an L trace to specification traces."

$$\frac{}{\theta^*_{\delta_{\mathtt{sCCT}}}([\cdot]) = \underline{[\cdot]}}\ \text{(i-scct-filter-empty)} \qquad \frac{\theta_{\delta_{\mathtt{sCCT}}}(a) = \underline{a} \quad \theta^*_{\delta_{\mathtt{sCCT}}}(\overline{a}) = \overline{a^{\mathrm{scct}}} \quad \underline{a} \neq \underline{\varepsilon}}{\theta^*_{\delta_{\mathtt{sCCT}}}(a \cdot \overline{a}) = \underline{a} \cdot \overline{a^{\mathrm{scct}}}}\ \text{(i-scct-filter-cons-relevant)}$$

$$\frac{\theta_{\delta_{\mathtt{sCCT}}}(a) = \underline{\varepsilon} \quad \theta^*_{\delta_{\mathtt{sCCT}}}(\overline{a}) = \overline{a^{\mathrm{scct}}}}{\theta^*_{\delta_{\mathtt{sCCT}}}(a \cdot \overline{a}) = \overline{a^{\mathrm{scct}}}}\ \text{(i-scct-filter-cons-relevant)}$$

Figure 63: Projection of L events to specification events.

Lemma 76 (Injective Compiler (for final expressions))**.** *If*

(a) $[\![\mathbf{f}_{\lightning}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}} = e_0$

(b) $[\![\mathbf{f}_{\lightning}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}} = e_1$

Then

(i) $e_0 = e_1$

Proof. Case analysis on f. □

Lemma 77 (Subst $[\![\bullet]\!]^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}}$ Compatibility)**.**

(i) $[\![\mathbf{e}\gamma]\!]^{\mathrm{L}_{ms} \to \mathrm{L}} = [\![\mathbf{e}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}} [\![\gamma]\!]^{\mathrm{L}_{ms} \to \mathrm{L}}$

Proof. Induction on $\mathbf{e}$. □

Lemma 78 (Prim. Forward Simulation)**.** *If*

(a) $\mathbf{\Omega} \triangleright \mathbf{e}\gamma \xrightarrow{\mathbf{a}} \mathbf{\Omega}' \triangleright \mathbf{e}'\gamma'$

(b) $\mathbf{\Omega} = \Omega = \mathbf{\Psi}; \mathbf{comp}; \mathbf{\Phi}$

(c) $\vdash T_{SMS} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{SMS}{}'$

(d) $T_{SMS} \simeq_{\delta} \mathbf{\Omega}$

(e) $\theta_{\delta}(\mathbf{a}) \cong \boldsymbol{a}$

(f) $\mathbf{a} = a$

(g) $\mathbf{\Omega}' = \Omega'$

(h) $\Omega.\Delta \Vdash \gamma$

(i) $\gamma \prec \gamma$

Then $\exists \gamma'$,

(i) $\Omega \triangleright [\![\mathbf{e}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}}\gamma \xrightarrow{a}{}^{*}_{ctx} \Omega' \triangleright [\![\mathbf{e}']\!]^{\mathrm{L}_{ms} \to \mathrm{L}}\gamma'$

(ii) $T_{SMS}{}' \simeq_{\delta} \mathbf{\Omega}'$

(iii) $\Omega'.\Delta \Vdash \gamma'$

(iv) $\gamma' \prec \gamma'$

Proof. Induction on Assumption (a).

Case $e - \oplus$**:** If

(H_1) $n_3 = n_1 \oplus n_2$

(H_2) $\gamma = \gamma' = [\cdot]$

(H_3) $\mathbf{\Omega} \triangleright \mathbf{n_1} + \mathbf{n_2} \xrightarrow{\varepsilon} \mathbf{\Omega} \triangleright \mathbf{n_3}$

(H_4) $\mathbf{\Omega} = \Omega = \mathbf{\Psi}; \mathbf{comp}; \mathbf{\Phi}$

(H_5) $\vdash T_{\mathrm{SMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\mathrm{SMS}}{}'$

(H_6) $T_{\mathrm{SMS}} \simeq_{\delta} \mathbf{\Omega}$

(H_7) $\varepsilon = a$

(H_8) $\theta_{\delta}(\varepsilon) \cong \boldsymbol{a}$

(H_9) $\Omega.\Delta \Vdash \gamma$

(H_{10}) $\gamma \prec \gamma$

Then $\exists \gamma'$,

(i) $\Omega \triangleright [\![\mathbf{n_1} \oplus \mathbf{n_2}]\!]^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}}\gamma \xrightarrow{a}{}^{*}_{\mathrm{ctx}} \Omega \triangleright [\![\mathbf{n_3}]\!]^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}}\gamma'$

(ii) $T_{\mathrm{SMS}}{}' \simeq_{\delta} \mathbf{\Omega}$

(iii) $\Omega.\Delta \Vdash \gamma'$

(iv) $\gamma' \prec \gamma'$

Just instantiate $\gamma' = [\cdot]$ and note that no substitutions are applicable to $[\![\mathbf{n_1} \oplus \mathbf{n_2}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} = n_1 \oplus n_2$ as well as $[\![\mathbf{n_3}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} = n_3$. Induce on $\gamma \prec \gamma$ and see that no substitution changes the expression. Now, all goals follow immediately from assumptions.

Case $e - \mathbf{get} - \in$**:** If

(H_1) $\mathbf{\Psi} = \mathbf{H^{ctx}}; \mathbf{H^{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{comp}; \rho; \mathbf{m}), \mathbf{\Delta_2}$

(H_2) $\ell + \mathbf{n} \in \mathrm{dom}\, \mathbf{H^{comp}}$

(H_3) $\gamma = \gamma' = [\cdot]$

(H_4) $\mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi} \triangleright \mathbf{x}[\mathbf{n}] \xrightarrow{\mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{comp}} \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi} \triangleright \mathbf{H^{comp}}(\ell + \mathbf{n})$

(H_5) $\mathbf{\Omega} = \Omega = \mathbf{\Psi}; \mathbf{comp}; \mathbf{\Phi}$

(H_6) $\vdash T_{\mathrm{SMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\mathrm{SMS}}{}'$

(H_7) $T_{\mathrm{SMS}} \simeq_\delta \mathbf{\Omega}$

(H_8) $\mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{comp} = a$

(H_9) $\theta_\delta\, (\mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{comp}) \cong \boldsymbol{a}$

(H_{10}) $\Omega.\Delta \Vdash \gamma$

(H_{11}) $\gamma \prec \gamma$

Then $\exists \gamma'$,

(i) $\Omega \triangleright [\![\mathbf{x}[\mathbf{n}]]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma \xrightarrow{a}{}^*_{\mathrm{ctx}} \Omega \triangleright [\![\mathbf{H^{comp}}(\ell + \mathbf{n})]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma'$

(ii) $T_{\mathrm{SMS}}{}' \simeq_\delta \mathbf{\Omega}$

(iii) $\Omega.\Delta \Vdash \gamma$

(iv) $\gamma' \prec \gamma'$

Instantiate $\gamma' = [\cdot]$.

Note that by inversion on Assumptions (H_6), (H_7) and (H_9) we have:

(H_1) $\boldsymbol{a} = \boldsymbol{Use}\ \underline{\ell}\ \boldsymbol{n}$

(H_2) $\mathbf{n} = \boldsymbol{n}$

(H_3) $(\underline{\ell}, \boldsymbol{m}) \in T_{\mathrm{SMS}}$

(H_4) $\boldsymbol{n} < \boldsymbol{m}$

(H_5) $\boldsymbol{m} = \mathbf{m}$

(H_6) $T_{\mathrm{SMS}}{}' = T_{\mathrm{SMS}}$

From $\mathbf{n} = \boldsymbol{n}$, $\boldsymbol{n} < \boldsymbol{m}$, and $\boldsymbol{m} = \mathbf{m}$, we conclude $\mathbf{n} < \mathbf{m}$ by transitivity of $<$ and equality. Since $\mathbf{m} = m$ by the state relation, we have $\mathbf{n} < m$.

Note that $[\![\mathbf{x}[\mathbf{n}]]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} = \mathit{let}\ x_{ACCESS} = [\![\mathbf{n}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}\ \mathit{in}\ \mathit{if}\ 0 \leq x_{ACCESS} \leq x_{SIZE}\ \mathit{then}\ x[x_{ACCESS}]\ \mathit{else}\ \mathit{abort}()$.

Induce on Assumption (H_{10}). The base case is spurious, since we have at least $x \mapsto (\ell; \mathit{comp}; \rho; m)$ in $\Omega.\Delta$. The inductive case splits into two other cases: Either we have a substitution with x_{SIZE} or not. If not, make

use of the inductive hypothesis. If yes, then we know by definition that $x_{SIZE} = m$.

Thus, we can execute to the desired state Ω with just $Get\ \ell\ n; comp$. The other goals follow by assumption or are an immediate consequence.

Case $e - \mathbf{get} - \notin$**:** If

(H_1) $\mathbf{\Psi} = \mathbf{H^{ctx}}; \mathbf{H^{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{comp}; \rho; \mathbf{m}), \mathbf{\Delta_2}$

(H_2) $\gamma = \gamma' = [\cdot]$

(H_3) $\ell + \mathbf{n} \notin \operatorname{dom} \mathbf{H^{comp}}$

(H_4) $\mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi} \triangleright \mathbf{x}[\mathbf{n}] \xrightarrow{\mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{comp}} \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi} \triangleright \mathbf{H^{comp}}(\ell + \mathbf{n})$

(H_5) $\mathbf{\Omega} = \Omega = \mathbf{\Psi}; \mathbf{comp}; \mathbf{\Phi}$

(H_6) $\vdash T_{\text{SMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{SMS}}{}'$

(H_7) $T_{\text{SMS}} \simeq_\delta \mathbf{\Omega}$

(H_8) $\mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{comp} = a$

(H_9) $\theta_\delta\,(\mathbf{Get}\ \ell\ \mathbf{n}; \mathbf{comp}) \cong \boldsymbol{a}$

(H_{10}) $\Omega.\Delta \Vdash \gamma$

(H_{11}) $\gamma \prec \gamma$

Then $\exists \gamma'$,

(i) $\Omega \triangleright [\![\mathbf{x}[\mathbf{n}]]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \xrightarrow{a}{}^*_{\text{ctx}} \Omega \triangleright [\![\mathbf{H^{comp}}(\ell + \mathbf{n})]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}$

(ii) $T_{\text{SMS}}{}' \simeq_\delta \mathbf{\Omega}$

(iii) $\Omega.\Delta \Vdash \gamma'$

(iv) $\gamma' \prec \gamma'$

Immediate contradiction of Assumptions (H_3) and (H_7).

Case $e - \mathbf{set} - comp$**:** If

(H_1) $\mathbf{\Psi} = \mathbf{H^{ctx}}; \mathbf{H^{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{comp}; \rho; \mathbf{m}), \mathbf{\Delta_2}$

(H_2) $\mathbf{H^{comp'}} = \mathbf{H^{comp}}\,(\ell + \mathbf{n} \mapsto \mathbf{v})$

(H_3) $\mathbf{\Psi'} = \mathbf{H^{ctx}}; \mathbf{H^{comp'}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{comp}; \rho; \mathbf{m}), \mathbf{\Delta_2}$

(H_4) $\gamma = \gamma' = [\cdot]$

(H_5) $\mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi} \triangleright \mathbf{x}[\mathbf{n}] \leftarrow \mathbf{v} \xrightarrow{\mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{v}; \mathbf{comp}} \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi'} \triangleright \mathbf{v}$

(H_6) $\mathbf{\Omega} = \Omega = \mathbf{\Psi}; \mathbf{comp}; \mathbf{\Phi}$

(H_7) $\vdash T_{\text{SMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{SMS}}{}'$

(H_8) $\theta_\delta\,(\mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{v}; \mathbf{comp}) \cong \boldsymbol{a}$

(H_9) $T_{\text{SMS}} \simeq_\delta \mathbf{\Omega}$

(H_{10}) $\Omega' = \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi'}$

(H_{11}) $a = \mathbf{Set}\ \ell\ \mathbf{n}\ \mathbf{v}; \mathbf{comp}$

(H_{12}) $\Omega.\Delta \Vdash \gamma$

(H_{13}) $\gamma \prec \gamma$

Then $\exists \gamma'$,

(i) $\Omega \triangleright [\![\mathbf{x}[\mathbf{n}] \leftarrow \mathbf{v}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \xrightarrow{a}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright [\![\mathbf{v}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}$

(ii) ${T_{\mathrm{SMS}}}' \simeq_{\delta} \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi'}$

(iii) $\gamma' \Vdash \Omega.\Delta$

(iv) $\gamma' \prec \gamma'$

Instantiate $\gamma' = [\cdot]$.

Note that by inversion on Assumptions (H_7) to (H_9) we have:

(H_1) $\boldsymbol{a} = \boldsymbol{Use}\ \underline{\ell}\ \boldsymbol{n}$

(H_2) $\mathbf{n} = \boldsymbol{n}$

(H_3) $(\underline{\ell}, \boldsymbol{m}) \in T_{\mathrm{SMS}}$

(H_4) $\boldsymbol{n} < \boldsymbol{m}$

(H_5) $\boldsymbol{m} = \mathbf{m}$

(H_6) ${T_{\mathrm{SMS}}}' = T_{\mathrm{SMS}}$

From $\mathbf{n} = \boldsymbol{n}$, $\boldsymbol{n} < \boldsymbol{m}$, and $\boldsymbol{m} = \mathbf{m}$, we conclude $\mathbf{n} < \mathbf{m}$ by transitivity of $<$ and equality. Since $\mathbf{m} = m$ by the state relation, we have $\mathbf{n} < m$.

Note that $[\![\mathbf{x}[\mathbf{n}] \leftarrow \mathbf{v}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} = \mathit{let}\ x_{ACCESS} = [\![\mathbf{n}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}\ \mathit{in}\ \mathit{ifz}\ 0 \leq x_{ACCESS} \leq x_{SIZE}\ \mathit{then}\ x[x_{ACCESS}] \leftarrow [\![\mathbf{v}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}\ \mathit{else}\ \mathit{abort}()$.

Induce on Assumption (H_{12}). The base case is spurious, since we have at least $x \mapsto (\ell; \mathit{comp}; \rho; m)$ in $\Omega.\Delta$. The inductive case splits into two other cases: Either we have a substitution with x_{SIZE} or not. If not, make use of the inductive hypothesis.If yes, then we know by definition that $x_{SIZE} = m$.

Thus, we can execute to the desired state Ω with just $\mathit{Set}\ \ell\ n\ v; \mathit{comp}$. The other goals follow by assumption or are an immediate consequence.

Case $e - \mathbf{new} - \mathit{comp}$**:** If

(H_1) $\mathbf{\Psi} = \mathbf{H^{ctx}}; \mathbf{H^{comp}}; \mathbf{\Delta}$

(H_2) $\mathbf{\Delta} \vdash \ell\ \mathit{fresh}$

(H_3) $\mathbf{\Delta} \vdash \mathbf{z}\ \mathit{fresh}$

(H_4) $\mathbf{H^{comp'}} = \mathbf{H^{comp}} \ll \mathbf{n}$

(H_5) $\mathbf{\Psi'} = \mathbf{H^{ctx}}; \mathbf{H^{comp'}}; \mathbf{z} \mapsto (\ell; \mathbf{ctx}; \square; \mathbf{n}), \mathbf{\Delta}$

(H_6) $\gamma = [\cdot]$

(H_7) $\gamma' = [\mathbf{z}/\mathbf{x}], [\cdot]$

(H_8) $\mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi} \triangleright \mathbf{let\ x = new\ n\ in\ e} \xrightarrow{\mathbf{Alloc}\ \ell\ \mathbf{n}; \mathbf{comp}} \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi'} \triangleright \mathbf{e}[\mathbf{z}/\mathbf{x}]$

(H_9) $\mathbf{\Omega} = \Omega$

(H_{10}) $\vdash T_{\mathrm{SMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} {T_{\mathrm{SMS}}}'$

(H_{11}) $\theta_{\delta}\,(\mathbf{Alloc}\ \ell\ \mathbf{n}; \mathbf{comp}) \cong \boldsymbol{a}$

(H_{12}) $T_{\mathrm{SMS}} \simeq_{\delta} \mathbf{\Omega}$

(H_{13}) $\Omega' = \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi'}$

(H_{14}) $a = \mathbf{Alloc}\ \ell\ \mathbf{n}; \mathbf{comp}$

(H_{15}) $\Omega.\Delta \Vdash \gamma$

(H_{16}) $\gamma \prec \gamma$

Then $\exists \gamma'$,

(i) $\Omega \triangleright [\![\mathbf{let\ x = new\ n\ in\ e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma \xrightarrow{a}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright [\![\mathbf{e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma'$

(ii) $T_{\mathrm{SMS}}{}' \simeq_{\delta} \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi'}$

(iii) $\gamma' \Vdash \Omega.\Delta$

(iv) $\gamma' \prec \gamma'$

Note that by inversion on Assumptions (H_7) to (H_9) we have:

(H_1) $\boldsymbol{a = Alloc\ \underline{\ell}\ n}$

(H_2) $\mathbf{n} = \boldsymbol{n}$

(H_3) $\underline{\ell} \notin \mathrm{dom}\, T_{\mathrm{SMS}}$

(H_4) $T_{\mathrm{SMS}} = (\underline{\ell}, \boldsymbol{n}) \cup T_{\mathrm{SMS}}$

Note that $[\![\mathbf{let\ x = new\ n\ in\ e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} = \begin{array}{l} let\ x_{SIZE} = [\![\mathbf{n}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}\ in \\ let\ x = new\ x_{SIZE}\ in\ [\![\mathbf{e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \end{array}$.

Instantiate $\gamma' = [[\![\mathbf{n}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}/x_{SIZE}], [z/x], \gamma$ and let $n = [\![\mathbf{n}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}$.

We make use of compatibility of substitution with the compiler: $[\![\mathbf{let\ x = new\ n\ in\ e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma = let\ x_{SIZE} = n\ in\ let\ x = new\ n\ in\ ([\![\mathbf{e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma)$

Now observe that we can perform the steps: (abusing compatiblity of substitution on-the-fly)

- $\Omega \triangleright let\ x_{SIZE} = n\ in\ let\ x = new\ n\ in\ ([\![\mathbf{e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma) \xrightarrow{\varepsilon; comp}{}^{*}_{\mathrm{ctx}} \Omega \triangleright let\ x = new\ n\ in\ ([\![\mathbf{e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} [n/x_{SIZE}], \gamma)$
- $\Omega \triangleright let\ x = new\ n\ in\ ([\![\mathbf{e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} [z/x], [n/x_{SIZE}], \gamma) \xrightarrow{Alloc\ \ell\ n; comp}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright [\![\mathbf{e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma'$

The first goal follows by composing the above execution steps using Rules $e-$**let**$-$**f** and $e-$**new**$-comp$: the first step produces an empty event, and the second produces $Alloc\ \ell\ n; comp$, so the combined trace is $Alloc\ \ell\ n; comp$ as required. The second goal (store agreement) follows since the monitor is updated to include $(\underline{\ell}, \boldsymbol{n})$ which corresponds to the newly allocated location in Ω'. The third goal ($\gamma' \Vdash \Omega.\Delta$) follows by Rules subst-subset-cons and subst-subset-x_{SIZE}, since $\gamma' = [n/x_{SIZE}], [z/x], \gamma$ extends γ with substitutions for x_{SIZE} and x that are consistent with the extended Δ. The fourth goal ($\gamma' \prec \gamma'$) follows since the substitution relation is preserved under extension with consistent bindings.

Case $e - \mathbf{new} - ctx$**:** Does not apply, since the sandboxtag is **comp**, but the rule requires **ctx**.

Case $e - \mathbf{set} - ctx$**:** Does not apply, since the sandboxtag is **comp**, but the rule requires **ctx**.

Case $e - \mathbf{delete}$**:** If

(H_1) $\mathbf{\Psi} = \mathbf{H^{ctx}}; \mathbf{H^{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{comp}; \square; \mathbf{m}), \mathbf{\Delta_2}$

(H_2) $\mathbf{\Psi'} = \mathbf{H^{ctx}}; \mathbf{H^{comp}}; \mathbf{\Delta_1}, \mathbf{x} \mapsto (\ell; \mathbf{comp}; \circledast; \mathbf{m}), \mathbf{\Delta_2}$

(H_3) $\gamma = \gamma' = [\cdot]$

(H_4) $\mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi} \triangleright \mathbf{delete\ x} \xrightarrow{\mathbf{Dealloc}\ \ell;\mathbf{comp}} \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi'} \triangleright \mathbf{0}$

(H_5) $\mathbf{\Omega} = \Omega = \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi}$

(H_6) $\vdash T_{\text{SMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{SMS}}{}'$

(H_7) $T_{\text{SMS}} \simeq_\delta \mathbf{\Omega}$

(H_8) $\theta_\delta\,(\mathbf{Dealloc}\ \ell; \mathbf{comp}) \cong \boldsymbol{a}$

(H_9) $\Omega' = \mathbf{\Omega'} = \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi'}$

(H_{10}) $a = \mathbf{Dealloc}\ \ell; \mathbf{comp}$

(H_{11}) $\Omega.\Delta \Vdash \gamma$

(H_{12}) $\gamma \prec \gamma$

Then $\exists \gamma'$,

(i) $\Omega \triangleright [\![\mathbf{delete\ x}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \xrightarrow{a}{}^{*}_{\text{ctx}} \Omega' \triangleright [\![\mathbf{0}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}$

(ii) $T_{\text{SMS}}{}' \simeq_\delta \mathbf{\Omega'}$

(iii) $\gamma' \Vdash \Omega.\Delta$

(iv) $\gamma' \prec \gamma'$

Just pick $\gamma' = [\cdot]$.

Since $[\![\mathbf{delete\ x}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} = \mathit{delete}\ x$, this case is entirely trivial up to an induction on Assumption (H_{11}).

Case $e - \mathbf{let} - \mathbf{f}$**:** If

(H_1) $\mathbf{\Omega} = \Omega = \mathbf{\Phi}; \mathbf{comp}; \mathbf{\Psi}$

(H_2) $\gamma = [\cdot]$

(H_3) $\gamma' = [\mathbf{v}/\mathbf{x}], [\cdot]$

(H_4) $\mathbf{\Omega} \triangleright \mathbf{let\ x = v\ in\ e} \xrightarrow{\varepsilon;\mathbf{comp}} \mathbf{\Omega} \triangleright \mathbf{e}[\mathbf{v}/\mathbf{x}]$

(H_5) $\vdash T_{\text{SMS}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\text{SMS}}{}'$

(H_6) $T_{\text{SMS}} \simeq_\delta \mathbf{\Omega}$

(H_7) $\theta_\delta\,(\varepsilon; \mathbf{comp}) \cong \boldsymbol{a}$

(H_8) $a = \varepsilon; \mathbf{comp}$

(H_9) $\Omega.\Delta \Vdash \gamma$

(H_{10}) $\gamma \prec \gamma$

Then $\exists \gamma'$,

(i) $\Omega \triangleright [\![\mathbf{let\ x = v\ in\ e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma \xrightarrow{a}{}^{*}_{\text{ctx}} \Omega \triangleright [\![\mathbf{e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} \gamma'$

(ii) $T_{\text{SMS}} \simeq_\delta \mathbf{\Omega}$

(iii) $\gamma' \Vdash \Omega.\Delta$

(iv) $\gamma' \prec \gamma'$

Let $v = [\![\mathbf{v}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}$. Pick $\gamma = [v/x], [\cdot]$.

Note that $[\![\mathbf{let\ x = v\ in\ e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}} = \mathit{let}\ x = [\![\mathbf{v}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}\ \mathit{in}\ [\![\mathbf{e}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}$.

Induce on Assumption (H_9). In the base-case, we can immediately step to the desired term and all goals go through trivially. The inductive cases go through easily.

Case $e - \mathbf{abort}$**:** This case and all following are entirely similar to case $e - \mathbf{delete}$ or $e - \oplus$.

Case $e - \mathbf{x\ is}\ ☣ -$yes**:**

Case $e - \mathbf{x\ is}\ ☣ -$no**:**

Case $e - \pi_1$**:**

Case $e - \pi_2$**:**

Case $e - \mathbf{x\ has}\ \mathbb{N}$**:**

Case $e - \mathbf{n\ has}\ \mathbb{N}$**:**

Case $e - \mathit{pair} - \mathbf{has}\ \mathbb{N}$**:**

Case $e - \mathbf{x\ has}\ \mathbb{N} \times \mathbb{N}$**:**

Case $e - \mathbf{n\ has}\ \mathbb{N} \times \mathbb{N}$**:**

Case $e - \mathit{pair} - \mathbf{has}\ \mathbb{N} \times \mathbb{N}$**:**

Case $e - \mathbf{ifz} -$true**:**

Case $e - \mathbf{ifz} -$false**:**

□

Lemma 79 (Ctx. Forward Simulation)**.** *If*

(a) $\mathbf{\Omega} \triangleright \mathbf{e} \xrightarrow{\bar{\mathbf{a}}}_{ctx} \mathbf{\Omega}' \triangleright \mathbf{e}'$

(b) $\mathbf{\Omega} = \Omega = \Psi; \mathit{comp}; \Phi$

(c) $\vdash T_{SMS} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{SMS}'$

(d) $T_{SMS} \simeq_\delta \mathbf{\Omega}$

(e) $\theta_\delta\,(\mathbf{a}) \cong \boldsymbol{a}$

(f) $a = \mathbf{a}$

(g) $\mathbf{\Omega}' = \Omega'$

Then

(i) $\Omega \triangleright [\![\mathbf{e}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}} \xrightarrow{a}{}^*_{ctx} \Omega' \triangleright [\![\mathbf{e}']\!]^{\mathrm{L}_{ms} \to \mathrm{L}}$

(ii) $T_{SMS}' \simeq_\delta \mathbf{\Omega}'$

Proof. Unfolding and using Lemma 78. □

Lemma 80 (Forward Simulation)**.** *If*

(a) $\mathbf{\Omega} \triangleright \mathbf{e} \xrightarrow{\overline{\mathbf{a}}}{}^{*}_{ctx} \mathbf{\Omega}' \triangleright \mathbf{f}$

(b) $\mathbf{\Omega} = \Omega$

(c) $\overline{\mathbf{a}} = \overline{a}$

(d) $\mathbf{\Omega}' = \Omega'$

(e) $\vdash T_{SMS} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^{*} T_{SMS}{}'$

(f) $\theta^{*}_{\delta}(\overline{\mathbf{a}}) \cong \overline{\boldsymbol{a}}$

(g) $T_{SMS} \simeq_{\delta} \mathbf{\Omega}$

Then

(i) $\Omega \triangleright [\![\mathbf{e}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}} \xrightarrow{\overline{a}}{}^{*}_{ctx} \Omega' \triangleright [\![\mathbf{f}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}}$

(ii) $T_{SMS}{}' \simeq_{\delta} \mathbf{\Omega}'$

Proof. Proof is similar to Lemma 44 (Steps TMS via Monitor). □

Lemma 81 (Backward Simulation)**.** *If*

(a) $\Omega \triangleright [\![\mathbf{e}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}} \xrightarrow{\overline{a}}{}^{*}_{ctx} \Omega' \triangleright [\![\mathbf{f}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}}$

(b) $\mathbf{\Omega} = \Omega$

(c) $\overline{\mathbf{a}} = \overline{a}$

(d) $\mathbf{\Omega}' = \Omega'$

Then

(i) $\mathbf{\Omega} \triangleright \mathbf{e} \xrightarrow{\overline{\mathbf{a}}}{}^{*}_{ctx} \mathbf{\Omega}' \triangleright \mathbf{f}$

Proof. Case analysis on wether $\exists \mathbf{f}', \mathbf{\Omega} \triangleright \mathbf{e} \xrightarrow{\overline{\mathbf{a}}}{}^{*}_{\mathrm{ctx}} \mathbf{\Omega}' \triangleright \mathbf{f}'$ holds or not.

Case $\exists \mathbf{f}', \mathbf{\Omega} \triangleright \mathbf{e} \xrightarrow{\overline{\mathbf{a}}}{}^{*}_{\mathrm{ctx}} \mathbf{\Omega}' \triangleright \mathbf{f}'$**:** By Lemma 80 (Forward Simulation) on what we assume in this case:

(H_1) $\Omega \triangleright [\![\mathbf{e}]\!]^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}} \xrightarrow{\overline{a}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright [\![\mathbf{f}']\!]^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}}$

(H_2) $\overline{\mathbf{a}} = \overline{a}$

(H_3) $\mathbf{\Omega}' = \Omega'$

By Lemma 48 (Determinism of Steps):

(H_4) $[\![\mathbf{f}]\!]^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}} = [\![\mathbf{f}']\!]^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}}$

Subsequently, by Lemma 76 (Injective Compiler (for final expressions)):

(H_5) $\mathbf{f} = \mathbf{f}'$

Rewrite Assumption (H_1) using Assumption (H_5), solving our goal.

Case $\neg(\exists \mathbf{f}', \mathbf{\Omega} \triangleright \mathbf{e} \xrightarrow{\overline{\mathbf{a}}}{}^{*}_{\mathrm{ctx}} \mathbf{\Omega}' \triangleright \mathbf{f}')$**:** We can rewrite the assumption done in this case as:

(H_1) $\forall \mathbf{f}', \mathbf{\Omega} \triangleright \mathbf{e} \not\xrightarrow{\overline{\mathbf{a}}}{}^{*}_{\mathrm{ctx}} \mathbf{\Omega}' \triangleright \mathbf{f}'$

We prove the goal by contradiction, so we have to prove $\bot$, assuming:

(H_2) $\mathbf{\Omega} \triangleright \mathbf{e} \xrightarrow{\overline{\mathbf{a}}}{}^{*}_{\mathrm{ctx}} \mathbf{\Omega}' \triangleright \mathbf{f}$

Assumption (H_2) contradicts Assumption (H_1).

□

Lemma 82 (Top-Level Backward Simulation)**.** *If*

(a) $\Xi = \Xi_{ctx} \bowtie [\![\mathbf{\Xi_{comp}}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}}$

(b) $\Xi_{ctx} = \mathbf{\Xi_{ctx}}$

(c) $\xi = [\![\xi]\!]^{\mathrm{L}_{ms} \to \mathrm{L}} = \mathrm{dom}[\![\mathbf{\Xi_{comp}}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}}$

(d) $\xi; \Xi; [\cdot]; comp; [\cdot]; [\cdot]; [\cdot] \triangleright call\ \ main\ \ 0 \xrightarrow{\overline{a}}{}^{*}_{ctx} \xi; \Xi; \overline{K}; ctx; H^{ctx}; H^{comp}; \Delta \triangleright f$

(e) $\xi = \xi$

Then $\exists \overline{\mathbf{a}}\ \overline{\mathbf{K}}\ \mathbf{H^{ctx}}\ \mathbf{H^{comp}}\ \mathbf{\Delta}$,

(i) $\xi; \mathbf{\Xi}; [\cdot]; \mathbf{comp}; [\cdot]; [\cdot]; [\cdot] \triangleright \mathbf{call\ \ main\ \ 0} \xrightarrow{\overline{\mathbf{a}}}{}^{*}_{ctx} \xi; \mathbf{\Xi}; \overline{\mathbf{K}}; \mathbf{ctx}; \mathbf{H^{ctx}}; \mathbf{H^{comp}}; \mathbf{\Delta} \triangleright \mathbf{f}$

(ii) $\overline{K} = \overline{\mathbf{K}}$

(iii) $H^{ctx} = \mathbf{H^{ctx}}$

(iv) $H^{comp} = \mathbf{H^{comp}}$

(v) $\Delta = \mathbf{\Delta}$

(vi) $\overline{a} = \overline{\mathbf{a}}$

Proof. Suppose control is never handled over to $[\![\mathbf{\Xi_{comp}}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}$; that is, there is no $Call\ ?\ foo\ n$ in $\overline{a}$. In this case, the goals follow easily, since the semantics of $\mathrm{L_{ms}}$ and L are identical and $\Xi_{ctx} = \mathbf{\Xi_{ctx}}$.

If control is handled over, we have, at some point in the reduction by transitivty, $\Omega_0 \triangleright [\![\mathbf{e_{foo}}]\!]^{\mathrm{L_{ms}} \to \mathrm{L}}[v/x] \xrightarrow{\overline{a}}{}^{*}_{\mathrm{ctx}} \xi; \Xi; \overline{K}; ctx; H^{ctx}; H^{comp}; \Delta \triangleright f$. Thus, we can apply Lemma 81 (Backward Simulation) and are done. □

Lemma 83 (Program Backward Simulation)**.** *If*

(a) $prog\ \ \Xi_{ctx}\ \ [\![\mathbf{\Xi_{comp}}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}} \overset{\overline{a}}{\Longrightarrow} \Omega \triangleright f_t$

(b) $\Xi_{ctx} = \mathbf{\Xi_{ctx}}$

(c) $\Omega = \mathbf{\Omega}$

(d) $\overline{a} = \overline{\mathbf{a}}$

(e) $f = \mathbf{f}$

Then

(i) $\mathbf{prog}\ \Xi_{\mathbf{ctx}}\ \Xi_{\mathbf{comp}} \overset{\bar{\mathbf{a}}}{\Longrightarrow} \Omega \triangleright \mathbf{f}_{\lightning}$

Proof. Immediate consequence of Lemma 82 (Top-Level Backward Simulation). □

Theorem 4 (TMS Relation Correctness for $[\![\bullet]\!]^{\mathsf{L}_{\mathrm{ms}} \to \mathsf{L}}$).

$$\text{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{tms})\right) = \mathrm{tms} \ \text{ and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{tms})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{tms})$$
$$\text{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{tms}))\right) = \mathrm{tms}$$

Proof. By unfolding Definition 15 and applying the first assumption to rewrite the inner σ. Since $\mathsf{L} = \mathsf{L}_{\mathrm{ms}}$, the trace relation is identity, so $\sigma_{\approx^*_{\delta;\mathsf{X}}}$ preserves the property. The second assumption ensures well-formedness of the composition. □

Theorem 5 (SMS Relation Correctness for $[\![\bullet]\!]^{\mathsf{L}_{\mathrm{ms}} \to \mathsf{L}}$).

$$\text{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms})\right) = \mathrm{sms} \ \text{ and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms})$$
$$\text{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms}))\right) = \mathrm{sms}$$

Proof. By unfolding Definitions 15 and 32 and applying the first assumption. The property is preserved because memory accesses must be in bounds (Definition 32), and the filter preserves allocation sizes and access offsets. □

Theorem 6 (sCCT Relation Correctness for $[\![\bullet]\!]^{\mathsf{L}_{\mathrm{ms}} \to \mathsf{L}}$).

$$\text{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct})\right) = \mathrm{scct} \ \text{ and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct})$$
$$\text{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct}))\right) = \mathrm{scct}$$

Proof. By unfolding Definitions 15 and 36 and applying the first assumption. The property is preserved because the filter and trace relation preserve the structure of branching and memory access events relevant to sCCT. □

Theorem 7 (SS Relation Correctness for $[\![\bullet]\!]^{\mathsf{L}_{\mathrm{ms}} \to \mathsf{L}}$).

$$\text{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec})\right) = \mathrm{spec} \ \text{ and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec})$$
$$\text{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec}))\right) = \mathrm{spec}$$

Proof. By unfolding Definitions 15 and 38 and applying the first assumption. The property is preserved because the filter and trace relation preserve speculative execution barriers and transient events. □

Theorem 8 (SMS Relation Correctness for $[\![\bullet]\!]^{\mathsf{L} \to \mathsf{L}_{\mathrm{scct}}}$).

$$\text{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms})\right) = \mathrm{sms} \ \text{ and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms})$$
$$\text{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms}))\right) = \mathrm{sms}$$

Proof. By unfolding Definitions 15 and 32 and applying the first assumption. The cross-language relation preserves allocation sizes and access offsets, ensuring spatial memory safety (Definition 32) is maintained. □

Theorem 9 (sCCT Relation Correctness for $[\![\bullet]\!]^{\mathtt{L}\to\mathtt{L}_{\mathrm{scct}}}$).

$$\begin{aligned} &\textit{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct})\right) = \mathrm{scct} \;\; \textit{and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct}) \\ &\textit{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct}))\right) = \mathrm{scct} \end{aligned}$$

Proof. By unfolding Definitions 15 and 36 and applying the first assumption. The cross-language relation ensures no leaks occur: branching and memory access patterns are preserved, maintaining strict cryptographic constant time (Definition 36). □

Theorem 10 (SS Relation Correctness for $[\![\bullet]\!]^{\mathtt{L}\to\mathtt{L}_{\mathrm{scct}}}$).

$$\begin{aligned} &\textit{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec})\right) = \mathrm{spec} \;\; \textit{and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec}) \\ &\textit{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec}))\right) = \mathrm{spec} \end{aligned}$$

Proof. By unfolding Definitions 15 and 38 and applying the first assumption. The cross-language relation preserves speculative execution barriers and transient events, maintaining speculation safety (Definition 38). □

Definition 43 ($\mathtt{L}$ Robust Satisfaction). *We write* $\Xi_{ctx} \vDash_R \pi$ *for*

$$\forall \Xi_{comp}\ \overline{a}\ \Omega\ f_{\lightning}, \mathit{prog}\ \Xi_{ctx}\ \Xi_{comp} \overset{\overline{a}}{\Longrightarrow} \Omega \triangleright f_{\lightning} \implies \exists \delta_{MS}, \theta_{\delta_{MS}}(\overline{a}) \in \pi$$

Theorem 11 (Robust Memory Safety Preservation).

(i) $\vdash [\![\bullet]\!]^{\mathtt{L}_{ms}\to\mathtt{L}}$: msafe

Proof. Unfolding the definition of $\vdash [\![\bullet]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}}$: msafe: If

(H_1) $\Xi_{\mathbf{comp}} \vDash_R$ msafe

then

(i) $[\![\Xi_{\mathbf{comp}}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \vDash_R$ msafe

By Definition 34, msafe = tmssafe ∩ smsafe.
So Assumption (H_1) gives us:

(H_2) $\Xi_{\mathbf{comp}} \vDash_R$ tmssafe

(H_3) $\Xi_{\mathbf{comp}} \vDash_R$ smsafe

Unfold Assumption (i): If

(H_4) $\mathit{prog}\ \Xi_{ctx}\ [\![\Xi_{\mathbf{comp}}]\!]^{\mathtt{L}_{\mathrm{ms}}\to\mathtt{L}} \overset{\overline{a}}{\Longrightarrow} \Omega \triangleright f_{\lightning}$

then $\exists \delta_{MS}$

(i) $\theta_{\delta_{MS}}(\overline{a}) \in$ msafe

We must show $\overline{a} \in$ tmssafe $\cap$ smsafe, i.e., two sub-goals.

Use Lemma 83 (Program Backward Simulation) on Assumption (H_4): (remember that $\overline{a} = \overline{\mathbf{a}}$, since $\mathrm{L} = \mathrm{L}_{\mathrm{ms}}$)

(H_5) $\mathbf{prog}\ \Xi_{\mathbf{ctx}}\ \Xi_{\mathbf{comp}} \overset{\overline{\mathbf{a}}}{\Longrightarrow} \Omega \triangleright \mathbf{f}_{\ell}$

Sub-goal 1: $\overline{a} \in$ smsafe

By Assumption (H_3), $\overline{\mathbf{a}} \in$ smsafe. Since $\overline{a} = \overline{\mathbf{a}}$, we have $\overline{a} \in$ smsafe.

Sub-goal 2: $\overline{a} \in$ tmssafe

By Assumption (H_2), $\overline{\mathbf{a}} \in$ tmssafe. Since $\overline{a} = \overline{\mathbf{a}}$, we have $\overline{a} \in$ tmssafe.

Conclusion: Combining sub-goals 1 and 2, $\overline{a} \in$ tmssafe $\cap$ smsafe $=$ msafe. □

Theorem 12 (Robust Memory Safety Preservation (Composed))**.**

(i) $\vdash_{\sigma_\sim} [\![[\![\bullet]\!]^{\mathrm{L}_{tms} \to \mathrm{L}_{ms}}]\!]^{\mathrm{L}_{ms} \to \mathrm{L}}$: msafe

Proof. By Theorem 3 (Robust Memory Safety Preservation), we have $\vdash_{\sigma_\sim} [\![\bullet]\!]^{\mathrm{L}_{\mathrm{tms}} \to \mathrm{L}_{\mathrm{ms}}}$: msafe.

By Theorem 11 (Robust Memory Safety Preservation), we have $\vdash [\![\bullet]\!]^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}}$: msafe.

Since $\mathrm{L} = \mathrm{L}_{\mathrm{ms}}$, the trace relation for the second compiler is identity.

By Lemma 7 (Sequential Composition with RTPσ), composing the two compilers: $\vdash_{\sigma_\sim} [\![[\![\bullet]\!]^{\mathrm{L}_{\mathrm{tms}} \to \mathrm{L}_{\mathrm{ms}}}]\!]^{\mathrm{L}_{\mathrm{ms}} \to \mathrm{L}}$: msafe. □

Theorem 13 (TMS Relation Correctness for $[\![\bullet]\!]^{\mathrm{L}_{\mathrm{tms}} \to \mathrm{L}_{\mathrm{ms}}}$)**.**

$$\textit{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{tms})\right) = \mathrm{tms} \ \textit{ and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{tms})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{tms})$$
$$\textit{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{tms}))\right) = \mathrm{tms}$$

Proof. By unfolding Definition 15 and applying the first assumption to rewrite the inner σ. The cross-language relation $\approx^*_{\delta;\mathsf{X}}$ preserves allocation and deallocation structure, maintaining temporal memory safety. By Lemma 63 (Filters Equal), related traces have equal filtered forms, so the property is preserved. □

Theorem 14 (SMS Relation Correctness for $[\![\bullet]\!]^{\mathrm{L}_{\mathrm{tms}} \to \mathrm{L}_{\mathrm{ms}}}$)**.**

$$\textit{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms})\right) = \mathrm{sms} \ \textit{ and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms})$$
$$\textit{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{sms}))\right) = \mathrm{sms}$$

Proof. By unfolding Definitions 15 and 32 and applying the first assumption. By Lemma 64 (Cross-Language Relation Preserves SMS), the cross-language relation preserves allocation sizes and access offsets, maintaining spatial memory safety (Definition 32). By Lemma 63 (Filters Equal), related traces have equal filtered forms. □

Theorem 15 (sCCT Relation Correctness for $[\![\bullet]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$).

$$\textit{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct})\right) = \mathrm{scct} \;\; \textit{and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct})$$
$$\textit{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{scct}))\right) = \mathrm{scct}$$

Proof. By unfolding Definitions 15 and 36 and applying the first assumption. The cross-language relation preserves branching and memory access patterns, maintaining strict cryptographic constant time. By Lemma 63 (Filters Equal), related traces have equal filtered forms. □

Theorem 16 (SS Relation Correctness for $[\![\bullet]\!]^{\mathrm{L_{tms}} \to \mathrm{L_{ms}}}$).

$$\textit{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec})\right) = \mathrm{spec} \;\; \textit{and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec})$$
$$\textit{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\mathrm{spec}))\right) = \mathrm{spec}$$

Proof. By unfolding Definitions 15 and 38 and applying the first assumption. The cross-language relation preserves speculative execution barriers and transient events, maintaining speculation safety. By Lemma 63 (Filters Equal), related traces have equal filtered forms. □

4.6 Optimising Passes

4.6.1 DCE

$\boxed{[\![e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} = e'}$ „Compile $\mathtt{L}$ expression e to $\mathtt{L}$ expression e'."

$\boxed{[\![[v/x]]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} = [v/x]}$ „Compile $\mathtt{L}$ substitution to $\mathtt{L}$ substitution."

$\boxed{[\![\xi]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} = \xi'}$ „Compile $\mathtt{L}$ component library to $\mathtt{L}$ component library."

$$
\begin{array}{rcl}
[\![f_t]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & f_t \\
[\![call\ foo\ e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & call\ [\![foo]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}\ [\![e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
[\![return\ e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & return\ [\![e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
[\![e_1 \oplus e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & [\![e_1]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \oplus [\![e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
[\![x[e]]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & x[[\![e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}] \\
[\![let\ x = e_1\ in\ e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & let\ x = [\![e_1]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}\ in\ [\![e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
[\![x[e_1] \leftarrow e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & x[[\![e_1]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}] \leftarrow [\![e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
[\![let\ x = new\ e_1\ in\ e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & let\ x = new\ [\![e_1]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}\ in\ [\![e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
[\![delete\ x]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & delete\ x \\
[\![x\ is\ \text{☣}]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & x\ is\ \text{☣} \\
[\![(e_1; e_2)]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & ([\![e_1]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}; [\![e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}) \\
[\![\pi_1\ e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & \pi_1\ [\![e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
[\![\pi_2\ e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & \pi_2\ [\![e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
[\![e\ has\ \tau]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & [\![e]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}\ has\ \tau \\
[\![ifz\ true\ then\ e_2\ else\ e_3]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & e_2 \\
[\![ifz\ true\ then\ e_3\ else\ e_3]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & e_3 \\
[\![ifz\ e_1\ then\ e_2\ else\ e_3]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & ifz\ [\![e_1]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
 & & then\ [\![e_2]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
 & & else\ [\![e_3]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} \\
 & & \\
[\![[v/x]]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & [\,[\![v]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}/[\![x]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}\,] \\
 & & \\
[\![[\cdot]]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & [\cdot] \\
[\![foo, \xi]\!]^{\mathtt{L}\to\mathtt{L}}_{dce} & = & foo, [\![\xi]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}
\end{array}
$$

Figure 64: Compiler from $\mathtt{L}$ to $\mathtt{L}$, performing dead code elimination.

Since the traces do not change after compiling with $[\![\bullet]\!]^{\mathtt{L}\to\mathtt{L}}_{dce}$, the security proof follows easily using a context-based backtranslation that is essentially just the identity. Anything else is similar to Section 4.5.

4.6.2 CF

$\boxed{[\![e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} = e'}$ „Compile $\mathtt{L}$ expression e to $\mathtt{L}$ expression e'."

$\boxed{[\![\xi, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} = \xi'}$ „Compile $\mathtt{L}$ component library to $\mathtt{L}$ component library."

$$
\begin{array}{rcl}
[\![x, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & v \quad \text{if } [v \text{ for } x] \in \overline{\rho} \\
[\![x, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & x \quad \text{if } [v \text{ for } x] \notin \overline{\rho} \\
[\![f_\ell, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & f_\ell \\
[\![\mathit{call}\ \mathit{foo}\ e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & \mathit{call}\ [\![\mathit{foo}, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}\ [\![e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
[\![\mathit{return}\ e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & \mathit{return}\ [\![e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
[\![n \oplus m, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & k \quad \text{where } k = n \oplus m \\
[\![e_1 \oplus e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & [\![e_1, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \oplus [\![e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
[\![x[e], \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & x[[\![e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}] \\
[\![\mathit{let}\ x = v\ \mathit{in}\ e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & [\![e_2, [v \text{ for } x] \cdot \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
[\![\mathit{let}\ x = e_1\ \mathit{in}\ e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & \mathit{let}\ x = [\![e_1, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}\ \mathit{in}\ [\![e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
[\![x[e_1] \leftarrow e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & x[[\![e_1, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}] \leftarrow [\![e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
[\![\mathit{let}\ x = \mathit{new}\ e_1\ \mathit{in}\ e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & \mathit{let}\ x = \mathit{new}\ [\![e_1, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}\ \mathit{in}\ [\![e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
[\![\mathit{delete}\ x, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & \mathit{delete}\ x \\
[\![x\ \mathit{is}\ \text{☣}, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & x\ \mathit{is}\ \text{☣} \\
[\![(e_1; e_2), \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & ([\![e_1, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}; [\![e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}) \\
[\![\pi_1\ e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & \pi_1\ [\![e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
[\![\pi_2\ e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & \pi_2\ [\![e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
[\![e\ \mathit{has}\ \tau, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & [\![e, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}\ \mathit{has}\ \tau \\
[\![\mathit{ifz}\ e_1\ \mathit{then}\ e_2\ \mathit{else}\ e_3, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & \mathit{ifz}\ [\![e_1, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
 & & \mathit{then}\ [\![e_2, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
 & & \mathit{else}\ [\![e_3, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} \\
 & & \\
[\![[\cdot], \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & [\cdot] \\
[\![\mathit{foo}, \xi, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}} & = & \mathit{foo}, [\![\xi, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}
\end{array}
$$

Figure 65: Compiler from $\mathtt{L}$ to $\mathtt{L}$, performing constant folding.

Since the traces do not change after compiling with $[\![\bullet, \overline{\rho}]\!]_{cf}^{\mathtt{L}\to\mathtt{L}}$, the security proof follows easily using a context-based backtranslation that is essentially just the identity. Anything else is similar to Section 4.5.

4.7 Object Language

$$
\begin{array}{rl}
\textit{Final Result } \mathtt{f} ::= & \mathtt{v}^{\sigma} \mid \mathtt{x}^{\sigma} \qquad \textit{May be a Result } \mathtt{f}_{\lightning} ::= \mathtt{f} \mid \mathtt{stuck} \\
\textit{Expressions } \mathtt{e} ::= & \mathtt{f}_{\lightning} \mid \mathtt{e}_1 \oplus \mathtt{e}_2 \mid \mathtt{x}[\mathtt{e}] \mid \mathtt{getDIT\ x\ in\ e} \mid \mathtt{setDIT\ e} \\
& \mid \mathtt{let\ x} = \mathtt{e}_1 \mathtt{\ in\ e}_2 \mid \mathtt{x}[\mathtt{e}_1] \leftarrow \mathtt{e}_2 \\
& \mid \mathtt{let\ x} = \mathtt{new\ e}_1 \mathtt{\ in\ e}_2 \mid \mathtt{delete\ x} \\
& \mid \mathtt{return\ e} \mid \mathtt{call\ foo\ e} \mid \mathtt{ifz\ e}_1 \mathtt{\ then\ e}_2 \mathtt{\ else\ e}_3 \\
& \mid \mathtt{abort}() \mid \mathtt{x\ is} \ \text{☣} \\
& \mid \langle \mathtt{e}_1, \mathtt{e}_2 \rangle \mid \pi_1 \mathtt{e} \mid \pi_2 \mathtt{e} \mid \mathtt{e\ has}\ \tau \\
& \text{where } \oplus \in \{+, -, \times, <, /\} \\
\textit{Functions } \mathtt{F} ::= & \mathtt{let\ foo\ x} := \mathtt{e} \qquad \textit{Types } \tau ::= \mathbb{N} \mid \mathbb{N} \times \mathbb{N} \\
\textit{Values } \mathtt{v} ::= & \mathtt{n} \in \mathbb{N} \qquad \textit{References } \ell \in \mathbb{N} \\
\textit{Eval.Ctx. } \mathtt{K} ::= & [\cdot] \mid \mathtt{K} \oplus \mathtt{e} \mid \mathtt{v} \oplus \mathtt{K} \mid \mathtt{x}[\mathtt{K}] \mid \mathtt{let\ x} = \mathtt{K\ in\ e} \\
& \mid \mathtt{x}[\mathtt{K}] \leftarrow \mathtt{e} \mid \mathtt{x}[\mathtt{v}] \leftarrow \mathtt{K} \mid \mathtt{let\ x} = \mathtt{new\ K\ in\ e} \\
& \mid \mathtt{ifz\ K\ then\ e}_1 \mathtt{\ else\ e}_2 \mid \mathtt{call\ foo\ K} \mid \mathtt{return\ K} \\
& \mid \langle \mathtt{K}, \mathtt{e} \rangle \mid \langle \mathtt{v}, \mathtt{K} \rangle \mid \pi_1 \mathtt{K} \mid \pi_2 \mathtt{K} \mid \mathtt{K\ has}\ \tau \\
& \mid \mathtt{ifz\ K\ then\ e}_1 \mathtt{\ else\ e}_2 \mid \mathtt{setDIT\ K} \\
\textit{Variables } & \mathtt{x} \mid \mathtt{y} \mid \mathtt{foo} \mid \ldots \qquad \textit{Poison } \rho ::= \square \mid \text{☣} \\
\textit{Sandbox Tag } & \mathtt{t} ::= \mathtt{ctx} \mid \mathtt{comp} \\
\textit{Typing.Env. } \Gamma ::= & [\cdot] \mid \Gamma, \mathtt{x} : \tau \qquad \textit{Store } \Delta ::= [\cdot] \mid \mathtt{x} \mapsto (\ell; \mathtt{t}; \rho; \mathtt{n}), \Delta \\
\textit{Communication } \mathbf{c} ::= & ? \mid ! \mid \varnothing \qquad \textit{Heaps } \mathtt{H} ::= [\cdot] \mid \mathtt{H} :: \mathtt{n} \\
\textit{Cont. Stack } \overline{\mathtt{K}} ::= & [\cdot] \mid (\mathtt{K}; \mathtt{foo}), \overline{\mathtt{K}} \qquad \textit{Library } \Xi ::= [\cdot] \mid \mathtt{F}, \Xi \\
\textit{Relevant } \xi ::= & [\cdot] \mid \mathtt{foo}, \xi \qquad \textit{State } \Omega ::= \Phi; \mathtt{t}; \mathtt{n}; \Psi \\
\textit{Flow State } \Phi ::= & \xi; \Xi; \overline{\mathtt{K}} \qquad \textit{Memory State } \Psi ::= \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta \\
\textit{Programs } & \mathtt{prog}\ \Xi_{\mathtt{ctx}}\ \Xi_{\mathtt{comp}} \qquad \textit{Substitutions } \gamma ::= [\mathtt{v}/\mathtt{x}], \gamma \mid [\cdot] \\
\textit{Security Tag } \sigma ::= & \text{🔒} \mid \text{🔓}
\end{array}
$$

Figure 66: Syntax of $\mathtt{L}_{\mathtt{scct}}$

The only difference between $\mathtt{L}$ and $\mathtt{L}_{\mathtt{scct}}$ is that we have access to model specific registers by means of the `getDIT` and `setDIT` instructions. Thus, we also extend the dynamic state with a special purpose register representing a model specific register for a data operand independent timing mode). Whenever this flag is set, certain instructions will run in a „constant-time mode". Usually, access to model specific registers requires privileged execution. Since we want to model the strongest possible attacker, we disregard privilege and, thus, grant contexts access to that variable.

The language also has security tags to model annotations to variables (and, by taint analysis, to expressions) as high or low. 🔒 means it's high, while 🔓 means it's low. We have an $\cdot \leq \cdot$ such that 🔒 $\leq$ 🔓. Final results are tagged with the respected taint and the semantics keeps track of this taint during execution.

4.7.1 Dynamic Semantics

$\boxed{\text{dom } \Xi = \texttt{foo}, \ldots, \texttt{bar}}$ „Collect function names."

$$\frac{}{\text{dom } [\cdot] = [\cdot]}\ (\Xi\text{-dom-}[\cdot]) \qquad \frac{\text{dom } \Xi = \texttt{D}}{\text{dom } (\texttt{let foo x} : \tau_\lambda := \texttt{e}), \Xi = \texttt{foo}, \texttt{D}}\ (\Xi\text{-dom-cons})$$

$\boxed{\Xi \equiv \Xi_1 \cup \Xi_2}$ „Merging $\mathrm{L}_{\mathrm{scct}}$ libraries."

$$\frac{\Xi = \Xi_1}{\Xi \equiv \Xi_1 \cup [\cdot]}\ (\text{lib-merge-empty}) \qquad \frac{\Xi \equiv \Xi_1 \cup \Xi_2}{\texttt{F}, \Xi \equiv \Xi_1 \cup \texttt{F}, \Xi_2}\ (\text{lib-merge-cons})$$

$\boxed{\Xi_1 \bowtie \Xi_2 = \Xi}$ „Syntactically linking $\mathrm{L}_{\mathrm{scct}}$ libraries."

$$\frac{\text{dom}\, \Xi_1 \cap \text{dom}\, \Xi_2 = \emptyset \qquad \Xi \equiv \Xi_1 \cup \Xi_2}{\Xi_1 \bowtie \Xi_2 = \Xi}\ (\text{syntactic-plugging})$$

Figure 67: $\mathrm{L}_{\mathrm{scct}}$ plugging of libraries and collecting of function names.

$$\begin{aligned}
\textit{Base Events}\ \mathtt{a_b} ::= {} & \texttt{Alloc}\ \ell\ \texttt{v} \mid \texttt{Dealloc}\ \ell \mid \texttt{Get}\ \ell\ \texttt{v} \mid \texttt{Set}\ \ell\ \texttt{v}\ \texttt{v}' \mid \lightning \\
& \mid \texttt{Call}\ \mathbf{c}\ \texttt{foo}\ \texttt{v} \mid \texttt{Ret}\ \mathbf{c}\ \texttt{v} \mid \texttt{Start} \mid \texttt{End}\ \texttt{v} \\
& \mid \texttt{Branch}\ \texttt{n} \mid \texttt{Binop}\ \texttt{n}\ \texttt{m} \\
& \mid \texttt{iGet}\ \ell\ \texttt{v} \mid \texttt{iSet}\ \ell\ \texttt{v}\ \texttt{v}' \\
\textit{Events}\ \mathtt{a} ::= {} & \varepsilon \mid \mathtt{a_b}; \mathtt{t}; \sigma
\end{aligned}$$

Figure 68: Events of $\mathrm{L}_{\mathrm{scct}}$.

$\boxed{\cdot \leq \cdot}$ „Order of security labels."

$$\frac{}{🔓 \leq 🔓}\ (\text{leq-🔓-refl}) \qquad \frac{}{🔒 \leq \sigma}\ (\text{leq-lock})$$

$\boxed{\cdot \sqcap \cdot}$ „Meet of security labels."

$$\frac{}{\sigma \sqcap \sigma = \sigma}\ (\text{meet-refl}) \qquad \frac{\sigma \neq \sigma'}{\sigma \sqcap \sigma' = 🔓}\ (\text{meet-ineq})$$

Figure 69: Lattice operations for security labels.

$\boxed{\Omega \triangleright \mathtt{e} \xrightarrow{\mathtt{a}} \Omega' \triangleright \mathtt{e}'}$ „Expression $\mathtt{e}$ evaluates under Configuration Ω to $\mathtt{e}'$ and new Configuration Ω', emitting event $\mathtt{a}$."

$$\frac{\begin{array}{c}\mathtt{n}_1 \oplus \mathtt{n}_2 = \mathtt{n}_3 \qquad \oplus \notin \{*, /, <\} \\ \sigma'' = \sigma \sqcap \sigma'\end{array}}{\Omega \triangleright \mathtt{n}_1^{\sigma} \oplus \mathtt{n}_2^{\sigma'} \xrightarrow{\varepsilon} \Omega \triangleright \mathtt{n}_3^{\sigma''}}\ (e - \oplus\text{−noleak-op})$$

$$\frac{\begin{array}{c}\mathtt{n}_1 \oplus \mathtt{n}_2 = \mathtt{n}_3 \qquad \oplus \in \{*, /, <\} \\ \sigma'' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{t}; 0; \Psi \triangleright \mathtt{n}_1^{\sigma} \oplus \mathtt{n}_2^{\sigma'} \xrightarrow{\mathtt{Binop\ n_1\ n_2; t}; \sigma''} \Phi; \mathtt{t}; 0; \Psi \triangleright \mathtt{n}_3^{\sigma''}}\ (e - \oplus\text{−leak})$$

$$\frac{\begin{array}{c}\mathtt{n}_1 \oplus \mathtt{n}_2 = \mathtt{n}_3 \qquad \oplus \in \{*, /, <\} \\ \sigma'' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{t}; \mathtt{m}+1; \Psi \triangleright \mathtt{n}_1^{\sigma} \oplus \mathtt{n}_2^{\sigma'} \xrightarrow{\varepsilon} \Phi; \mathtt{t}; \mathtt{m}+1; \Psi \triangleright \mathtt{n}_3^{\sigma''}}\ (e - \oplus\text{−noleak-doit})$$

$$\frac{\begin{array}{c}\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{t}; \rho; \mathtt{m}), \Delta_2 \qquad \ell + \mathtt{n} \in \mathrm{dom}\ \mathtt{H}^{\mathtt{t}} \\ \sigma'' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{t}'; 0; \Psi \triangleright \mathtt{x}^{\sigma}[\mathtt{n}^{\sigma'}] \xrightarrow{\mathtt{Get}\ \ell\ \mathtt{n;t}; \sigma''} \Phi; \mathtt{t}'; 0; \Psi \triangleright (\mathtt{H}^{\mathtt{t}}(\ell + \mathtt{n}))^{\sigma''}}\ (e - \mathtt{get} - \in\text{−leak})$$

$$\frac{\begin{array}{c}\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{t}; \rho; \mathtt{m}), \Delta_2 \qquad \ell \notin \mathrm{dom}\ \mathtt{H}^{\mathtt{t}} \\ \sigma'' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{t}'; 0; \Psi \triangleright \mathtt{x}^{\sigma}[\mathtt{n}^{\sigma'}] \xrightarrow{\mathtt{Get}\ \ell\ \mathtt{n;t}; \sigma''} \Phi; \mathtt{t}'; 0; \Psi \triangleright 1729^{\text{🔓}}}\ (e - \mathtt{get} - \notin\text{−leak})$$

$$\frac{\begin{array}{c}\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{t}; \rho; \mathtt{m}), \Delta_2 \qquad \ell + \mathtt{n} \in \mathrm{dom}\ \mathtt{H}^{\mathtt{t}} \\ \sigma'' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{t}'; \mathtt{m}+1; \Psi \triangleright \mathtt{x}^{\sigma}[\mathtt{n}^{\sigma'}] \xrightarrow{\mathtt{iGet}\ \ell\ \mathtt{n;t}; \sigma''} \Phi; \mathtt{t}'; \mathtt{m}+1; \Psi \triangleright (\mathtt{H}^{\mathtt{t}}(\ell + \mathtt{n}))^{\sigma''}}\ (e - \mathtt{get} - \in\text{−noleak})$$

$$\frac{\begin{array}{c}\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{t}; \rho; \mathtt{m}), \Delta_2 \qquad \ell \notin \mathrm{dom}\ \mathtt{H}^{\mathtt{t}} \\ \sigma'' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{t}'; \mathtt{m}+1; \Psi \triangleright \mathtt{x}^{\sigma}[\mathtt{n}^{\sigma'}] \xrightarrow{\mathtt{iGet}\ \ell\ \mathtt{n;t}; \sigma''} \Phi; \mathtt{t}'; \mathtt{m}+1; \Psi \triangleright 1729^{\text{🔓}}}\ (e - \mathtt{get} - \notin\text{−noleak})$$

$$\frac{\begin{array}{c}\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{ctx}; \rho; \mathtt{n}), \Delta_2 \qquad \mathtt{H}^{\mathtt{ctx}'} = \mathtt{H}^{\mathtt{ctx}}(\ell + \mathtt{n} \mapsto \mathtt{v}) \\ \Psi' = \mathtt{H}^{\mathtt{ctx}'}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{ctx}; \rho; \mathtt{n}), \Delta_2 \qquad \sigma''' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{ctx}; 0; \Psi \triangleright \mathtt{x}^{\sigma}[\mathtt{n}^{\sigma'}] \leftarrow \mathtt{v}^{\sigma''} \xrightarrow{\mathtt{Set}\ \ell\ \mathtt{n\ v;ctx}; \sigma'''} \Phi; \mathtt{ctx}; 0; \Psi' \triangleright \mathtt{v}^{\sigma''}}\ (e - \mathtt{set} - ctx\text{−leak})$$

$$\frac{\begin{array}{c}\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{comp}; \rho; \mathtt{n}), \Delta_2 \qquad \mathtt{H}^{\mathtt{comp}'} = \mathtt{H}^{\mathtt{comp}}(\ell + \mathtt{n} \mapsto \mathtt{v}) \\ \Psi' = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}'}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{comp}; \rho; \mathtt{n}), \Delta_2 \qquad \sigma''' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{comp}; 0; \Psi \triangleright \mathtt{x}^{\sigma}[\mathtt{n}^{\sigma'}] \leftarrow \mathtt{v}^{\sigma''} \xrightarrow{\mathtt{Set}\ \ell\ \mathtt{n\ v;comp}; \sigma'''} \Phi; \mathtt{comp}; 0; \Psi' \triangleright \mathtt{v}^{\sigma''}}\ (e - \mathtt{set} - comp\text{−leak})$$

Figure 70: Evaluation of $\mathrm{L}_{\mathrm{scct}}$ expressions.

$$\frac{\begin{array}{c}\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{ctx}; \rho; \mathtt{n}), \Delta_2 \qquad \mathtt{H}^{\mathtt{ctx}'} = \mathtt{H}^{\mathtt{ctx}}(\ell + \mathtt{n} \mapsto \mathtt{v}) \\ \Psi' = \mathtt{H}^{\mathtt{ctx}'}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{ctx}; \rho; \mathtt{n}), \Delta_2 \qquad \sigma''' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{ctx}; \mathtt{m}+1; \Psi \triangleright \mathtt{x}^{\sigma}[\mathtt{n}^{\sigma'}] \leftarrow \mathtt{v}^{\sigma''} \xrightarrow{\mathtt{iSet}\ \ell\ \mathtt{n}\ \mathtt{v}; \mathtt{ctx}; \sigma'''} \Phi; \mathtt{ctx}; \mathtt{m}+1; \Psi' \triangleright \mathtt{v}^{\sigma''}}\ (e - \mathtt{set} - ctx - \text{noleak})$$

$$\frac{\begin{array}{c}\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{comp}; \rho; \mathtt{n}), \Delta_2 \qquad \mathtt{H}^{\mathtt{comp}'} = \mathtt{H}^{\mathtt{comp}}(\ell + \mathtt{n} \mapsto \mathtt{v}) \\ \Psi' = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}'}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{comp}; \rho; \mathtt{n}), \Delta_2 \qquad \sigma''' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{comp}; \mathtt{m}+1; \Psi \triangleright \mathtt{x}^{\sigma}[\mathtt{n}^{\sigma'}] \leftarrow \mathtt{v}^{\sigma''} \xrightarrow{\mathtt{iSet}\ \ell\ \mathtt{n}\ \mathtt{v}; \mathtt{comp}; \sigma'''} \Phi; \mathtt{comp}; \mathtt{m}+1; \Psi' \triangleright \mathtt{v}^{\sigma''}}\ (e - \mathtt{set} - comp - \text{noleak})$$

$$\frac{\sigma'' = \sigma \sqcap \sigma'}{\Omega \triangleright \mathtt{let}\ \mathtt{x}^{\sigma} = \mathtt{f}^{\sigma'}\ \mathtt{in}\ \mathtt{e} \xrightarrow{\varepsilon} \Omega \triangleright \mathtt{e}[\mathtt{f}^{\sigma''}/\mathtt{x}]}\ (e - \mathtt{let} - \mathtt{f})$$

$$\frac{\begin{array}{c}\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{t}; \square; \mathtt{n}), \Delta_2 \\ \Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{t}; \cancel{\mathtt{v}}; \mathtt{n}), \Delta_2\end{array}}{\Phi; \mathtt{t}; \mathtt{m}; \Psi \triangleright \mathtt{delete}\ \mathtt{x}^{\sigma} \xrightarrow{\mathtt{Dealloc}\ \ell; \mathtt{t}; \sigma} \Phi; \mathtt{t}; \mathtt{m}; \Psi' \triangleright 0^{\text{🔓}}}\ (e - \mathtt{delete})$$

$$\frac{\begin{array}{c}\Delta \vdash \ell\ \mathit{fresh} \qquad \Delta \vdash \mathtt{z}\ \mathit{fresh} \qquad \mathtt{H}^{\mathtt{ctx}'} = \mathtt{H}^{\mathtt{ctx}} \ll \mathtt{n} \\ \Psi = \mathtt{H}^{\mathtt{ctx}'}; \mathtt{H}^{\mathtt{comp}}; \mathtt{z} \mapsto (\ell; \mathtt{ctx}; \square; \mathtt{n}), \Delta \qquad \sigma'' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{ctx}; \mathtt{m}; \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta \triangleright \mathtt{let}\ \mathtt{x}^{\sigma} = \mathtt{new}\ \mathtt{n}^{\sigma'}\ \mathtt{in}\ \mathtt{e} \xrightarrow{\mathtt{Alloc}\ \ell\ \mathtt{n}; \mathtt{ctx}; \sigma''} \Phi; \mathtt{ctx}; \mathtt{m}; \Psi \triangleright \mathtt{e}[\mathtt{z}^{\sigma''}/\mathtt{x}]}\ (e - \mathtt{new} - ctx)$$

$$\frac{\begin{array}{c}\Delta \vdash \ell\ \mathit{fresh} \qquad \Delta \vdash \mathtt{z}\ \mathit{fresh} \qquad \mathtt{H}^{\mathtt{comp}'} = \mathtt{H}^{\mathtt{comp}} \ll \mathtt{n} \\ \Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}'}; \mathtt{z} \mapsto (\ell; \mathtt{ctx}; \square; \mathtt{n}), \Delta \qquad \sigma'' = \sigma \sqcap \sigma'\end{array}}{\Phi; \mathtt{comp}; \mathtt{m}; \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta \triangleright \mathtt{let}\ \mathtt{x}^{\sigma} = \mathtt{new}\ \mathtt{n}^{\sigma'}\ \mathtt{in}\ \mathtt{e} \xrightarrow{\mathtt{Alloc}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma''} \Phi; \mathtt{comp}; \mathtt{m}; \Psi \triangleright \mathtt{e}[\mathtt{z}^{\sigma''}/\mathtt{x}]}\ (e - \mathtt{new} - comp)$$

$$\frac{}{\Phi; \mathtt{t}; 0; \Psi \triangleright \mathtt{ifz}\ 0^{\sigma}\ \mathtt{then}\ \mathtt{e}_1\ \mathtt{else}\ \mathtt{e}_2 \xrightarrow{\mathtt{Branch}\ 0; \mathtt{t}; \sigma} \Phi; \mathtt{t}; 0; \Psi \triangleright \mathtt{e}_1}\ (e - \mathtt{ifz} - \text{false-noleak})$$

$$\frac{}{\Phi; \mathtt{t}; 0; \Psi \triangleright \mathtt{ifz}\ (\mathtt{S}\ \mathtt{n})^{\sigma}\ \mathtt{then}\ \mathtt{e}_1\ \mathtt{else}\ \mathtt{e}_2 \xrightarrow{\mathtt{Branch}\ \mathtt{S}(\mathtt{n}); \mathtt{t}; \sigma} \Phi; \mathtt{t}; 0; \Psi \triangleright \mathtt{e}_2}\ (e - \mathtt{ifz} - \text{true-leak})$$

$$\frac{}{\Phi; \mathtt{t}; \mathtt{m}+1; \Psi \triangleright \mathtt{ifz}\ 0^{\sigma}\ \mathtt{then}\ \mathtt{e}_1\ \mathtt{else}\ \mathtt{e}_2 \xrightarrow{\varepsilon} \Phi; \mathtt{t}; \mathtt{m}+1; \Psi \triangleright \mathtt{e}_1}\ (e - \mathtt{ifz} - \text{false-leak})$$

$$\frac{}{\Phi; \mathtt{t}; \mathtt{m}+1; \Psi \triangleright \mathtt{ifz}\ (\mathtt{S}\ \mathtt{n})^{\sigma}\ \mathtt{then}\ \mathtt{e}_1\ \mathtt{else}\ \mathtt{e}_2 \xrightarrow{\varepsilon} \Phi; \mathtt{t}; \mathtt{m}+1; \Psi \triangleright \mathtt{e}_2}\ (e - \mathtt{ifz} - \text{true-leak})$$

$$\frac{}{\Phi; \mathtt{t}; \mathtt{m}; \Psi \triangleright \mathtt{abort}() \xrightarrow{\lightning; \mathtt{t}; \text{🔓}} \Phi; \mathtt{t}; \mathtt{m}; \Psi \triangleright \mathtt{stuck}}\ (e - \mathtt{abort})$$

Figure 71: Evaluation of $\mathrm{L}_{\mathrm{scct}}$ expressions, continued.

$$\frac{}{\Omega \triangleright \pi_1\langle \mathtt{n}_1^{\sigma}; \mathtt{n}_2^{\sigma'}\rangle \xrightarrow{\varepsilon} \Omega \triangleright \mathtt{n}_1^{\sigma}}\ (e-\pi_1) \qquad \frac{}{\Omega \triangleright \pi_2\langle \mathtt{n}_1^{\sigma}; \mathtt{n}_2^{\sigma'}\rangle \xrightarrow{\varepsilon} \Omega \triangleright \mathtt{n}_2^{\sigma'}}\ (e-\pi_2)$$

$$\frac{\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta \qquad \Delta = \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{t}; \text{☣}; \mathtt{n}), \Delta_2}{\Phi; \mathtt{t}'; \mathtt{m}; \Psi \triangleright \mathtt{x}^{\sigma}\ \mathtt{is}\ \text{☣} \xrightarrow{\varepsilon} \Phi; \mathtt{t}'; \mathtt{m}; \Psi \triangleright 0^{🔓}}\ (e-\mathtt{x}\ \mathtt{is}\ \text{☣}-\mathsf{yes})$$

$$\frac{\Psi = \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta \qquad \Delta = \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{t}; \square; \mathtt{n}), \Delta_2}{\Phi; \mathtt{t}'; \mathtt{m}; \Psi \triangleright \mathtt{x}^{\sigma}\ \mathtt{is}\ \text{☣} \xrightarrow{\varepsilon} \Phi; \mathtt{t}'; \mathtt{m}; \Psi \triangleright 1^{🔓}}\ (e-\mathtt{x}\ \mathtt{is}\ \text{☣}-\mathsf{no})$$

$$\frac{}{\Omega \triangleright \mathtt{n}^{\sigma}\ \mathtt{has}\ \mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 0^{🔓}}\ (e-\mathtt{n}\ \mathtt{has}\ \mathbb{N}) \qquad \frac{}{\Omega \triangleright \langle \mathtt{n}_1^{\sigma}, \mathtt{n}_2^{\sigma'}\rangle\ \mathtt{has}\ \mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 1^{🔓}}\ (e-pair-\mathtt{has}\ \mathbb{N})$$

$$\frac{}{\Omega \triangleright \mathtt{x}^{\sigma}\ \mathtt{has}\ \mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 1^{🔓}}\ (e-\mathtt{x}\ \mathtt{has}\ \mathbb{N}) \qquad \frac{}{\Omega \triangleright \mathtt{n}^{\sigma}\ \mathtt{has}\ \mathbb{N}\times\mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 1^{🔓}}\ (e-\mathtt{n}\ \mathtt{has}\ \mathbb{N}\times\mathbb{N})$$

$$\frac{}{\Omega \triangleright \langle \mathtt{n}_1^{\sigma}, \mathtt{n}_2^{\sigma'}\rangle\ \mathtt{has}\ \mathbb{N}\times\mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 0^{🔓}}\ (e-pair-\mathtt{has}\ \mathbb{N}\times\mathbb{N}) \qquad \frac{}{\Omega \triangleright \mathtt{x}^{\sigma}\ \mathtt{has}\ \mathbb{N}\times\mathbb{N} \xrightarrow{\varepsilon} \Omega \triangleright 1^{🔓}}\ (e-\mathtt{x}\ \mathtt{has}\ \mathbb{N}\times\mathbb{N})$$

$$\frac{}{\Phi; \mathtt{t}; \mathtt{m}; \Psi \triangleright \mathtt{setDIT}\ \mathtt{n}^{\sigma} \xrightarrow{\varepsilon} \Phi; \mathtt{t}; \mathtt{n}; \Psi \triangleright \mathtt{n}^{\sigma}}\ (e-\mathtt{setDIT}\ \mathtt{n})$$

$$\frac{}{\Phi; \mathtt{t}; \mathtt{n}; \Psi \triangleright \mathtt{getDIT}\ \mathtt{x}^{\sigma}\ \mathtt{in}\ \mathtt{e} \xrightarrow{\varepsilon} \Phi; \mathtt{t}; \mathtt{n}; \Psi \triangleright \mathtt{e}[\mathtt{n}^{\sigma}/\mathtt{x}]}\ (e-\mathtt{getDIT}\ \mathtt{x}\ \mathtt{in}\ \mathtt{e})$$

Figure 72: Evaluation of $\mathrm{L}_{\mathrm{scct}}$ expressions, continued.

$\boxed{\Omega \triangleright \mathtt{K}[\mathtt{e}] \xrightarrow{\mathtt{a}}_{\mathrm{ctx}} \Omega' \triangleright \mathtt{K}[\mathtt{e}']}$ „Given an evaluation context $\mathtt{K}$ and an expression $\mathtt{e}$, it evaluates under configuration Ω to $\mathtt{e}'$ and new configuration Ω' in context $\mathtt{K}$, emitting event $\mathtt{a}$."

$$\frac{\Omega \triangleright \mathtt{e} \xrightarrow{\mathtt{a}} \Omega' \triangleright \mathtt{e}'}{\Omega \triangleright \mathtt{K}[\mathtt{e}] \xrightarrow{\mathtt{a}}_{\mathrm{ctx}} \Omega' \triangleright \mathtt{K}[\mathtt{e}']}\ (e - \mathsf{ctx})$$

$$\frac{\Omega \triangleright \mathtt{e} \xrightarrow{\lightning} \Omega' \triangleright \mathtt{stuck}}{\Omega \triangleright \mathtt{K}[\mathtt{e}] \xrightarrow{\lightning}_{\mathrm{ctx}} \Omega' \triangleright \mathtt{stuck}}\ (e - \mathsf{ctx} - \mathtt{stuck})$$

$$\frac{\begin{array}{c}\Omega = \xi; \Xi; [\cdot]; \mathtt{comp}; 0; \Psi \qquad \Xi = \Xi_1, (\mathtt{let\ main\ x} : \tau_\lambda := \mathtt{e}), \Xi_2 \\ \Omega' = \xi; \Xi; \mathtt{K}^{\mathtt{main}}, [\cdot]; \mathtt{ctx}; 0; \Psi\end{array}}{\Omega \triangleright \mathtt{K}[\mathtt{call\ main}^{\sigma}\ \mathtt{v}^{\sigma'}] \xrightarrow{\mathtt{Start;comp;}\sigma'}_{\mathrm{ctx}} \Omega' \triangleright \mathtt{e}[\mathtt{v}^{\sigma'}/\mathtt{x}]}\ (e - \mathsf{ctx} - \mathtt{call} - \mathsf{main})$$

$$\frac{\begin{array}{c}\Omega = \xi; \Xi; \overline{\mathtt{K}}; \mathtt{t}; \mathtt{m}; \Psi \qquad \Xi = \Xi_1, (\mathtt{let\ foo\ x} : \tau_\lambda := \mathtt{e}), \Xi_2 \\ \mathtt{foo} \in_{\neg \mathtt{t}} \xi \qquad \rho(\neg \mathtt{t}) = \mathbf{c} \qquad \Omega' = \xi; \Xi; \mathtt{K}^{\mathtt{foo}}, \overline{\mathtt{K}}; \neg \mathtt{t}; \mathtt{m}; \Psi\end{array}}{\Omega \triangleright \mathtt{K}[\mathtt{call\ foo}^{\sigma}\ \mathtt{v}^{\sigma'}] \xrightarrow{\mathtt{Call\ c\ foo\ v;t;}\sigma'}_{\mathrm{ctx}} \Omega' \triangleright \mathtt{e}[\mathtt{v}^{\sigma'}/\mathtt{x}]}\ (e - \mathsf{ctx} - \mathtt{call} - \mathsf{notsame})$$

$$\frac{\begin{array}{c}\Omega = \xi; \Xi; \overline{\mathtt{K}}; \mathtt{t}; \mathtt{m}; \Psi \qquad \Xi = \Xi_1, (\mathtt{let\ foo\ x} : \tau_\lambda := \mathtt{e}), \Xi_2 \\ \mathtt{foo} \in_{\mathtt{t}} \xi \qquad \Omega' = \xi; \Xi; \mathtt{K}^{\mathtt{foo}}, \overline{\mathtt{K}}; \mathtt{t}; \mathtt{m}; \Psi\end{array}}{\Omega \triangleright \mathtt{K}[\mathtt{call\ foo}^{\sigma}\ \mathtt{v}^{\sigma'}] \xrightarrow{\mathtt{Call}\ \varnothing\ \mathtt{foo\ v;t;}\sigma'}_{\mathrm{ctx}} \Omega' \triangleright \mathtt{e}[\mathtt{v}^{\sigma'}/\mathtt{x}]}\ (e - \mathsf{ctx} - \mathtt{call} - \mathsf{same})$$

$$\frac{}{\xi; \Xi; [\cdot]^{\mathtt{main}}, [\cdot]; \mathtt{ctx}; \mathtt{m}; \Psi \triangleright \mathtt{K}'[\mathtt{return\ v}^{\sigma}] \xrightarrow{\mathtt{End\ v;t;}\sigma}_{\mathrm{ctx}} \xi; \Xi; [\cdot]; \mathtt{comp}; 1; \Psi \triangleright \mathtt{v}^{\sigma}}\ (e - \mathsf{ctx} - \mathtt{return} - \mathsf{main})$$

$$\frac{\mathtt{foo} \in_{\neg \mathtt{t}} \xi \qquad \rho(\neg \mathtt{t}) = \mathbf{c}; \mathtt{t}}{\xi; \Xi; \mathtt{K}^{\mathtt{foo}}, \overline{\mathtt{K}}; \mathtt{t}; \mathtt{m}; \Psi \triangleright \mathtt{K}'[\mathtt{return\ v}^{\sigma}] \xrightarrow{\mathtt{Ret\ c\ v;t;}\sigma}_{\mathrm{ctx}} \xi; \Xi; \overline{\mathtt{K}}; \neg \mathtt{t}; \mathtt{m}; \Psi \triangleright \mathtt{K}[\mathtt{v}^{\sigma}]}\ (e - \mathsf{ctx} - \mathtt{return} - \mathsf{notsame})$$

$$\frac{\mathtt{foo} \in_{\mathtt{t}} \xi}{\xi; \Xi; \mathtt{K}^{\mathtt{foo}}, \overline{\mathtt{K}}; \mathtt{t}; \mathtt{m}; \Psi \triangleright \mathtt{K}'[\mathtt{return\ v}^{\sigma}] \xrightarrow{\mathtt{Ret}\ \varnothing\ \mathtt{v;t;}\sigma}_{\mathrm{ctx}} \xi; \Xi; \overline{\mathtt{K}}; \mathtt{t}; \mathtt{m}; \Psi \triangleright \mathtt{K}[\mathtt{v}^{\sigma}]}\ (e - \mathsf{ctx} - \mathtt{return} - \mathsf{same})$$

$\boxed{\rho(\mathtt{t}) = \mathbf{c}}$ „Returns either ? or ! depending on $\mathtt{t}$."

$$\frac{}{\rho(\mathtt{ctx}) = ?}\ (\mathsf{comm\text{-}ctxtocomp}) \qquad \frac{}{\rho(\mathtt{comp}) = !}\ (\mathsf{comm\text{-}comptoctx})$$

$\boxed{\neg \mathtt{t} = \mathtt{t}'}$ „Negation of $\mathtt{t}$."

$$\frac{}{\neg \mathtt{ctx} = \mathtt{comp}}\ (\mathsf{neg\text{-}ctx}) \qquad \frac{}{\neg \mathtt{comp} = \mathtt{ctx}}\ (\mathsf{neg\text{-}comp})$$

Figure 73: Contextual Evaluation of $\mathrm{L}_{\mathrm{scct}}$ expressions.

$\boxed{\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}'}$ „Expression e evaluates under configuration Ω to e' and new configuration Ω', emitting list of events $\bar{\mathsf{a}}$."

$$\frac{}{\Omega \triangleright \mathsf{f}_{\lightning} \xrightarrow{[\cdot]}{}^{*}_{\mathrm{ctx}} \Omega \triangleright \mathsf{f}_{\lightning}} \quad (et - \mathsf{refl})$$

$$\frac{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}' \qquad \Omega' \triangleright \mathsf{e}' \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}'' \qquad \mathsf{a} \neq \varepsilon}{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}\cdot\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}''} \quad (et - \mathsf{trans-important})$$

$$\frac{\Omega \triangleright \mathsf{e} \xrightarrow{\varepsilon}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}' \qquad \Omega' \triangleright \mathsf{e}' \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}''}{\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega'' \triangleright \mathsf{e}''} \quad (et - \mathsf{trans-unimportant})$$

$\boxed{\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}{}^{n}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}'}$ „Expression e evaluates in n steps under configuration Ω to e' and new configuration Ω', emitting list of events $\bar{\mathsf{a}}$."

$$\frac{}{\Omega \triangleright \mathsf{e} \xrightarrow{\varepsilon}_{\mathrm{ctx}}{}^{0} \Omega \triangleright \mathsf{e}} \quad (en - \mathsf{refl})$$

$$\frac{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}' \qquad \Omega' \triangleright \mathsf{e}' \xrightarrow{\bar{\mathsf{a}}}_{\mathrm{ctx}}{}^{n} \Omega'' \triangleright \mathsf{e}'' \qquad \mathsf{a} \neq \varepsilon}{\Omega \triangleright \mathsf{e} \xrightarrow{\mathsf{a}\cdot\bar{\mathsf{a}}}_{\mathrm{ctx}}{}^{(n+1)} \Omega'' \triangleright \mathsf{e}''} \quad (en - \mathsf{trans-important})$$

$$\frac{\Omega \triangleright \mathsf{e} \xrightarrow{\varepsilon}_{\mathrm{ctx}} \Omega' \triangleright \mathsf{e}' \qquad \Omega' \triangleright \mathsf{e}' \xrightarrow{\bar{\mathsf{a}}}_{\mathrm{ctx}}{}^{n} \Omega'' \triangleright \mathsf{e}''}{\Omega \triangleright \mathsf{e} \xrightarrow{\bar{\mathsf{a}}}_{\mathrm{ctx}}{}^{(n+1)} \Omega'' \triangleright \mathsf{e}''} \quad (en - \mathsf{trans-unimportant})$$

Figure 74: Trace prefix generation given a $\mathrm{L}_{\mathrm{scct}}$ program using the reflexive-transitive closure.

$\boxed{\mathtt{prog}\ \Xi_{\mathtt{ctx}}\ \Xi_{\mathtt{comp}} \overset{\bar{\mathsf{a}}}{\Longrightarrow} \Omega \triangleright \mathsf{f}}$ „Run $\mathrm{L}_{\mathrm{scct}}$ program $\mathtt{prog}\ \Xi_{\mathtt{ctx}}\ \Xi_{\mathtt{comp}}$, giving dynamic state Ω and emitting trace $\bar{\mathsf{a}}$."

$$\frac{\Xi = \Xi_{\mathtt{ctx}} \bowtie \Xi_{\mathtt{comp}} \qquad \mathtt{main} \notin \xi \qquad \xi = \mathrm{dom}\ \Xi_{\mathtt{comp}} \qquad \xi; \Xi; [\cdot]; \mathtt{t}; 1; [\cdot]; [\cdot]; [\cdot] \triangleright \mathtt{call}\ \mathtt{main}\ 0 \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{\mathrm{ctx}} \Omega \triangleright \mathsf{f}_{\lightning}}{\mathtt{prog}\ \Xi_{\mathtt{ctx}}\ \Xi_{\mathtt{comp}} \overset{\bar{\mathsf{a}}}{\Longrightarrow} \Omega \triangleright \mathsf{f}_{\lightning}} \quad (e - \mathsf{wprog})$$

$$\frac{\Xi = \Xi_{\mathtt{ctx}} \bowtie \Xi_{\mathtt{comp}} \qquad \mathtt{main} \notin \xi \qquad \xi = \mathrm{dom}\ \Xi_{\mathtt{comp}} \qquad \xi; \Xi; [\cdot]; \mathtt{t}; 1; [\cdot]; [\cdot]; [\cdot] \triangleright \mathtt{call}\ \mathtt{main}\ 0 \xrightarrow{\bar{\mathsf{a}}\cdot\lightning;\mathtt{t}}{}^{*}_{\mathrm{ctx}} \Omega \triangleright \mathsf{f}_{\lightning}}{\mathtt{prog}\ \Xi_{\mathtt{ctx}}\ \Xi_{\mathtt{comp}} \overset{\bar{\mathsf{a}}\cdot\lightning;\mathtt{t}}{\Longrightarrow} \Omega \triangleright \mathsf{f}_{\lightning}} \quad (e - \mathsf{wprog} - \lightning)$$

Figure 75: Running a whole $\mathrm{L}_{\mathrm{scct}}$ program.

4.7.2 Translation to Specification Events

$\boxed{\delta_{MS}(\ell) = \underline{\ell}}$ „A map from $\mathtt{L}_{\mathrm{scct}}$ memory locations ℓ to specification locations $\underline{\ell}$."

$\boxed{\theta_{\delta_{MS}}(\mathtt{a}) = a^{\mathrm{ms}}}$ „Project an $\mathtt{L}_{\mathrm{scct}}$ event to specification events."

(o-filter-context)
$$\frac{\mathtt{a_b} \neq \lightning}{\theta_{\delta_{MS}}(\mathtt{a_b}; \mathtt{ctx}; \sigma) = \underline{\varepsilon}}$$

(o-filter-comp-start)
$$\frac{}{\theta_{\delta_{MS}}(\mathtt{Start}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-filter-comp-end)
$$\frac{}{\theta_{\delta_{MS}}(\mathtt{End}\ \mathtt{v}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-filter-comp-alloc)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{MS}}(\mathtt{Alloc}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma) = \underline{Alloc\ \ell\ n}}$$

(o-filter-comp-dealloc)
$$\frac{\delta_{MS}(\ell) = \underline{\ell}}{\theta_{\delta_{MS}}(\mathtt{Dealloc}\ \ell; \mathtt{comp}; \sigma) = \underline{Dealloc\ \ell}}$$

(o-filter-comp-get-noleak)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{MS}}(\mathtt{iGet}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma) = \underline{Use\ \ell\ n}}$$

(o-filter-comp-set-noleak)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{MS}}(\mathtt{iSet}\ \ell\ \mathtt{n}\ \mathtt{v}; \mathtt{comp}; \sigma) = \underline{Use\ \ell\ n}}$$

(o-filter-comp-get-leak)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{MS}}(\mathtt{Get}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma) = \underline{Use\ \ell\ n}}$$

(o-filter-comp-set-leak)
$$\frac{\delta_{MS}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{MS}}(\mathtt{Set}\ \ell\ \mathtt{n}\ \mathtt{v}; \mathtt{comp}; \sigma) = \underline{Use\ \ell\ n}}$$

(o-filter-comp-call)
$$\frac{}{\theta_{\delta_{MS}}(\mathtt{Call}\ \mathbf{c}\ \mathtt{foo}\ \mathtt{v}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-filter-comp-ret)
$$\frac{}{\theta_{\delta_{MS}}(\mathtt{Ret}\ \mathbf{c}\ \mathtt{v}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-filter-comp-binop)
$$\frac{}{\theta_{\delta_{MS}}(\mathtt{Binop}\ \mathtt{n_1}\ \mathtt{n_2}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-filter-comp-branch)
$$\frac{}{\theta_{\delta_{MS}}(\mathtt{Branch}\ \mathtt{n}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-filter-abort)
$$\frac{}{\theta_{\delta_{MS}}(\lightning; \mathtt{t}; \sigma) = \underline{\lightning}}$$

$\boxed{\theta^*_{\delta_{MS}}(\overline{\mathtt{a}}) = \overline{a^{\mathrm{ms}}}}$ „Project an $\mathtt{L}_{\mathrm{scct}}$ trace to specification traces."

(o-filter-empty)
$$\frac{}{\theta^*_{\delta_{MS}}([\cdot]) = \underline{[\cdot]}}$$

(o-filter-cons-relevant)
$$\frac{\theta_{\delta_{MS}}(\mathtt{a}) = \underline{a} \qquad \theta^*_{\delta_{MS}}(\overline{\mathtt{a}}) = \overline{a^{\mathrm{ms}}} \qquad \underline{a} \neq \underline{\varepsilon}}{\theta^*_{\delta_{MS}}(\mathtt{a} \cdot \overline{\mathtt{a}}) = \underline{a} \cdot \overline{a^{\mathrm{ms}}}}$$

(o-filter-cons-relevant)
$$\frac{\theta_{\delta_{MS}}(\mathtt{a}) = \underline{\varepsilon} \qquad \theta^*_{\delta_{MS}}(\overline{\mathtt{a}}) = \overline{a^{\mathrm{ms}}}}{\theta^*_{\delta_{MS}}(\mathtt{a} \cdot \overline{\mathtt{a}}) = \overline{a^{\mathrm{ms}}}}$$

Figure 76: Projection of $\mathtt{L}_{\mathrm{scct}}$ events to specification events, but for memory safety. Since the security tag of specification events for memory safety doesn't play any role whatsoever, we leave them out for brevity.

$\boxed{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell}}$ „A map from $\mathtt{L}_{\mathtt{scct}}$ memory locations ℓ to specification locations $\underline{\ell}$."

$\boxed{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{a}) = a^{\mathtt{scct}}}$ „Project an $\mathtt{L}_{\mathtt{scct}}$ event to specification events."

(o-scct-filter-context)
$$\frac{\mathtt{a_b} \neq \lightning}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{a_b}; \mathtt{ctx}; \sigma) = \underline{\varepsilon}}$$

(o-scct-filter-comp-start)
$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{Start}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-scct-filter-comp-end)
$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{End\ v}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-scct-filter-comp-alloc)
$$\frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{Alloc}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma) = \underline{\mathit{Alloc}\ \ell\ n; 🔓}}$$

(o-scct-filter-comp-dealloc)
$$\frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell}}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{Dealloc}\ \ell; \mathtt{comp}; \sigma) = \underline{\mathit{Dealloc}\ \ell; 🔓}}$$

(o-scct-filter-comp-get-noleak)
$$\frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{iGet}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-scct-filter-comp-set-noleak)
$$\frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{iSet}\ \ell\ \mathtt{n}\ \mathtt{v}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-scct-filter-comp-get-leak)
$$\frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{Get}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma) = \underline{\mathit{Use}\ \ell\ n; \sigma}}$$

(o-scct-filter-comp-set-leak)
$$\frac{\delta_{\mathtt{sCCT}}(\ell) = \underline{\ell} \qquad \underline{n} = \mathtt{n}}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{Set}\ \ell\ \mathtt{n}\ \mathtt{v}; \mathtt{comp}; \sigma) = \underline{\mathit{Use}\ \ell\ n; \sigma}}$$

(o-scct-filter-comp-call)
$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{Call}\ \mathbf{c}\ \mathtt{foo}\ \mathtt{v}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-scct-filter-comp-ret)
$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{Ret}\ \mathbf{c}\ \mathtt{v}; \mathtt{comp}; \sigma) = \underline{\varepsilon}}$$

(o-scct-filter-comp-binop)
$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{Binop}\ \mathtt{n_1}\ \mathtt{n_2}; \mathtt{comp}; \sigma) = \underline{\mathit{Binop}\ n_1\ n_2; \sigma}}$$

(o-scct-filter-comp-branch)
$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{Branch}\ \mathtt{n}; \mathtt{comp}; \sigma) = \underline{\mathit{Branch}\ n; \sigma}}$$

(o-scct-filter-abort)
$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\lightning; \mathtt{t}; \sigma) = \underline{\lightning; \sigma}}$$

(o-scct-filter-emptyevent)
$$\frac{}{\theta_{\delta_{\mathtt{sCCT}}}(\varepsilon) = \underline{\varepsilon}}$$

$\boxed{\theta^*_{\delta_{\mathtt{sCCT}}}(\overline{\mathtt{a}}) = \overline{a^{\mathtt{scct}}}}$ „Project an $\mathtt{L}_{\mathtt{scct}}$ trace to specification traces."

(o-scct-filter-empty)
$$\frac{}{\theta^*_{\delta_{\mathtt{sCCT}}}([\cdot]) = \underline{[\cdot]}}$$

(o-scct-filter-cons-relevant)
$$\frac{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{a}) = \underline{a} \qquad \theta^*_{\delta_{\mathtt{sCCT}}}(\overline{\mathtt{a}}) = \overline{a^{\mathtt{scct}}} \qquad \underline{a} \neq \underline{\varepsilon}}{\theta^*_{\delta_{\mathtt{sCCT}}}(\mathtt{a} \cdot \overline{\mathtt{a}}) = \underline{a} \cdot \overline{a^{\mathtt{scct}}}}$$

(o-scct-filter-cons-relevant)
$$\frac{\theta_{\delta_{\mathtt{sCCT}}}(\mathtt{a}) = \underline{\varepsilon} \qquad \theta^*_{\delta_{\mathtt{sCCT}}}(\overline{\mathtt{a}}) = \overline{a^{\mathtt{scct}}}}{\theta^*_{\delta_{\mathtt{sCCT}}}(\mathtt{a} \cdot \overline{\mathtt{a}}) = \overline{a^{\mathtt{scct}}}}$$

Figure 77: Projection of $\mathtt{L}_{\mathtt{scct}}$ events to specification events.

$\boxed{\delta(\ell) = \ell}$ „Map from $\mathtt{L}$ locations ℓ to $\mathtt{L}_{\mathrm{scct}}$ locations ℓ."

$\boxed{\overline{a} \approx^{*} \overline{\mathtt{a}}}$ „The $\mathtt{L}$ trace $\overline{a}$ describes the same actions as $\mathtt{L}_{\mathrm{scct}}$ trace $\overline{\mathtt{a}}$."

(scct-empty-trace-eq)
$$\frac{}{[\cdot] \approx^{*} [\cdot]}$$

(scct-cons-trace-eq)
$$\frac{t = \mathtt{t} \qquad a_b \approx_{\delta} \mathtt{a_b} \qquad \overline{a} \approx^{*} \overline{\mathtt{a}}}{a_b; t \cdot \overline{a} \approx^{*} \mathtt{a_b}; \mathtt{t}; \sigma \cdot \overline{\mathtt{a}}}$$

$\boxed{a_b \approx_{\delta} \mathtt{a_b}}$ „The $\mathtt{L}$ event a_b describes the same action as $\mathtt{L}$ event $\mathtt{a_b}$."

(scct-start-event-eq)
$$\frac{}{\mathit{Start} \approx_{\delta} \mathtt{Start}}$$

(scct-end-event-eq)
$$\frac{[\![v]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{v}}{\mathit{End}\ v \approx_{\delta} \mathtt{End}\ \mathtt{v}}$$

(scct-alloc-event-eq)
$$\frac{[\![n]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{n} \qquad \delta(\ell) = \ell}{\mathit{Alloc}\ \ell\ n \approx_{\delta} \mathtt{Alloc}\ \ell\ \mathtt{n}}$$

(scct-dealloc-event-eq)
$$\frac{\delta(\ell) = \ell}{\mathit{Dealloc}\ \ell \approx_{\delta} \mathtt{Dealloc}\ \ell}$$

(scct-get-event-eq-noleak)
$$\frac{[\![n]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{n} \qquad \delta(\ell) = \ell}{\mathit{Set}\ \ell\ n\ v \approx_{\delta} \mathtt{iGet}\ \ell\ \mathtt{n}}$$

(scct-get-event-eq-leak)
$$\frac{[\![n]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{n} \qquad \delta(\ell) = \ell}{\mathit{Get}\ \ell\ n \approx_{\delta} \mathtt{Get}\ \ell\ \mathtt{n}}$$

(scct-set-event-eq-leak)
$$\frac{[\![n]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{n} \qquad [\![v]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{v} \qquad \delta(\ell) = \ell}{\mathit{Set}\ \ell\ n\ v \approx_{\delta} \mathtt{Set}\ \ell\ \mathtt{n}\ \mathtt{v}}$$

(scct-set-event-eq-noleak)
$$\frac{[\![n]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{n} \qquad [\![v]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{v} \qquad \delta(\ell) = \ell}{\mathit{Set}\ \ell\ n\ v \approx_{\delta} \mathtt{iSet}\ \ell\ \mathtt{n}\ \mathtt{v}}$$

(scct-call-event-eq)
$$\frac{[\![n]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{n} \qquad [\![\mathit{foo}]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{foo} \qquad \mathbf{c} = \mathbf{c}}{\mathit{Call}\ \mathbf{c}\ \mathit{foo}\ n \approx_{\delta} \mathtt{Call}\ \mathbf{c}\ \mathtt{foo}\ \mathtt{n}}$$

(scct-ret-event-eq)
$$\frac{[\![n]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{n} \qquad \mathbf{c} = \mathbf{c}}{\mathit{Ret}\ \mathbf{c}\ n \approx_{\delta} \mathtt{Ret}\ \mathbf{c}\ \mathtt{n}}$$

(scct-ε-event-eq)
$$\frac{}{\varepsilon \approx_{\delta} \varepsilon}$$

(scct-$\lightning$-event-eq)
$$\frac{}{\lightning \approx_{\delta} \lightning}$$

(scct-binop-event-eq-leak)
$$\frac{}{\varepsilon \approx_{\delta} \mathtt{Binop}\ \mathtt{n}_1\ \mathtt{n}_2}$$

(scct-branch-event-eq-leak)
$$\frac{}{\varepsilon \approx_{\delta} \mathtt{Branch}\ \mathtt{n}}$$

$\boxed{\mathbf{c} = \mathbf{c}}$ „Communications are equal."

$$\frac{(? = ?)}{? = ?} \qquad \frac{(! = !)}{! = !} \qquad \frac{(\varnothing = \varnothing)}{\varnothing = \varnothing}$$

Figure 78: Trace Relation from $\mathtt{L}$ to $\mathtt{L}_{\mathrm{scct}}$.

$\boxed{\delta(\ell) = \ell}$ „The $\mathtt{L}$ memory location ℓ corresponds to the $\mathtt{L}_{\mathrm{scct}}$ memory location ℓ."

$\boxed{\Omega \approx_{\delta_{scct}} \Omega}$ „The $\mathtt{L}$ state Ω agrees with $\mathtt{L}$ state Ω."

(scct-state-eq)

$$\frac{\Omega = \Phi; t; \Psi \qquad \Omega = \Phi; \mathtt{t}; \mathtt{n}; \Psi \qquad \mathtt{n} \neq 0 \qquad t = \mathtt{t} \qquad \Phi \approx \Phi \qquad \Psi \approx_{\delta_{scct}} \Psi}{\Omega \approx_{\delta_{scct}} \Omega}$$

(scct-abort-state-eq)

$$\frac{}{\lightning \approx_{\delta_{scct}} \lightning}$$

$\boxed{\Psi \approx_{\delta_{scct}} \Psi}$ „The $\mathtt{L}$ memory-state Ψ agrees with $\mathtt{L}_{\mathrm{scct}}$ one Ψ. L contains locations introduced by the backtranslation wrapper."

(scct-empty-memstate-eq)

$$\frac{}{[\cdot]; [\cdot]; [\cdot] \approx_{\delta_{scct}} [\cdot]; [\cdot]; [\cdot]}$$

(scct-comp-cons-memstate-eq)

$$\frac{\ell \notin L \qquad n = \mathtt{n} \qquad \rho = \rho \qquad \delta(\ell) = \ell \qquad \ell, n \vdash_\delta H^{comp} \approx \mathtt{H}^{\mathtt{comp}} \qquad H^{ctx}; H^{comp}; \Delta \approx_{\delta_{scct}} \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \Delta_2}{H^{ctx}; H^{comp}; x \mapsto (\ell; comp; \rho; n), \Delta \approx_{\delta_{scct}} \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta_1, \mathtt{x} \mapsto (\ell; \mathtt{comp}; \rho; \mathtt{n}), \Delta_2}$$

(scct-ctx-cons-memstate-eq)

$$\frac{\ell \notin L \qquad H^{ctx}; H^{comp}; \Delta \approx_{\delta_{scct}} \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta}{H^{ctx}; H^{comp}; x \mapsto (\ell; ctx; \rho; n), \Delta \approx_{\delta_{scct}} \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta}$$

(scct-whatever-cons-memstate-eq)

$$\frac{\ell \in L \qquad H^{ctx}; H^{comp}; \Delta \approx_{\delta_{scct}} \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta}{H^{ctx}; H^{comp}; x \mapsto (\ell; t; \rho; n), \Delta \approx_{\delta_{scct}} \mathtt{H}^{\mathtt{ctx}}; \mathtt{H}^{\mathtt{comp}}; \Delta}$$

$\boxed{\ell, n \vdash_\delta H^{comp} \approx \mathtt{H}^{\mathtt{comp}}}$ „The heaps H^{comp} and $\mathtt{H}^{\mathtt{comp}}$ are related at ℓ for n memory cells."

(scct-heap-related)

$$\frac{[\![n]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} = \mathtt{n} \qquad \delta(\ell) = \ell \qquad \forall 0 \leq m < n,\ 0 \leq \mathtt{m} < \mathtt{n}, H^{comp}(\ell + m) = \mathtt{H}^{\mathtt{comp}}(\ell + \mathtt{m})}{\ell, n \vdash_\delta H^{comp} \approx \mathtt{H}^{\mathtt{comp}}}$$

$\boxed{\Phi \approx \Phi}$ „The $\mathtt{L}$ control-flow-state Φ agrees with $\mathtt{L}_{\mathrm{scct}}$ one Φ."

(scct-cfstate-eq)

$$\frac{\Phi = \xi; \Xi; \overline{K} \qquad \Phi = \xi; \Xi; \overline{\mathtt{K}} \qquad \Xi \approx \Xi \qquad \overline{K} \approx_\xi \overline{\mathtt{K}} \qquad \xi = [\![\xi]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}}}{\Phi \approx \Phi}$$

$\boxed{\rho = \rho}$ „$\mathtt{L}$ poison equals $\mathtt{L}_{\mathrm{scct}}$ one ρ."

(☣-equal)

$$\frac{}{☣ = ☣}$$

(□-equal)

$$\frac{}{\square = \square}$$

Figure 79: State Relation from $\mathtt{L}$ to $\mathtt{L}_{\mathrm{scct}}$. This is meant to relate the states whenever we are „inside" a component. Note that the DOIT flag is required to be non-zero.

$\boxed{\Xi \approx \Xi}$ „The procedures of L agree with $\mathtt{L}_{\mathrm{scct}}$ ones."

(scct-empty-commlib-lib-eq)

$$\frac{}{[\cdot] \approx \Xi}$$

(scct-cons-commlib-eq)

$$\frac{[\![\mathit{let}\ \mathit{foo}\ x : \tau_\lambda := e]\!]^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = \mathtt{let\ foo\ x := e} \qquad \Xi \approx \Xi_1, \Xi_2}{(\mathit{let}\ \mathit{foo}\ x : \tau_\lambda := e), \Xi \approx \Xi_1, (\mathtt{let\ foo\ x := e}), \Xi_2}$$

$\boxed{\overline{K} \approx_\xi \overline{\mathtt{K}}}$ „The stack of L continuations $\overline{K}$ agrees with $\mathtt{L}_{\mathrm{scct}}$ one $\overline{\mathtt{K}}$. Only checks $\mathit{foo} \in \xi$"

(scct-empty-kontstack-eq)

$$\frac{}{[\cdot] \approx_\xi [\cdot]}$$

(scct-ign-cons-kontstack-eq)

$$\frac{\mathit{foo} \notin \xi \qquad \mathtt{foo} = [\![\mathit{foo}]\!]^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} \qquad \overline{K} \approx_\xi \overline{\mathtt{K}}}{\overline{K} \approx_\xi (\mathtt{K}; \mathtt{foo}), \overline{\mathtt{K}}}$$

(cons-kontstack-eq)

$$\frac{\mathit{foo} \in \xi \qquad \mathit{foo} = \mathtt{foo} \qquad \overline{K} \approx_\xi \overline{\mathtt{K}}}{(K; \mathit{foo}), \overline{K} \approx_\xi (\mathtt{K}; \mathtt{foo}), \overline{\mathtt{K}}}$$

Figure 80: Memory-State Relations from L to $\mathtt{L}_{\mathrm{scct}}$.

$\boxed{\delta(\ell) = \ell}$ „The L memory location ℓ corresponds to the $\mathtt{L}_{\mathrm{scct}}$ memory location ℓ."

$\boxed{\Omega \multimap_{\delta_{scct}} \mathtt{\Omega}}$ „The L state Ω agrees with $\mathtt{L}_{\mathrm{scct}}$ state $\mathtt{\Omega}$. L contains locations introduced by the backtranslation wrapper, which are subsequently ignored."

(state-qe)

$$\frac{\Omega = \Phi; t; \Psi \qquad \mathtt{\Omega} = \mathtt{\Phi}; \mathtt{t}; \mathtt{n}; \mathtt{\Psi} \qquad \Phi \multimap \mathtt{\Phi} \qquad \Psi \multimap_{\delta_{scct}} \mathtt{\Psi}}{\Omega \multimap_{\delta_{scct}} \mathtt{\Omega}}$$

(abort-state-qe)

$$\frac{}{\lightning \multimap_{\delta_{scct}} \lightning}$$

$\boxed{\Psi \multimap_{\delta_{scct}} \mathtt{\Psi}}$ „The L memory-state Ψ agrees with $\mathtt{L}_{\mathrm{scct}}$ one $\mathtt{\Psi}$. L contains locations introduced by the backtranslation wrapper."

(memstate-qe)

$$\frac{\Psi \approx_{\delta_{scct}} \mathtt{\Psi}}{\Psi \multimap_{\delta_{scct}} \mathtt{\Psi}}$$

$\boxed{\Phi \multimap \mathtt{\Phi}}$ „The L control-flow-state Φ agrees with $\mathtt{L}_{\mathrm{scct}}$ one $\mathtt{\Phi}$."

(cfstate-qe)

$$\frac{\Phi \approx \mathtt{\Phi}}{\Phi \multimap \mathtt{\Phi}}$$

Figure 81: State Relation from L to $\mathtt{L}_{\mathrm{scct}}$. This is meant to relate the states whenever we are „inside" a context.

4.8 Robustly Preserving sCCT

4.8.1 Compiler

$\boxed{[\![e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} = \mathtt{e}'}$ „Compile L expression e to $\mathsf{L}_{\mathsf{scct}}$ expression $\mathtt{e}'$."

$\boxed{[\![[v/x]]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} = [\mathtt{v}/\mathtt{x}]}$ „Compile L substitution to $\mathsf{L}_{\mathsf{scct}}$ substitution."

$\boxed{[\![\xi]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} = \xi'}$ „Compile L component library to $\mathsf{L}_{\mathsf{scct}}$ component library."

$$
\begin{array}{rcl}
[\![f_i]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{f}_i \\
[\![\mathit{call}\ \mathit{foo}\ e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{call}\ [\![\mathit{foo}]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}\ [\![e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}};\mathtt{setDIT\ 1} \\
[\![\mathit{return}\ e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{return}\ [\![e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
[\![e_1 \oplus e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & [\![e_1]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \oplus [\![e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
[\![x[e]]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{x}[[\![e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}] \\
[\![\mathit{let}\ x = e_1\ \mathit{in}\ e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{let\ x} = [\![e_1]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}\ \mathtt{in}\ [\![e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
[\![x[e_1] \leftarrow e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{x}[[\![e_1]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}] \leftarrow [\![e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
[\![\mathit{let}\ x = \mathit{new}\ e_1\ \mathit{in}\ e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{let\ x} = \mathtt{new}\ [\![e_1]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}\ \mathtt{in}\ [\![e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
[\![\mathit{delete}\ x]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{delete\ x} \\
[\![x\ \mathit{is}\ \text{☣}]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{x\ is}\ \text{☣} \\
[\![(e_1; e_2)]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & ([\![e_1]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}};[\![e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}) \\
[\![\pi_1\ e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \pi_1\ [\![e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
[\![\pi_2\ e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \pi_2\ [\![e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
[\![e\ \mathit{has}\ \tau]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & [\![e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}\ \mathtt{has}\ \tau \\
[\![\mathit{ifz}\ e_1\ \mathit{then}\ e_2\ \mathit{else}\ e_3]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{ifz}\ [\![e_1]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
 & & \mathtt{then}\ [\![e_2]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
 & & \mathtt{else}\ [\![e_3]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \\
\\
[\![[v/x]]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & [[\![v]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}/[\![x]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}] \\
\\
[\![[\cdot]]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & [\cdot] \\
[\![\mathit{foo}, \xi]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & \mathtt{foo}, [\![\xi]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}
\end{array}
$$

Figure 82: Compiler from L to $\mathsf{L}_{\mathsf{scct}}$.

$\boxed{[\![F]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} = \mathtt{F}}$ „Compile L procedures to $\mathsf{L}_{\mathsf{scct}}$ procedures."

$$
[\![\mathit{let}\ \mathit{foo}\ x := e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} \quad = \quad \mathtt{let\ foo\ x} := \mathtt{setDIT\ 1};[\![e]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}
$$

$\boxed{[\![\Xi]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} = \Xi}$ „Compile L libraries to $\mathsf{L}_{\mathsf{scct}}$ libraries."

$$
\begin{array}{rcl}
[\![[\cdot]]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & [\cdot] \\
[\![F, \Xi]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}} & = & [\![F]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}, [\![\Xi]\!]^{\mathsf{L}\to\mathsf{L}_{\mathsf{scct}}}
\end{array}
$$

Figure 83: Compiler from L components to $\mathsf{L}_{\mathsf{scct}}$ components.

4.8.2 Proofs and Auxiliary Lemmas

Lemma 84 (Trace Equality transfer Safety)**.** *If*

(a) $\forall \ell\ \ell, \delta(\ell) = \ell \implies \delta_{scct}(\ell) = \delta_{scct}(\delta^{-1}(\ell))$

(b) $\overline{a} \approx^*_\delta \overline{\mathtt{a}}$

(c) $\theta^*_{\delta_{scct} \circ \delta^{-1}}(\overline{\mathtt{a}}) \cong \overline{\boldsymbol{a}}$

(d) $\vdash T_{sCCT} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{sCCT}{}'$

Then

(i) $\theta_{\delta_{scct}}(\overline{a}) \cong \overline{\boldsymbol{a}}$

Proof. Induction on Assumption (b).

Case scct-empty-trace-eq**:** By Rule i-scct-filter-empty we know $\theta_{\delta_{scct}}([\cdot]) = \underline{[\cdot]}$ and similarily Rule o-scct-filter-empty we know $\theta_{\delta_{scct} \circ \delta^{-1}}([\cdot]) = \underline{[\cdot]}$. Invert Assumption (c) to know that $\overline{\boldsymbol{a}} = [\cdot]$. Rule scct-None resolves the goal.

Case scct-cons-trace-eq**:**

For readability, we repeat the proof context: If

(H_1) $\forall \ell\ \ell, \delta(\ell) = \ell \implies \delta_{scct}(\ell) = \delta_{scct}(\delta^{-1}(\ell))$

(H_2) $a_b; t \cdot \overline{a} \approx^*_\delta \mathtt{a_b}; \mathtt{t}; \sigma \cdot \overline{\mathtt{a}}$

(H_3) $\theta^*_{\delta_{scct} \circ \delta^{-1}}(\mathtt{a_b}; \mathtt{t}; \sigma \cdot \overline{\mathtt{a}}) \cong \overline{\boldsymbol{a}}$

(H_4) $\vdash T_{\mathrm{sCCT}} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{\mathrm{sCCT}}{}'$

(IH_1) Given $\overline{\boldsymbol{a}}_{IH}, T_{\mathrm{sCCT}\,IH}, T_{\mathrm{sCCT}}{}'_{IH}$,
if $\overline{a} \approx^*_\delta \overline{\mathtt{a}}$
and $\theta^*_{\delta_{scct} \circ \delta^{-1}}(\overline{\mathtt{a}}) = \overline{\boldsymbol{a}}_{IH}$
and $\vdash T_{\mathrm{sCCT}\,IH} \overset{\overline{\boldsymbol{a}}_{IH}}{\rightsquigarrow}{}^* T_{\mathrm{sCCT}}{}'_{IH}$
then $\theta^*_{\delta_{scct}}(\overline{a}) = \overline{\boldsymbol{a}}_{IH}$

Then

(i) $\theta^*_{\delta_{scct}}(a_b; t \cdot \overline{a}) \cong \overline{\boldsymbol{a}}$

Invert Assumption (H_2) to get:

(H_5) $t = \mathtt{t}$

(H_6) $a_b \approx_\delta \mathtt{a_b}$

(H_7) $\overline{a} \approx^*_\delta \overline{\mathtt{a}}$

Note that $\theta^*_{\delta_{scct} \circ \delta^{-1}}(\mathtt{a} \cdot \overline{\mathtt{a}}) = \theta^*_{\delta_{scct} \circ \delta^{-1}}(\mathtt{a}) \underline{:} \theta^*_{\delta_{scct} \circ \delta^{-1}}(\overline{\mathtt{a}})$ and rewrite Assumption (c) with that.

So, by inversion on Assumption (c) we know:

(H_8) $\overline{\boldsymbol{a}} = \boldsymbol{a} \cdot \overline{\boldsymbol{a}}'$

(H_9) $\theta^*_{\delta_{scct} \circ \delta^{-1}}(\mathtt{a}) \cong \boldsymbol{a}$

(H_{10}) $\theta^*_{\delta_{scct} \circ \delta^{-1}}(\overline{\mathtt{a}}) \cong^* \overline{\boldsymbol{a}}'$

Invert Assumption (H_4):

(H_{11}) $\vdash T_{\mathrm{sCCT}} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{\mathrm{sCCT0}}$

(H_{12}) $\vdash T_{\mathrm{sCCT0}} \overset{\overline{\boldsymbol{a}}}{\rightsquigarrow}{}^* T_{\mathrm{sCCT}}{}'$

Rewrite Goal (i), exploiting distributivity of the filter:

(i) $\theta^*_{\delta_{scct}}(a_b; t) \cong \boldsymbol{a}$

(ii) $\theta^*_{\delta_{scct}}(\overline{a}) \cong \overline{\boldsymbol{a}}'$

We are ready to solve Goal (ii) with Assumption (IH_1) using Assumptions (H_7), (H_{10}) and (H_{12}).

For Goal (i), perform case analysis on Assumption (H_6). For cases with $t = ctx$: by Rule i-scct-filter-context, we have $\theta^*_{\delta_{scct}}(a_b; ctx) = \underline{\varepsilon}$. Furthermore, inverting Assumption (H_9) with a context sandboxtag yields $\boldsymbol{a} = \boldsymbol{\varepsilon}$. Thus, Goal (i) reduces to showing $\underline{\varepsilon} \cong \boldsymbol{\varepsilon}$, which follows by Rule `sCCT`-None-Authentic. So, in the following, let $t = comp$ and $\mathtt{t} = \mathtt{comp}$.

Case scct-start-event-eq**:** Invert Assumption (H_9) to get $\boldsymbol{a} = \boldsymbol{\varepsilon}$. Note Rule i-scct-filter-comp-start, so Goal (i) follows from Rule `sCCT`-None-Authentic.

Case scct-end-event-eq**:** Invert Assumption (H_9) to get $\boldsymbol{a} = \boldsymbol{\varepsilon}$. Note Rule i-scct-filter-comp-end, so Goal (i) follows from Rule `sCCT`-None-Authentic.

Case scct-alloc-event-eq**:** Invert Assumption (H_9) to get $\boldsymbol{a} = \boldsymbol{Any}; 🔓$. Note Rule i-scct-filter-comp-alloc, so Goal (i) follows from Rule `sCCT`-Public-Authentic.

Case scct-dealloc-event-eq**:** Invert Assumption (H_9) to get $\boldsymbol{a} = \boldsymbol{Any}; 🔓$. Note Rule i-scct-filter-comp-dealloc, so Goal (i) follows from Rule `sCCT`-Public-Authentic.

Case scct-get-event-eq-noleak**:** Invert Assumption (H_9) to get $\boldsymbol{a} = \boldsymbol{\varepsilon}$. Note Rule i-scct-filter-comp-get-noleak, so Goal (i) follows from Rule `sCCT`-None-Authentic.

Case scct-set-event-eq-noleak**:** Invert Assumption (H_9) to get $\boldsymbol{a} = \boldsymbol{\varepsilon}$. Note Rule i-scct-filter-comp-set-noleak, so Goal (i) follows from Rule `sCCT`-None-Authentic.

Case scct-get-event-eq-leak**:** Case analysis on $\boldsymbol{\sigma}$.

Case $\boldsymbol{\sigma} = 🔓$**:** Invert Assumption (H_9) to get:

(F_1) $\boldsymbol{a} = \boldsymbol{\varepsilon}$.

Note Rule i-scct-filter-comp-get-noleak, so Goal (i) follows from Rule `sCCT`-None-Authentic.

Case $\boldsymbol{\sigma} = 🔒$**:** Inverting Assumption (H_{11}) immediately gives us a contradiction, because there is no matching rule that steps on 🔒.

Case scct-set-event-eq-leak**:** Case analysis on $\boldsymbol{\sigma}$.

Case $\boldsymbol{\sigma} = 🔓$**:** Invert Assumption (H_9) to get:

(F_1) $\boldsymbol{a} = \varepsilon$.

Note Rule i-scct-filter-comp-set-noleak, so Goal (i) follows from Rule sCCT-None-Authentic.

Case $\boldsymbol{\sigma} = 🔒$**:** Inverting Assumption (H_{11}) immediately gives us a contradiction, because there is no matching rule that steps on 🔒.

Case scct-call-event-eq**:** Invert Assumption (H_9) to get $\boldsymbol{a} = \varepsilon$. Note Rule i-scct-filter-comp-call, so Goal (i) follows from Rule sCCT-None-Authentic.

Case scct-ret-event-eq**:** Invert Assumption (H_9) to get $\boldsymbol{a} = \varepsilon$. Note Rule i-scct-filter-comp-ret, so Goal (i) follows from Rule sCCT-None-Authentic.

Case scct-$\underline{\sigma}$-event-eq**:** Invert Assumption (H_9) to get $\boldsymbol{a} = \varepsilon$. Note Rule i-scct-filter-emptyevent, so Goal (i) follows from Rule sCCT-None-Authentic.

Case scct-$\lightning$-event-eq**:** Invert Assumption (H_9) to get:

(F_1) $\underline{\sigma} = \boldsymbol{\sigma}$

(F_2) $\boldsymbol{a} = \lightning; \boldsymbol{\sigma}$.

Case analysis on $\boldsymbol{\sigma}$.

Case $\boldsymbol{\sigma} = 🔓$**:** Note Rule i-scct-filter-abort, so Goal (i) follows from Rule sCCT-Abort-Authentic.

Case $\boldsymbol{\sigma} = 🔒$**:** Inverting Assumption (H_{11}) immediately gives us a contradiction, because there is no matching rule that steps on 🔒.

Case scct-binop-event-eq-leak**:** This case is very similar to the next case.

Case scct-branch-event-eq-leak**:** Case analysis on $\boldsymbol{\sigma}$.

Case $\boldsymbol{\sigma} = 🔓$**:** Invert Assumption (H_9) to get:

(F_1) $\boldsymbol{a} = \varepsilon$.

Note Rule i-scct-filter-emptyevent, so Goal (i) follows from Rule sCCT-None-Authentic.

Case $\boldsymbol{\sigma} = 🔒$**:** Inverting Assumption (H_{11}) immediately gives us a contradiction, because there is no matching rule that steps on 🔒.

□

Lemma 85 (Primitive Forward Simulation)**.** *If*

(a) $\Omega \triangleright e\gamma \xrightarrow{a} \Omega' \triangleright e'\gamma'$

(b) $\vdash T_{sCCT} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{sCCT}{}'$

(c) $\theta_\delta(a) \cong \boldsymbol{a}$

(d) $\Omega \approx_{\delta_{scct}} \Omega$

Then $\exists \delta'_{scct}\ \mathtt{a}\ \Omega'$,

(i) $\Omega \triangleright [\![e]\!]^{\mathrm{L}\to\mathrm{L}_{scct}}[\![\gamma]\!]^{\mathrm{L}\to\mathrm{L}_{scct}} \xrightarrow{\mathtt{a}} \Omega' \triangleright [\![e']\!]^{\mathrm{L}\to\mathrm{L}_{scct}}[\![\gamma']\!]^{\mathrm{L}\to\mathrm{L}_{scct}}$

(ii) $a \approxeq_{\delta'_{scct}} \mathtt{a}$

(iii) $\Omega' \approx_{\delta'_{scct}} \Omega'$

(iv) $\theta_{\delta'_{scct}}(\mathsf{a}) \cong \boldsymbol{a}$

Proof. Case analysis on Assumption (a). □

Lemma 86 (Contextual Forward Simulation)**.** *If*

(a) $\Omega \triangleright e\gamma \xrightarrow{a}_{ctx} \Omega' \triangleright e'\gamma'$

(b) $\vdash T_{sCCT} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{sCCT}{}'$

(c) $\theta_{\delta}(a) \cong \boldsymbol{a}$

(d) $\Omega \approx_{\delta_{scct}} \Omega$

Then $\exists \delta'_{scct}\ \mathsf{a}\ \Omega'$,

(i) $\Omega \triangleright [\![e]\!]^{\mathrm{L}\to\mathrm{L}_{scct}}[\![\gamma]\!]^{\mathrm{L}\to\mathrm{L}_{scct}} \xrightarrow{\mathsf{a}}_{ctx} \Omega' \triangleright [\![e']\!]^{\mathrm{L}\to\mathrm{L}_{scct}}[\![\gamma']\!]^{\mathrm{L}\to\mathrm{L}_{scct}}$

(ii) $a \approxeq_{\delta'_{scct}} \mathsf{a}$

(iii) $\Omega' \approx_{\delta'_{scct}} \Omega'$

(iv) $\theta_{\delta'_{scct}}(\mathsf{a}) \cong \boldsymbol{a}$

Proof. Case analysis on Assumption (a) and using Lemma 85 (Primitive Forward Simulation). □

Lemma 87 (Forward Simulation)**.** *If*

(a) $\Omega \triangleright e\gamma \xrightarrow{a}{}^{*}_{ctx} \Omega' \triangleright e'\gamma'$

(b) $\vdash T_{sCCT} \overset{\boldsymbol{a}}{\rightsquigarrow} T_{sCCT}{}'$

(c) $\theta_{\delta}(a) \cong \boldsymbol{a}$

(d) $\Omega \approx_{\delta_{scct}} \Omega$

Then $\exists \delta'_{scct}\ \mathsf{a}\ \Omega'$,

(i) $\Omega \triangleright [\![e]\!]^{\mathrm{L}\to\mathrm{L}_{scct}}[\![\gamma]\!]^{\mathrm{L}\to\mathrm{L}_{scct}} \xrightarrow{\mathsf{a}}{}^{*}_{ctx} \Omega' \triangleright [\![e']\!]^{\mathrm{L}\to\mathrm{L}_{scct}}[\![\gamma']\!]^{\mathrm{L}\to\mathrm{L}_{scct}}$

(ii) $a \approxeq_{\delta'_{scct}} \mathsf{a}$

(iii) $\Omega' \approx_{\delta'_{scct}} \Omega'$

(iv) $\theta_{\delta'_{scct}}(\mathsf{a}) \cong \boldsymbol{a}$

Proof. Case analysis on Assumption (a) and using Lemma 86 (Contextual Forward Simulation). □

Lemma 88 (Backward Simulation)**.** *If*

(a) $\Omega \triangleright [\![e]\!]^{\mathrm{L}\to\mathrm{L}_{scct}}[\![\gamma]\!]^{\mathrm{L}\to\mathrm{L}_{scct}} \xrightarrow{\bar{\mathsf{a}}}{}^{*}_{ctx} \Omega' \triangleright [\![e']\!]^{\mathrm{L}\to\mathrm{L}_{scct}}[\![\gamma']\!]^{\mathrm{L}\to\mathrm{L}_{scct}}$

(b) $\Omega \approx_{\delta_{scct}} \Omega$

(c) $\vdash T_{sCCT} \overset{\boldsymbol{a}}{\rightsquigarrow}{}^{*} T_{sCCT}{}'$

(d) $\theta_\delta(a) \cong \boldsymbol{a}$

Then $\exists \delta'_{scct}\ \Omega'\ \overline{a}$

(i) $\delta_{scct} \subseteq \delta'_{scct}$

(ii) $\Omega \triangleright e\gamma \xrightarrow{\overline{a}}{}^*_{ctx}\ \Omega' \triangleright e'\gamma'$

(iii) $\Omega' \approx_{\delta'_{scct}} \Omega'$

(iv) $\overline{a} \approx^*_{\delta'_{scct}} \overline{\mathsf{a}}$

(v) $\theta_{\delta'_{scct}}(\overline{\mathsf{a}}) \cong \overline{\boldsymbol{a}}$

Proof. Follows from Lemma 87 and determinism. □

4.8.3 Backtranslation

$\boxed{\theta_\bullet(\mathtt{a}) = \mathtt{a}}$ „Filter an $\mathrm{L}_{\mathrm{scct}}$ event."

(obj-filter-context)
$$\frac{\begin{array}{c}\mathtt{a_b} = \mathtt{Alloc}\ \ell\ \mathtt{n} \lor \mathtt{a_b} = \mathtt{Dealloc}\ \ell \lor \\ \mathtt{a_b} = \mathtt{Get}\ \ell\ \mathtt{n} \lor \mathtt{a_b} = \mathtt{Set}\ \ell\ \mathtt{n}\ \mathtt{m} \lor \\ \mathtt{a_b} = \mathtt{Call}\ \emptyset\ \mathtt{foo}\ \mathtt{n} \lor \mathtt{a_b} = \mathtt{Ret}\ \emptyset\ \mathtt{n} \lor \\ \mathtt{a_b} = \mathtt{iGet}\ \ell\ \mathtt{n} \lor \mathtt{a_b} = \mathtt{iSet}\ \ell\ \mathtt{n}\ \mathtt{m} \lor \\ \mathtt{a_b} = \mathtt{Branch}\ \mathtt{n} \lor \mathtt{a_b} = \mathtt{Binop}\ \mathtt{n_1}\ \mathtt{n_2}\end{array}}{\theta_\bullet(\mathtt{a_b}; \mathtt{ctx}; \sigma) = \varepsilon}$$

(obj-filter-unimportant)
$$\frac{}{\theta_\bullet(\varepsilon) = \varepsilon}$$

(obj-filter-comp-start)
$$\frac{}{\theta_\bullet(\mathtt{Start}; \mathtt{comp}; \sigma) = \varepsilon}$$

(obj-filter-comp-alloc)
$$\frac{}{\theta_\bullet(\mathtt{Alloc}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma) = \mathtt{Alloc}\ \ell\ \mathtt{n}}$$

(obj-filter-comp-dealloc)
$$\frac{}{\theta_\bullet(\mathtt{Dealloc}\ \ell; \mathtt{comp}; \sigma) = \mathtt{Dealloc}\ \ell}$$

(obj-filter-comp-get-noleak)
$$\frac{}{\theta_\bullet(\mathtt{iGet}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma) = \mathtt{iGet}\ \ell\ \mathtt{n}}$$

(obj-filter-comp-set-noleak)
$$\frac{}{\theta_\bullet(\mathtt{iSet}\ \ell\ \mathtt{n}\ \mathtt{v}; \mathtt{comp}; \sigma) = \mathtt{iSet}\ \ell\ \mathtt{n}\ \mathtt{v}}$$

(obj-filter-comp-get-leak)
$$\frac{}{\theta_\bullet(\mathtt{Get}\ \ell\ \mathtt{n}; \mathtt{comp}; \sigma) = \mathtt{Get}\ \ell\ \mathtt{n}}$$

(obj-filter-comp-set-leak)
$$\frac{}{\theta_\bullet(\mathtt{Set}\ \ell\ \mathtt{n}\ \mathtt{v}; \mathtt{comp}; \sigma) = \mathtt{Set}\ \ell\ \mathtt{n}\ \mathtt{v}}$$

(obj-filter-comp-call)
$$\frac{}{\theta_\bullet(\mathtt{Call}\ \mathbf{c}\ \mathtt{foo}\ \mathtt{v}; \mathtt{t}; \sigma) = \mathtt{Call}\ \mathbf{c}\ \mathtt{foo}\ \mathtt{v}; \mathtt{t}}$$

(obj-filter-comp-ret)
$$\frac{}{\theta_\bullet(\mathtt{Ret}\ \mathbf{c}\ \mathtt{v}; \mathtt{t}; \sigma) = \mathtt{Ret}\ \mathbf{c}\ \mathtt{v}; \mathtt{t}}$$

(obj-filter-branch)
$$\frac{}{\theta_\bullet(\mathtt{Branch}\ \mathtt{n}; \mathtt{comp}; \sigma) = \mathtt{Branch}\ \mathtt{n}}$$

(obj-filter-binop)
$$\frac{}{\theta_\bullet(\mathtt{Binop}\ \mathtt{n_1}\ \mathtt{n_2}; \mathtt{comp}; \sigma) = \mathtt{Binop}\ \mathtt{n_1}\ \mathtt{n_2}}$$

(obj-filter-abort)
$$\frac{}{\theta_\bullet(\lightning; \mathtt{t}; \sigma) = \lightning}$$

$\boxed{\theta^*_\bullet(\overline{\mathtt{a}}) = \overline{\mathtt{a}}'}$ „Filter an $\mathrm{L}_{\mathrm{scct}}$ trace."

(obj-filter-empty)
$$\frac{}{\theta^*_\bullet([\cdot]) = [\cdot]}$$

(obj-filter-cons-relevant)
$$\frac{\theta_\bullet(\mathtt{a}) = \mathtt{a} \neq \varepsilon \qquad \theta^*_\bullet(\overline{\mathtt{a}}) = \overline{\mathtt{a}}}{\theta^*_\bullet(\mathtt{a} \cdot \overline{\mathtt{a}}) = \mathtt{a} \cdot \overline{\mathtt{a}}}$$

(obj-filter-cons-relevant)
$$\frac{\theta_\bullet(\mathtt{a}) = \varepsilon \qquad \theta^*_\bullet(\overline{\mathtt{a}}) = \overline{\mathtt{a}}}{\theta^*_\bullet(\mathtt{a} \cdot \overline{\mathtt{a}}) = \overline{\mathtt{a}}}$$

Figure 84: Filtering of $\mathrm{L}_{\mathrm{scct}}$ events, getting rid of unimportant ones, for backtranslation.

$\boxed{\langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = v}$ „Map an $\mathrm{L}_{\mathrm{scct}}$ value to an L value."

(scct-backtrans-value)
$$\frac{}{\langle\!\langle\!\langle \mathtt{n} \rangle\!\rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = n}$$

Figure 85: Backtranslation of $\mathrm{L}_{\mathrm{scct}}$ values to L values.

$\boxed{\langle\!\langle \bar{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}$ „Backtranslate a non-interface trace $\bar{\mathtt{a}}$ to an L expression."

(scct-backtrans-empty)
$$\frac{}{\langle\!\langle [\cdot] \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = 42}$$

(scct-backtrans-alloc)
$$\frac{\langle\!\langle \bar{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}{\langle\!\langle \mathtt{Alloc}\ \ell\ \mathtt{s}; \mathtt{t}; \sigma \cdot \bar{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}$$

(scct-backtrans-dealloc)
$$\frac{\langle\!\langle \bar{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}{\langle\!\langle \mathtt{Dealloc}\ \ell; \mathtt{t}; \sigma \cdot \bar{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}$$

(scct-backtrans-set)
$$\frac{\langle\!\langle \bar{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}{\langle\!\langle \mathtt{Set}\ \ell\ \mathtt{n}\ \mathtt{v}; \mathtt{t}; \sigma \cdot \bar{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}$$

(scct-backtrans-get)
$$\frac{\langle\!\langle \bar{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}{\langle\!\langle \mathtt{Get}\ \ell\ \mathtt{n}; \mathtt{t}; \sigma \cdot \bar{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}$$

(scct-backtrans-abort)
$$\frac{}{\langle\!\langle \lightning; \mathtt{t}; \sigma \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = abort()}$$

Figure 86: Trace-Based Backtranslation from $\mathrm{L}_{\mathrm{scct}}$ backtranslation-events to L terms.

$\boxed{{}^{!}\langle\!\langle \mathtt{a} \rangle\!\rangle_{\supset}^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}$ „Yield an L expression from target !-interaction-event a."

(scct-backtrans-start)
$$\frac{}{{}^{!}\langle\!\langle \mathtt{Start} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = 42}$$

(scct-backtrans-ret-comptoctx)
$$\frac{}{{}^{!}\langle\!\langle \mathtt{Ret}\ !\ \mathtt{v} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}}}$$

$\boxed{{}^{?}\langle\!\langle \mathtt{a} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e}$ „Create an $\mathrm{L}_{\mathrm{scct}}$ expression from target ?-interaction-event a."

(scct-backtrans-call-ctxtocomp)
$$\frac{}{{}^{?}\langle\!\langle \mathtt{Call}\ ?\ \mathtt{foo}\ \mathtt{v} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}}}$$

(scct-backtrans-end-ctxtocomp)
$$\frac{}{{}^{?}\langle\!\langle \mathtt{End}\ \mathtt{v} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}}}$$

Figure 87: scct-backtranslation of interaction-events from $\mathrm{L}_{\mathrm{scct}}$ specification events to L terms.

(scct-non-interface-ε)
$$\frac{}{\vdash \varepsilon\ \text{non-int-}\mathtt{a}}$$

(scct-non-interface-Alloc)
$$\frac{}{\vdash \mathtt{Alloc}\ \ell\ \mathtt{n}; \mathtt{t}; \sigma\ \text{non-int-}\mathtt{a}}$$

(scct-non-interface-Dealloc)
$$\frac{}{\vdash \mathtt{Dealloc}\ \ell; \mathtt{t}; \sigma\ \text{non-int-}\mathtt{a}}$$

(scct-non-interface-Get)
$$\frac{}{\vdash \mathtt{Get}\ \ell\ \mathtt{n}; \mathtt{t}; \sigma\ \text{non-int-}\mathtt{a}}$$

(scct-non-interface-Set)
$$\frac{}{\vdash \mathtt{Set}\ \ell\ \mathtt{n}; \mathtt{t}; \sigma\ \text{non-int-}\mathtt{a}}$$

(scct-non-interface-Binop)
$$\frac{}{\vdash \mathtt{Binop}\ \mathtt{n}_1\ \mathtt{n}_2; \mathtt{t}; \sigma\ \text{non-int-}\mathtt{a}}$$

(scct-non-interface-Branch)
$$\frac{}{\vdash \mathtt{Branch}\ \mathtt{n}; \mathtt{t}; \sigma\ \text{non-int-}\mathtt{a}}$$

(scct-non-interface-$\lightning$)
$$\frac{}{\vdash \lightning; \mathtt{t}; \sigma\ \text{non-int-}\mathtt{a}}$$

(scct-non-interface-trace-empty)
$$\frac{}{\vdash [\cdot]\ \text{non-int-}\bar{\mathtt{a}}}$$

(scct-non-interface-trace-cons)
$$\frac{\vdash \mathtt{a}\ \text{non-int-}\bar{\mathtt{a}} \qquad \vdash \bar{\mathtt{a}}\ \text{non-int-}\bar{\mathtt{a}}}{\vdash \mathtt{a} \cdot \bar{\mathtt{a}}\ \text{non-int-}\bar{\mathtt{a}}}$$

Figure 88: Non-Interfacing events.

$\boxed{{}^{!?}\langle\!\langle \mathtt{a}_1 \cdot \overline{\mathtt{a}} \cdot \mathtt{a}_2 \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e_1 ; e_2 ; e_3}$ „Construct an L ensemble of expressions from target-trace $\overline{\mathtt{a}}$ which starts with a ? event and ends in a !."

(scct-subtoplevel-backtrans)

$$\frac{\vdash \overline{\mathtt{a}} \text{ non-int-}\overline{\mathtt{a}} \qquad {}^{!}\langle\!\langle \mathtt{a}_1 \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e_1 \qquad \langle\!\langle \overline{\mathtt{a}} \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e_2 \qquad {}^{?}\langle\!\langle \mathtt{a}_2 \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e_3}{{}^{!?}\langle\!\langle \mathtt{a}_1 \cdot \overline{\mathtt{a}} \cdot \mathtt{a}_2 \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e_1 ; e_2 ; e_3}$$

Figure 89: Interaction-Trace-Based Backtranslation from $\mathrm{L}_{\mathrm{scct}}$ specification events to L terms.

$\boxed{\langle\!\langle\!\langle \overline{\mathtt{a}^{\mathrm{ms}}} \rangle\!\rangle\!\rangle^{\mathrm{L}_{\mathrm{scct}} \to \mathrm{L}} = \Xi}$ „Top-Level backtranslation of $\overline{\mathtt{a}}$ to L library Ξ."

(scct-backtrans-top-level)

$$\frac{\begin{array}{c} \overline{\mathtt{a}} = \overline{\mathtt{a}_0} \cdot \overline{\mathtt{a}^{\mathtt{comp}}} \cdot \overline{\mathtt{a}_1} \qquad \vdash \overline{\mathtt{a}^{\mathtt{comp}}} \text{ non-int-}\overline{\mathtt{a}} \\ \overline{\mathtt{a}_0} = \mathtt{Start} \cdot \overline{\mathtt{a}_0'} \cdot \mathtt{Call}\ ?\ \mathtt{foo}\ \mathtt{v}_1 \qquad \overline{\mathtt{a}_1} = \mathtt{Ret}\ !\ \mathtt{v}_2 \cdot \overline{\mathtt{a}_0'} \cdot \mathtt{End}\ \mathtt{v}_3 \\ {}^{!?}\langle\!\langle \theta_\bullet^* (\overline{\mathtt{a}_0}) \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e_0 ; e_0' ; e_0'' \\ {}^{!?}\langle\!\langle \theta_\bullet^* (\overline{\mathtt{a}_1}) \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{\mathrm{scct}}} = e_1 ; e_1' ; e_1'' \\ e = e_0 ; e_0' ; e_0'' ; e_1 ; e_1' ; e_1'' \end{array}}{\langle\!\langle\!\langle \overline{\mathtt{a}} \rangle\!\rangle\!\rangle^{\mathrm{L}_{\mathrm{scct}} \to \mathrm{L}} = \mathit{let}\ \mathit{main}\ x : \mathbb{N} \to \mathbb{N} := e, [\cdot]}$$

Figure 90: Top-Level trace-based Backtranslation from $\mathrm{L}_{\mathrm{scct}}$ trace $\overline{\mathtt{a}}$ to L context Ξ.

Lemma 89 (Backtranslation Correctness of `Start`)**.** *If*

(a) $\Xi = \Xi_{\mathtt{ctx}} \bowtie [\![\Xi_{comp}]\!]^{\mathrm{L} \to \mathrm{L}_{scct}}$

(b) $\xi = [\![\xi]\!]^{\mathrm{L} \to \mathrm{L}_{scct}} = \mathrm{dom}\ [\![\Xi_{comp}]\!]^{\mathrm{L} \to \mathrm{L}_{scct}}$

(c) $\sigma'' = \sigma \sqcap \sigma'$

(d) $\xi; \Xi; [\cdot]; \mathtt{comp}; 1; [\cdot]; [\cdot]; [\cdot] \triangleright \mathtt{call}\ \mathtt{main}^{\sigma}\ 0^{\sigma'} \xrightarrow{\mathtt{Start};\mathtt{comp};\sigma''}_{ctx} \xi; \Xi; ([\cdot]; \mathtt{main}), [\cdot]; \mathtt{ctx}; 1; [\cdot]; [\cdot]; [\cdot] \triangleright \mathtt{e}_{\mathtt{main}}[0^{\sigma'}/\mathtt{x}]$

(e) ${}^{!}\langle\!\langle \theta_\bullet (\mathtt{Start}; \mathtt{comp}; \sigma'') \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{scct}} = 42$

(f) $\xi; \Xi; [\cdot]; comp; [\cdot]; [\cdot]; [\cdot] \approx_{\emptyset;[\cdot]} \xi; \Xi; [\cdot]; \mathtt{comp}; 1; [\cdot]; [\cdot]; [\cdot]$

(g) $\Xi = \{main \mapsto 42; e_{main}\} \bowtie \Xi_{comp}$

Then $\exists n$

(i) $\xi; \Xi; [\cdot]; comp; [\cdot]; [\cdot]; [\cdot] \triangleright call\ main\ 0 \xrightarrow{Start;comp}_{ctx}{}^{n} \xi; \Xi; ([\cdot]; main), [\cdot]; ctx; [\cdot]; [\cdot]; [\cdot] \triangleright e_{main}[0/x]$

(ii) $\xi; \Xi; ([\cdot]; main), [\cdot]; ctx; [\cdot]; [\cdot]; [\cdot] \multimap_{\emptyset} \xi; \Xi; ([\cdot]; \mathtt{main}), [\cdot]; \mathtt{ctx}; 1; [\cdot]; [\cdot]$

(iii) $Start \approx^*_{\emptyset} \mathtt{Start}$

Proof. With Rules $e-\mathrm{ctx}-\mathbf{call}-\mathrm{main}$, $en-\mathrm{refl}$ and $en-\mathrm{trans}-$important one can conclude:

(H_1) $\xi; \Xi; [\cdot]; comp; [\cdot]; [\cdot]; [\cdot] \triangleright call\ main\ 0 \xrightarrow{Start;comp}_{\mathrm{ctx}}{}^{1} \xi; \Xi; ([\cdot]; main), [\cdot]; ctx; [\cdot]; [\cdot]; [\cdot] \triangleright let\ _ = 42\ in\ e_{main}[0/x]$

Now following with Rules $e-\textbf{let}-\textbf{f}$, $en-$ refl and $en-$ trans$-$unimportant:

(H_2) $\xi; \Xi; ([\cdot]\,; main), [\cdot]\,; comp; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright let\ _ = 42\ in\ e_{main}[0/x] \xrightarrow{\varepsilon}_{\mathrm{ctx}}{}^{1} \xi; \Xi; ([\cdot]\,; main), [\cdot]\,; ctx; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright e_{main}[0/x]$

Combine Assumptions (H_1) and (H_2) with Rules $en-$refl to $en-$trans$-$unimportant:

(H_3) $\xi; \Xi; [\cdot]\,; comp; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright call\ main\ 0 \xrightarrow{Start;comp}_{\mathrm{ctx}}{}^{2} \xi; \Xi; ([\cdot]\,; main), [\cdot]\,; ctx; [\cdot]\,; [\cdot]\,; [\cdot] \triangleright e_{main}[0/x]$

Instantiate $n = 2$ and Assumption (H_3) solves Goal (i).
Goal (ii) by Rules scct-empty-memstate-eq, scct-empty-kontstack-eq, state-qe and cfstate-qe and assumptions (b) and (f).
Goal (iii) by Rule scct-start-event-eq. □

Lemma 90 (Backtranslation Correctness of `Ret`)**.** *If*

(a) $\Omega = \xi; \Xi; (\mathtt{K}; \mathtt{foo}), \overline{\mathtt{K}}; \mathtt{comp}; \mathtt{1}; \Psi$

(b) $\Xi = \Xi_{\mathtt{ctx}} \bowtie [\![\Xi_{comp}]\!]^{\mathrm{L}\to\mathrm{L}_{scct}}$

(c) $\xi = [\![\xi]\!]^{\mathrm{L}\to\mathrm{L}_{scct}} = \mathrm{dom}\ [\![\Xi_{comp}]\!]^{\mathrm{L}\to\mathrm{L}_{scct}}$

(d) $\Omega \triangleright \mathtt{K}_{\mathtt{component}}[\mathtt{return}\ \mathtt{v}^{\sigma}] \xrightarrow{\mathtt{Ret\ !\ v;comp;}\sigma}_{ctx} \Omega' \triangleright \mathtt{K}[\mathtt{v}^{\sigma}]$

(e) $\mathtt{K}_{\mathtt{component}} = [\![K_{component}]\!]^{\mathrm{L}\to\mathrm{L}_{scct}}$

(f) $\llbracket\!\langle\!\langle \theta_{\bullet}\,(\mathtt{Ret\ !\ v;comp;}\sigma)\rangle\!\rangle\!\rrbracket^{\mathrm{L}\to\mathrm{L}_{scct}} = \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}$

(g) $\Omega \approx_{\delta_{scct}} \Omega$

(h) $\Omega = \xi; \Xi; (K, foo), \overline{K}; comp; \Psi$

(i) $\Xi = \{main \mapsto e_{main}\} \bowtie \Xi_{comp}$

then $\exists n\ \overline{a}$,

(i) $\Omega \triangleright K_{component}[return\ \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}] \xrightarrow{\overline{a}}{}^{n}_{ctx} \xi; \Xi; \overline{K}; ctx; \Psi \triangleright K[\langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}]$

(ii) $\xi; \Xi; \overline{K}; ctx; \Psi \multimap_{\delta_{scct}} \Omega'$

(iii) $\overline{a} \approxeq^{*}_{\delta} \mathtt{Ret\ !\ v}$

Proof. Instantiate the existentials with $n = 1$ and $\overline{a} = Ret\ !\ \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}_{\circlearrowleft}$. Let $v = \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}_{\circlearrowleft}$. For Goal (i) use Rule $e-$ctx$-$**return**$-$notsame.

(H_1) $\xi; \Xi; (K; foo), \overline{K}; comp; \Psi \triangleright K_{component}[return\ v] \xrightarrow{Ret\ ?\ v}_{\mathrm{ctx}}{}^{1} \xi; \Xi; \overline{K}; comp; \Psi \triangleright K[v]$

This solves Goal (i).

By inverting Assumption (d) we have $\Omega.\Psi = \Omega'.\Psi$. Invert Assumption (g):

(H_2) $\mathtt{1} \neq \mathtt{0}$

(H_3) $comp = \mathtt{comp}$

(H_4) $\Psi \approx_{\delta_{scct}} \Psi$

(H_5) $\xi; \Xi; (K; foo), \overline{K} \approx_{\delta_{scct}} \xi; \Xi; (\mathtt{K}; \mathtt{foo}), \overline{\mathtt{K}}$

From this, Goal (ii) follows easily.

Goal (iii) by Rule scct-ret-event-eq. □

Lemma 91 (Middle of Backtranslation Correctness)**.** *If*

(a) $\Omega = \xi; \Xi; \overline{\mathtt{K}}; \mathtt{ctx}; \mathtt{m}; \Psi$

(b) $\Xi = \Xi_{\mathtt{ctx}} \bowtie [\![\Xi_{comp}]\!]^{\mathrm{L} \to \mathrm{L}_{scct}}$

(c) $\xi = [\![\xi]\!]^{\mathrm{L} \to \mathrm{L}_{scct}} = \mathrm{dom}\ [\![\Xi_{comp}]\!]^{\mathrm{L} \to \mathrm{L}_{scct}}$

(d) $\Omega \triangleright \mathtt{e}\gamma \xrightarrow{\overline{\mathtt{a}}}_{ctx}{}^{n} \Omega' \triangleright \mathtt{e}'\gamma'$

(e) $\langle\!\langle \theta^*_{\bullet}(\overline{\mathtt{a}}) \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{scct}} = e$

(f) $\vdash \overline{\mathtt{a}}$ non-int- $\overline{\mathtt{a}}$

(g) $\Omega \multimap_{\delta_{scct}} \Omega$

(h) $\Omega = \xi; \Xi; \overline{K}; ctx; \Psi$

(i) $\Xi = \Xi_{ctx} \bowtie \Xi_{comp}$

then $\exists n'\ \overline{a}$,

(i) $\Omega \triangleright K[e] \xrightarrow{\overline{a}}_{ctx}{}^{n'} \xi; \Xi; \overline{K}; \Psi' \triangleright K[42]$

(ii) $\Omega' \multimap_{\delta_{scct}} \xi; \Xi; \overline{K}; \Psi'$

(iii) $\overline{a} \approx^*_{\delta_{scct}} [\cdot]$

Proof. Similar to Lemma 72. □

Lemma 92 (Backtranslation Correctness of `Call ?`)**.** *If*

(a) $\Omega = \xi; \Xi; \overline{\mathtt{K}}; \mathtt{ctx}; \mathtt{n}; \Psi$

(b) $\overline{\mathtt{K}} = (\mathtt{K}'; \mathtt{bar}), \overline{\mathtt{K}}'$

(c) $\Xi = \Xi_{\mathtt{ctx}} \bowtie [\![\Xi_{comp}]\!]^{\mathrm{L} \to \mathrm{L}_{scct}}$

(d) $\xi = [\![\xi]\!]^{\mathrm{L} \to \mathrm{L}_{scct}} = \mathrm{dom}\ [\![\Xi_{comp}]\!]^{\mathrm{L} \to \mathrm{L}_{scct}}$

(e) $\Omega \triangleright \mathtt{K}\left[\mathtt{call\ foo}^{\sigma}\ \mathtt{v}^{\sigma'}\right] \xrightarrow{\mathtt{Call\ ?\ foo\ v;ctx;}\sigma'}_{ctx}{}^{2} \xi; \Xi; (\mathtt{K}; \mathtt{foo}), \overline{\mathtt{K}}; \mathtt{comp}; 1; \Psi \triangleright [\![\mathtt{e_{foo}}]\!]^{\mathrm{L} \to \mathrm{L}_{scct}}[\mathtt{v}^{\sigma'}/\mathtt{y}]$

(f) ${}^{?}\langle\!\langle \theta_{\bullet}(\mathtt{Call\ ?\ foo\ v}) \rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{scct}} = call\ foo\ \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L} \to \mathrm{L}_{scct}}$

(g) $\Omega \multimap_{\delta} \Omega$

(h) $\Omega = \xi; \Xi; \overline{K}; ctx; \Psi$

(i) $let\ foo\ y := e_{foo} \in \Xi_{comp}$

(j) $\Xi = \Xi_{ctx} \bowtie \Xi_{comp}$

then $\exists n\ \overline{a}$,

(i) $\Omega \triangleright K[call\ foo\ \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}] \xrightarrow{\overline{a}}_{ctx}{}^{n} \xi; \Xi; (K; foo), \overline{K}; comp; \Psi \triangleright e_{foo}[v/y]$

(ii) $\xi; \Xi; (K, foo), \overline{K}; comp; \Psi \approx_{\delta} \xi; \Xi; (\mathtt{K}; \mathtt{foo}), \overline{\mathtt{K}}; \mathtt{comp}; \mathtt{1}; \Psi$

(iii) $\overline{a} \approx^{*}_{\delta} \mathtt{Call}\ ?\ \mathtt{foo}\ \mathtt{v}; \mathtt{ctx}; \sigma'$

Proof. Let $v = \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}_{\supset}$. Instantiate the existentials of the goal $n = 1$ and $\overline{a} = Call\ ?\ foo\ v; ctx, [\cdot]$.
Goal (i) follows immediately from Rule $e-$ctx$-$**call**$-$notsame.
Goal (ii) follows immediately from Rule scct-state-eq.
Goal (iii) follows immediately from Rules scct-cons-trace-eq and scct-call-event-eq. □

Lemma 93 (Backtranslation Correctness of End)**.** *If*

(a) $\Xi = \Xi_{\mathtt{ctx}} \bowtie [\![\Xi_{comp}]\!]^{\mathrm{L}\to\mathrm{L}_{scct}}$

(b) $\xi = [\![\xi]\!]^{\mathrm{L}\to\mathrm{L}_{scct}} = \mathrm{dom}\ [\![\Xi_{comp}]\!]^{\mathrm{L}\to\mathrm{L}_{scct}}$

(c) $\xi; \Xi; ([\cdot]; \mathtt{main}), [\cdot]; \mathtt{ctx}; \mathtt{n}; \Psi \triangleright \mathtt{K}[\mathtt{return}\ \mathtt{v}^{\sigma}] \xrightarrow{\mathtt{End}\ \mathtt{v};\mathtt{ctx};\sigma}_{ctx}{}^{2} \xi; \Xi; [\cdot]; \mathtt{comp}; \mathtt{1}; \Psi \triangleright \mathtt{v}^{\sigma}$

(d) ${}^{?}\langle\!\langle \theta_{\bullet}\ (\mathtt{End}\ \mathtt{v}; \mathtt{ctx}; \sigma) \rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}} = return\ \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}$

(e) $\xi; \Xi; ([\cdot]; main), [\cdot]; ctx; \Psi \multimap_{\delta} \xi; \Xi; ([\cdot]; \mathtt{main}), [\cdot]; \mathtt{ctx}; \mathtt{n}; \Psi$

(f) $\Xi = \Xi_{ctx} \bowtie \Xi_{comp}$

then $\exists n\ \overline{a}$,

(i) $\Omega \triangleright K[return\ \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}] \xrightarrow{\overline{a}}_{ctx}{}^{n} \xi; \Xi; [\cdot]; comp; \Psi \triangleright \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{scct}}$

(ii) $\xi; \Xi; [\cdot]; comp; \Psi \approx_{\delta'} \xi; \Xi; [\cdot]; \mathtt{comp}; \mathtt{1}; \Psi$

(iii) $\overline{a} \approx^{*}_{\delta'} \mathtt{End}\ \mathtt{v}; \mathtt{ctx}; \sigma$

Proof. Instantiate the goal with $n = 1$ and $\overline{a} = End\ v; ctx$. Goal (i) by Rule $e-$ctx$-$**return**$-$main.
Goal (ii) by inversion on Assumption (e).
Goal (iii) by Rules scct-cons-trace-eq and scct-end-event-eq. □

Lemma 94 (Backtranslation Correctness)**.** *If*

(a) $\Omega \triangleright \mathtt{call}\ \mathtt{main}\ \mathtt{0} \xrightarrow{\overline{\mathtt{a}}}{}^{*}_{ctx} \Omega' \triangleright \mathtt{f}_{\not{i}}$

(b) $\langle\!\langle\!\langle \overline{\mathtt{a}} \rangle\!\rangle\!\rangle^{\mathrm{L}_{scct}\to\mathrm{L}} = \Xi_{ctx}$

(c) $\Omega \approx_{\emptyset} \Omega$

(d) $\Omega = \mathtt{foo}, [\cdot]; \Xi; [\cdot]; \mathtt{comp}; \mathtt{1}; [\cdot]; [\cdot]; [\cdot]$

(e) $\Omega = \xi; \Xi; \overline{K}; comp; \Psi$

(f) $\Xi = \Xi_{ctx} \bowtie \Xi_{comp}$

then $\exists \delta_e\ \Omega'_e\ \overline{a}_e$,

(i) $\Omega \triangleright \mathit{call}\ \mathit{main}\ 0 \xrightarrow{\overline{a}_e}{}^{*}_{ctx}\ \Omega'_e \triangleright \langle\!\langle\!\langle \mathtt{f}\ \lightning \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}}$

(ii) $\Omega'_e \approx_{\delta_e} \Omega'$

(iii) $\overline{a}_e \approxeq^{*}_{\delta_e} \overline{\mathtt{a}}$

Proof. Inverting Assumption (b) gives:

(H_1) $\overline{\mathtt{a}} = \overline{\mathtt{a}_0} \cdot \overline{\mathtt{a}_{\mathtt{comp}}} \cdot \overline{\mathtt{a}_1}$

(H_2) $\overline{\mathtt{a}_0} = \mathtt{Start} \cdot \overline{\mathtt{a}'_0} \cdot \mathtt{Call}\ ?\ \mathtt{foo}\ \mathtt{v}$

(H_3) $\overline{\mathtt{a}_1} = \mathtt{Ret}\ !\ \mathtt{v}' \cdot \overline{\mathtt{a}_1}' \cdot \mathtt{End}\ \mathtt{v}''$

(H_4) $\vdash \overline{\mathtt{a}_{\mathtt{comp}}}$ non-int-$\overline{\mathtt{a}}$

(H_5) ${}^{!?}\langle\!\langle \theta^{*}_{\bullet}(\overline{\mathtt{a}_0}) \rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}} = e_0; e'_0; e''_0$

(H_6) ${}^{!?}\langle\!\langle \theta^{*}_{\bullet}(\overline{\mathtt{a}_1}) \rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}} = e_1; e'_1; e''_1$

By definition of $\theta_{\bullet}(\cdot)$, we have:

(H_7) $\theta^{*}_{\bullet}(\overline{\mathtt{a}_0}) = \mathtt{Start} \cdot \mathtt{Call}\ ?\ \mathtt{foo}\ \mathtt{v}$

(H_8) $\theta^{*}_{\bullet}(\overline{\mathtt{a}_1}) = \mathtt{Ret}\ !\ \mathtt{v}' \cdot \mathtt{End}\ \mathtt{v}''$

Invert Assumptions (H_5) and (H_6):

(H_9) ${}^{!}\langle\!\langle \mathtt{Start} \rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}} = e_0$

(H_{10}) $\langle\!\langle [\cdot] \rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}} = e'_0$

(H_{11}) ${}^{?}\langle\!\langle \mathtt{Call}\ ?\ \mathtt{foo}\ \mathtt{v} \rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}} = e''_0$

(H_{12}) ${}^{!}\langle\!\langle \mathtt{Ret}\ !\ \mathtt{v}' \rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}} = e_1$

(H_{13}) $\langle\!\langle [\cdot] \rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}} = e'_1$

(H_{14}) ${}^{?}\langle\!\langle \mathtt{End}\ \mathtt{v}'' \rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}} = e''_1$

Invert Assumptions (H_9) to (H_{14}):

(H_{15}) $e_0 = 42$

(H_{16}) $e'_0 = 42$

(H_{17}) $e''_0 = \mathit{call}\ \mathit{foo}\langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}}$

(H_{18}) $e_1 = \langle\!\langle\!\langle \mathtt{v}' \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}}$

(H_{19}) $e'_1 = 42$

(H_{20}) $e''_1 = \langle\!\langle\!\langle \mathtt{v}'' \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}}$

Let $v = \langle\!\langle\!\langle \mathtt{v} \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}}$ and $v' = \langle\!\langle\!\langle \mathtt{v}' \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}}$ and $v'' = \langle\!\langle\!\langle \mathtt{v}'' \rangle\!\rangle\!\rangle^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}}$.

By Rule $e-\mathrm{ctx}-\mathbf{call}-\mathrm{main}$, we have

(H_{21}) $\Omega \triangleright \mathit{call}\ \mathit{main}\ 0 \xrightarrow{\mathit{Start}}_{\mathrm{ctx}} \mathit{foo}, [\cdot]; \Xi; (\mathit{main}; [\cdot]), [\cdot]; \mathit{comp}; \Psi \triangleright 42; 42; \mathit{call}\ \mathit{foo}\ v; v'; 42; v''$

This reduces further in three steps: the first two steps evaluate the leading $42; 42$ pairs using the let-binding and projection rules, each producing an empty event since values in sequence position are discarded. The third step evaluates $call\ foo\ v$ using Rule $e-\text{ctx}-\textbf{call}-\text{notsame}$, producing the $Call\ ?\ foo\ v$ event since $foo \neq main$. Thus:

$$(H_{22})\ \ \Omega \triangleright 42; 42; call\ foo\ v; v'; 42; v'' \xrightarrow{Call\ ?\ foo\ v}{}_{\text{ctx}}{}^{3}\ foo, [\cdot]; \Xi; ([\cdot]; v'; 42; v''), (main; [\cdot]), [\cdot]; comp; \Psi \triangleright e_{foo}$$

Now by backward simulation
then the other BT correctness lemmas □

Theorem 17 (TMS Relation Correctness for $[\![\bullet]\!]^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}}$).

$$\text{if } \theta^*_{\delta_{MS}}\left(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\text{tms})\right) = \text{tms} \ \text{ and } \sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\text{tms})) \approx^*_{\delta;\mathsf{X}} \sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\text{tms})$$
$$\text{then } \theta^*_{\delta_{MS}}\left(\sigma_{\approx^*_{\delta;\mathsf{X}}}(\sigma_{\theta^*_{\delta_{MS}}(\bullet)}(\text{tms}))\right) = \text{tms}$$

Proof. By unfolding Definition 15 and applying the first assumption to rewrite the inner σ. The cross-language relation $\approx^*$ preserves allocation and deallocation structure, maintaining temporal memory safety. The second assumption ensures well-formedness of the composition. □

Definition 44 (L Robust Satisfaction). *We write* $\Xi_{comp} \vDash_R \pi$ *for If*

(a) $prog\ \Xi_{ctx}\ \Xi_{comp} \overset{\overline{a}}{\Longrightarrow} \Omega \triangleright f_{\lightning}$

Then $\exists \delta_{sCCT}$

(i) $\theta_{\delta_{sCCT}}(\overline{a}) \in \pi$

Definition 45 ($\mathrm{L}_{\mathrm{scct}}$ Robust Satisfaction). *We write* $\Xi_{\mathtt{comp}} \vDash_R \pi$ *for If*

(a) $\mathtt{prog}\ \Xi_{\mathtt{ctx}}\ \Xi_{\mathtt{comp}} \overset{\overline{\mathtt{a}}}{\Longrightarrow} \Omega \triangleright \mathtt{f}_{\lightning}$

Then $\exists \delta_{sCCT}$

(i) $\theta_{\delta_{sCCT}}(\overline{\mathtt{a}}) \in \pi$

Theorem 18 (Robust Strict Cryptographic Constant Time Preservation).

(i) $\vdash_{\sigma_\sim} [\![\bullet]\!]^{\mathrm{L}\to\mathrm{L}_{scct}} : \mathtt{sCCT}$

Proof. Analogous to Theorem 3. Unfolding $\vdash_{\sigma_\sim} [\![\bullet]\!]^{\mathrm{L}\to\mathrm{L}_{\mathrm{scct}}} : \mathtt{sCCT}$, the source assumption is $\Xi_{comp} \vDash_R \sigma_\sim(\mathtt{sCCT})$. The cross-language trace relation $\sim$ is defined by $\overline{a} \approx^* \overline{\mathtt{a}}$. By backtranslation correctness, we get a source trace related by $\sim$. By $\sigma_\sim$ definition, we conclude the target trace satisfies $\mathtt{sCCT}$. □

Theorem 19 (Robust Memory Safety and Strict Cryptographic Constant Time Preservation).

(i) $\vdash_{\sigma_{\sim_1 \circ \sim_2}} [\![[\![[\![\bullet]\!]^{\mathrm{L}_{tms}\to\mathrm{L}_{ms}}]\!]^{\mathrm{L}_{ms}\to\mathrm{L}}]\!]^{\mathrm{L}\to\mathrm{L}_{scct}} : \text{msafe} \cap \mathtt{sCCT}$

Proof. Note from Theorem 12 (Robust Memory Safety Preservation (Composed)) and Theorem 18 (Robust Strict Cryptographic Constant Time Preservation):

$(H_1)\ \vdash_{\sigma_{\sim_1}} [\![[\![\bullet]\!]^{\mathtt{L}_{\mathrm{tms}} \to \mathtt{L}_{\mathrm{ms}}}]\!]^{\mathtt{L}_{\mathrm{ms}} \to \mathtt{L}} : \mathrm{msafe}$

$(H_2)\ \vdash_{\sigma_{\sim_2}} [\![\bullet]\!]^{\mathtt{L} \to \mathtt{L}_{\mathrm{scct}}} : \mathtt{sCCT}$

By Lemma 7 (Sequential Composition with RTPσ) using Assumptions (H_1) and (H_2), composing the cross-language relations $\sim_1$ and $\sim_2$, we have the result. □

4.9 Low-Level Language

The next language is „low-level", because it extends the previous one with speculation.

$$
\begin{array}{rl}
\textit{Final Result}\ \mathsf{f} ::= & \mathsf{v}^{\sigma} \mid \mathsf{x}^{\sigma} \qquad \textit{May be a Result}\ \mathsf{f}_{\lightning} ::= \mathsf{f} \mid \mathsf{stuck} \\
\textit{Expressions}\ \mathsf{e} ::= & \mathsf{f}_{\lightning} \mid \mathsf{e}_1 \oplus \mathsf{e}_2 \mid \mathsf{x}[\mathsf{e}] \mid \mathsf{getDIT}\ \mathsf{x}\ \mathsf{in}\ \mathsf{e} \mid \mathsf{setDIT}\ \mathsf{e} \\
& \mid \mathsf{let}\ \mathsf{x} = \mathsf{e}_1\ \mathsf{in}\ \mathsf{e}_2 \mid \mathsf{x}[\mathsf{e}_1] \leftarrow \mathsf{e}_2 \\
& \mid \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{e}_1\ \mathsf{in}\ \mathsf{e}_2 \mid \mathsf{delete}\ \mathsf{x} \\
& \mid \mathsf{return}\ \mathsf{e} \mid \mathsf{call}\ \mathsf{foo}\ \mathsf{e} \mid \mathsf{ifz}\ \mathsf{e}_1\ \mathsf{then}\ \mathsf{e}_2\ \mathsf{else}\ \mathsf{e}_3 \\
& \mid \mathsf{abort}() \mid \mathsf{x}\ \mathsf{is}\ \text{☣} \\
& \mid \langle \mathsf{e}_1, \mathsf{e}_2 \rangle \mid \pi_1 \mathsf{e} \mid \pi_2 \mathsf{e} \mid \mathsf{e}\ \mathsf{has}\ \tau \\
& \text{where } \oplus \in \{+, -, \times, <, /\} \\
\textit{Functions}\ \mathsf{F} ::= & \mathsf{let}\ \mathsf{foo}\ \mathsf{x} := \mathsf{e} \qquad \textit{Types}\ \tau ::= \mathbb{N} \mid \mathbb{N} \times \mathbb{N} \\
\textit{Values}\ \mathsf{v} ::= & \mathsf{n} \in \mathbb{N} \qquad \textit{References}\ \ell \in \mathbb{N} \\
\textit{Eval.Ctx.}\ \mathsf{K} ::= & [\cdot] \mid \mathsf{K} \oplus \mathsf{e} \mid \mathsf{v} \oplus \mathsf{K} \mid \mathsf{x}[\mathsf{K}] \mid \mathsf{let}\ \mathsf{x} = \mathsf{K}\ \mathsf{in}\ \mathsf{e} \\
& \mid \mathsf{x}[\mathsf{K}] \leftarrow \mathsf{e} \mid \mathsf{x}[\mathsf{v}] \leftarrow \mathsf{K} \mid \mathsf{let}\ \mathsf{x} = \mathsf{new}\ \mathsf{K}\ \mathsf{in}\ \mathsf{e} \\
& \mid \mathsf{ifz}\ \mathsf{K}\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2 \mid \mathsf{call}\ \mathsf{foo}\ \mathsf{K} \mid \mathsf{return}\ \mathsf{K} \\
& \mid \langle \mathsf{K}, \mathsf{e} \rangle \mid \langle \mathsf{v}, \mathsf{K} \rangle \mid \pi_1 \mathsf{K} \mid \pi_2 \mathsf{K} \mid \mathsf{K}\ \mathsf{has}\ \tau \mid \mathsf{barrier} \\
& \mid \mathsf{ifz}\ \mathsf{K}\ \mathsf{then}\ \mathsf{e}_1\ \mathsf{else}\ \mathsf{e}_2 \mid \mathsf{setDIT}\ \mathsf{K} \\
\textit{Variables}\ \mathsf{x} \mid \mathsf{y} \mid \mathsf{foo} \mid \ldots & \qquad \textit{Poison}\ \rho ::= \square \mid \text{☣} \\
\textit{Sandbox Tag}\ \mathsf{t} ::= & \mathsf{ctx} \mid \mathsf{comp} \\
\textit{Typing.Env.}\ \Gamma ::= & [\cdot] \mid \Gamma, \mathsf{x} : \tau \qquad \textit{Store}\ \Delta ::= [\cdot] \mid \mathsf{x} \mapsto (\ell; \mathsf{t}; \rho; \mathsf{n}), \Delta \\
\textit{Communication}\ \mathbf{c} ::= & ? \mid ! \mid \varnothing \qquad \textit{Heaps}\ \mathsf{H} ::= [\cdot] \mid \mathsf{H} :: \mathsf{n} \\
\textit{Cont. Stack}\ \overline{\mathsf{K}} ::= & [\cdot] \mid (\mathsf{K}; \mathsf{foo}), \overline{\mathsf{K}} \qquad \textit{Library}\ \Xi ::= [\cdot] \mid \mathsf{F}, \Xi \\
\textit{Relevant}\ \xi ::= & [\cdot] \mid \mathsf{foo}, \xi \qquad \textit{State}\ \Omega ::= \Phi; \mathsf{t}; \mathsf{n}; \Psi \\
\textit{Flow State}\ \Phi ::= & \xi; \Xi; \overline{\mathsf{K}} \qquad \textit{Memory State}\ \Psi ::= \mathsf{H}^{\mathsf{ctx}}; \mathsf{H}^{\mathsf{comp}}; \Delta \\
\textit{Programs}\ \mathsf{prog}\ \Xi_{\mathsf{ctx}}\ \Xi_{\mathsf{comp}} & \qquad \textit{Substitutions}\ \gamma ::= [\mathsf{v}/\mathsf{x}], \gamma \mid [\cdot] \\
\textit{Security Tag}\ \sigma ::= & \text{🔒}_{\mathsf{vX}} \mid \text{🔓} \\
\textit{Leak Tag}\ \mathsf{vX} ::= & \mathsf{NONE} \mid \mathsf{PHT}
\end{array}
$$

Figure 91: Syntax of $\mathtt{L}_{\text{🔔}}$

The only difference between $\mathtt{L}_{\mathrm{scct}}$ and $\mathtt{L}_{\text{🔔}}$ is speculative semantics.

4.9.1 Dynamic Semantics

$$\boxed{\text{dom}\ \Xi = \mathsf{foo}, \ldots, \mathsf{bar}}\ \text{„Collect function names."}$$

$$\frac{}{\text{dom}\ [\cdot] = [\cdot]}\ (\Xi\text{-dom-}[\cdot]) \qquad \frac{\text{dom}\ \Xi = \mathsf{D}}{\text{dom}\ (\mathsf{let\ foo\ x} : \tau_\lambda := \mathsf{e}), \Xi = \mathsf{foo}, \mathsf{D}}\ (\Xi\text{-dom-cons})$$

$$\boxed{\Xi \equiv \Xi_1 \cup \Xi_2}\ \text{„Merging } \mathsf{L}_{\text{🔔}} \text{ libraries."}$$

$$\frac{\Xi = \Xi_1}{\Xi \equiv \Xi_1 \cup [\cdot]}\ (\text{lib-merge-empty}) \qquad \frac{\Xi \equiv \Xi_1 \cup \Xi_2}{\mathsf{F}, \Xi \equiv \Xi_1 \cup \mathsf{F}, \Xi_2}\ (\text{lib-merge-cons})$$

$$\boxed{\Xi_1 \bowtie \Xi_2 = \Xi}\ \text{„Syntactically linking } \mathsf{L}_{\text{🔔}} \text{ libraries."}$$

$$\frac{\text{dom}\ \Xi_1 \cap \text{dom}\ \Xi_2 = \emptyset \qquad \Xi \equiv \Xi_1 \cup \Xi_2}{\Xi_1 \bowtie \Xi_2 = \Xi}\ (\text{syntactic-plugging})$$

Figure 92: $\mathsf{L}_{\text{🔔}}$ plugging of libraries and collecting of function names.

$$\begin{aligned}
\textit{Base Events}\ \mathsf{a_b} ::= {} & \mathsf{Alloc}\ \ell\ \mathsf{v} \mid \mathsf{Dealloc}\ \ell \mid \mathsf{Get}\ \ell\ \mathsf{v} \mid \mathsf{Set}\ \ell\ \mathsf{v}\ \mathsf{v}' \mid \lightning \\
& \mid \mathsf{Call}\ \mathbf{c}\ \mathsf{foo}\ \mathsf{v} \mid \mathsf{Ret}\ \mathbf{c}\ \mathsf{v} \mid \mathsf{Start} \mid \mathsf{End}\ \mathsf{v} \\
& \mid \mathsf{Branch}\ \mathsf{n} \mid \mathsf{Binop}\ \mathsf{n}\ \mathsf{m} \\
& \mid \mathsf{iGet}\ \ell\ \mathsf{v} \mid \mathsf{iSet}\ \ell\ \mathsf{v}\ \mathsf{v}' \\
& \mid \mathsf{Spec} \mid \mathsf{Rlb} \mid \mathsf{Barrier} \\
\textit{Events}\ \mathsf{a} ::= {} & \varepsilon \mid \mathsf{a_b}; \mathsf{t}; \sigma
\end{aligned}$$

Figure 93: Events of $\mathsf{L}_{\text{🔔}}$.

Rest similar to [**?**].

4.10 Robustly Preserving spec

4.10.1 Compiler

$\boxed{[\![\mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} = \mathtt{e}'}$ „Compile $\mathtt{L}_{\mathrm{scct}}$ expression $\mathtt{e}$ to $\mathtt{L}_{\bell}$ expression $\mathtt{e}'$."

$\boxed{[\![[\mathtt{v}/\mathtt{x}]]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} = [\mathtt{v}/\mathtt{x}]}$ „Compile $\mathtt{L}_{\mathrm{scct}}$ substitution to $\mathtt{L}_{\bell}$ substitution."

$\boxed{[\![\xi]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} = \xi'}$ „Compile $\mathtt{L}_{\mathrm{scct}}$ component library to $\mathtt{L}_{\bell}$ component library."

$$
\begin{array}{rcl}
[\![\mathtt{f}_{\lightning}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathtt{f}_{\lightning} \\
[\![\mathtt{call\ foo\ e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathsf{call}\ [\![\mathtt{foo}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}\ [\![\mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
[\![\mathtt{return\ e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathsf{return}\ [\![\mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
[\![\mathtt{e}_1 \oplus \mathtt{e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & [\![\mathtt{e}_1]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \oplus [\![\mathtt{e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
[\![\mathtt{x}[\mathtt{e}]]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathtt{x}[[\![\mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}] \\
[\![\mathtt{let\ x} = \mathtt{e}_1\ \mathtt{in\ e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathsf{let}\ \mathtt{x} = [\![\mathtt{e}_1]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}\ \mathsf{in}\ [\![\mathtt{e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
[\![\mathtt{x}[\mathtt{e}_1] \leftarrow \mathtt{e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathtt{x}[[\![\mathtt{e}_1]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}] \leftarrow [\![\mathtt{e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
[\![\mathtt{let\ x} = \mathtt{new\ e}_1\ \mathtt{in\ e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathsf{let}\ \mathtt{x} = \mathsf{new}\ [\![\mathtt{e}_1]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}\ \mathsf{in}\ [\![\mathtt{e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
[\![\mathtt{delete\ x}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathsf{delete}\ \mathtt{x} \\
[\![\mathtt{x\ is}\ \text{☣}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathtt{x}\ \mathsf{is}\ \text{☣} \\
[\![(\mathtt{e}_1; \mathtt{e}_2)]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & ([\![\mathtt{e}_1]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}; [\![\mathtt{e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}) \\
[\![\pi_1\ \mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \pi_1\ [\![\mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
[\![\pi_2\ \mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \pi_2\ [\![\mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
[\![\mathtt{e\ has}\ \tau]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & [\![\mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}\ \mathsf{has}\ \tau \\
[\![\mathtt{ifz\ e}_1\ \mathtt{then\ e}_2\ \mathtt{else\ e}_3]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathsf{ifz}\ [\![\mathtt{e}_1]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
 & & \mathsf{then\ barrier}; [\![\mathtt{e}_2]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
 & & \mathsf{else\ barrier}; [\![\mathtt{e}_3]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \\
 & & \\
[\![[\mathtt{v}/\mathtt{x}]]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & [\,[\![\mathtt{v}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} / [\![\mathtt{x}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}\,] \\
 & & \\
[\![[\cdot]]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & [\cdot] \\
[\![\mathtt{foo}, \xi]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & \mathtt{foo}, [\![\xi]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}
\end{array}
$$

Figure 94: Compiler from $\mathtt{L}_{\mathrm{scct}}$ to $\mathtt{L}_{\bell}$.

$\boxed{[\![\mathtt{F}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} = \mathtt{F}}$ „Compile $\mathtt{L}_{\mathrm{scct}}$ procedures to $\mathtt{L}_{\bell}$ procedures."

$$
[\![\mathtt{let\ foo\ x} := \mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} \quad = \quad [\![\mathtt{e}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}
$$

$\boxed{[\![\Xi]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} = \Xi}$ „Compile $\mathtt{L}_{\mathrm{scct}}$ libraries to $\mathtt{L}_{\bell}$ libraries."

$$
\begin{array}{rcl}
[\![[\cdot]]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & [\cdot] \\
[\![\mathtt{F}, \Xi]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}} & = & [\![\mathtt{F}]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}, [\![\Xi]\!]^{\mathtt{L}_{\mathrm{scct}}\to\mathtt{L}_{\bell}}
\end{array}
$$

Figure 95: Compiler from $\mathtt{L}_{\mathrm{scct}}$ components to $\mathtt{L}_{\bell}$ components.

Theorem 20 (Robust Speculative Safety Preservation)**.**

(i) $\vdash_{\sigma_\sim} [\![\bullet]\!]^{\mathtt{L}_{scct} \to \mathtt{L}_{\text{🔔}}} : \text{spec}$

Proof. Analogous to Theorems 3 and 18. Unfolding $\vdash_{\sigma_\sim} [\![\bullet]\!]^{\mathtt{L}_{\mathrm{scct}} \to \mathtt{L}_{\text{🔔}}} : \text{spec}$, the source assumption is $\Xi_{\mathsf{comp}} \vDash_R \sigma_\sim(\text{spec})$. By backtranslation correctness, we get a source trace related by $\sim$. By $\sigma_\sim$ definition, we conclude the target trace satisfies spec. □

Theorem 21 (Robust Memory Safety, Strict Cryptographic Constant Time, and Speculation Safety Preservation)**.**

(i) $\vdash_{\sigma_{\sim_1 \circ \sim_2 \circ \sim_3}} [\![\![\![\![\bullet]\!]^{\mathtt{L}_{tms} \to \mathtt{L}_{ms}}]\!]^{\mathtt{L}_{ms} \to \mathtt{L}}]\!]^{\mathtt{L} \to \mathtt{L}_{scct}}]\!]^{\mathtt{L}_{scct} \to \mathtt{L}_{\text{🔔}}} : \text{msafe} \cap \text{sCCT} \cap \text{spec}$

Proof. By Lemma 7 (Sequential Composition with RTPσ), composing the cross-language relations, using Theorem 12 (Robust Memory Safety Preservation (Composed)), Theorem 18 (Robust Strict Cryptographic Constant Time Preservation), and Theorem 20 (Robust Speculative Safety Preservation). □